Mathematics

Class 7 (English Medium)

Solved Exercises — Sindh Textbook Board

13 lessons — 1612 questions

1. Sets

Question: Practical Work: Write the following sets in descriptive form. (i) D = {3, 6, 9, 12}

Answer: The members of this set are 3, 6, 9, 12, which are the first four multiples of 3. So in descriptive form: D = the set of the first four multiples of the number 3.

Question: Practical Work: (ii) O = {a, e, i, o, u}

Answer: The members of this set are a, e, i, o, u, which are the vowels of the English alphabet. So in descriptive form: O = the set of the vowels of the English alphabet.

Question: Practical Work: (iii) W = {0, 1, 2, 3, 4, 5, 6, 7, 8}

Answer: The members of this set are all the whole numbers from 0 to 8. So in descriptive form: W = the set of whole numbers up to 8.

Question: Practical Work: (iv) M = {March, May}

Answer: The members of this set are March and May, which are the months of the year whose names begin with 'M'. So in descriptive form: M = the set of the names of the months of the year beginning with 'M'.

Question: Practical Work: (v) B = {ت، ر، ص، ب، و، خ}

Answer: The members of this set are the distinct letters of the word "خوبصورت". So in descriptive form: B = the set of the letters of the word "خوبصورت".

Question: Practical Work: Match each set in column A with its correct set-builder form in column B. (1) The set of the first five even numbers

Answer: The first five even numbers are 2, 4, 6, 8, 10, i.e. the even numbers from 2 to 10. Set-builder form: {x / x ∈ E ∧ 2 ≤ x ≤ 10}.

Question: Practical Work: (2) {1, 3, 5, 7, 9}

Answer: 1, 3, 5, 7, 9 are the odd numbers from 1 to 9. Set-builder form: {x / x ∈ O, 1 ≤ x ≤ 9}.

Question: Practical Work: (3) The set of the first ten whole numbers

Answer: The first ten whole numbers are 0 to 9. Set-builder form: {x / x ∈ W ∧ 0 ≤ x ≤ 9}.

Question: Practical Work: (4) {−2, −1, 0, 1, 2}

Answer: −2, −1, 0, 1, 2 are the integers from −2 to 2. Set-builder form: {x / x ∈ Z ∧ −2 ≤ x ≤ 2}.

Question: Practical Work: (5) The set of all natural numbers

Answer: All natural numbers are 1, 2, 3, ... Set-builder form: {x / x ∈ N}.

Question: Exercise 1.1, Part I: Write the following sets in descriptive form. (1) N = {1, 2, 3, 4, 5, 6, ...}

Answer: This set contains all natural numbers starting from 1. Descriptive form: N = the set of all natural numbers.

Question: Exercise 1.1, Part I: (2) E = {2, 4, 6, 8}

Answer: This set contains 2, 4, 6, 8, which are the first four even numbers. Descriptive form: E = the set of the first four even numbers.

Question: Exercise 1.1, Part I: (3) F = {1, 2, 5, 10}

Answer: This set contains 1, 2, 5, 10, which are all the factors of 10 (10 is exactly divisible by each of them). Descriptive form: F = the set of the factors of the number 10.

Question: Exercise 1.1, Part I: (4) J = {2, 3, 5, 7, 11}

Answer: This set contains 2, 3, 5, 7, 11, which are all the prime numbers up to 11. Descriptive form: J = the set of the prime numbers up to 11.

Question: Exercise 1.1, Part I: (5) W = {0, 1, 2, 3, 4, 5}

Answer: This set contains all whole numbers from 0 to 5. Descriptive form: W = the set of the first six whole numbers.

Question: Exercise 1.1, Part I: (6) S = {5, 10, 15, 20, 25}

Answer: This set contains 5, 10, 15, 20, 25, which are the first five multiples of 5. Descriptive form: S = the set of the first five multiples of the number 5.

Question: Exercise 1.1, Part I: (7) T = {3, 6, 9, 12, ...}

Answer: This set contains the multiples of 3: 3, 6, 9, 12, ... without end. Descriptive form: T = the set of the multiples of the number 3.

Question: Exercise 1.1, Part II: Write the following sets in tabular form. (1) A = the set of the first seven natural numbers

Answer: The first seven natural numbers are 1, 2, 3, 4, 5, 6, 7. Tabular form: A = {1, 2, 3, 4, 5, 6, 7}.

Question: Exercise 1.1, Part II: (2) B = the set of the first eight positive even numbers

Answer: The first eight positive even numbers are 2, 4, 6, 8, 10, 12, 14, 16. Tabular form: B = {2, 4, 6, 8, 10, 12, 14, 16}.

Question: Exercise 1.1, Part II: (3) C = the set of integers less than 2 and greater than −4

Answer: The integers greater than −4 and less than 2 are −3, −2, −1, 0, 1. Tabular form: C = {−3, −2, −1, 0, 1}.

Question: Exercise 1.1, Part II: (4) D = the set of the letters of the English alphabet used in the word 'Pakistan'

Answer: The letters of the word Pakistan are p, a, k, i, s, t, a, n; a occurs twice but in a set each member is written only once. Tabular form: D = {p, a, k, i, s, t, n}.

Question: Exercise 1.1, Part II: (5) E = the set of the first eight multiples of the number 2

Answer: The first eight multiples of 2 are 2, 4, 6, 8, 10, 12, 14, 16. Tabular form: E = {2, 4, 6, 8, 10, 12, 14, 16}.

Question: Exercise 1.1, Part II: (6) F = the set of the first ten odd numbers

Answer: The first ten odd numbers are 1, 3, 5, 7, 9, 11, 13, 15, 17, 19.

Tabular form: F = {1, 3, 5, 7, 9, 11, 13, 15, 17, 19}.

Question: Exercise 1.1, Part II: (7) G = the set of all the factors of the number 15

Answer: The factors of 15 are 1, 3, 5, 15 (15 is exactly divisible by each of them). Tabular form: G = {1, 3, 5, 15}.

Question: Exercise 1.1, Part III: Write the following sets in set-builder form. (1) A = {1, 2, 3, ..., 10}

Answer: This set contains all natural numbers from 1 to 10. Set-builder form: A = {x / x ∈ N ∧ x ≤ 10}.

Question: Exercise 1.1, Part III: (2) B = {−2, −1, 0, +1, +2, +3, +4, +5}

Answer: This set contains all integers from −2 to 5. Set-builder form: B = {x / x ∈ Z ∧ −2 ≤ x ≤ 5}.

Question: Exercise 1.1, Part III: (3) C = {Ravi, Chenab, Jhelum, Sutlej, Indus}

Answer: The members of this set are the five rivers of Pakistan. Set-builder form: C = {x / x is a river of Pakistan}.

Question: Exercise 1.1, Part III: (4) E = {2, 4, 6, 8, 10, 12, ..., 20}

Answer: This set contains all even numbers from 2 to 20. Set-builder form: E = {x / x ∈ E ∧ 2 ≤ x ≤ 20}.

Question: Exercise 1.1, Part III: (5) D = {1, 3, 5, 7, 9, ...}

Answer: This set contains all odd natural numbers. Set-builder form: D = {x / x is an odd natural number}.

Question: Exercise 1.1, Part III: (6) S = the set of the names of the days of the week beginning with 'J'

Answer: Of the days of the week, only Friday begins with 'J'. Set-builder form: S = {x / x is the name of a day of the week beginning with 'J'}, i.e. S = {Friday}.

Question: Exercise 1.1, Part III: (7) Z = the set of all integers

Answer: All integers are ..., −2, −1, 0, 1, 2, ... Set-builder form: Z = {x / x ∈ Z}.

Question: Exercise 1.1, Part III: (8) L = {4, 8, 12, 16, 20, ...}

Answer: This set contains all multiples of 4. Set-builder form: L = {x / x is a multiple of the number 4}.

Question: Exercise 1.1, Part IV: Write the following sets in tabular form. (1) A = {x / x ∈ N ∧ x ≤ 5}

Answer: The natural numbers less than or equal to 5 are 1, 2, 3, 4, 5. Tabular form: A = {1, 2, 3, 4, 5}.

Question: Exercise 1.1, Part IV: (2) B = {y / y ∈ E ∧ 4 ≤ y ≤ 20}

Answer: The even numbers from 4 to 20 are 4, 6, 8, 10, 12, 14, 16, 18, 20. Tabular form: B = {4, 6, 8, 10, 12, 14, 16, 18, 20}.

Question: Exercise 1.1, Part IV: (3) C = {a / a ∈ Z ∧ −3 < a < 3}

Answer: The integers greater than −3 and less than 3 are −2, −1, 0, 1, 2. Tabular form: C = {−2, −1, 0, 1, 2}.

Question: Exercise 1.1, Part IV: (4) D = {z / z ∈ W ∧ z < 5}

Answer: The whole numbers less than 5 are 0, 1, 2, 3, 4. Tabular form: D = {0, 1, 2, 3, 4}.

Question: Exercise 1.1, Part V: Write the following sets in the three forms. (A) A = {a, e, i, o, u}

Answer: Descriptive form: A = the set of the vowels of the English alphabet. Tabular form: A = {a, e, i, o, u}. Set-builder form: A = {x / x is a vowel of the English alphabet}.

Question: Exercise 1.1, Part V: (B) B = {2, 4, 6, 8, ...}

Answer: Descriptive form: B = the set of even natural numbers. Tabular form: B = {2, 4, 6, 8, ...}. Set-builder form: B = {x / x ∈ E}.

Question: Exercise 1.1, Part V: (C) C = {1, 2, 3, 4, 5}

Answer: Descriptive form: C = the set of the first five natural numbers. Tabular form: C = {1, 2, 3, 4, 5}. Set-builder form: C = {x / x ∈ N ∧ x ≤ 5}.

Question: Exercise 1.1, Part V: (D) D = {x / x ∈ N ∧ x > 50}

Answer: Descriptive form: D = the set of natural numbers greater than 50. Tabular form: D = {51, 52, 53, ...}. Set-builder form: D = {x / x ∈ N ∧ x > 50}.

Question: Exercise 1.1, Part V: (E) E = {p, a, k, i, s, t, n}

Answer: Descriptive form: E = the set of the letters of the word "pakistan". Tabular form: E = {p, a, k, i, s, t, n}. Set-builder form: E = {x / x is a letter of the word "pakistan"}.

Question: Exercise 1.1, Part V: (F) F = the set of the names of the days of the week

Answer: Descriptive form: F = the set of the names of the days of the week. Tabular form: F = {Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}. Set-builder form: F = {x / x is the name of a day of the week}.

Question: Exercise 1.2, Part I: Find the union of the following sets. (1) A = {4, 5} and B = {4, 6, 8, 9}

Answer: The union contains all members of both sets, each written once: A ∪ B = {4, 5} ∪ {4, 6, 8, 9} = {4, 5, 6, 8, 9}.

Question: Exercise 1.2, Part I: (2) A = {0, 1, 2, 3, 4, 5} and B = {2, 4, 6, 8}

Answer: The union contains all members of both sets, each written once: A ∪ B = {0, 1, 2, 3, 4, 5} ∪ {2, 4, 6, 8} = {0, 1, 2, 3, 4, 5, 6, 8}.

Question: Exercise 1.2, Part I: (3) A = {2, 4, 6, 8, 10} and B = {0, 1, 2, 3, 4, 5, 6, 7, 9, 10}

Answer: The union contains all members of both sets, each written once: A ∪ B = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}.

Question: Exercise 1.2, Part I: (4) X = the set of the first four odd numbers and Y = the set of the first 3 multiples of the number 3; find X ∪ Y

Answer: The first four odd numbers are X = {1, 3, 5, 7}, and the first 3 multiples of 3 are Y = {3, 6, 9}. Union: X ∪ Y = {1, 3, 5, 7} ∪ {3, 6, 9} = {1, 3, 5, 6, 7, 9}.

Question: Exercise 1.2, Part I: (5) D = {2, 4, 6}, E = {1, 3, 5, 6} and F = {1, 3, 6, 9, 10}; find D ∪ E ∪ F

Answer: The union of all three sets contains every member once: D ∪ E ∪ F = {2, 4, 6} ∪ {1, 3, 5, 6} ∪ {1, 3, 6, 9, 10} = {1, 2, 3, 4, 5, 6, 9, 10}.

Question: Exercise 1.2, Part I: (6) A = {1, 2, 3}, B = { } and C = {2, 3}; find (A ∪ B) ∪ C and A ∪ (B ∪ C)

Answer: First way: A ∪ B = {1, 2, 3} ∪ { } = {1, 2, 3}; then (A ∪ B) ∪ C = {1, 2, 3} ∪ {2, 3} = {1, 2, 3}. Second way: B ∪ C = { } ∪ {2, 3} = {2, 3}; then A ∪ (B ∪ C) = {1, 2, 3} ∪ {2, 3} = {1, 2, 3}. Both answers are equal.

Question: Exercise 1.2, Part II: Find the intersection of the following sets. (1) A = {1, 2, 3} and B = {3, 6, 9, 12}

Answer: The intersection contains only the members common to both sets: A ∩ B = {1, 2, 3} ∩ {3, 6, 9, 12} = {3}.

Question: Exercise 1.2, Part II: (2) D = {2, 4, 6, 8} and E = {2, 4, 6, 8, 10}

Answer: The members common to both are 2, 4, 6, 8: D ∩ E = {2, 4, 6, 8}.

Question: Exercise 1.2, Part II: (3) P = {a, b, c, d} and Q = {a, e, i, o, u}

Answer: The only member common to both is a: P ∩ Q = {a}.

Question: Exercise 1.2, Part II: (4) A = the set of integers and B = the set of positive integers; find A ∩ B

Answer: A contains all integers and B contains only the positive integers (1, 2, 3, ...). The common members are the positive integers: A ∩ B = {1, 2, 3, ...} = B.

Question: Exercise 1.2, Part II: (5) A = {a, b, c, d}, B = {d, e, f} and C =

{f, g, h}; find (A ∩ B) ∩ C and A ∩ (B ∩ C)

Answer: First way: A ∩ B = {d}; then (A ∩ B) ∩ C = {d} ∩ {f, g, h} = { }. Second way: B ∩ C = {f}; then A ∩ (B ∩ C) = {a, b, c, d} ∩ {f} = { }. Both answers are equal (the empty set).

Question: Exercise 1.2, Part II: (6) D = {2, 3}, E = {4, 5} and F = { }; find (D ∩ E) ∩ F and D ∩ (E ∩ F)

Answer: First way: D ∩ E = { } (no common member); then (D ∩ E) ∩ F = { } ∩ { } = { }. Second way: E ∩ F = { }; then D ∩ (E ∩ F) = {2, 3} ∩ { } = { }. Both answers are equal (the empty set).

Question: Exercise 1.2, Part III: From the following sets find A − B and B − A. (1) A = {4, 5, 6} and B = {5, 6, 7, 8}

Answer: A − B contains the members of A that are not in B: {4, 5, 6} − {5, 6, 7, 8} = {4}. B − A contains the members of B that are not in A: {5, 6, 7, 8} − {4, 5, 6} = {7, 8}.

Question: Exercise 1.2, Part III: (2) A = {1, 3, 5, 9, 11} and B = {1, 3, 5, 7, 11}

Answer: A − B = {1, 3, 5, 9, 11} − {1, 3, 5, 7, 11} = {9}. B − A = {1, 3, 5, 7, 11} − {1, 3, 5, 9, 11} = {7}.

Question: Exercise 1.2, Part III: (3) A = {0, 1, 2, 3, 4, 5, 6, ...} and B = {1, 2, 3, 4, 5, ...}

Answer: A contains all whole numbers starting from 0 and B contains all natural numbers starting from 1. A − B = {0} (only 0 is in A but not in B). B − A = { } (every member of B is also in A).

Question: Exercise 1.2, Part III: (4) A = the set of multiples of the number 3 and B = the set of multiples of the number 4; find A − B and B − A

Answer: A = {3, 6, 9, 12, 15, 18, 21, ...} and B = {4, 8, 12, 16, 20, 24, ...}. A − B contains the multiples of 3 that are not multiples of 4: {3, 6, 9, 15, 18, 21, 27, ...} (common multiples like 12, 24, ... are removed). B − A

contains the multiples of 4 that are not multiples of 3: {4, 8, 16, 20, 28, 32, ...}.

Question: Exercise 1.3, Part (a): Identify disjoint or overlapping sets. (1) F = {Δ, □, ○} and M = {ٹالهي، جگ، کوپ}

Answer: F and M have no common member (one has shapes, the other has names of things), so F ∩ M = { }. These are disjoint sets.

Question: Exercise 1.3, Part (a): (2) W = {0, 1, 2, 3} and N = {1, 2, 3, 4}

Answer: W and N share the members 1, 2, 3, but neither is a subset of the other (0 is not in N, and 4 is not in W). So these are overlapping sets.

Question: Exercise 1.3, Part (a): (3) C = {a, e, i, o, u} and D = {p, q, r}

Answer: C and D have no common member, so C ∩ D = { }. These are disjoint sets.

Question: Exercise 1.3, Part (a): (4) C = {a, c, d, f} and D = {1, 2, 3}

Answer: C and D have no common member (one has letters, the other has numbers), so C ∩ D = { }. These are disjoint sets.

Question: Exercise 1.3, Part (a): (5) N = {1, 2, 4, 6, 8, 12, 24, 48} and M = {6, 8, 12, 14, 16, 18}

Answer: N and M share the members 6, 8, 12, but neither is a subset of the other (1 is not in M, and 14 is not in N). So these are overlapping sets.

Question: Exercise 1.3, Part (a): (6) A = the set of natural numbers from 10 to 40 and W = the set of whole numbers up to 40

Answer: A = {10, 11, ..., 40} and W = {0, 1, ..., 40}. Every member of A is in W, i.e. A ⊆ W. So these are neither disjoint (they have common members) nor overlapping (overlapping requires that neither set be a subset of the other) — A is a subset of W.

Question: Exercise 1.3, Part (a): (7) E = {2, 4, 6, ...} and O = {5, 10, 15, ...}

Answer: E contains the even numbers and O contains the multiples of 5.

The common members are 10, 20, 30, …, but neither is a subset of the other (2 is not in O, and 5 is not in E). So these are overlapping sets.

Question: Exercise 1.3, Part (b): Identify the following pairs as disjoint or overlapping. (i) F = {2, 3} and A = {1, 4, 6, 8}

Answer: F and A have no common member, so F ∩ A = { }. These are disjoint sets.

Question: Exercise 1.3, Part (b): (ii) H = {2, 3, 5, 7, 9} and C = {1, 3, 5, …}

Answer: H and C share the members 3, 5, 9, but neither is a subset of the other (2 is not in C, and 11 is not in H). So these are overlapping sets.

Question: Exercise 1.3, Part (b): (iii) N = {a, e, i, o, u} and G = {h, o, c, k, y}

Answer: N and G share only the member o, but neither is a subset of the other (a is not in G, and h is not in N). So these are overlapping sets.

Question: Exercise 1.3, Part (b): (iv) W = {0, 1, 2, 3, …} and B = {x, y, z}

Answer: W and B have no common member (one has numbers, the other has letters), so W ∩ B = { }. These are disjoint sets.

Question: Exercise 1.3, Part (b): (v) P = {ٹالی، جگ، کوپ} and D = {f, e, s, l}

Answer: P and D have no common member, so P ∩ D = { }. These are disjoint sets.

Question: Exercise 1.3, Part (b): (vi) Q = {★, Δ, □, ○} and E = {1, 3, 5, 7, 9}

Answer: Q and E have no common member (one has shapes, the other has numbers), so Q ∩ E = { }. These are disjoint sets.

Question: Exercise 1.3, Part (b): (vii) A = {10, 20, 30, …} and B = {4, 8, 12, …}

Answer: A and B share the members 20, 40, 60, … (the common multiples

of 10 and 4), but neither is a subset of the other (10 is not in B, and 4 is not in A). So these are overlapping sets.

Question: Exercise 1.4, Question 1: Suppose U = {a, b, c, d, e, f} and A = {a, e, f}; find: (i) A′

Answer: A′ = U − A = {a, b, c, d, e, f} − {a, e, f} = {b, c, d}.

Question: Exercise 1.4, Question 1: (ii) A ∪ A′

Answer: A ∪ A′ = {a, e, f} ∪ {b, c, d} = {a, b, c, d, e, f} = U.

Question: Exercise 1.4, Question 1: (iii) A ∩ A′

Answer: A and A′ have no common member: A ∩ A′ = {a, e, f} ∩ {b, c, d} = { } = ∅.

Question: Exercise 1.4, Question 1: (iv) U′

Answer: U′ = U − U = { } = ∅.

Question: Exercise 1.4, Question 1: (v) ∅′

Answer: ∅′ = U − ∅ = {a, b, c, d, e, f} = U.

Question: Exercise 1.4, Question 2: Suppose U = {1, 2, 3, ..., 10} and E = {2, 4, 6, 8, 10}; find: (i) E′

Answer: E′ = U − E = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} − {2, 4, 6, 8, 10} = {1, 3, 5, 7, 9}.

Question: Exercise 1.4, Question 2: (ii) E ∪ E′

Answer: E ∪ E′ = {2, 4, 6, 8, 10} ∪ {1, 3, 5, 7, 9} = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} = U.

Question: Exercise 1.4, Question 2: (iii) E ∩ E′

Answer: E and E′ have no common member: E ∩ E′ = {2, 4, 6, 8, 10} ∩ {1, 3, 5, 7, 9} = { } = ∅.

Question: Exercise 1.4, Question 2: (iv) U′

Answer: U′ = U − U = { } = ∅.

Question: Exercise 1.4, Question 2: (v) ∅′

Answer: ∅′ = U − ∅ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} = U.

Question: Exercise 1.4, Question 3: If U = {0, 1, 2, 3, ..., 12} and P = {1, 3, 5, 7, 9, 11}; find: (i) P′

Answer: P′ = U − P = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12} − {1, 3, 5, 7, 9, 11} = {0, 2, 4, 6, 8, 10, 12}.

Question: Exercise 1.4, Question 3: (ii) P ∪ P′

Answer: P ∪ P′ = {1, 3, 5, 7, 9, 11} ∪ {0, 2, 4, 6, 8, 10, 12} = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12} = U.

Question: Exercise 1.4, Question 3: (iii) P′ ∩ P

Answer: P′ and P have no common member: P′ ∩ P = {0, 2, 4, 6, 8, 10, 12} ∩ {1, 3, 5, 7, 9, 11} = { } = ∅.

Question: Exercise 1.4, Question 3: (iv) U ∩ U′

Answer: U′ = ∅, so U ∩ U′ = U ∩ ∅ = { } = ∅.

Question: Exercise 1.4, Question 4: If U = {u, v, w, x, y, z} and B = {x, y, z}; find: (i) B′

Answer: B′ = U − B = {u, v, w, x, y, z} − {x, y, z} = {u, v, w}.

Question: Exercise 1.4, Question 4: (ii) B′ ∪ B

Answer: B′ ∪ B = {u, v, w} ∪ {x, y, z} = {u, v, w, x, y, z} = U.

Question: Exercise 1.4, Question 4: (iii) B′ ∩ B

Answer: B′ and B have no common member: B′ ∩ B = {u, v, w} ∩ {x, y, z} = { } = ∅.

Question: Exercise 1.4, Question 4: (iv) U ∪ U′

Answer: U′ = ∅, so U ∪ U′ = U ∪ ∅ = {u, v, w, x, y, z} = U.

Question: Exercise 1.4, Question 5: If U = {1, 2, 3, 4, ..., 10}, A = {1, 2, 3, 5, 6, 7} and B = {2, 4, 6, 8, 10}; find: (i) A′

Answer: A′ = U − A = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} − {1, 2, 3, 5, 6, 7} = {4, 8, 9, 10}.

Question: Exercise 1.4, Question 5: (ii) B′

Answer: B′ = U − B = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} − {2, 4, 6, 8, 10} = {1,

3, 5, 7, 9}.

Question: Exercise 1.4, Question 5: (iii) A′ ∩ A

Answer: A′ and A have no common member: A′ ∩ A = {4, 8, 9, 10} ∩ {1, 2, 3, 5, 6, 7} = { } = ∅.

Question: Exercise 1.4, Question 5: (iv) A ∪ A′

Answer: A ∪ A′ = {1, 2, 3, 5, 6, 7} ∪ {4, 8, 9, 10} = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} = U.

Question: Exercise 1.4, Question 5: (v) B ∪ B′

Answer: B ∪ B′ = {2, 4, 6, 8, 10} ∪ {1, 3, 5, 7, 9} = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} = U.

Question: Exercise 1.4, Question 5: (vi) B ∩ B′

Answer: B and B′ have no common member: B ∩ B′ = {2, 4, 6, 8, 10} ∩ {1, 3, 5, 7, 9} = { } = ∅.

Question: Exercise 1.5, Part (a): If A = {1, −2, −3}, B = {−1, −4, −5, −6} and C = {0, −1, −2, −3, −4}, verify: (1) A ∪ B = B ∪ A

Answer: Left side: A ∪ B = {1, −2, −3} ∪ {−1, −4, −5, −6} = {1, −2, −3, −1, −4, −5, −6}. Right side: B ∪ A = {−1, −4, −5, −6} ∪ {1, −2, −3} = {−1, −4, −5, −6, 1, −2, −3} = the same set. So A ∪ B = B ∪ A is verified. ✓

Question: Exercise 1.5, Part (a): (2) B − C ≠ C − B

Answer: B − C = {−1, −4, −5, −6} − {0, −1, −2, −3, −4} = {−5, −6}. C − B = {0, −1, −2, −3, −4} − {−1, −4, −5, −6} = {0, −2, −3}. {−5, −6} ≠ {0, −2, −3}, so B − C ≠ C − B is verified. ✓

Question: Exercise 1.5, Part (a): (3) A ∪ C = C ∪ A

Answer: Left side: A ∪ C = {1, −2, −3} ∪ {0, −1, −2, −3, −4} = {1, −2, −3, 0, −1, −4}. Right side: C ∪ A = {0, −1, −2, −3, −4} ∪ {1, −2, −3} = the same set. So A ∪ C = C ∪ A is verified. ✓

Question: Exercise 1.5, Part (a): (4) (A ∪ B) ∪ C = (B ∪ C) ∪ A

Answer: A ∪ B = {1, −2, −3, −1, −4, −5, −6}; then (A ∪ B) ∪ C = {1, −2, −3, −1, −4, −5, −6, 0}. B ∪ C = {−1, −4, −5, −6, 0, −2, −3}; then (B ∪ C) ∪ A = {−1, −4, −5, −6, 0, −2, −3, 1} = the same set. So the equality is verified. ✓

Question: Exercise 1.5, Part (a): (5) (A ∪ C) ∪ B = (C ∪ B) ∪ A

Answer: A ∪ C = {1, −2, −3, 0, −1, −4}; then (A ∪ C) ∪ B = {1, −2, −3, 0, −1, −4, −5, −6}. C ∪ B = {0, −1, −2, −3, −4, −5, −6}; then (C ∪ B) ∪ A = {0, −1, −2, −3, −4, −5, −6, 1} = the same set. So the equality is verified. ✓

Question: Exercise 1.5, Part (a): (6) A ∩ B = B ∩ A

Answer: A ∩ B = {1, −2, −3} ∩ {−1, −4, −5, −6} = { } (no common member). B ∩ A = {−1, −4, −5, −6} ∩ {1, −2, −3} = { }. Both are the empty set, so A ∩ B = B ∩ A is verified. ✓

Question: Exercise 1.5, Part (a): (7) B ∩ C = C ∩ B

Answer: B ∩ C = {−1, −4, −5, −6} ∩ {0, −1, −2, −3, −4} = {−1, −4}. C ∩ B = {0, −1, −2, −3, −4} ∩ {−1, −4, −5, −6} = {−1, −4}. Both are equal, so B ∩ C = C ∩ B is verified. ✓

Question: Exercise 1.5, Part (a): (8) A − C ≠ C − A

Answer: A − C = {1, −2, −3} − {0, −1, −2, −3, −4} = {1}. C − A = {0, −1, −2, −3, −4} − {1, −2, −3} = {0, −1, −4}. {1} ≠ {0, −1, −4}, so A − C ≠ C − A is verified. ✓

Question: Exercise 1.5, Part (a): (9) (A ∩ B) ∩ C = (B ∩ C) ∩ A

Answer: A ∩ B = { }, so (A ∩ B) ∩ C = { } ∩ C = { }. B ∩ C = {−1, −4}, so (B ∩ C) ∩ A = {−1, −4} ∩ {1, −2, −3} = { }. Both are the empty set, so the equality is verified. ✓

Question: Exercise 1.5, Part (a): (10) (A ∩ C) ∩ B = (A ∩ B) ∩ C

Answer: A ∩ C = {−2, −3}, so (A ∩ C) ∩ B = {−2, −3} ∩ {−1, −4, −5, −6} = { }. A ∩ B = { }, so (A ∩ B) ∩ C = { } ∩ C = { }. Both are the

empty set, so the equality is verified. ✓

Question: Exercise 1.5, Part (b): If U = {10, 11, 12, 13, 14, ..., 20}, A = {10, 12, 14, 16, 18} and B = {10, 11, 12, 13, 14, 15}, verify: (i) U′ = ∅

Answer: U′ = U − U = { } = ∅. So U′ = ∅ is verified. ✓

Question: Exercise 1.5, Part (b): (ii) ∅′ = U

Answer: ∅′ = U − ∅ = {10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20} = U. So ∅′ = U is verified. ✓

Question: Exercise 1.5, Part (b): (iii) A ∪ A′ = U

Answer: A′ = U − A = {11, 13, 15, 17, 19, 20}. Then A ∪ A′ = {10, 12, 14, 16, 18} ∪ {11, 13, 15, 17, 19, 20} = {10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20} = U. So A ∪ A′ = U is verified. ✓

Question: Exercise 1.5, Part (b): (iv) B ∩ B′ = ∅

Answer: B′ = U − B = {16, 17, 18, 19, 20}. Then B ∩ B′ = {10, 11, 12, 13, 14, 15} ∩ {16, 17, 18, 19, 20} = { } = ∅. So B ∩ B′ = ∅ is verified. ✓

Question: Exercise 1.5, Part (b): (v) (A ∪ B)′ = A′ ∩ B′

Answer: A ∪ B = {10, 11, 12, 13, 14, 15, 16, 18}, so (A ∪ B)′ = U − (A ∪ B) = {17, 19, 20}. A′ = {11, 13, 15, 17, 19, 20} and B′ = {16, 17, 18, 19, 20}, so A′ ∩ B′ = {17, 19, 20}. Both are equal, so (A ∪ B)′ = A′ ∩ B′ is verified. ✓

Question: Exercise 1.5, Part (b): (vi) (A ∩ B)′ = A′ ∪ B′

Answer: A ∩ B = {10, 12, 14}, so (A ∩ B)′ = U − {10, 12, 14} = {11, 13, 15, 16, 17, 18, 19, 20}. A′ ∪ B′ = {11, 13, 15, 17, 19, 20} ∪ {16, 17, 18, 19, 20} = {11, 13, 15, 16, 17, 18, 19, 20}. Both are equal, so (A ∩ B)′ = A′ ∪ B′ is verified. ✓

Question: Exercise 1.6, Part (a): Show the following operations by means of Venn diagrams. (1) Suppose A = {1, 3, 5, 7} and B = {2, 4, 6, 8}: (i) A ∪ B

Answer: A and B have no common member, so in the Venn diagram two

separate (non-overlapping) circles are drawn. The shaded part is the whole region of both circles: A ∪ B = {1, 2, 3, 4, 5, 6, 7, 8}.

Question: Exercise 1.6, Part (a): (1) (ii) A ∩ B

Answer: In the Venn diagram there are two separate circles with no common part, so there is no region to shade: A ∩ B = { } = ∅.

Question: Exercise 1.6, Part (a): (1) (iii) A − B

Answer: In the Venn diagram the whole region of the A circle is shaded (since no member of A is in B): A − B = {1, 3, 5, 7} = A.

Question: Exercise 1.6, Part (a): (1) (iv) B − A

Answer: In the Venn diagram the whole region of the B circle is shaded (since no member of B is in A): B − A = {2, 4, 6, 8} = B.

Question: Exercise 1.6, Part (a): (2) Suppose W = {a, b, c} and X = {u, v, w}: (i) W ∪ X

Answer: W and X have no common member, so in the Venn diagram two separate circles are drawn and the whole region of both is shaded: W ∪ X = {a, b, c, u, v, w}.

Question: Exercise 1.6, Part (a): (2) (ii) W ∩ X

Answer: There are two separate circles with no common part, so there is no region to shade: W ∩ X = { } = ∅.

Question: Exercise 1.6, Part (a): (2) (iii) W − X

Answer: The whole region of the W circle is shaded: W − X = {a, b, c} = W.

Question: Exercise 1.6, Part (a): (2) (iv) X − W

Answer: The whole region of the X circle is shaded: X − W = {u, v, w} = X.

Question: Exercise 1.6, Part (a): (3) Suppose Y = {Ali, Raza, Sameer} and Z = {Sabir, Talha, Zahid}: (i) Y ∪ Z

Answer: Y and Z have no common member, so in the Venn diagram two separate circles are drawn and the whole region of both is shaded: Y ∪ Z =

{Ali, Raza, Sameer, Sabir, Talha, Zahid}.

Question: Exercise 1.6, Part (a): (3) (ii) Y ∩ Z

Answer: There are two separate circles with no common part, so there is no region to shade: Y ∩ Z = { } = ∅.

Question: Exercise 1.6, Part (a): (3) (iii) Y − Z

Answer: The whole region of the Y circle is shaded: Y − Z = {Ali, Raza, Sameer} = Y.

Question: Exercise 1.6, Part (a): (3) (iv) Z − Y

Answer: The whole region of the Z circle is shaded: Z − Y = {Sabir, Talha, Zahid} = Z.

Question: Exercise 1.6, Part (a): (4) Suppose Y = {1, 2, 3, ..., 12} and Z = {2, 4, 6, ..., 16}: (i) Y ∪ Z

Answer: Y and Z are shown with overlapping circles; the common part is {2, 4, 6, 8, 10, 12}. The whole region of both circles is shaded: Y ∪ Z = {1, 2, 3, ..., 16}.

Question: Exercise 1.6, Part (a): (4) (ii) Y ∩ Z

Answer: Only the common (middle) part of the overlapping circles is shaded: Y ∩ Z = {2, 4, 6, 8, 10, 12}.

Question: Exercise 1.6, Part (a): (4) (iii) Y − Z

Answer: The part of the Y circle that does not overlap Z is shaded: Y − Z = {1, 3, 5, 7, 9, 11}.

Question: Exercise 1.6, Part (a): (4) (iv) Z − Y

Answer: The part of the Z circle that does not overlap Y is shaded: Z − Y = {14, 16}.

Question: Exercise 1.6, Part (b): If U = {8, 10, 11, 12, 13, 14, 15, 16}, A = {8, 10, 12, 14, 16} and B = {8, 15}, find and show by Venn diagram: (1) A′

Answer: In the Venn diagram U is a rectangle, and A and B are overlapping

circles (common member 8). The part of the rectangle outside A is shaded: A′ = U − A = {11, 13, 15}.

Question: Exercise 1.6, Part (b): (2) B′

Answer: The part of the rectangle outside B is shaded: B′ = U − B = {10, 11, 12, 13, 14, 16}.

Question: Exercise 1.6, Part (b): (3) A − B

Answer: The part of the A circle that does not overlap B is shaded: A − B = {10, 12, 14, 16}.

Question: Exercise 1.6, Part (b): (4) (A − B)′

Answer: A − B = {10, 12, 14, 16}, so (A − B)′ = U − (A − B) = {8, 11, 13, 15}. In the Venn diagram the whole rectangle outside the {10, 12, 14, 16} region is shaded.

Question: Exercise 1.6, Part (b): (5) A ∪ B′

Answer: B′ = {10, 11, 12, 13, 14, 16}, so A ∪ B′ = {8, 10, 12, 14, 16} ∪ {10, 11, 12, 13, 14, 16} = {8, 10, 11, 12, 13, 14, 16}. In the Venn diagram the A circle and the part outside B are shaded.

Question: Exercise 1.6, Part (b): (6) A′ ∪ B′

Answer: A′ = {11, 13, 15} and B′ = {10, 11, 12, 13, 14, 16}, so A′ ∪ B′ = {10, 11, 12, 13, 14, 15, 16}. In the Venn diagram the whole rectangle outside the common part of A and B (only 8) is shaded.

Question: Exercise 1.6, Part (b): (7) A′ − B′

Answer: A′ − B′ = {11, 13, 15} − {10, 11, 12, 13, 14, 16} = {15}. In the Venn diagram only the part of the B circle that does not overlap A is shaded.

Question: Exercise 1.6, Part (b): (8) (A ∩ B)′

Answer: A ∩ B = {8}, so (A ∩ B)′ = U − {8} = {10, 11, 12, 13, 14, 15, 16}. In the Venn diagram the whole rectangle outside the common part (8) is shaded.

Question: Exercise 1.6, Part (b): (9) A′ ∩ B

Answer: A′ ∩ B = {11, 13, 15} ∩ {8, 15} = {15}. In the Venn diagram only the part of the B circle that does not overlap A is shaded.

Question: Exercise 1.6, Part (b): (10) (A ∪ B)′

Answer: A ∪ B = {8, 10, 12, 14, 15, 16}, so (A ∪ B)′ = U − (A ∪ B) = {11, 13}. In the Venn diagram the part of the rectangle outside both circles is shaded.

Question: Exercise 1.6, Part (b): (11) A′ ∪ A

Answer: A′ ∪ A = {11, 13, 15} ∪ {8, 10, 12, 14, 16} = {8, 10, 11, 12, 13, 14, 15, 16} = U. In the Venn diagram the whole rectangle is shaded.

Question: Exercise 1.6, Part (b): (12) B′ ∩ B

Answer: B′ and B have no common member: B′ ∩ B = {10, 11, 12, 13, 14, 16} ∩ {8, 15} = { } = ∅. In the Venn diagram there is no region to shade.

Question: Review Exercise 1, Question 1: State the names of the three forms of expressing a set.

Answer: The three forms of expressing a set are: (i) descriptive form, (ii) tabular (roster) form, (iii) set-builder form.

Question: Review Exercise 1, Question 2: (a) Express the set of the first five whole numbers in tabular form.

Answer: The first five whole numbers are 0, 1, 2, 3, 4. In tabular form: {0, 1, 2, 3, 4}.

Question: Review Exercise 1, Question 2: (b) Express A = {a, b, c, d, ...} in descriptive form.

Answer: This set contains all the letters of the English alphabet starting from a. In descriptive form: A = the set of the letters of the English alphabet.

Question: Review Exercise 1, Question 3: Give an example of your own of sets in set-builder form.

Answer: Example: {x / x ∈ N ∧ x < 5} = {1, 2, 3, 4} — this is the set of natural numbers less than 5.

Question: Review Exercise 1, Question 4: Explain disjoint sets and overlapping sets with the help of examples.

Answer: Disjoint sets: {1, 2} and {3, 4} — they have no common member, so they are disjoint. Overlapping sets: {1, 2, 3} and {2, 3, 4} — they share the members 2, 3, but neither set is a subset of the other, so they are overlapping.

Question: Review Exercise 1, Question 5: If U = {1, 2, 3, ..., 12}, A = {1, 2, 3, 10}, B = {2, 4, 6, ..., 12} and C = {1, 3, 5, ..., 11}, find: (i) A ∪ B

Answer: A ∪ B = {1, 2, 3, 10} ∪ {2, 4, 6, 8, 10, 12} = {1, 2, 3, 4, 6, 8, 10, 12}.

Question: Review Exercise 1, Question 5: (ii) A ∩ B

Answer: The only member common to A and B is 2: A ∩ B = {1, 2, 3, 10} ∩ {2, 4, 6, 8, 10, 12} = {2}.

Question: Review Exercise 1, Question 5: (iii) A − B

Answer: A − B = {1, 2, 3, 10} − {2, 4, 6, 8, 10, 12} = {1, 3, 10}.

Question: Review Exercise 1, Question 5: (iv) (A ∩ C)′

Answer: A ∩ C = {1, 2, 3, 10} ∩ {1, 3, 5, 7, 9, 11} = {1, 3}. Then (A ∩ C)′ = U − {1, 3} = {2, 4, 5, 6, 7, 8, 9, 10, 11, 12}.

Question: Review Exercise 1, Question 5: (v) (B ∪ C)′

Answer: B ∪ C = {2, 4, 6, 8, 10, 12} ∪ {1, 3, 5, 7, 9, 11} = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12} = U. Then (B ∪ C)′ = U − U = { } = ∅.

Question: Review Exercise 1, Question 5: (vi) B′ ∩ C′

Answer: B′ = U − B = {1, 3, 5, 7, 9, 11} and C′ = U − C = {2, 4, 6, 8, 10, 12}. Then B′ ∩ C′ = {1, 3, 5, 7, 9, 11} ∩ {2, 4, 6, 8, 10, 12} = { } = ∅.

Question: Review Exercise 1, Question 5: (vii) A − B′

Answer: B′ = {1, 3, 5, 7, 9, 11}, so A − B′ = {1, 2, 3, 10} − {1, 3, 5, 7, 9,

11} = {2, 10}.

Question: Review Exercise 1, Question 5: (viii) B − A′

Answer: A′ = U − A = {4, 5, 6, 7, 8, 9, 11, 12}, so B − A′ = {2, 4, 6, 8, 10, 12} − {4, 5, 6, 7, 8, 9, 11, 12} = {2, 10}.

Question: Review Exercise 1, Question 5: (ix) A′ − C

Answer: A′ = {4, 5, 6, 7, 8, 9, 11, 12}, so A′ − C = {4, 5, 6, 7, 8, 9, 11, 12} − {1, 3, 5, 7, 9, 11} = {4, 6, 8, 12}.

Question: Review Exercise 1, Question 5: (x) (B − C)′

Answer: B − C = {2, 4, 6, 8, 10, 12} − {1, 3, 5, 7, 9, 11} = {2, 4, 6, 8, 10, 12} = B. Then (B − C)′ = U − B = {1, 3, 5, 7, 9, 11} = B′.

Question: Review Exercise 1, Question 5: (xi) (C − A)′

Answer: C − A = {1, 3, 5, 7, 9, 11} − {1, 2, 3, 10} = {5, 7, 9, 11}. Then (C − A)′ = U − {5, 7, 9, 11} = {1, 2, 3, 4, 6, 8, 10, 12}.

Question: Review Exercise 1, Question 5: (xii) A′ − C′

Answer: A′ = {4, 5, 6, 7, 8, 9, 11, 12} and C′ = {2, 4, 6, 8, 10, 12}, so A′ − C′ = {4, 5, 6, 7, 8, 9, 11, 12} − {2, 4, 6, 8, 10, 12} = {5, 7, 9, 11}.

Question: Review Exercise 1, Question 6: If A = {a, b, c, d}, B = {a, c, d} and C = {b, d, e}, prove: (i) A ∪ B = B ∪ A

Answer: A ∪ B = {a, b, c, d} ∪ {a, c, d} = {a, b, c, d}. B ∪ A = {a, c, d} ∪ {a, b, c, d} = {a, b, c, d}. Both are equal, so A ∪ B = B ∪ A is proved. ✓

Question: Review Exercise 1, Question 6: (ii) B ∩ C = C ∩ B

Answer: B ∩ C = {a, c, d} ∩ {b, d, e} = {d}. C ∩ B = {b, d, e} ∩ {a, c, d} = {d}. Both are equal, so B ∩ C = C ∩ B is proved. ✓

Question: Review Exercise 1, Question 6: (iii) C ∪ A = A ∪ C

Answer: C ∪ A = {b, d, e} ∪ {a, b, c, d} = {a, b, c, d, e}. A ∪ C = {a, b, c, d} ∪ {b, d, e} = {a, b, c, d, e}. Both are equal, so C ∪ A = A ∪ C is proved. ✓

Question: Review Exercise 1, Question 6: (iv) (A ∪ B) ∪ C = A ∪ (B ∪ C)

Answer: A ∪ B = {a, b, c, d}, so (A ∪ B) ∪ C = {a, b, c, d} ∪ {b, d, e} = {a, b, c, d, e}. B ∪ C = {a, b, c, d, e}, so A ∪ (B ∪ C) = {a, b, c, d} ∪ {a, b, c, d, e} = {a, b, c, d, e}. Both are equal, so the equality is proved. ✓

Question: Review Exercise 1, Question 6: (v) (A ∩ C) ∩ B = A ∩ (C ∩ B)

Answer: A ∩ C = {b, d}, so (A ∩ C) ∩ B = {b, d} ∩ {a, c, d} = {d}. C ∩ B = {d}, so A ∩ (C ∩ B) = {a, b, c, d} ∩ {d} = {d}. Both are equal, so the equality is proved. ✓

Question: Review Exercise 1, Question 6: (vi) (B ∪ A) ∪ C = B ∪ (A ∪ C)

Answer: B ∪ A = {a, b, c, d}, so (B ∪ A) ∪ C = {a, b, c, d} ∪ {b, d, e} = {a, b, c, d, e}. A ∪ C = {a, b, c, d, e}, so B ∪ (A ∪ C) = {a, c, d} ∪ {a, b, c, d, e} = {a, b, c, d, e}. Both are equal, so the equality is proved. ✓

Question: Review Exercise 1, Question 7: Put a (✓) mark on the correct answer. (i) If U = {a, b, c, d, e} and A = {a, b, c}, then A′ is: (a) {a, b} (b) {b, c} (c) {c, d} (d) {d, e}

Answer: A′ = U − A = {a, b, c, d, e} − {a, b, c} = {d, e}. So the correct answer is (d) {d, e}. ✓

Question: Review Exercise 1, Question 7: (ii) If A = {l, m, n} and B = {l, m, n, o}, then B − A is: (a) {l} (b) {o} (c) {n} (d) {m}

Answer: B − A = {l, m, n, o} − {l, m, n} = {o}. So the correct answer is (b) {o}. ✓

Question: Review Exercise 1, Question 7: (iii) If P = {1, 2, 3} and Q = {2, 3, 4}, then P ∩ Q is: (a) {2, 3} (b) {1, 3} (c) {3, 4} (d) {1, 4}

Answer: P ∩ Q = {1, 2, 3} ∩ {2, 3, 4} = {2, 3}. So the correct answer is (a) {2, 3}. ✓

Question: Review Exercise 1, Question 7: (iv) If U = {1, 2, 3, 4, 5, 6}, X = {2, 3, 4} and Y = {4, 5, 6}, then (X ∪ Y)′ is: (a) ∅ (b) {2, 3} (c) {1} (d) {6}

Answer: X ∪ Y = {2, 3, 4, 5, 6}, so (X ∪ Y)′ = U − (X ∪ Y) = {1, 2, 3, 4, 5,

6} − {2, 3, 4, 5, 6} = {1}. So the correct answer is (c) {1}. ✓

Question: Review Exercise 1, Question 7: (v) If A = {x, y, z} and B = {y, z, x}, then A − B is: (a) A (b) B (c) ∅ (d) A ∩ B

Answer: A and B have the same members, so A − B = {x, y, z} − {y, z, x} = { } = ∅. So the correct answer is (c) ∅. ✓

Question: Review Exercise 1, Question 8: Show the following by means of Venn diagrams. (i) C = {c, d, e} and D = {d, e, f}; find C ∪ D and C ∩ D

Answer: C and D are shown with overlapping circles; the common part is {d, e}. Shading the whole region of both circles gives C ∪ D = {c, d, e, f}, and shading only the common part gives C ∩ D = {d, e}.

Question: Review Exercise 1, Question 8: (ii) U = {l, m, n, o, r}, P = {l, m, n} and Q = {n, o, r}; find Q′ and P′

Answer: Q′ = U − Q = {l, m, n, o, r} − {n, o, r} = {l, m}. P′ = U − P = {l, m, n, o, r} − {l, m, n} = {o, r}. In the Venn diagram U is a rectangle, P and Q are overlapping circles (common member n); shading the part outside each set gives its complement.

Question: Review Exercise 1, Question 8: (iii) X = {x, z} and Y = {x, y, z}; find X ∪ Y and X ∩ Y

Answer: Every member of X is in Y, so in the Venn diagram the X circle is drawn inside the Y circle. X ∪ Y = {x, y, z} = Y (the whole region of the larger circle) and X ∩ Y = {x, z} = X (the region of the smaller circle).

2. Rational Numbers

Question: Exercise 2.1, Question 1: Write true or false for each of the following statements. (i) 0/5 is a rational number.

Answer: In 0/5 the denominator is 5, which is not zero, and 0/5 = 0. So this statement is true. ✓

Question: Exercise 2.1, Question 1: (ii) 5/0 is a rational number.

Answer: In 5/0 the denominator is zero, and no number can be divided by zero. So 5/0 is not a rational number; this statement is false. ✗

Question: Exercise 2.1, Question 1: (iii) 0 is not a rational number.

Answer: 0 can be written in the form 0/1, whose denominator is not zero. So 0 is a rational number; this statement is false. ✗

Question: Exercise 2.1, Question 1: (iv) 5 and 2 1/5 are rational numbers.

Answer: 5 = 5/1 and 2 1/5 = 11/5 can be written, both with non-zero denominators. So both are rational numbers; this statement is true. ✓

Question: Exercise 2.1, Question 1: (v) All whole numbers are rational numbers.

Answer: Every whole number a can be written in the form a/1. So all whole numbers are rational numbers; this statement is true. ✓

Question: Exercise 2.1, Question 1: (vi) Positive integers are rational numbers.

Answer: Every positive integer a can be written in the form a/1. So positive integers are rational numbers; this statement is true. ✓

Question: Exercise 2.1, Question 1: (vii) Every integer is also a rational number.

Answer: Every integer a can be written in the form a/1. So every integer is a rational number; this statement is true. ✓

Question: Exercise 2.1, Question 1: (viii) Every natural number is also a rational number.

Answer: Every natural number a can be written in the form a/1. So every natural number is a rational number; this statement is true. ✓

Question: Exercise 2.1, Question 1: (ix) In a rational number p/q, q is always even.

Answer: q can be any integer that is not zero; for example in 1/3, q = 3 is not even. So this statement is false. ✗

Question: Exercise 2.1, Question 1: (x) There is only one rational number between 2 and 3.

Answer: There are infinitely many rational numbers between 2 and 3, e.g. 5/2, 9/4, 11/4, ... So this statement is false. ✗

Question: Exercise 2.1, Question 1: (xi) Rational numbers are in the form p/q, where p and q are always natural numbers.

Answer: In rational numbers p and q are integers, not natural numbers; e.g. in −2/3, p = −2 is not a natural number, and q can also be negative. So this statement is false. ✗

Question: Exercise 2.1, Question 2: Write a rational number between the numbers 2 and 3.

Answer: A number between two numbers is found by their average: (2 + 3)/2 = 5/2 = 2 1/2. Since 2 < 5/2 < 3, 5/2 is a rational number between 2 and 3.

Question: Exercise 2.1, Question 3: Show the following rational numbers on the number line. (i) 1/4

Answer: 1/4 = 0.25, which lies between 0 and 1. On the number line, divide the part between 0 and 1 into 4 equal parts; the point at the end of the first part is 1/4.

Question: Exercise 2.1, Question 3: (ii) 7/2

Answer: 7/2 = 3 1/2, which lies between 3 and 4. On the number line, divide the part between 3 and 4 into 2 equal parts; the point at the end of the first part is 7/2.

Question: Exercise 2.1, Question 3: (iii) 11/3

Answer: 11/3 = 3 2/3, which lies between 3 and 4. On the number line, divide the part between 3 and 4 into 3 equal parts; the point at the end of the second part is 11/3.

Question: Exercise 2.1, Question 3: (iv) −2/5

Answer: −2/5 = −0.4, which lies between 0 and −1. On the number line, divide the part between 0 and −1 into 5 equal parts; the point at the end of the second part from 0 is −2/5.

Question: Exercise 2.1, Question 3: (v) −12/5

Answer: −12/5 = −2 2/5, which lies between −2 and −3. On the number line, divide the part between −2 and −3 into 5 equal parts; the point at the end of the second part from −2 is −12/5.

Question: Exercise 2.1, Question 3: (vi) 2 5/6

Answer: 2 5/6 = 17/6, which lies between 2 and 3. On the number line, divide the part between 2 and 3 into 6 equal parts; the point at the end of the fifth part is 2 5/6.

Question: Exercise 2.1, Question 3: (vii) −3 4/7

Answer: −3 4/7 = −25/7, which lies between −3 and −4. On the number line, divide the part between −3 and −4 into 7 equal parts; the point at the end of the fourth part from −3 is −3 4/7.

Question: Exercise 2.1, Question 3: (viii) −13/4

Answer: −13/4 = −3 1/4, which lies between −3 and −4. On the number line, divide the part between −3 and −4 into 4 equal parts; the point at the end of the first part from −3 is −13/4.

Question: Exercise 2.1, Question 3: (ix) 15/8

Answer: 15/8 = 1 7/8, which lies between 1 and 2. On the number line, divide the part between 1 and 2 into 8 equal parts; the point at the end of the seventh part is 15/8.

Question: Exercise 2.1, Question 3: (x) 4 2/3

Answer: 4 2/3 = 14/3, which lies between 4 and 5. On the number line, divide the part between 4 and 5 into 3 equal parts; the point at the end of the second part is 4 2/3.

Question: Exercise 2.1, Question 4: According to the following number line, write true or false. (i) The value of x is 0.

Answer: On the number line the point x lies between 1 and 3, not at 0. So this statement is false. ✗

Question: Exercise 2.1, Question 4: (ii) The value of x is 2.

Answer: On the number line the part from 1 to 3 is divided into six equal parts; the point x is at 2. So this statement is true. ✓

Question: Exercise 2.1, Question 4: (iii) The value of y is 2 1/6.

Answer: The point y is one-sixth of a part ahead of x (2), i.e. 2 + 1/6 = 2 1/6. So this statement is true. ✓

Question: Exercise 2.1, Question 4: (iv) The value of y is 3 2/5.

Answer: The point y is at 2 1/6, not at 3 2/5. So this statement is false. ✗

Question: Exercise 2.1, Question 4: (v) The value of z is 4.

Answer: The point z is at the position of 4, after 3. So this statement is true. ✓

Question: Exercise 2.1, Question 4: (vi) The value of z is −4.

Answer: The point z is at 4 on the positive side, not at −4 on the negative side. So this statement is false. ✗

Question: Exercise 2.2, Question 1: Find the reciprocals of the following rational numbers. (i) 19/17

Answer: The reciprocal of a rational number is found by interchanging its

numerator and denominator: the reciprocal of 19/17 = 17/19. Check: (19/17) × (17/19) = 1.

Question: Exercise 2.2, Question 1: (ii) −13

Answer: −13 = −13/1, so its reciprocal = −1/13. Check: (−13) × (−1/13) = 1.

Question: Exercise 2.2, Question 1: (iii) 1/10

Answer: The reciprocal of 1/10 = 10/1 = 10. Check: (1/10) × 10 = 1.

Question: Exercise 2.2, Question 1: (iv) −6/25

Answer: The reciprocal of −6/25 = −25/6. Check: (−6/25) × (−25/6) = 1.

Question: Exercise 2.2, Question 2: Simplify. (i) 9/7 + 15/7

Answer: The denominators are the same, so the numerators are added: (9 + 15)/7 = 24/7 = 3 3/7.

Question: Exercise 2.2, Question 2: (ii) 5/9 − 2/9

Answer: The denominators are the same, so the numerators are subtracted: (5 − 2)/9 = 3/9 = 1/3.

Question: Exercise 2.2, Question 2: (iii) 3/5 − (−7/5)

Answer: The denominators are the same; subtracting a negative number is the same as adding a positive one: 3/5 − (−7/5) = (3 + 7)/5 = 10/5 = 2.

Question: Exercise 2.2, Question 2: (iv) 5/6 + 2/6 − 1/6

Answer: The denominators are the same: (5 + 2 − 1)/6 = 6/6 = 1.

Question: Exercise 2.2, Question 2: (v) 5/8 − 4/5

Answer: The denominators differ; the LCM of 8 and 5 = 40. So 5/8 − 4/5 = (5 × 5 − 4 × 8)/40 = (25 − 32)/40 = −7/40.

Question: Exercise 2.2, Question 2: (vi) −2/7 + 7/6

Answer: The LCM of 7 and 6 = 42. So −2/7 + 7/6 = (−2 × 6 + 7 × 7)/42 = (−12 + 49)/42 = 37/42.

Question: Exercise 2.2, Question 2: (vii) 2/3 + 5/7 − 1/6

Answer: The LCM of 3, 7 and 6 = 42. So 2/3 + 5/7 − 1/6 = (2 × 14 + 5 × 6

− 1 × 7)/42 = (28 + 30 − 7)/42 = 51/42 = 17/14 = 1 3/14.

Question: Exercise 2.2, Question 2: (viii) 4/5 − 1/7 + 3/70

Answer: The LCM of 5, 7 and 70 = 70. So 4/5 − 1/7 + 3/70 = (4 × 14 − 1 × 10 + 3)/70 = (56 − 10 + 3)/70 = 49/70 = 7/10.

Question: Exercise 2.2, Question 2: (ix) 2/9 − (−3/6) + 5/8

Answer: −(−3/6) = +3/6 = +1/2. The LCM of 9, 2 and 8 = 72. So 2/9 + 1/2 + 5/8 = (2 × 8 + 1 × 36 + 5 × 9)/72 = (16 + 36 + 45)/72 = 97/72 = 1 25/72.

Question: Exercise 2.2, Question 2: (x) 17/10 + 11/20 − 4/5

Answer: The LCM of 10, 20 and 5 = 20. So 17/10 + 11/20 − 4/5 = (17 × 2 + 11 − 4 × 4)/20 = (34 + 11 − 16)/20 = 29/20 = 1 9/20.

Question: Exercise 2.2, Question 3: Find the additive inverse and the multiplicative inverse of each of the following. (i) 17/15

Answer: The additive inverse is found by changing the sign: −17/15. Check: (17/15) + (−17/15) = 0. The multiplicative inverse is the reciprocal: 15/17. Check: (17/15) × (15/17) = 1.

Question: Exercise 2.2, Question 3: (ii) −26

Answer: Additive inverse: 26 (since (−26) + 26 = 0). Multiplicative inverse: −1/26 (since (−26) × (−1/26) = 1).

Question: Exercise 2.2, Question 3: (iii) −1/20

Answer: Additive inverse: 1/20 (since (−1/20) + 1/20 = 0). Multiplicative inverse: −20 (since (−1/20) × (−20) = 1).

Question: Exercise 2.2, Question 3: (iv) −5/19

Answer: Additive inverse: 5/19 (since (−5/19) + 5/19 = 0). Multiplicative inverse: −19/5 (since (−5/19) × (−19/5) = 1).

Question: Exercise 2.2, Question 3: (v) 0

Answer: Additive inverse: 0 (since 0 + 0 = 0). Multiplicative inverse: none — 0 has no multiplicative inverse, because multiplying 0 by any number

never gives 1 (0 × anything = 0).

Question: Exercise 2.2, Question 3: (vi) 8

Answer: Additive inverse: −8 (since 8 + (−8) = 0). Multiplicative inverse: 1/8 (since 8 × 1/8 = 1).

Question: Exercise 2.2, Question 3: (vii) 1/9

Answer: Additive inverse: −1/9 (since 1/9 + (−1/9) = 0). Multiplicative inverse: 9 (since (1/9) × 9 = 1).

Question: Exercise 2.2, Question 3: (viii) 200/7

Answer: Additive inverse: −200/7 (since (200/7) + (−200/7) = 0). Multiplicative inverse: 7/200 (since (200/7) × (7/200) = 1).

Question: Exercise 2.2, Question 4: Simplify. (i) 5/8 × 3/2

Answer: Multiply numerator by numerator and denominator by denominator: (5 × 3)/(8 × 2) = 15/16.

Question: Exercise 2.2, Question 4: (ii) 6/7 ÷ 2/5

Answer: Change the division to multiplication and invert the second number: 6/7 ÷ 2/5 = 6/7 × 5/2 = 30/14 = 15/7 = 2 1/7.

Question: Exercise 2.2, Question 4: (iii) 2/3 × 1/7 × 4/5

Answer: Multiply all three in the same way: (2 × 1 × 4)/(3 × 7 × 5) = 8/105.

Question: Exercise 2.2, Question 4: (iv) −6/7 ÷ (−7/6)

Answer: Change the division to multiplication: −6/7 ÷ (−7/6) = (−6/7) × (−6/7) = 36/49. (Two negative signs give a positive.)

Question: Exercise 2.2, Question 4: (v) 14/5 ÷ 7/25

Answer: 14/5 ÷ 7/25 = 14/5 × 25/7 = (14 × 25)/(5 × 7) = 350/35 = 10.

Question: Exercise 2.2, Question 4: (vi) 9/4 × (−4/9) × (2/3)

Answer: (9 × (−4) × 2)/(4 × 9 × 3) = −72/108 = −2/3. (There is one negative sign, so the answer is negative.)

Question: Exercise 2.2, Question 4: (vii) 1/8 × 2/5 × −15/4

Answer: (1 × 2 × (−15))/(8 × 5 × 4) = −30/160 = −3/16.

Question: Exercise 2.2, Question 4: (viii) (−2/3) × (−4/6) × (−3/4)

Answer: There are three negative signs, so the answer is negative: (−2 × −4 × −3)/(3 × 6 × 4) = −24/72 = −1/3.

Question: Exercise 2.3, Question 1: Verify the operations on the following rational numbers. (i) 2/3 + (−4/3) = (−4/3) + 2/3

Answer: Left side (LHS): 2/3 + (−4/3) = (2 − 4)/3 = −2/3. Right side (RHS): (−4/3) + 2/3 = (−4 + 2)/3 = −2/3. LHS = RHS, so the commutative property of addition is verified. ✓

Question: Exercise 2.3, Question 1: (ii) 4/7 × 2/5 = 2/5 × 4/7

Answer: Left side: 4/7 × 2/5 = 8/35. Right side: 2/5 × 4/7 = 8/35. LHS = RHS, so the commutative property of multiplication is verified. ✓

Question: Exercise 2.3, Question 1: (iii) 3/4 + (5/6 + 4/8) = (3/4 + 5/6) + 4/8

Answer: Left side: 5/6 + 4/8 = 5/6 + 1/2 = (5 + 3)/6 = 8/6 = 4/3; then 3/4 + 4/3 = (9 + 16)/12 = 25/12. Right side: 3/4 + 5/6 = (9 + 10)/12 = 19/12; then 19/12 + 4/8 = 19/12 + 1/2 = 25/12. LHS = RHS, so the associative property of addition is verified. ✓

Question: Exercise 2.3, Question 1: (iv) 7/15 × (3/4 × 5/7) = (7/15 × 3/4) × 5/7

Answer: Left side: 3/4 × 5/7 = 15/28; then 7/15 × 15/28 = 105/420 = 1/4. Right side: 7/15 × 3/4 = 21/60 = 7/20; then 7/20 × 5/7 = 35/140 = 1/4. LHS = RHS, so the associative property of multiplication is verified. ✓

Question: Exercise 2.3, Question 1: (v) 2/7 × (3/4 − 1/2) = 2/7 × 3/4 − 2/7 × 1/2

Answer: Left side: 3/4 − 1/2 = (3 − 2)/4 = 1/4; then 2/7 × 1/4 = 2/28 = 1/14. Right side: 2/7 × 3/4 = 6/28 = 3/14; 2/7 × 1/2 = 2/14; then 3/14 − 2/14 = 1/14. LHS = RHS, so the distributive property of multiplication over

subtraction is verified. ✓

Question: Exercise 2.3, Question 1: (vi) 4/5 × (7/8 + 3/5) = 4/5 × 7/8 + 4/5 × 3/5

Answer: Left side: 7/8 + 3/5 = (35 + 24)/40 = 59/40; then 4/5 × 59/40 = 236/200 = 59/50. Right side: 4/5 × 7/8 = 28/40 = 7/10; 4/5 × 3/5 = 12/25; then 7/10 + 12/25 = (35 + 24)/50 = 59/50. LHS = RHS, so the distributive property of multiplication over addition is verified. ✓

Question: Exercise 2.3, Question 2: Verify the associative property of addition for the rational numbers 1/2, 1/3 and 1/4.

Answer: Left side: 1/2 + (1/3 + 1/4) = 1/2 + (4 + 3)/12 = 1/2 + 7/12 = (6 + 7)/12 = 13/12. Right side: (1/2 + 1/3) + 1/4 = (3 + 2)/6 + 1/4 = 5/6 + 1/4 = (10 + 3)/12 = 13/12. LHS = RHS, so the associative property of addition is verified. ✓

Question: Exercise 2.3, Question 3: Verify the associative property of multiplication for the rational numbers 3/4, 4/5 and 2/7.

Answer: Left side: 3/4 × (4/5 × 2/7) = 3/4 × 8/35 = 24/140 = 6/35. Right side: (3/4 × 4/5) × 2/7 = (12/20) × 2/7 = 3/5 × 2/7 = 6/35. LHS = RHS, so the associative property of multiplication is verified. ✓

Question: Exercise 2.3, Question 4: Verify the distributive property of multiplication over addition for the rational numbers 7/8, 4/5 and 3/4.

Answer: Left side: 7/8 × (4/5 + 3/4) = 7/8 × (16 + 15)/20 = 7/8 × 31/20 = 217/160. Right side: 7/8 × 4/5 + 7/8 × 3/4 = 28/40 + 21/32 = 7/10 + 21/32 = (112 + 105)/160 = 217/160. LHS = RHS, so the distributive property of multiplication over addition is verified. ✓

Question: Exercise 2.3, Question 5: Confirm the distributive property of multiplication over subtraction for the rational numbers 1/5, 3/4 and 5/7.

Answer: Left side: 1/5 × (3/4 − 5/7) = 1/5 × (21 − 20)/28 = 1/5 × 1/28 = 1/140. Right side: 1/5 × 3/4 − 1/5 × 5/7 = 3/20 − 1/7 = (21 − 20)/140 =

1/140. LHS = RHS, so the distributive property of multiplication over subtraction is verified. ✓

Question: Exercise 2.3, Question 6: Write the following rational numbers in ascending order. (i) 1/2, 2/3, 8/9

Answer: The LCM of all three = 18. So 1/2 = 9/18, 2/3 = 12/18, 8/9 = 16/18. Comparing numerators: 9 < 12 < 16. So in ascending order: 1/2 < 2/3 < 8/9.

Question: Exercise 2.3, Question 6: (ii) 1/6, 3/4, 1/2

Answer: The LCM of all three = 12. So 1/6 = 2/12, 3/4 = 9/12, 1/2 = 6/12. Comparing numerators: 2 < 6 < 9. So in ascending order: 1/6 < 1/2 < 3/4.

Question: Exercise 2.3, Question 6: (iii) 4/7, 1/3, 5/6

Answer: The LCM of all three = 42. So 4/7 = 24/42, 1/3 = 14/42, 5/6 = 35/42. Comparing numerators: 14 < 24 < 35. So in ascending order: 1/3 < 4/7 < 5/6.

Question: Exercise 2.3, Question 7: Write the following rational numbers in descending order. (i) 1/2, 1/3, 1/4

Answer: The LCM of all three = 12. So 1/2 = 6/12, 1/3 = 4/12, 1/4 = 3/12. Comparing numerators: 6 > 4 > 3. So in descending order: 1/2 > 1/3 > 1/4.

Question: Exercise 2.3, Question 7: (ii) 4/5, 1/10, 2/15

Answer: The LCM of all three = 30. So 4/5 = 24/30, 1/10 = 3/30, 2/15 = 4/30. Comparing numerators: 24 > 4 > 3. So in descending order: 4/5 > 2/15 > 1/10.

Question: Exercise 2.3, Question 7: (iii) 3/8, 1/4, 5/6

Answer: The LCM of all three = 24. So 3/8 = 9/24, 1/4 = 6/24, 5/6 = 20/24. Comparing numerators: 20 > 9 > 6. So in descending order: 5/6 > 3/8 > 1/4.

Question: Exercise 2.3, Question 8: Write the following rational numbers in

ascending or descending order. (i) 4/3, 5/7, 1/4

Answer: The LCM of all three = 84. So 4/3 = 112/84, 5/7 = 60/84, 1/4 = 21/84. Ascending order: 1/4 < 5/7 < 4/3. Descending order: 4/3 > 5/7 > 1/4.

Question: Exercise 2.3, Question 8: (ii) 1/2, 2/5, 1/11

Answer: The LCM of all three = 110. So 1/2 = 55/110, 2/5 = 44/110, 1/11 = 10/110. Ascending order: 1/11 < 2/5 < 1/2. Descending order: 1/2 > 2/5 > 1/11.

Question: Exercise 2.3, Question 8: (iii) 2/9, 5/3, 1/2, 5/6

Answer: The LCM of all four = 18. So 2/9 = 4/18, 5/3 = 30/18, 1/2 = 9/18, 5/6 = 15/18. Ascending order: 2/9 < 1/2 < 5/6 < 5/3. Descending order: 5/3 > 5/6 > 1/2 > 2/9.

Question: Exercise 2.3, Question 8: (iv) 4/20, 20/15, 1/2, 9/7

Answer: First simplify: 4/20 = 1/5 and 20/15 = 4/3. The LCM of all four = 210. So 1/5 = 42/210, 4/3 = 280/210, 1/2 = 105/210, 9/7 = 270/210. Ascending order: 4/20 < 1/2 < 20/15 < 9/7. Descending order: 9/7 > 20/15 > 1/2 > 4/20.

Question: Review Exercise 2, Question 1: Answer the following questions. (i) Define rational numbers and give examples of them.

Answer: A rational number is a number that can be written in the form p/q, where p and q are integers and q ≠ 0. Examples: 3/4, −2/5, 7, 0.

Question: Review Exercise 2, Question 1: (ii) Write the additive inverse of the rational number 'a'.

Answer: The additive inverse of the rational number a is −a, because a + (−a) = 0.

Question: Review Exercise 2, Question 1: (iii) What is the reciprocal of the rational number p/q, q ≠ 0?

Answer: The reciprocal of the rational number p/q is q/p (when p ≠ 0),

because (p/q) × (q/p) = 1.

Question: Review Exercise 2, Question 1: (iv) Write the sum of the two rational numbers p/q and r/s (where q and s are non-zero numbers).

Answer: p/q + r/s = (ps + rq)/(qs). Method: write both with the same denominator qs, then add the numerators.

Question: Review Exercise 2, Question 1: (v) What is the rule for finding the product of two rational numbers?

Answer: To multiply two rational numbers, multiply numerator by numerator and denominator by denominator: (a/b) × (c/d) = ac/bd, then simplify the answer.

Question: Review Exercise 2, Question 1: (vi) What are the inverse operations of addition and multiplication of rational numbers?

Answer: For rational numbers, the inverse operation of addition is subtraction (a − b = a + (−b)) and the inverse operation of multiplication is division (a ÷ b = a × 1/b, where b ≠ 0).

Question: Review Exercise 2, Question 2: Fill in the blanks. (i) Every improper ____ number.

Answer: Every improper fraction is a rational number. Blank: rational.

Question: Review Exercise 2, Question 2: (ii) The rational numbers p/q and −p/q are ____ inverses of each other.

Answer: p/q + (−p/q) = 0, so they are additive inverses of each other. Blank: additive.

Question: Review Exercise 2, Question 2: (iii) A number that can be shown in the form p/q is called ____, where p and q are integers.

Answer: Such a number is called a rational number. Blank: rational number.

Question: Review Exercise 2, Question 2: (iv) The rational number 0 has no ____.

Answer: Multiplying 0 by any number never gives 1, so 0 has no reciprocal (multiplicative inverse). Blank: reciprocal (multiplicative inverse).

Question: Review Exercise 2, Question 2: (v) The ____ inverse of a rational number is its reciprocal.

Answer: The multiplicative inverse of a rational number is its reciprocal. Blank: multiplicative.

Question: Review Exercise 2, Question 3: Put a (✓) mark on the correct answer. (i) If p, q ∈ Z and q ≠ 0, then a number that can be shown in the form p/q is called: (a) integer (b) rational number (c) whole number (d) positive number

Answer: A number in the form p/q is called a rational number. So the correct answer is (b) rational number. ✓

Question: Review Exercise 2, Question 3: (ii) The additive inverse of 2/3 is: (a) −2/3 (b) 3/2 (c) 1/3 (d) −3/2

Answer: (2/3) + (−2/3) = 0, so the additive inverse of 2/3 is −2/3. The correct answer is (a) −2/3. ✓

Question: Review Exercise 2, Question 3: (iii) The multiplicative inverse of −4/7 is: (a) 4/7 (b) 7/4 (c) −7/4 (d) 1/7

Answer: (−4/7) × (−7/4) = 1, so the multiplicative inverse of −4/7 is −7/4. The correct answer is (c) −7/4. ✓

Review Exercise 2, Question 3: (iv) 1/3 + 1/2 = _____. (a) 1/5 (b) 1/6 (c) 2/5 (d) 5/6

Answer: (2 + 3)/6 = 5/6. The correct answer is

Review Exercise 2, Question 3: (v) 2/5 + (−4/5) = _____. (a) 2 (b) −2 (c) −1/2 (d) −2/5

Answer: (2 − 4)/5 = −2/5. The correct answer is

Question: Review Exercise 2, Question 4: Show the following rational numbers on the number lines. (i) 1 1/2

Answer: 1 1/2 = 3/2, which lies between 1 and 2. On the number line, divide the part between 1 and 2 into 2 equal parts; the point at the end of the first part is 1 1/2.

Question: Review Exercise 2, Question 4: (ii) 3 1/3

Answer: 3 1/3 = 10/3, which lies between 3 and 4. On the number line, divide the part between 3 and 4 into 3 equal parts; the point at the end of the first part is 3 1/3.

Question: Review Exercise 2, Question 4: (iii) −1/4

Answer: −1/4 = −0.25, which lies between 0 and −1. On the number line, divide the part between 0 and −1 into 4 equal parts; the point at the end of the first part from 0 is −1/4.

Question: Review Exercise 2, Question 4: (iv) −1 4/5

Answer: −1 4/5 = −9/5, which lies between −1 and −2. On the number line, divide the part between −1 and −2 into 5 equal parts; the point at the end of the fourth part from −1 is −1 4/5.

Question: Review Exercise 2, Question 5: Find the additive and multiplicative inverses of the following rational numbers. (i) −14

Answer: Additive inverse: 14 (since (−14) + 14 = 0). Multiplicative inverse: −1/14 (since (−14) × (−1/14) = 1).

Question: Review Exercise 2, Question 5: (ii) 1/5

Answer: Additive inverse: −1/5 (since 1/5 + (−1/5) = 0). Multiplicative inverse: 5 (since (1/5) × 5 = 1).

Question: Review Exercise 2, Question 5: (iii) −2/3

Answer: Additive inverse: 2/3 (since (−2/3) + 2/3 = 0). Multiplicative inverse: −3/2 (since (−2/3) × (−3/2) = 1).

Question: Review Exercise 2, Question 5: (iv) −11/27

Answer: Additive inverse: 11/27 (since (−11/27) + 11/27 = 0). Multiplicative inverse: −27/11 (since (−11/27) × (−27/11) = 1).

Question: Review Exercise 2, Question 6: Put the correct sign > or < between the pairs of like rational numbers. (i) 2/3, 1/5

Answer: The LCM of 3 and 5 = 15. 2/3 = 10/15 and 1/5 = 3/15. Since 10 > 3, 2/3 > 1/5.

Question: Review Exercise 2, Question 6: (ii) 10/13, 11/14

Answer: The LCM of 13 and 14 = 182. 10/13 = 140/182 and 11/14 = 143/182. Since 140 < 143, 10/13 < 11/14.

Question: Review Exercise 2, Question 6: (iii) 1/4, −1/2

Answer: Every positive number is greater than every negative number, so 1/4 > −1/2.

Question: Review Exercise 2, Question 6: (iv) −11/17, 3/8

Answer: A negative number is always less than a positive number, so −11/17 < 3/8.

Question: Review Exercise 2, Question 6: (v) −4/9, −2/5

Answer: The LCM of 9 and 5 = 45. −4/9 = −20/45 and −2/5 = −18/45. Since −20 < −18, −4/9 < −2/5.

Question: Review Exercise 2, Question 6: (vi) −5/22, −11/25

Answer: The LCM of 22 and 25 = 550. −5/22 = −125/550 and −11/25 = −242/550. Since −125 > −242, −5/22 > −11/25.

Question: Review Exercise 2, Question 7: Solve the operations on the following rational numbers. (i) 1/2 + 1/3 + 1/6

Answer: LCM = 6. (3 + 2 + 1)/6 = 6/6 = 1.

Question: Review Exercise 2, Question 7: (ii) 2/7 + 1/2 − 3/14

Answer: LCM = 14. (4 + 7 − 3)/14 = 8/14 = 4/7.

Question: Review Exercise 2, Question 7: (iii) 5/8 − 1/5 − 3/4

Answer: LCM = 40. (25 − 8 − 30)/40 = −13/40.

Question: Review Exercise 2, Question 7: (iv) (−1/3) + (−1/4) + 1/2

Answer: LCM = 12. (−4 − 3 + 6)/12 = −1/12.

Question: Review Exercise 2, Question 7: (v) (−11/15) + (−3/5) + 5/4

Answer: LCM = 60. (−44 − 36 + 75)/60 = −5/60 = −1/12.

Question: Review Exercise 2, Question 7: (vi) (−19/55) + 51/55 + (−21/55)

Answer: The denominators are the same: (−19 + 51 − 21)/55 = 11/55 = 1/5.

Question: Review Exercise 2, Question 8: Simplify the operations on the following rational numbers. (i) (−1/100) ÷ (−1/10)

Answer: Change the division to multiplication: (−1/100) ÷ (−1/10) = (−1/100) × (−10/1) = 10/100 = 1/10.

Question: Review Exercise 2, Question 8: (ii) 1/5 × 2/3 × (−30/44)

Answer: (1 × 2 × (−30))/(5 × 3 × 44) = −60/660 = −1/11.

Question: Review Exercise 2, Question 8: (iii) 2/3 ÷ 16/21 × 27/49

Answer: From left to right: 2/3 ÷ 16/21 = 2/3 × 21/16 = 42/48 = 7/8; then 7/8 × 27/49 = 189/392 = 27/56.

Question: Review Exercise 2, Question 8: (iv) 8/21 ÷ 7/12

Answer: 8/21 ÷ 7/12 = 8/21 × 12/7 = 96/147 = 32/49.

Question: Review Exercise 2, Question 8: (v) 1/6 × (−2/3) ÷ (−11/63)

Answer: From left to right: 1/6 × (−2/3) = −2/18 = −1/9; then (−1/9) ÷ (−11/63) = (−1/9) × (−63/11) = 63/99 = 7/11.

Question: Review Exercise 2, Question 8: (vi) −2/7 ÷ 3/4 × 63/100

Answer: From left to right: −2/7 ÷ 3/4 = −2/7 × 4/3 = −8/21; then −8/21 × 63/100 = −504/2100 = −6/25.

Question: Review Exercise 2, Question 9: Prove on the following rational numbers. (i) (−1) + 35/54 = 35/54 + (−1)

Answer: Left side: (−1) + 35/54 = (−54 + 35)/54 = −19/54. Right side: 35/54 + (−1) = (35 − 54)/54 = −19/54. LHS = RHS, so the commutative property of addition is proved. ✓

Question: Review Exercise 2, Question 9: (ii) (−121/169) × (−13/11) =

(−13/11) × (−121/169)

Answer: Left side: (−121/169) × (−13/11) = (121 × 13)/(169 × 11) = 1573/1859 = 11/13. Right side: (−13/11) × (−121/169) = the same = 11/13. LHS = RHS, so the commutative property of multiplication is proved. ✓

Question: Review Exercise 2, Question 9: (iii) 4/9 × (2/3 × 5/7) = (4/9 × 2/3) × 5/7

Answer: Left side: 2/3 × 5/7 = 10/21; then 4/9 × 10/21 = 40/189. Right side: 4/9 × 2/3 = 8/27; then 8/27 × 5/7 = 40/189. LHS = RHS, so the associative property of multiplication is proved. ✓

Question: Review Exercise 2, Question 9: (iv) −4/5 × (1/8 + 11/12) = (−4/5 × 1/8) + (−4/5 × 11/12)

Answer: Left side: 1/8 + 11/12 = (3 + 22)/24 = 25/24; then −4/5 × 25/24 = −100/120 = −5/6. Right side: −4/5 × 1/8 = −4/40 = −1/10; −4/5 × 11/12 = −44/60 = −11/15; then −1/10 + (−11/15) = (−3 − 22)/30 = −25/30 = −5/6. LHS = RHS, so the distributive property of multiplication over addition is proved. ✓

Question: Review Exercise 2, Question 9: (v) −1/4 + (1/6 + 3/5) = (−1/4 + 1/6) + 3/5

Answer: Left side: 1/6 + 3/5 = (5 + 18)/30 = 23/30; then −1/4 + 23/30 = (−15 + 46)/60 = 31/60. Right side: −1/4 + 1/6 = (−3 + 2)/12 = −1/12; then −1/12 + 3/5 = (−5 + 36)/60 = 31/60. LHS = RHS, so the associative property of addition is proved. ✓

Question: Review Exercise 2, Question 9: (vi) 5/12 × (−2/7 − 2) = (5/12 × −2/7) − (5/12 × 2)

Answer: Left side: −2/7 − 2 = (−2 − 14)/7 = −16/7; then 5/12 × (−16/7) = −80/84 = −20/21. Right side: 5/12 × (−2/7) = −10/84 = −5/42; 5/12 × 2 = 10/12 = 5/6; then −5/42 − 5/6 = (−5 − 35)/42 = −40/42 = −20/21.

LHS = RHS, so the distributive property of multiplication over subtraction is proved. ✓

Question: Review Exercise 2, Question 10: Write the names of the properties used in the following rational numbers. (i) (−1/2) + 1/3 = 1/3 + (−1/2)

Answer: The order of the two numbers is changed and the answer stays the same. Property: commutative property of addition.

Question: Review Exercise 2, Question 10: (ii) (−10/11) × (−5/44) = (−5/44) × (−10/11)

Answer: The order of the two numbers is changed and the answer stays the same. Property: commutative property of multiplication.

Question: Review Exercise 2, Question 10: (iii) −2/3 × (7/8 × 9/14) = (−2/3 × 7/8) × 9/14

Answer: In the multiplication of three numbers the brackets are changed. Property: associative property of multiplication.

Question: Review Exercise 2, Question 10: (iv) −3/5 + (1/2 + 7/10) = (−3/5 + 1/2) + 7/10

Answer: In the addition of three numbers the brackets are changed. Property: associative property of addition.

Question: Review Exercise 2, Question 10: (v) −1/2 + (3/5 + 1/4) = (−1/2 + 3/5) + 1/4

Answer: In the addition of three numbers the brackets are changed. Property: associative property of addition.

Question: Review Exercise 2, Question 10: (vi) 2/3 × (1/2 + 5/6) = (2/3 × 1/2) + (2/3 × 5/6)

Answer: One number is multiplied by the sum of two numbers and the multiplication is distributed over each. Property: distributive property of multiplication over addition.

Question: Review Exercise 2, Question 10: (vii) (−12/105) × (−15/84) = (−15/84) × (−12/105)

Answer: The order of the two numbers is changed and the answer stays the same. Property: commutative property of multiplication.

Question: Review Exercise 2, Question 10: (viii) 1/4 × (8/9 − 12/15) = (1/4 × 8/9) − (1/4 × 12/15)

Answer: One number is multiplied by the difference of two numbers and the multiplication is distributed over each. Property: distributive property of multiplication over subtraction.

Question: Review Exercise 2, Question 10: (ix) −5/8 × (4/7 − 2/3) = (−5/8 × 4/7) − (−5/8 × 2/3)

Answer: One number is multiplied by the difference of two numbers and the multiplication is distributed over each. Property: distributive property of multiplication over subtraction.

Question: Review Exercise 2, Question 10: (x) 24/49 × (7/8 + 14/6) = (24/49 × 7/8) + (24/49 × 14/6)

Answer: One number is multiplied by the sum of two numbers and the multiplication is distributed over each. Property: distributive property of multiplication over addition.

Question: Fun Store: Look at the interesting facts about the number 37 and fill in the boxes. 37 × 3 = 111, 37 × 6 = 222, 37 × 9 = 333, 37 × (3 × 4) = 444, 37 × (3 × 5) = 555, 37 × □ = 666, 37 × □ = 777, 37 × □ = 888, 37 × □ = 999

Answer: The pattern is: 37 × 3 × n = 111 × n. So for 666: 37 × (3 × 6) = 37 × 18 = 666; for 777: 37 × (3 × 7) = 37 × 21 = 777; for 888: 37 × (3 × 8) = 37 × 24 = 888; for 999: 37 × (3 × 9) = 37 × 27 = 999. Boxes: 18, 21, 24, 27.

3. Decimal Fractions

Question: Exercise 3.1, Part (A): Convert the decimals into rational numbers and write the answer in lowest form. (1) 0.2

Answer: 0.2 = 2/10. Divide 2 and 10 by 2: 2/10 = 1/5. Answer: 1/5.

Question: Exercise 3.1, Part (A): (2) 0.4

Answer: 0.4 = 4/10. Divide 4 and 10 by 2: 4/10 = 2/5. Answer: 2/5.

Question: Exercise 3.1, Part (A): (3) 0.6

Answer: 0.6 = 6/10. Divide 6 and 10 by 2: 6/10 = 3/5. Answer: 3/5.

Question: Exercise 3.1, Part (A): (4) 0.8

Answer: 0.8 = 8/10. Divide 8 and 10 by 2: 8/10 = 4/5. Answer: 4/5.

Question: Exercise 3.1, Part (A): (5) 0.9

Answer: 0.9 = 9/10. 9 and 10 have no common divisor. Answer: 9/10.

Question: Exercise 3.1, Part (B): Convert the decimals into rational numbers and write the answer in lowest form. (1) 0.15

Answer: 0.15 = 15/100. Divide 15 and 100 by 5: 15/100 = 3/20. Answer: 3/20.

Question: Exercise 3.1, Part (B): (2) 0.35

Answer: 0.35 = 35/100. Divide 35 and 100 by 5: 35/100 = 7/20. Answer: 7/20.

Question: Exercise 3.1, Part (B): (3) 0.48

Answer: 0.48 = 48/100. Divide 48 and 100 by 4: 48/100 = 12/25. Answer: 12/25.

Question: Exercise 3.1, Part (B): (4) 0.75

Answer: 0.75 = 75/100. Divide 75 and 100 by 25: 75/100 = 3/4. Answer: 3/4.

Question: Exercise 3.1, Part (B): (5) 0.95

Answer: 0.95 = 95/100. Divide 95 and 100 by 5: 95/100 = 19/20. Answer: 19/20.

Question: Exercise 3.1, Part (B): (6) −1.30

Answer: −1.30 = −130/100. Divide 130 and 100 by 10: −130/100 = −13/10 = −1 3/10. Answer: −13/10 = −1 3/10.

Question: Exercise 3.1, Part (B): (7) −2.50

Answer: −2.50 = −250/100. Divide 250 and 100 by 50: −250/100 = −5/2 = −2 1/2. Answer: −5/2 = −2 1/2.

Question: Exercise 3.1, Part (B): (8) 5.05

Answer: 5.05 = 505/100. Divide 505 and 100 by 5: 505/100 = 101/20 = 5 1/20. Answer: 101/20 = 5 1/20.

Question: Exercise 3.1, Part (B): (9) 5.60

Answer: 5.60 = 560/100. Divide 560 and 100 by 20: 560/100 = 28/5 = 5 3/5. Answer: 28/5 = 5 3/5.

Question: Exercise 3.1, Part (B): (10) 6.85

Answer: 6.85 = 685/100. Divide 685 and 100 by 5: 685/100 = 137/20 = 6 17/20. Answer: 137/20 = 6 17/20.

Question: Exercise 3.1, Part (C): Convert the decimals into rational numbers and write the answer in lowest form. (1) 0.024

Answer: 0.024 = 24/1000. Divide 24 and 1000 by 8: 24/1000 = 3/125. Answer: 3/125.

Question: Exercise 3.1, Part (C): (2) 3.125

Answer: 3.125 = 3125/1000. Divide 3125 and 1000 by 125: 3125/1000 = 25/8 = 3 1/8. Answer: 25/8 = 3 1/8.

Question: Exercise 3.1, Part (C): (3) 0.375

Answer: 0.375 = 375/1000. Divide 375 and 1000 by 125: 375/1000 = 3/8. Answer: 3/8.

Question: Exercise 3.1, Part (C): (4) 4.065

Answer: 4.065 = 4065/1000. Divide 4065 and 1000 by 5: 4065/1000 = 813/200 = 4 13/200. Answer: 813/200 = 4 13/200.

Question: Exercise 3.1, Part (C): (5) 5.625

Answer: 5.625 = 5625/1000. Divide 5625 and 1000 by 125: 5625/1000 = 45/8 = 5 5/8. Answer: 45/8 = 5 5/8.

Question: Exercise 3.1, Part (C): (6) −0.001

Answer: −0.001 = −1/1000. 1 and 1000 have no common divisor. Answer: −1/1000.

Question: Exercise 3.1, Part (C): (7) −0.0005

Answer: −0.0005 = −5/10000. Divide 5 and 10000 by 5: −5/10000 = −1/2000. Answer: −1/2000.

Question: Exercise 3.1, Part (C): (8) −0.005

Answer: −0.005 = −5/1000. Divide 5 and 1000 by 5: −5/1000 = −1/200. Answer: −1/200.

Question: Exercise 3.1, Part (C): (9) −5.005

Answer: −5.005 = −5005/1000. Divide 5005 and 1000 by 5: −5005/1000 = −1001/200 = −5 1/200. Answer: −1001/200 = −5 1/200.

Question: Exercise 3.1, Part (C): (10) −11.375

Answer: −11.375 = −11375/1000. Divide 11375 and 1000 by 125: −11375/1000 = −91/8 = −11 3/8. Answer: −91/8 = −11 3/8.

Question: Exercise 3.1, Part (C): (11) −0.00125

Answer: −0.00125 = −125/100000. Divide 125 and 100000 by 125: −125/100000 = −1/800. Answer: −1/800.

Question: Exercise 3.1, Part (C): (12) −15.00625

Answer: −15.00625 = −1500625/100000. Divide 1500625 and 100000 by 625: −1500625/100000 = −2401/160 = −15 1/160. Answer: −2401/160 = −15 1/160.

Question: Practical Work: In the given table, divide each number and sort

it into terminating or non-terminating decimals. (i) 4/9

Answer: 4 ÷ 9 = 0.444… — the decimal never ends; 4 keeps repeating. So 4/9 = 0.444…, a non-terminating (recurring) decimal.

Question: Practical Work: (ii) 25/7

Answer: 25 ÷ 7 = 3.571428571428… — the decimal never ends; the group 571428 repeats. So 25/7 = 3.571428…, a non-terminating (recurring) decimal.

Question: Practical Work: (iii) 3/11

Answer: 3 ÷ 11 = 0.272727… — the decimal never ends; the group 27 repeats. So 3/11 = 0.2727…, a non-terminating (recurring) decimal.

Question: Exercise 3.2, Part I: Convert each given number into a decimal and state whether it is terminating or non-terminating. (1) 7/3

Answer: 7 ÷ 3 = 2.333… — the decimal never ends; 3 repeats. So 7/3 = 2.333…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (2) 11/6

Answer: 11 ÷ 6 = 1.8333… — the decimal never ends; 3 repeats. So 11/6 = 1.8333…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (3) −13/9

Answer: −13 ÷ 9 = −1.444… — the decimal never ends; 4 repeats. So −13/9 = −1.444…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (4) 15/7

Answer: 15 ÷ 7 = 2.142857142857… — the decimal never ends; the group 142857 repeats. So 15/7 = 2.142857…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (5) 14/9

Answer: 14 ÷ 9 = 1.555… — the decimal never ends; 5 repeats. So 14/9 = 1.555…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (6) −17/11

Answer: −17 ÷ 11 = −1.545454… — the decimal never ends; the group

54 repeats. So −17/11 = −1.5454..., a non-terminating decimal.

Question: Exercise 3.2, Part I: (7) 19/12

Answer: 19 ÷ 12 = 1.58333... — the decimal never ends; 3 repeats. So 19/12 = 1.58333..., a non-terminating decimal.

Question: Exercise 3.2, Part I: (8) 21/13

Answer: 21 ÷ 13 = 1.615384615384... — the decimal never ends; the group 615384 repeats. So 21/13 = 1.615384..., a non-terminating decimal.

Question: Exercise 3.2, Part I: (9) −23/14

Answer: −23 ÷ 14 = −1.642857142857... — the decimal never ends; the group 642857 repeats. So −23/14 = −1.642857..., a non-terminating decimal.

Question: Exercise 3.2, Part I: (10) −29/15

Answer: −29 ÷ 15 = −1.9333... — the decimal never ends; 3 repeats. So −29/15 = −1.9333..., a non-terminating decimal.

Question: Exercise 3.2, Part I: (11) 31/17

Answer: 31 ÷ 17 = 1.823529411764... — the decimal never ends; the group 8235294117647058 repeats. So 31/17 = 1.823529..., a non-terminating decimal.

Question: Exercise 3.2, Part I: (12) −35/18

Answer: −35 ÷ 18 = −1.9444... — the decimal never ends; 4 repeats. So −35/18 = −1.9444..., a non-terminating decimal.

Question: Exercise 3.2, Part I: (13) 47/19

Answer: 47 ÷ 19 = 2.473684210526... — the decimal never ends; the group 4736842105263157 repeats. So 47/19 = 2.473684..., a non-terminating decimal.

Question: Exercise 3.2, Part I: (14) 35/23

Answer: 35 ÷ 23 = 1.521739130434... — the decimal never ends; the group 5217391304347826 repeats. So 35/23 = 1.521739..., a

non-terminating decimal.

Question: Exercise 3.2, Part I: (15) −20/21

Answer: −20 ÷ 21 = −0.952380952380… — the decimal never ends; the group 952380 repeats. So −20/21 = −0.952380…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (16) 39/22

Answer: 39 ÷ 22 = 1.77272727… — the decimal never ends; the group 72 repeats. So 39/22 = 1.772727…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (17) −41/29

Answer: −41 ÷ 29 = −1.413793103448… — the decimal never ends; the group 4137931034482758 repeats. So −41/29 = −1.413793…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (18) 40/31

Answer: 40 ÷ 31 = 1.290322580645… — the decimal never ends; the group 2903225806451612 repeats. So 40/31 = 1.290322…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (19) −50/37

Answer: −50 ÷ 37 = −1.351351351… — the decimal never ends; the group 351 repeats. So −50/37 = −1.351351…, a non-terminating decimal.

Question: Exercise 3.2, Part I: (20) 101/41

Answer: 101 ÷ 41 = 2.463414634146… — the decimal never ends; the group 46341 repeats. So 101/41 = 2.463414…, a non-terminating decimal.

Question: Exercise 3.2, Part II: Convert each given number into a decimal and state whether it is recurring or not. (1) 19/20

Answer: 19 ÷ 20 = 0.95 — the decimal ends; no digit repeats. So 19/20 = 0.95, a terminating decimal, not recurring.

Question: Exercise 3.2, Part II: (2) 17/6

Answer: 17 ÷ 6 = 2.8333… — the decimal never ends; 3 repeats. So 17/6

= 2.8333..., a recurring decimal (recurring digit 3).

Question: Exercise 3.2, Part II: (3) 15/8

Answer: 15 ÷ 8 = 1.875 — the decimal ends; no digit repeats. So 15/8 = 1.875, a terminating decimal, not recurring.

Question: Exercise 3.2, Part II: (4) 25/10

Answer: 25 ÷ 10 = 2.5 — the decimal ends; no digit repeats. So 25/10 = 2.5, a terminating decimal, not recurring.

Question: Exercise 3.2, Part II: (5) 2/11

Answer: 2 ÷ 11 = 0.181818... — the decimal never ends; the group 18 repeats. So 2/11 = 0.1818..., a recurring decimal (recurring group 18).

Question: Exercise 3.2, Part II: (6) 10/3

Answer: 10 ÷ 3 = 3.333... — the decimal never ends; 3 repeats. So 10/3 = 3.333..., a recurring decimal (recurring digit 3).

Question: Exercise 3.2, Part II: (7) 11/12

Answer: 11 ÷ 12 = 0.91666... — the decimal never ends; 6 repeats. So 11/12 = 0.91666..., a recurring decimal (recurring digit 6).

Question: Exercise 3.2, Part II: (8) 16/7

Answer: 16 ÷ 7 = 2.285714285714... — the decimal never ends; the group 285714 repeats. So 16/7 = 2.285714..., a recurring decimal (recurring group 285714).

Question: Exercise 3.2, Part II: (9) 2/13

Answer: 2 ÷ 13 = 0.153846153846... — the decimal never ends; the group 153846 repeats. So 2/13 = 0.153846..., a recurring decimal (recurring group 153846).

Question: Exercise 3.2, Part II: (10) 4/25

Answer: 4 ÷ 25 = 0.16 — the decimal ends; no digit repeats. So 4/25 = 0.16, a terminating decimal, not recurring.

Question: Exercise 3.3, Question 1: Without dividing, classify the following

as terminating and non-terminating decimals. (1) 12/25

Answer: First simplify: 12/25 is already in lowest form. Denominator 25 = 5 × 5, which has only 5. So 12/25 is terminating: 12/25 = 0.48.

Question: Exercise 3.3, Question 1: (2) 43/21

Answer: 43/21 is already in lowest form. Denominator 21 = 3 × 7, which has primes other than 2 or 5. So 43/21 is non-terminating: 43/21 = 2.047619... .

Question: Exercise 3.3, Question 1: (3) 5/12

Answer: 5/12 is already in lowest form. Denominator 12 = 2 × 2 × 3, which includes 3. So 5/12 is non-terminating: 5/12 = 0.41666... .

Question: Exercise 3.3, Question 1: (4) 17/20

Answer: 17/20 is already in lowest form. Denominator 20 = 2 × 2 × 5, which has only 2 and 5. So 17/20 is terminating: 17/20 = 0.85.

Question: Exercise 3.3, Question 1: (5) 117/125

Answer: 117/125 is already in lowest form. Denominator 125 = 5 × 5 × 5, which has only 5. So 117/125 is terminating: 117/125 = 0.936.

Question: Exercise 3.3, Question 1: (6) 5/7

Answer: 5/7 is already in lowest form. Denominator 7, which is neither 2 nor 5. So 5/7 is non-terminating: 5/7 = 0.714285... .

Question: Exercise 3.3, Question 1: (7) −9/40

Answer: −9/40 is already in lowest form. Denominator 40 = 2 × 2 × 2 × 5, which has only 2 and 5. So −9/40 is terminating: −9/40 = −0.225.

Question: Exercise 3.3, Question 1: (8) 10/33

Answer: 10/33 is already in lowest form. Denominator 33 = 3 × 11, which has primes other than 2 or 5. So 10/33 is non-terminating: 10/33 = 0.3030... .

Question: Exercise 3.3, Question 1: (9) 23/60

Answer: 23/60 is already in lowest form. Denominator 60 = 2 × 2 × 3 × 5,

which includes 3. So 23/60 is non-terminating: 23/60 = 0.38333... .

Question: Exercise 3.3, Question 1: (10) 40/35

Answer: First simplify: 40/35 = 8/7. Denominator 7, which is neither 2 nor 5. So 40/35 is non-terminating: 40/35 = 1.142857... .

Question: Exercise 3.3, Question 1: (11) 55/75

Answer: First simplify: 55/75 = 11/15. Denominator 15 = 3 × 5, which includes 3. So 55/75 is non-terminating: 55/75 = 0.7333... .

Question: Exercise 3.3, Question 1: (12) −96/100

Answer: First simplify: −96/100 = −24/25. Denominator 25 = 5 × 5, which has only 5. So −96/100 is terminating: −96/100 = −0.96.

Question: Exercise 3.3, Question 1: (13) 101/125

Answer: 101/125 is already in lowest form. Denominator 125 = 5 × 5 × 5, which has only 5. So 101/125 is terminating: 101/125 = 0.808.

Question: Exercise 3.3, Question 1: (14) 111/96

Answer: First simplify: 111/96 = 37/32. Denominator 32 = 2 × 2 × 2 × 2 × 2, which has only 2. So 111/96 is terminating: 111/96 = 1.15625.

Question: Exercise 3.3, Question 1: (15) 125/200

Answer: First simplify: 125/200 = 5/8. Denominator 8 = 2 × 2 × 2, which has only 2. So 125/200 is terminating: 125/200 = 0.625.

Question: Exercise 3.3, Question 1: (16) 141/144

Answer: First simplify: 141/144 = 47/48. Denominator 48 = 2 × 2 × 2 × 2 × 3, which includes 3. So 141/144 is non-terminating: 141/144 = 0.979166... .

Question: Exercise 3.3, Question 1: (17) 200/201

Answer: 200/201 is already in lowest form. Denominator 201 = 3 × 67, which has primes other than 2 or 5. So 200/201 is non-terminating: 200/201 = 0.995024... .

Question: Exercise 3.3, Question 1: (18) −210/147

Answer: First simplify: −210/147 = −10/7. Denominator 7, which is neither 2 nor 5. So −210/147 is non-terminating: −210/147 = −1.428571… .

Question: Exercise 3.3, Question 1: (19) 372/400

Answer: First simplify: 372/400 = 93/100. Denominator 100 = 2 × 2 × 5 × 5, which has only 2 and 5. So 372/400 is terminating: 372/400 = 0.93.

Question: Exercise 3.3, Question 1: (20) −401/333

Answer: −401/333 is already in lowest form. Denominator 333 = 3 × 3 × 37, which has primes other than 2 or 5. So −401/333 is non-terminating: −401/333 = −1.204204… .

Question: Exercise 3.4, Question 1: Convert the following rational numbers into decimals and state whether each is terminating or non-terminating recurring. (1) 5/14

Answer: 5 ÷ 14 = 0.357142857142… — the decimal never ends; the group 571428 repeats. So 5/14 = 0.357142…, a non-terminating recurring decimal.

Question: Exercise 3.4, Question 1: (2) −13/18

Answer: −13 ÷ 18 = −0.7222… — the decimal never ends; 2 repeats. So −13/18 = −0.7222…, a non-terminating recurring decimal.

Question: Exercise 3.4, Question 1: (3) 19/20

Answer: 19 ÷ 20 = 0.95 — the decimal ends. So 19/20 = 0.95, a terminating decimal.

Question: Exercise 3.4, Question 1: (4) −3 21/22

Answer: −3 21/22 = −87/22. −87 ÷ 22 = −3.954545… — the decimal never ends; the group 54 repeats. So −3 21/22 = −3.954545…, a non-terminating recurring decimal.

Question: Exercise 3.4, Question 1: (5) 6 17/30

Answer: 6 17/30 = 197/30. 197 ÷ 30 = 6.5666… — the decimal never ends; 6 repeats. So 6 17/30 = 6.5666…, a non-terminating recurring

decimal.

Question: Exercise 3.4, Question 1: (6) 23/40

Answer: 23 ÷ 40 = 0.575 — the decimal ends. So 23/40 = 0.575, a terminating decimal.

Question: Exercise 3.4, Question 1: (7) 3 27/32

Answer: 3 27/32 = 123/32. 123 ÷ 32 = 3.84375 — the decimal ends. So 3 27/32 = 3.84375, a terminating decimal.

Question: Exercise 3.4, Question 1: (8) −3 49/50

Answer: −3 49/50 = −199/50. −199 ÷ 50 = −3.98 — the decimal ends. So −3 49/50 = −3.98, a terminating decimal.

Question: Exercise 3.4, Question 1: (9) 2 45/48

Answer: 2 45/48 = 141/48 = 47/16. 47 ÷ 16 = 2.9375 — the decimal ends. So 2 45/48 = 2.9375, a terminating decimal.

Question: Exercise 3.4, Question 1: (10) 91/65

Answer: 91/65 = 7/5. 7 ÷ 5 = 1.4 — the decimal ends. So 91/65 = 1.4, a terminating decimal.

Question: Exercise 3.4, Question 1: (11) −153/85

Answer: −153/85 = −9/5. −9 ÷ 5 = −1.8 — the decimal ends. So −153/85 = −1.8, a terminating decimal.

Question: Exercise 3.4, Question 1: (12) 266/171

Answer: 266/171 = 14/9. 14 ÷ 9 = 1.555… — the decimal never ends; 5 repeats. So 266/171 = 1.555…, a non-terminating recurring decimal.

Question: Exercise 3.4, Question 1: (13) 272/288

Answer: 272/288 = 17/18. 17 ÷ 18 = 0.9444… — the decimal never ends; 4 repeats. So 272/288 = 0.9444…, a non-terminating recurring decimal.

Question: Practical Work: In the given table, round each decimal to 2 decimal places, 3 decimal places and 4 decimal places. (2) 2.421875

Answer: Method: look at the digit after the required place; if it is 5 or more,

increase the last kept digit by 1, otherwise it stays the same. 2.421875 → 2 dp: 2.42; 3 dp: 2.422; 4 dp: 2.4219.

Question: Practical Work: (3) 0.83333

Answer: 0.83333 → 2 dp: 0.83; 3 dp: 0.833; 4 dp: 0.8333.

Question: Practical Work: (4) 1.76794

Answer: 1.76794 → 2 dp: 1.77; 3 dp: 1.768; 4 dp: 1.7679.

Question: Practical Work: (5) 0.9916667

Answer: 0.9916667 → 2 dp: 0.99; 3 dp: 0.992; 4 dp: 0.9917.

Question: Practical Work: (6) 0.96257

Answer: 0.96257 → 2 dp: 0.96; 3 dp: 0.963; 4 dp: 0.9626.

Question: Practical Work: (7) 1.61538

Answer: 1.61538 → 2 dp: 1.62; 3 dp: 1.615; 4 dp: 1.6154.

Question: Practical Work: (8) 8.08547

Answer: 8.08547 → 2 dp: 8.09; 3 dp: 8.085; 4 dp: 8.0855.

Question: Practical Work: (9) 6.98677

Answer: 6.98677 → 2 dp: 6.99; 3 dp: 6.987; 4 dp: 6.9868.

Question: Practical Work: (10) 15.47648

Answer: 15.47648 → 2 dp: 15.48; 3 dp: 15.476; 4 dp: 15.4765.

Question: Exercise 3.5, Part I: Convert each given number into a decimal, state whether it is terminating or non-terminating, and round it to 3 decimal places. (1) 23/400

Answer: 23 ÷ 400 = 0.0575 — the decimal ends, so it is terminating. To 3 decimal places: 0.058 (because the fourth digit is 5, the third digit increases by 1).

Question: Exercise 3.5, Part I: (2) 27/64

Answer: 27 ÷ 64 = 0.421875 — the decimal ends, so it is terminating. To 3 decimal places: 0.422 (because the fourth digit is 8, the third digit increases by 1).

Question: Exercise 3.5, Part I: (3) 53/60

Answer: 53 ÷ 60 = 0.88333… — the decimal never ends, so it is non-terminating. To 3 decimal places: 0.883 (because the fourth digit is 3, the third digit stays the same).

Question: Exercise 3.5, Part I: (4) −46/75

Answer: −46 ÷ 75 = −0.61333… — the decimal never ends, so it is non-terminating. To 3 decimal places: −0.613 (because the fourth digit is 3, the third digit stays the same).

Question: Exercise 3.5, Part I: (5) 69/70

Answer: 69 ÷ 70 = 0.985714… — the decimal never ends, so it is non-terminating. To 3 decimal places: 0.986 (because the fourth digit is 7, the third digit increases by 1).

Question: Exercise 3.5, Part I: (6) 77/80

Answer: 77 ÷ 80 = 0.9625 — the decimal ends, so it is terminating. To 3 decimal places: 0.963 (because the fourth digit is 5, the third digit increases by 1).

Question: Exercise 3.5, Part I: (7) −43/90

Answer: −43 ÷ 90 = −0.4777… — the decimal never ends, so it is non-terminating. To 3 decimal places: −0.478 (because the fourth digit is 7, the third digit increases by 1).

Question: Exercise 3.5, Part I: (8) 119/120

Answer: 119 ÷ 120 = 0.99166… — the decimal never ends, so it is non-terminating. To 3 decimal places: 0.992 (because the fourth digit is 6, the third digit increases by 1).

Question: Exercise 3.5, Part I: (9) 246/300

Answer: 246 ÷ 300 = 0.82 — the decimal ends, so it is terminating. To 3 decimal places: 0.820 (because there is no fourth digit, the third digit stays 0).

Question: Exercise 3.5, Part I: (10) −81/96

Answer: −81 ÷ 96 = −0.84375 — the decimal ends, so it is terminating. To 3 decimal places: −0.844 (because the fourth digit is 7, the third digit increases by 1).

Question: Exercise 3.5, Part II: By division, convert each given number into a decimal, state whether it is terminating or non-terminating, and round it to 2 decimal places. (1) 4/15

Answer: 4 ÷ 15 = 0.2666… — the decimal never ends, so it is non-terminating. To 2 decimal places: 0.27 (because the third digit is 6, the second digit increases by 1).

Question: Exercise 3.5, Part II: (2) −11/12

Answer: −11 ÷ 12 = −0.91666… — the decimal never ends, so it is non-terminating. To 2 decimal places: −0.92 (because the third digit is 6, the second digit increases by 1).

Question: Exercise 3.5, Part II: (3) 8/9

Answer: 8 ÷ 9 = 0.888… — the decimal never ends, so it is non-terminating. To 2 decimal places: 0.89 (because the third digit is 8, the second digit increases by 1).

Question: Exercise 3.5, Part II: (4) 22/7

Answer: 22 ÷ 7 = 3.142857… — the decimal never ends, so it is non-terminating. To 2 decimal places: 3.14 (because the third digit is 2, the second digit stays the same).

Question: Exercise 3.5, Part II: (5) −21/16

Answer: −21 ÷ 16 = −1.3125 — the decimal ends, so it is terminating. To 2 decimal places: −1.31 (because the third digit is 2, the second digit stays the same).

Question: Exercise 3.5, Part II: (6) 3/40

Answer: 3 ÷ 40 = 0.075 — the decimal ends, so it is terminating. To 2

decimal places: 0.08 (because the third digit is 5, the second digit increases by 1).

Question: Exercise 3.5, Part II: (7) −18/45

Answer: −18 ÷ 45 = −0.4 — the decimal ends, so it is terminating. To 2 decimal places: −0.40 (because there is no third digit, the second digit stays 0).

Question: Exercise 3.5, Part II: (8) 21/66

Answer: 21/66 = 7/22. 7 ÷ 22 = 0.31818… — the decimal never ends, so it is non-terminating. To 2 decimal places: 0.32 (because the third digit is 8, the second digit increases by 1).

Question: Exercise 3.5, Part II: (9) 33/90

Answer: 33/90 = 11/30. 11 ÷ 30 = 0.3666… — the decimal never ends, so it is non-terminating. To 2 decimal places: 0.37 (because the third digit is 6, the second digit increases by 1).

Question: Exercise 3.5, Part II: (10) −81/101

Answer: −81 ÷ 101 = −0.801980… — the decimal never ends, so it is non-terminating. To 2 decimal places: −0.80 (because the third digit is 1, the second digit stays the same).

Question: Exercise 3.5, Part III: Round the following non-terminating recurring decimals to 3 decimal places. (1) 10/11

Answer: 10 ÷ 11 = 0.9090… — non-terminating recurring (recurring group 90). To 3 decimal places: 0.909 (because the fourth digit is 0, the third digit stays the same).

Question: Exercise 3.5, Part III: (2) −41/90

Answer: −41 ÷ 90 = −0.4555… — non-terminating recurring (recurring digit 5). To 3 decimal places: −0.456 (because the fourth digit is 5, the third digit increases by 1).

Question: Exercise 3.5, Part III: (3) 21/13

Answer: 21 ÷ 13 = 1.615384… — non-terminating recurring (recurring group 615384). To 3 decimal places: 1.615 (because the fourth digit is 3, the third digit stays the same).

Question: Exercise 3.5, Part III: (4) 17/19

Answer: 17 ÷ 19 = 0.894736… — non-terminating recurring (recurring group 894736842105263 1). To 3 decimal places: 0.895 (because the fourth digit is 7, the third digit increases by 1).

Question: Exercise 3.5, Part III: (5) 9/7

Answer: 9 ÷ 7 = 1.285714… — non-terminating recurring (recurring group 285714). To 3 decimal places: 1.286 (because the fourth digit is 7, the third digit increases by 1).

Question: Exercise 3.5, Part III: (6) 15/23

Answer: 15 ÷ 23 = 0.652173… — non-terminating recurring (recurring group 652173913043478 2). To 3 decimal places: 0.652 (because the fourth digit is 1, the third digit stays the same).

Question: Exercise 3.5, Part III: (7) −111/96

Answer: −111/96 = −37/32 = −1.15625 — the decimal ends, so it is terminating (not recurring). To 3 decimal places: −1.156 (because the fourth digit is 2, the third digit stays the same).

Question: Exercise 3.5, Part III: (8) 22/29

Answer: 22 ÷ 29 = 0.758620… — non-terminating recurring (recurring group 7586206896551724). To 3 decimal places: 0.759 (because the fourth digit is 6, the third digit increases by 1).

Question: Exercise 3.5, Part IV: Round the following decimals to 2 decimal places. (1) 0.4555

Answer: The third digit is 5, so the second digit increases by 1: 0.4555 → 0.46.

Question: Exercise 3.5, Part IV: (2) 0.3648

Answer: The third digit is 4, so the second digit stays the same: 0.3648 → 0.36.

Question: Exercise 3.5, Part IV: (3) 0.6666

Answer: The third digit is 6, so the second digit increases by 1: 0.6666 → 0.67.

Question: Exercise 3.5, Part IV: (4) 0.1054

Answer: The third digit is 5, so the second digit increases by 1: 0.1054 → 0.11.

Question: Exercise 3.5, Part IV: (5) 1.9091

Answer: The third digit is 9, so the second digit increases by 1: 1.9091 → 1.91.

Question: Exercise 3.5, Part IV: (6) 3.2479

Answer: The third digit is 7, so the second digit increases by 1: 3.2479 → 3.25.

Question: Exercise 3.5, Part IV: (7) 2.0518

Answer: The third digit is 1, so the second digit stays the same: 2.0518 → 2.05.

Question: Exercise 3.5, Part IV: (8) 8.3381

Answer: The third digit is 8, so the second digit increases by 1: 8.3381 → 8.34.

Question: Exercise 3.5, Part IV: (9) 10.1036

Answer: The third digit is 3, so the second digit stays the same: 10.1036 → 10.10.

Question: Exercise 3.5, Part IV: (10) 3.9995

Answer: The third digit is 9, so the second digit increases by 1 (9 + 1 = 10, then carry): 3.9995 → 4.00.

Question: Exercise 3.5, Part IV: (11) 13.0967

Answer: The third digit is 6, so the second digit increases by 1: 13.0967 →

13.10.

Question: Exercise 3.5, Part IV: (12) 15.8159

Answer: The third digit is 5, so the second digit increases by 1: 15.8159 → 15.82.

Question: Exercise 3.5, Part V: Round the following decimals to 3 decimal places. (1) 4.51689

Answer: The fourth digit is 8, so the third digit increases by 1: 4.51689 → 4.517.

Question: Exercise 3.5, Part V: (2) 6.7472

Answer: The fourth digit is 2, so the third digit stays the same: 6.7472 → 6.747.

Question: Exercise 3.5, Part V: (3) 0.0098

Answer: The fourth digit is 8, so the third digit increases by 1: 0.0098 → 0.010.

Question: Exercise 3.5, Part V: (4) 0.9861

Answer: The fourth digit is 1, so the third digit stays the same: 0.9861 → 0.986.

Question: Exercise 3.5, Part V: (5) 206.4176

Answer: The fourth digit is 6, so the third digit increases by 1: 206.4176 → 206.418.

Question: Exercise 3.5, Part V: (6) 3.4073

Answer: The fourth digit is 3, so the third digit stays the same: 3.4073 → 3.407.

Question: Exercise 3.5, Part V: (7) 102.99909

Answer: The fourth digit is 0, so the third digit stays the same: 102.99909 → 102.999.

Question: Exercise 3.5, Part V: (8) 11.1234

Answer: The fourth digit is 4, so the third digit stays the same: 11.1234 →

11.123.

Question: Exercise 3.5, Part V: (9) 55.12345

Answer: The fourth digit is 4, so the third digit stays the same: 55.12345 → 55.123.

Question: Exercise 3.5, Part V: (10) 4.10539

Answer: The fourth digit is 3, so the third digit stays the same: 4.10539 → 4.105.

Question: Exercise 3.5, Part V: (11) 66.3957

Answer: The fourth digit is 7, so the third digit increases by 1: 66.3957 → 66.396.

Question: Exercise 3.5, Part V: (12) 59.9196

Answer: The fourth digit is 6, so the third digit increases by 1: 59.9196 → 59.920.

Question: Review Exercise 3, Question 1: Convert the following decimals into rational numbers and write the answer in lowest form. (i) 0.325

Answer: 0.325 = 325/1000. Divide 325 and 1000 by 25: 325/1000 = 13/40. Answer: 13/40.

Question: Review Exercise 3, Question 1: (ii) 0.75

Answer: 0.75 = 75/100. Divide 75 and 100 by 25: 75/100 = 3/4. Answer: 3/4.

Question: Review Exercise 3, Question 1: (iii) 2.5

Answer: 2.5 = 25/10. Divide 25 and 10 by 5: 25/10 = 5/2. Answer: 5/2.

Question: Review Exercise 3, Question 1: (iv) 7.75

Answer: 7.75 = 775/100. Divide 775 and 100 by 25: 775/100 = 31/4. Answer: 31/4.

Question: Review Exercise 3, Question 1: (v) 0.78

Answer: 0.78 = 78/100. Divide 78 and 100 by 2: 78/100 = 39/50. Answer: 39/50.

Question: Review Exercise 3, Question 1: (vi) 1.297

Answer: 1.297 = 1297/1000. 1297 and 1000 have no common divisor. Answer: 1297/1000.

Question: Review Exercise 3, Question 1: (vii) 2.348

Answer: 2.348 = 2348/1000. Divide 2348 and 1000 by 4: 2348/1000 = 587/250. Answer: 587/250.

Question: Review Exercise 3, Question 2: Convert the following rational numbers into decimals and state whether each is terminating or non-terminating recurring. (i) 41/5

Answer: 41 ÷ 5 = 8.2 — the decimal ends. So 41/5 = 8.2, a terminating decimal.

Question: Review Exercise 3, Question 2: (ii) 111/12

Answer: 111/12 = 37/4. 37 ÷ 4 = 9.25 — the decimal ends. So 111/12 = 9.25, a terminating decimal.

Question: Review Exercise 3, Question 2: (iii) 28/9

Answer: 28 ÷ 9 = 3.111… — the decimal never ends; 1 repeats. So 28/9 = 3.111…, a non-terminating recurring decimal.

Question: Review Exercise 3, Question 2: (iv) 10/7

Answer: 10 ÷ 7 = 1.428571428571… — the decimal never ends; the group 428571 repeats. So 10/7 = 1.428571…, a non-terminating recurring decimal.

Question: Review Exercise 3, Question 2: (v) 22/7

Answer: 22 ÷ 7 = 3.142857142857… — the decimal never ends; the group 142857 repeats. So 22/7 = 3.142857…, a non-terminating recurring decimal.

Question: Review Exercise 3, Question 2: (vi) 21/16

Answer: 21 ÷ 16 = 1.3125 — the decimal ends. So 21/16 = 1.3125, a terminating decimal.

Question: Review Exercise 3, Question 2: (vii) 73/10

Answer: 73 ÷ 10 = 7.3 — the decimal ends. So 73/10 = 7.3, a terminating decimal.

Question: Review Exercise 3, Question 3: Round the following decimals to 2 decimal places. (i) 14.5828

Answer: The third digit is 2, so the second digit stays the same: 14.5828 → 14.58.

Question: Review Exercise 3, Question 3: (ii) 107.8287

Answer: The third digit is 8, so the second digit increases by 1: 107.8287 → 107.83. Note: the decimal point is missing in the printed book ("1078287" is printed); it was read as 107.8287, matching the pattern of the neighbouring numbers.

Question: Review Exercise 3, Question 3: (iii) 5.7895

Answer: The third digit is 9, so the second digit increases by 1: 5.7895 → 5.79.

Question: Review Exercise 3, Question 3: (iv) 6.7989

Answer: The third digit is 8, so the second digit increases by 1: 6.7989 → 6.80.

Question: Review Exercise 3, Question 3: (v) 25.4897

Answer: The third digit is 9, so the second digit increases by 1: 25.4897 → 25.49.

Question: Review Exercise 3, Question 4: Split the following rational numbers into terminating and non-terminating. (i) 13/4

Answer: 13 ÷ 4 = 3.25 — the decimal ends. So 13/4 = 3.25, a terminating decimal.

Question: Review Exercise 3, Question 4: (ii) 17/25

Answer: 17 ÷ 25 = 0.68 — the decimal ends. So 17/25 = 0.68, a terminating decimal.

Question: Review Exercise 3, Question 4: (iii) 80/3

Answer: 80 ÷ 3 = 26.666… — the decimal never ends. So 80/3 = 26.666…, a non-terminating decimal.

Question: Review Exercise 3, Question 4: (iv) 15/11

Answer: 15 ÷ 11 = 1.3636… — the decimal never ends. So 15/11 = 1.3636…, a non-terminating decimal.

Question: Review Exercise 3, Question 4: (v) 19/6

Answer: 19 ÷ 6 = 3.1666… — the decimal never ends. So 19/6 = 3.1666…, a non-terminating decimal.

Question: Review Exercise 3, Question 4: (vi) 25/15

Answer: 25/15 = 5/3. 5 ÷ 3 = 1.666… — the decimal never ends. So 25/15 = 1.666…, a non-terminating decimal.

Question: Review Exercise 3, Question 4: (vii) 22/7

Answer: 22 ÷ 7 = 3.142857… — the decimal never ends. So 22/7 = 3.142857…, a non-terminating decimal.

Question: Review Exercise 3, Question 4: (viii) 14/9

Answer: 14 ÷ 9 = 1.555… — the decimal never ends. So 14/9 = 1.555…, a non-terminating decimal.

Question: Review Exercise 3, Question 5: Convert the following decimals into rational numbers. (i) 0.21

Answer: 0.21 = 21/100. 21 and 100 have no common divisor. Answer: 21/100.

Question: Review Exercise 3, Question 5: (ii) 6.35

Answer: 6.35 = 635/100. Divide 635 and 100 by 5: 635/100 = 127/20. Answer: 127/20.

Question: Review Exercise 3, Question 5: (iii) 0.52

Answer: 0.52 = 52/100. Divide 52 and 100 by 4: 52/100 = 13/25. Answer: 13/25.

Question: Review Exercise 3, Question 5: (iv) 0.62

Answer: 0.62 = 62/100. Divide 62 and 100 by 2: 62/100 = 31/50. Answer: 31/50.

Question: Review Exercise 3, Question 5: (v) 0.145

Answer: 0.145 = 145/1000. Divide 145 and 1000 by 5: 145/1000 = 29/200. Answer: 29/200.

Question: Review Exercise 3, Question 5: (vi) 3.125

Answer: 3.125 = 3125/1000. Divide 3125 and 1000 by 125: 3125/1000 = 25/8. Answer: 25/8.

Question: Review Exercise 3, Question 5: (vii) 1.4375

Answer: 1.4375 = 14375/10000. Divide 14375 and 10000 by 625: 14375/10000 = 23/16. Answer: 23/16.

Question: Review Exercise 3, Question 5: (viii) 1.375

Answer: 1.375 = 1375/1000. Divide 1375 and 1000 by 125: 1375/1000 = 11/8. Answer: 11/8.

Question: Review Exercise 3, Question 5: (ix) 1.5

Answer: 1.5 = 15/10. Divide 15 and 10 by 5: 15/10 = 3/2. Answer: 3/2.

Question: Review Exercise 3, Question 6: Round the following non-terminating recurring decimals to 3 decimal places. (i) 40/3

Answer: 40 ÷ 3 = 13.333… — the fourth digit is 3, so the third digit stays the same: 40/3 → 13.333.

Question: Review Exercise 3, Question 6: (ii) 20/7

Answer: 20 ÷ 7 = 2.857142… — the fourth digit is 1, so the third digit stays the same: 20/7 → 2.857.

Question: Review Exercise 3, Question 6: (iii) 50/11

Answer: 50 ÷ 11 = 4.545454… — the fourth digit is 4, so the third digit stays the same: 50/11 → 4.545.

Question: Review Exercise 3, Question 6: (iv) 80/13

Answer: 80 ÷ 13 = 6.153846… — the fourth digit is 8, so the third digit increases by 1: 80/13 → 6.154.

Question: Review Exercise 3, Question 6: (v) 100/6

Answer: 100 ÷ 6 = 16.666… — the fourth digit is 6, so the third digit increases by 1: 100/6 → 16.667.

Question: Review Exercise 3, Question 6: (vi) 240/22

Answer: 240 ÷ 22 = 10.909090… — the fourth digit is 0, so the third digit stays the same: 240/22 → 10.909.

Question: Review Exercise 3, Question 6: (vii) 70/12

Answer: 70 ÷ 12 = 5.8333… — the fourth digit is 3, so the third digit stays the same: 70/12 → 5.833.

Question: Review Exercise 3, Question 6: (viii) 52/91

Answer: 52 ÷ 91 = 0.571428… — the fourth digit is 4, so the third digit stays the same: 52/91 → 0.571.

Question: Review Exercise 3, Question 7: Round the following decimals to 3 decimal places. (i) 5.71679

Answer: The fourth digit is 7, so the third digit increases by 1: 5.71679 → 5.717.

Question: Review Exercise 3, Question 7: (ii) 11.60365

Answer: The fourth digit is 6, so the third digit increases by 1: 11.60365 → 11.604.

Question: Review Exercise 3, Question 7: (iii) 0.92598

Answer: The fourth digit is 9, so the third digit increases by 1: 0.92598 → 0.926.

Question: Review Exercise 3, Question 7: (iv) 3.40855

Answer: The fourth digit is 5, so the third digit increases by 1: 3.40855 → 3.409.

Question: Review Exercise 3, Question 7: (v) 0.74396

Answer: The fourth digit is 9, so the third digit increases by 1: 0.74396 → 0.744.

Question: Review Exercise 3, Question 7: (vi) 23.15847

Answer: The fourth digit is 4, so the third digit stays the same: 23.15847 → 23.158.

Question: Review Exercise 3, Question 8: Answer the following questions. (i) State the difference between terminating decimals and non-terminating decimals.

Answer: In a terminating decimal the decimal part ends at some place, e.g. 0.75. In a non-terminating decimal the decimal part never ends, e.g. 1/3 = 0.333... .

Question: Review Exercise 3, Question 8: (ii) State the methods for converting decimals into rational numbers and give an example of each.

Answer: Method 1 (terminating): write the decimal with a denominator of 10, 100, 1000, etc., then simplify. Example: 0.75 = 75/100 = 3/4. Method 2 (recurring): use a denominator of 9s according to the number of recurring digits. Example: 0.333... = 3/9 = 1/3.

Question: Review Exercise 3, Question 8: (iii) Which non-terminating decimals are called recurring? Give an example.

Answer: When in a non-terminating decimal a digit or a group of digits keeps repeating, it is called a recurring decimal. Examples: 1/3 = 0.333... (3 repeats) and 1/7 = 0.142857142857... (the group 142857 repeats).

Question: Review Exercise 3, Question 8: (iv) State the rule by which it is known whether a decimal is terminating or non-terminating.

Answer: Rule: first write the number in lowest form. Then look at the prime factors of the denominator; if all prime factors of the denominator are only 2 and/or 5, the decimal is terminating, otherwise it is non-terminating. Examples: 3/8 (denominator 8 = 2 × 2 × 2) is terminating, 2/3

(denominator 3) is non-terminating.

Question: Review Exercise 3, Question 8: (v) What is meant by "rounding off / approximating"?

Answer: "Rounding off / approximating" means rounding a decimal to the required number of decimal places: if the next digit is 5 or more, increase the last kept digit by 1, otherwise it stays the same. Example: rounding 0.6666 to 2 decimal places gives 0.67.

Review Exercise 3, Question 9: Fill in the blanks. (i) 1/2 = 0.5, which is a _____.

Answer: terminating decimal, because its decimal part ends. Blank: terminating decimal.

Review Exercise 3, Question 9: (ii) 1/3 = 0.333..., which is a _____.

Answer: non-terminating recurring decimal, because 3 keeps repeating in it. Blank: non-terminating recurring decimal.

Review Exercise 3, Question 9: (iii) The two parts of a decimal are called _____ and _____.

Answer: the whole part, the decimal part. Blanks: whole part, decimal part.

Review Exercise 3, Question 9: (iv) In a terminating decimal, the division process _____ after a limited number of steps.

Answer: ends

Review Exercise 3, Question 9: (v) A fraction is called a terminating decimal if the prime factors of the denominator of _____ are 2 or 5 or both.

Answer: the fraction

Review Exercise 3, Question 9: (vi) In a recurring decimal, a digit or _____ of digits comes again and again.

Answer: group

Question: Review Exercise 3, Question 10: Put a (✓) mark on the correct

answer. (i) Which symbol do we use in a decimal to separate the whole part from the decimal part? (a) — (b) : (c) . (d) /

Answer: The whole part and the decimal part are separated by a point (.), e.g. in 3.5. The correct answer is (c) . ✓

Review Exercise 3, Question 10: (ii) If we round 3.8461 to two decimal places, we get _____: (a) 3.48 (b) 3.81 (c) 3.86 (d) 3.85

Answer: → 3.85. The correct answer is

Review Exercise 3, Question 10: (iii) A rational number will be a terminating decimal if the prime factors of its denominator are nothing except _____: (a) 2 and 5 (b) 3 and 6 (c) 4 and 5 (d) 5 and 7

Answer: only 2 and 5, then the decimal is terminating. The correct answer is

Question: Review Exercise 3, Question 10: (iv) When we convert 0.75 into a rational number, we get: (a) 1/2 (b) 1/3 (c) 3/4 (d) 1/7

Answer: 0.75 = 75/100 = 3/4. The correct answer is (c) 3/4. ✓

4. Powers

Question: Exercise 4.1, Question 1: Expand the following exponential forms and find their values. (i) 3^2

Answer: $3^2 = 3 \times 3 = 9$.

Question: Exercise 4.1, Question 1: (ii) 2^7

Answer: $2^7 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 128$.

Question: Exercise 4.1, Question 1: (iii) $2^3 \times 3^4$

Answer: $2^3 \times 3^4 = (2 \times 2 \times 2) \times (3 \times 3 \times 3 \times 3) = 8 \times 81 = 648$.

Question: Exercise 4.1, Question 1: (iv) $5^2 \times 7^2$

Answer: $5^2 \times 7^2 = (5 \times 5) \times (7 \times 7) = 25 \times 49 = 1225$.

Question: Exercise 4.1, Question 1: (v) $2^3 \times 3^2 \times 5^2$

Answer: $2^3 \times 3^2 \times 5^2 = (2 \times 2 \times 2) \times (3 \times 3) \times (5 \times 5) = 8 \times 9 \times 25 = 1800$.

Question: Exercise 4.1, Question 1: (vi) $2^4 \times 3^3 \times 7^2$

Answer: $2^4 \times 3^3 \times 7^2 = (2 \times 2 \times 2 \times 2) \times (3 \times 3 \times 3) \times (7 \times 7) = 16 \times 27 \times 49 = 21168$.

Question: Exercise 4.1, Question 2: In each of the following, identify the base and the exponent and also find the value of the expression. (i) 2^5

Answer: The base is 2 and the exponent is 5. Value: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

Question: Exercise 4.1, Question 2: (ii) 5^4

Answer: The base is 5 and the exponent is 4. Value: $5^4 = 5 \times 5 \times 5 \times 5 = 625$.

Question: Exercise 4.1, Question 2: (iii) 6^3

Answer: The base is 6 and the exponent is 3. Value: $6^3 = 6 \times 6 \times 6 = 216$.

Question: Exercise 4.1, Question 2: (iv) $(2/3)^6$

Answer: The base is 2/3 and the exponent is 6. Value: $(2/3)^6 = 64/729$.

Question: Exercise 4.1, Question 2: (v) $(-1/5)^5$

Answer: The base is $-1/5$ and the exponent is 5. Value: $(-1/5)^5 = -1/3125$ (the exponent is odd, so the answer is negative).

Question: Exercise 4.1, Question 3: Prove that: (i) $(-3)^3 = -27$

Answer: $(-3)^3 = (-3) \times (-3) \times (-3) = 9 \times (-3) = -27$. So $(-3)^3 = -27$ is proved. ✓

Question: Exercise 4.1, Question 3: (ii) $(1/2)^7 = 1/128$

Answer: $(1/2)^7 = 1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2 = 1/128$. So $(1/2)^7 = 1/128$ is proved. ✓

Question: Exercise 4.1, Question 3: (iii) $(-1/4)^4 = 1/256$

Answer: $(-1/4)^4 = (-1/4) \times (-1/4) \times (-1/4) \times (-1/4) = 1/256$ (the exponent is even, so the answer is positive). So $(-1/4)^4 = 1/256$ is proved. ✓

Question: Exercise 4.1, Question 3: (iv) $(-1/10)^6 = 1/1000000$

Answer: $(-1/10)^6 = 1/10^6 = 1/1000000$ (the exponent is even, so the answer is positive). So $(-1/10)^6 = 1/1000000$ is proved. ✓

Question: Exercise 4.1, Question 3: (v) $(-5/3)^4 = 625/81$

Answer: $(-5/3)^4 = 625/81$ (the exponent is even, so the answer is positive; $5^4 = 625$ and $3^4 = 81$). So $(-5/3)^4 = 625/81$ is proved. ✓

Question: Exercise 4.1, Question 3: (vi) $(4/5)^5 = 1024/3125$

Answer: $(4/5)^5 = 4^5/5^5 = 1024/3125$ ($4^5 = 1024$ and $5^5 = 3125$). So $(4/5)^5 = 1024/3125$ is proved. ✓

Question: Exercise 4.1, Question 4: Express each rational number in exponential form. (i) 1000

Answer: $1000 = 10 \times 10 \times 10 = 10^3$.

Question: Exercise 4.1, Question 4: (ii) 512

Answer: $512 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^9$.

Question: Exercise 4.1, Question 4: (iii) -343

Answer: $343 = 7 \times 7 \times 7 = 7^3$ and the exponent is odd, so $-343 = (-7)^3$.

Question: Exercise 4.1, Question 4: (iv) 625

Answer: $625 = 5 \times 5 \times 5 \times 5 = 5^4$.

Question: Exercise 4.1, Question 4: (v) $-1/100000$

Answer: $100000 = 10^5$ and the exponent is odd, so $-1/100000 = (-1/10)^5$.

Question: Exercise 4.1, Question 4: (vi) $32/243$

Answer: $32 = 2^5$ and $243 = 3^5$, so $32/243 = (2/3)^5$.

Question: Exercise 4.1, Question 4: (vii) $-125/343$

Answer: $125 = 5^3$ and $343 = 7^3$ and the exponent is odd, so $-125/343 = (-5/7)^3$.

Question: Exercise 4.1, Question 4: (viii) $729/64$

Answer: $729 = 9^3$ and $64 = 4^3$, so $729/64 = (9/4)^3$. (Another form: $(27/8)^2$ is also correct, since $27^2 = 729$ and $8^2 = 64$.)

Question: Exercise 4.2, Question 1: Simplify using the product law. (i) $(5/7)^2 \times (5/7)^4$

Answer: The base is the same, so the exponents add: $(5/7)^2 \times (5/7)^4 = (5/7)^{2+4} = (5/7)^6$.

Question: Exercise 4.2, Question 1: (ii) $(3/4)^6 \times (1/5)^6$

Answer: The exponent is the same, so multiply the bases: $(3/4)^6 \times (1/5)^6 = (3/4 \times 1/5)^6 = (3/20)^6$.

Question: Exercise 4.2, Question 1: (iii) $(1/7)^5 \times (1/7)^5 \times (-5/7)^8 \times (-5/7)^8$

Answer: $(1/7)^5 \times (1/7)^5 = (1/7)^{5+5} = (1/7)^{10}$ and $(-5/7)^8 \times (-5/7)^8 = (-5/7)^{8+8} = (-5/7)^{16} = (5/7)^{16}$ (the exponent is even). Answer: $(1/7)^{10} \times (5/7)^{16}$.

Question: Exercise 4.2, Question 1: (iv) $p^6 \times p^7$

Answer: The base is the same, so the exponents add: $p^6 \times p^7 = p^{6+7} = p^{13}$.

Question: Exercise 4.2, Question 1: (v) $y^5 \times x^{11} \times y^6$

Answer: $y^5 \times y^6 = y^{5+6} = y^{11}$, so $y^5 \times x^{11} \times y^6 = x^{11} \times y^{11} = (xy)^{11}$.

Question: Exercise 4.2, Question 1: (vi) $(4/3)^5 \times (4/3)^6 \times (4/3)^7$

Answer: The base is the same, so the exponents add: $(4/3)^5 \times (4/3)^6 \times (4/3)^7 = (4/3)^{5+6+7} = (4/3)^{18}$.

Question: Exercise 4.2, Question 1: (vii) $x^5 \times y^6 \times x^2 \times y^7$

Answer: $x^5 \times x^2 = x^7$ and $y^6 \times y^7 = y^{13}$, so $x^5 \times y^6 \times x^2 \times y^7 = x^7 \times y^{13}$.

Question: Exercise 4.2, Question 1: (viii) $(-1/3)^6 \times (-1/3)^5 \times (-1/3)^3$

Answer: The base is the same, so the exponents add: $(-1/3)^6 \times (-1/3)^5 \times (-1/3)^3 = (-1/3)^{6+5+3} = (-1/3)^{14} = (1/3)^{14}$ (the exponent is even).

Question: Exercise 4.2, Question 1: (ix) $x^5 \times y^5 \times z^5$

Answer: The exponent is the same, so multiply the bases: $x^5 \times y^5 \times z^5 = (xyz)^5$.

Question: Exercise 4.2, Question 1: (x) $p^2 \times q^7 \times p^5$

Answer: $p^2 \times p^5 = p^7$, so $p^2 \times q^7 \times p^5 = p^7 \times q^7 = (pq)^7$.

Question: Exercise 4.2, Question 2: Check that: (i) $2^5 \times 3^5 = 6^5$

Answer: Left side: $2^5 \times 3^5 = (2 \times 3)^5 = 6^5$ = right side. Checked. ✓

Question: Exercise 4.2, Question 2: (ii) $p^7 \times q^7 = (pq)^7$

Answer: Left side: $p^7 \times q^7 = (p \times q)^7 = (pq)^7$ = right side. Checked. ✓

Question: Exercise 4.2, Question 2: (iii) $(3/4)^4 \times (3/4)^8 = (3/4)^{12}$

Answer: Left side: $(3/4)^4 \times (3/4)^8 = (3/4)^{4+8} = (3/4)^{12}$ = right side. Checked. ✓

Question: Exercise 4.3, Question 1: Simplify using the division law. (i) $5^7 \div 5^4$

Answer: The base is the same, so subtract the exponents: $5^7 \div 5^4 = 5^{7-4} = 5^3 = 125$.

Question: Exercise 4.3, Question 1: (ii) $(5/7)^5 \div (5/7)^2$

Answer: $(5/7)^5 \div (5/7)^2 = (5/7)^{5-2} = (5/7)^3 = 125/343$.

Question: Exercise 4.3, Question 1: (iii) $(1/2)^6 \div (1/2)^2$

Answer: $(1/2)^6 \div (1/2)^2 = (1/2)^{6-2} = (1/2)^4 = 1/16$.

Question: Exercise 4.3, Question 1: (iv) $x^{10} \div x^4$

Answer: $x^{10} \div x^4 = x^{10-4} = x^6$.

Question: Exercise 4.3, Question 1: (v) $(5)^6 \div (3)^6$

Answer: The exponent is the same, so divide the bases: $(5)^6 \div (3)^6 = (5/3)^6$.

Question: Exercise 4.3, Question 1: (vi) $(10)^5 \div (2)^5$

Answer: $(10)^5 \div (2)^5 = (10/2)^5 = 5^5 = 3125$.

Question: Exercise 4.3, Question 1: (vii) $(-7)^4 \div (-5)^4$

Answer: $(-7)^4 \div (-5)^4 = (-7/-5)^4 = (7/5)^4 = 2401/625$.

Question: Exercise 4.3, Question 1: (viii) $(2/5)^6 \div (3/5)^6$

Answer: $(2/5)^6 \div (3/5)^6 = ((2/5) \div (3/5))^6 = (2/3)^6 = 64/729$.

Question: Exercise 4.3, Question 1: (ix) $(7/11)^6 \div (7/11)^5$

Answer: $(7/11)^6 \div (7/11)^5 = (7/11)^{6-5} = (7/11)^1 = 7/11$.

Question: Exercise 4.3, Question 1: (x) $x^4 \div y^4$

Answer: The exponent is the same, so: $x^4 \div y^4 = (x/y)^4$.

Question: Exercise 4.3, Question 2: Check that: (i) $(2/9)^5 \div (2/9)^2 = 2^3 \div 9^3$

Answer: Left side: $(2/9)^5 \div (2/9)^2 = (2/9)^3 = 8/729$. Right side: $2^3 \div 9^3 = 8/729$. Both sides are equal; checked. ✓

Question: Exercise 4.3, Question 2: (ii) $(-3)^8 \div (-5)^8 = (3/5)^{10} \div (3/5)^2$

Answer: Left side: $(-3)^8 \div (-5)^8 = (3/5)^8$ (the exponents are even). Right side: $(3/5)^{10} \div (3/5)^2 = (3/5)^8$. Both sides are equal; checked. ✓

Question: Exercise 4.3, Question 2: (iii) $(x/y)^7 \div (x/y)^5 = x^2 \div y^2$

Answer: Left side: $(x/y)^7 \div (x/y)^5 = (x/y)^2 = x^2/y^2$. Right side: $x^2 \div y^2 = x^2/y^2$. Both sides are equal; checked. ✓

Question: Exercise 4.3, Question 2: (iv) $5^{11} \div 5^8 = (20)^3 \div (4)^3$

Answer: Left side: $5^{11} \div 5^8 = 5^3 = 125$. Right side: $(20)^3 \div (4)^3 = (20/4)^3 = 5^3 = 125$. Both sides are equal; checked. ✓

Question: Exercise 4.4, Question 1: Simplify using the power law. (i) $(5^2)^7$

Answer: $(5^2)^7 = 5^{2\times7} = 5^{14}$ (the exponents multiply).

Question: Exercise 4.4, Question 1: (ii) $\{(1/3)^6\}^8$

Answer: $\{(1/3)^6\}^8 = (1/3)^{6\times8} = (1/3)^{48}$.

Question: Exercise 4.4, Question 1: (iii) $\{(-3)^2\}^{10}$

Answer: $\{(-3)^2\}^{10} = (-3)^{2\times10} = (-3)^{20} = 3^{20}$ (the exponent is even).

Question: Exercise 4.4, Question 1: (iv) $(x^5)^{20}$

Answer: $(x^5)^{20} = x^{5\times20} = x^{100}$.

Question: Exercise 4.4, Question 1: (v) $\{(4/7)^9\}^2$

Answer: $\{(4/7)^9\}^2 = (4/7)^{9\times2} = (4/7)^{18}$.

Question: Exercise 4.4, Question 1: (vi) $\{(-3/4)^{20}\}^2$

Answer: $\{(-3/4)^{20}\}^2 = (-3/4)^{20\times2} = (-3/4)^{40} = (3/4)^{40}$ (the exponent is even).

Question: Exercise 4.4, Question 2: Convert into negative exponent form. (i) $1/2^5$

Answer: $1/2^5 = 2^{-5}$ (since $a^{-m} = 1/a^m$).

Question: Exercise 4.4, Question 2: (ii) $1/3^7$

Answer: $1/3^7 = 3^{-7}$.

Question: Exercise 4.4, Question 2: (iii) $1/x^6$

Answer: $1/x^6 = x^{-6}$.

Question: Exercise 4.4, Question 2: (iv) $1/(4/5)^6$

Answer: $1/(4/5)^6 = (4/5)^{-6} = (5/4)^6$.

Question: Exercise 4.4, Question 3: Convert into positive exponent form. (i) 5^{-8}

Answer: $5^{-8} = 1/5^8$.

Question: Exercise 4.4, Question 3: (ii) $(1/3)^{-7}$

Answer: $(1/3)^{-7} = 1/(1/3)^7 = 3^7 = 2187$.

Question: Exercise 4.4, Question 3: (iii) x^{-5}

Answer: $x^{-5} = 1/x^5$.

Question: Exercise 4.4, Question 3: (iv) y^{-7}

Answer: $y^{-7} = 1/y^7$.

Question: Exercise 4.4, Question 4: Identify which of the following are positive and which are negative. (i) $(-7)^6$

Answer: Positive, because the exponent 6 is an even number. $(-7)^6 = 7^6$.

Question: Exercise 4.4, Question 4: (ii) $(-11/2)^{17}$

Answer: Negative, because the exponent 17 is an odd number.

Question: Exercise 4.4, Question 4: (iii) $(-1/5)^{-26}$

Answer: Positive, because the exponent −26 is an even number. $(-1/5)^{-26} = (-5)^{26}$.

Question: Exercise 4.4, Question 4: (iv) $(-3)^0$

Answer: Positive, because $(-3)^0 = 1$.

Question: Exercise 4.4, Question 4: (v) $(-1/7)^1$

Answer: Negative, because $(-1/7)^1 = -1/7$.

Question: Exercise 4.4, Question 4: (vi) $(-1/5)^{-100}$

Answer: Positive, because the exponent −100 is an even number. $(-1/5)^{-100} = (-5)^{100}$.

Question: Exercise 4.5, Question 1: Simplify and write in simplest form. (i) $(5/7)^6 \times (5/7)^4 \times (7/5)^{10}$

Answer: $(5/7)^6 \times (5/7)^4 = (5/7)^{10}$ and $(7/5)^{10} = (5/7)^{-10}$, so $(5/7)^{10} \times (5/7)^{-10} = (5/7)^0 = 1$.

Question: Exercise 4.5, Question 1: (ii) $\{(2/3)^2\}^3 \div (2/3)^4$

Answer: $\{(2/3)^2\}^3 = (2/3)^6$, so $(2/3)^6 \div (2/3)^4 = (2/3)^{6-4} = (2/3)^2 = 4/9$.

Question: Exercise 4.5, Question 1: (iii) $(7/9)^3 \times (9/7)^3 + (11/12)^5 \times (12/11)^5$

Answer: $(7/9)^3 \times (9/7)^3 = ((7/9) \times (9/7))^3 = 1^3 = 1$ and $(11/12)^5 \times (12/11)^5 = ((11/12) \times (12/11))^5 = 1^5 = 1$, so the answer is 1 + 1 = 2.

Question: Exercise 4.5, Question 1: (iv) $x^0 + x^5 \times x^{-5}$

Answer: $x^0 = 1$ and $x^5 \times x^{-5} = x^{5-5} = x^0 = 1$, so the answer is $1 + 1 = 2$.

Question: Exercise 4.5, Question 1: (v) $\{(4/5)^7 \times (4/5)^2\} \div (4/5)^8$

Answer: $(4/5)^7 \times (4/5)^2 = (4/5)^9$, so $(4/5)^9 \div (4/5)^8 = (4/5)^{9-8} = 4/5$.

Question: Exercise 4.5, Question 1: (vi) $\{(7/8)^6 \div (7/8)^4\} \times (8/7)^2$

Answer: $(7/8)^6 \div (7/8)^4 = (7/8)^2$, so $(7/8)^2 \times (8/7)^2 = ((7/8) \times (8/7))^2 = 1^2 = 1$.

Question: Exercise 4.5, Question 2: Find the value of x in the following. (i) $3^6 \times 3^4 = 3^x$

Answer: $3^6 \times 3^4 = 3^{6+4} = 3^{10}$, so $3^{10} = 3^x$, hence x = 10.

Question: Exercise 4.5, Question 2: (ii) $(11/9)^8 \div (11/9)^6 = (11/9)^{2x}$

Answer: $(11/9)^8 \div (11/9)^6 = (11/9)^{8-6} = (11/9)^2$, so $(11/9)^2 = (11/9)^{2x}$, hence 2x = 2, i.e. x = 1.

Question: Exercise 4.5, Question 2: (iii) $\{(4/7)^3\}^x = (4/7)^9$

Answer: $\{(4/7)^3\}^x = (4/7)^{3x}$, so $(4/7)^{3x} = (4/7)^9$, hence 3x = 9, i.e. x = 3.

Question: Exercise 4.5, Question 2: (iv) $(1/5)^x \times (1/5)^5 = (1/5)^{10}$

Answer: $(1/5)^x \times (1/5)^5 = (1/5)^{x+5}$, so $(1/5)^{x+5} = (1/5)^{10}$, hence x + 5 = 10, i.e. x = 5.

Question: Review Exercise 4, Question 1: Identify the base and the exponent and find the value of the following. (i) 3^5

Answer: The base is 3 and the exponent is 5. Value: $3^5 = 243$.

Question: Review Exercise 4, Question 1: (ii) 7^4

Answer: The base is 7 and the exponent is 4. Value: $7^4 = 2401$.

Question: Review Exercise 4, Question 1: (iii) $(3/4)^2$

Answer: The base is 3/4 and the exponent is 2. Value: $(3/4)^2 = 9/16$.

Question: Review Exercise 4, Question 1: (iv) $(1/4)^3$

Answer: The base is 1/4 and the exponent is 3. Value: $(1/4)^3 = 1/64$.

Question: Review Exercise 4, Question 1: (v) 8^0

Answer: The base is 8 and the exponent is 0. Value: $8^0 = 1$ (any number with exponent zero has value 1).

Question: Review Exercise 4, Question 1: (vi) y^0

Answer: The base is y and the exponent is 0. Value: $y^0 = 1$.

Question: Review Exercise 4, Question 2: Simplify using the exponent law. (i) $(-1/2)^4 \times (-1/2)^2$

Answer: $(-1/2)^4 \times (-1/2)^2 = (-1/2)^{4+2} = (-1/2)^6 = 1/64$ (the exponent is even).

Question: Review Exercise 4, Question 2: (ii) $(-4/5)^2 \times (-4/5)^2$

Answer: $(-4/5)^2 \times (-4/5)^2 = (-4/5)^{2+2} = (-4/5)^4 = 256/625$.

Question: Review Exercise 4, Question 2: (iii) $6^7 \div 6^4$

Answer: $6^7 \div 6^4 = 6^{7-4} = 6^3 = 216$.

Question: Review Exercise 4, Question 2: (iv) $(2/7)^5 \div (2/7)^3$

Answer: $(2/7)^5 \div (2/7)^3 = (2/7)^{5-3} = (2/7)^2 = 4/49$.

Question: Review Exercise 4, Question 2: (v) $(3^2)^5$

Answer: $(3^2)^5 = 3^{2\times5} = 3^{10} = 59049$.

Question: Review Exercise 4, Question 2: (vi) $\{(-4)^2\}^5$

Answer: $\{(-4)^2\}^5 = (-4)^{2\times5} = (-4)^{10} = 4^{10} = 1048576$.

Question: Review Exercise 4, Question 2: (vii) $(6^0)^2$

Answer: $(6^0)^2 = 1^2 = 1$ (since $6^0 = 1$).

Question: Review Exercise 4, Question 2: (viii) $(5^0)^2 \times (-5) \times (a^0)^2$

Answer: $(5^0)^2 = 1$ and $(a^0)^2 = 1$, so $(5^0)^2 \times (-5) \times (a^0)^2 = 1 \times (-5) \times 1 = -5$.

Question: Review Exercise 4, Question 3: Prove that: (i) $(1/5)^4 \times (1/5)^5 = (3/5)^9 \times (1/3)^9$

Answer: Left side: $(1/5)^4 \times (1/5)^5 = (1/5)^9$. Right side: $(3/5)^9 \times (1/3)^9 = ((3/5) \times (1/3))^9 = (1/5)^9$. Both sides are equal; proved. ✓

Question: Review Exercise 4, Question 3: (ii) $(6)^4 \div (2)^4 = 3^8 \div 3^4$

Answer: Left side: $(6)^4 \div (2)^4 = (6/2)^4 = 3^4 = 81$. Right side: $3^8 \div 3^4 = 3^4 = 81$. Both sides are equal; proved. ✓

Question: Review Exercise 4, Question 3: (iii) $6z^0 = 6 \times (z^0)^2$

Answer: $z^0 = 1$ (when $z \neq 0$), so the left side is $6 \times 1 = 6$ and the right side is $6 \times (1)^2 = 6$. Both sides are equal; proved. ✓

Question: Review Exercise 4, Question 4: Which is positive and which is negative? (i) $(-2)^7$

Answer: Negative, because the exponent 7 is an odd number.

Question: Review Exercise 4, Question 4: (ii) $(-5)^0$

Answer: Positive, because $(-5)^0 = 1$.

Question: Review Exercise 4, Question 4: (iii) $(-3)^8$

Answer: Positive, because the exponent 8 is an even number.

Question: Review Exercise 4, Question 4: (iv) $(-6)^{-7}$

Answer: Negative, because the exponent -7 is an odd number. $(-6)^{-7} = 1/(-6)^7$.

Question: Review Exercise 4, Question 4: (v) $(-8)^{-10}$

Answer: Positive, because the exponent -10 is an even number. $(-8)^{-10} = 1/(-8)^{10} = 1/8^{10}$.

Question: Review Exercise 4, Question 5: Simplify. (i) $6^2 \div 3^3$

Answer: $6^2 \div 3^3 = 36 \div 27 = 36/27 = 4/3$.

Question: Review Exercise 4, Question 5: (ii) $5^0 \div 5^{-2} \times 5^2$

Answer: $5^0 = 1$ and $5^0 \div 5^{-2} = 5^{0-(-2)} = 5^2 = 25$, so $25 \times 5^2 = 25 \times 25 = 625$.

Question: Review Exercise 4, Question 5: (iii) $[\{(2/5)^2\}^3 \div (2/5)^4] \times (2/5)^3$

Answer: $\{(2/5)^2\}^3 = (2/5)^6$, then $(2/5)^6 \div (2/5)^4 = (2/5)^2$, then $(2/5)^2 \times (2/5)^3 = (2/5)^5 = 32/3125$.

Question: Review Exercise 4, Question 6: Find the value of x. (i) $25^x = 125$

Answer: $25^x = (5^2)^x = 5^{2x}$ and $125 = 5^3$, so $5^{2x} = 5^3$, hence $2x = 3$, i.e. $x =$

3/2.

Question: Review Exercise 4, Question 6: (ii) $(2/3)^{x} \times (2/3)^{4} = (2/3)^{10} \div (2/3)^{2}$

Answer: The left side is $(2/3)^{x+4}$ and the right side is $(2/3)^{10-2} = (2/3)^{8}$, so x + 4 = 8, i.e. x = 4.

Question: Review Exercise 4, Question 6: (iii) $\{(2/7)^{x}\}^{5} = (2/7)^{10}$

Answer: $\{(2/7)^{x}\}^{5} = (2/7)^{5x}$, so $(2/7)^{5x} = (2/7)^{10}$, hence 5x = 10, i.e. x = 2.

Question: Brain Store: Using the easy method for the sum of the numbers from 1 to 10, find (i) 1 + 2 + 3 + ... + 20.

Answer: Sum = n(n + 1)/2, here n = 20: 20 × 21/2 = 210.

Question: Brain Store: (ii) Find 1 + 2 + 3 + ... + 25.

Answer: Sum = n(n + 1)/2, here n = 25: 25 × 26/2 = 325.

5. Square Root of Large Numbers

Question: Practical Work, row (i): Fill in the blanks of the following table — Number: 1, Square of the given number: $1^2 = 1 \times 1$, Perfect square (number): 1, Read as: The square of 1 is 1.

Answer: This row is already complete: $1^2 = 1 \times 1 = 1$, so the square of 1 is 1.

Practical Work, row (ii): Fill in the blanks of the table — Number: 2, Square of the given number: 2^2 = _____, Perfect square (number): 4, Read as: The square of 2 is 4.

Answer: 2×2 =

Practical Work, row (iii): Fill in the blanks of the table — Number: −3, Square of the given number: $(-3)^2 = (-3) \times (-3)$, Perfect square (number): 9, Read as: The square of _____ is _____.

Answer: −3, 9."

Practical Work, row (iv): Fill in the blanks of the table — Number: 5, Square of the given number: 5^2 = _____, Perfect square (number): 25, Read as: The square of _____ is 25.

Answer: take 5×5 and 5: "The, 5

Practical Work, row (v): Fill in the blanks of the table — Number: −6, Square of the given number: _____ = $(-6) \times (-6)$, Perfect square (number): _____, Read as: The square of (−6) is _____.

Answer: $(-6)^2$, $(-6) = 36$, so the blanks take $(-6)^2$, 36 and 36: "The, 36."

Practical Work, row (vi): Fill in the blanks of the table — Number: _____, Square of the given number: $7^2 = 7 \times 7$, Perfect square (number): _____, Read as: The square of _____ is _____.

Answer: take 7, 49,, and 49: "The, 7, 49."

Practical Work, row (vii): Fill in the blanks of the table — Number: 8, Square of the given number: _____ = 8 × 8, Perfect square (number): _____, Read as: The square of _____ is _____.

Answer: 8^2, 8 = 64, so the blanks take 8^2, 64, 8 and 64: "The, 8, 64."

Practical Work, row (viii): Fill in the blanks of the table — Number: _____, Square of the given number: _____ = (−9) × (−9), Perfect square (number): _____, Read as: The square of _____ is _____.

Answer: The number is, −9 and $(-9)^2$, (−9) = 81, so the blanks take −9, $(-9)^2$, 81, −9 and 81: "The, −9, 81."

Practical Work, row (ix): Fill in the blanks of the table — Number: _____, Square of the given number: _____ = _____ × _____, Perfect square (number): 100, Read as: The square of _____ is 100.

Answer: take 10,, 10^2, 10,, 10 and, 10: "The, 10

Question: Exercise 5.1, A: Find the squares of the following. (1) 11

Answer: $11^2 = 11 \times 11 = 121$.

Question: Exercise 5.1, A: (2) 19

Answer: $19^2 = 19 \times 19 = 361$.

Question: Exercise 5.1, A: (3) 25

Answer: $25^2 = 25 \times 25 = 625$.

Question: Exercise 5.1, A: (4) 50

Answer: $50^2 = 50 \times 50 = 2500$.

Question: Exercise 5.1, A: (5) 66

Answer: $66^2 = 66 \times 66 = 4356$.

Question: Exercise 5.1, A: (6) 78

Answer: $78^2 = 78 \times 78 = 6084$.

Question: Exercise 5.1, A: (7) 100

Answer: $100^2 = 100 \times 100 = 10000$.

Question: Exercise 5.1, A: (8) 500

Answer: $500^2 = 500 \times 500 = 250000$.

Question: Exercise 5.1, B: Check whether each of the following is a perfect square or not. (1) 81

Answer: $81 = 9 \times 9 = 9^2$, so 81 is a perfect square. ✓

Question: Exercise 5.1, B: (2) 95

Answer: $9^2 = 81$ and $10^2 = 100$, so 95 is not the square of any number. 95 is not a perfect square. ✗

Question: Exercise 5.1, B: (3) 121

Answer: $121 = 11 \times 11 = 11^2$, so 121 is a perfect square. ✓

Question: Exercise 5.1, B: (4) 169

Answer: $169 = 13 \times 13 = 13^2$, so 169 is a perfect square. ✓

Question: Exercise 5.1, B: (5) 224

Answer: $14^2 = 196$ and $15^2 = 225$, so 224 is not the square of any number. 224 is not a perfect square. ✗

Question: Exercise 5.1, B: (6) 3969

Answer: $3969 = 63 \times 63 = 63^2$, so 3969 is a perfect square. ✓

Question: Exercise 5.1, B: (7) 3872

Answer: $62^2 = 3844$ and $63^2 = 3969$, so 3872 is not the square of any number. 3872 is not a perfect square. ✗

Question: Exercise 5.1, B: (8) 9845

Answer: $99^2 = 9801$ and $100^2 = 10000$, so 9845 is not the square of any number. 9845 is not a perfect square. ✗

Question: Exercise 5.1, C: Separate the perfect squares of even and odd numbers. (i) 9

Answer: $9 = 3^2$ and 3 is an odd number, so 9 is a perfect square of an odd number.

Question: Exercise 5.1, C: (ii) 25

Answer: 25 = 5^2 and 5 is an odd number, so 25 is a perfect square of an odd number.

Question: Exercise 5.1, C: (iii) 196

Answer: 196 = 14^2 and 14 is an even number, so 196 is a perfect square of an even number.

Question: Exercise 5.1, C: (iv) 441

Answer: 441 = 21^2 and 21 is an odd number, so 441 is a perfect square of an odd number.

Question: Exercise 5.1, C: (v) 2704

Answer: 2704 = 52^2 and 52 is an even number, so 2704 is a perfect square of an even number.

Question: Exercise 5.1, C: (vi) 3600

Answer: 3600 = 60^2 and 60 is an even number, so 3600 is a perfect square of an even number.

Question: Exercise 5.1, C: (vii) 9216

Answer: 9216 = 96^2 and 96 is an even number, so 9216 is a perfect square of an even number.

Question: Exercise 5.1, C: (viii) 9801

Answer: 9801 = 99^2 and 99 is an odd number, so 9801 is a perfect square of an odd number.

Question: Exercise 5.1, D: Find the squares of the following fractions by multiplying each by itself. (1) 1/5

Answer: $(1/5)^2 = (1/5) \times (1/5) = 1/25$.

Question: Exercise 5.1, D: (2) 3/7

Answer: $(3/7)^2 = (3/7) \times (3/7) = 9/49$.

Question: Exercise 5.1, D: (3) 8/9

Answer: $(8/9)^2 = (8/9) \times (8/9) = 64/81$.

Question: Exercise 5.1, D: (4) 5/6

Answer: $(5/6)^2 = (5/6) \times (5/6) = 25/36$.

Question: Exercise 5.1, D: (5) 2/3

Answer: $(2/3)^2 = (2/3) \times (2/3) = 4/9$.

Question: Exercise 5.1, E: Find the squares of the following decimals by multiplying each by itself. (1) 0.1

Answer: $(0.1)^2 = 0.1 \times 0.1 = 0.01$.

Question: Exercise 5.1, E: (2) 0.5

Answer: $(0.5)^2 = 0.5 \times 0.5 = 0.25$.

Question: Exercise 5.1, E: (3) 0.07

Answer: $(0.07)^2 = 0.07 \times 0.07 = 0.0049$.

Question: Exercise 5.1, E: (4) 0.11

Answer: $(0.11)^2 = 0.11 \times 0.11 = 0.0121$.

Question: Exercise 5.1, E: (5) 0.15

Answer: $(0.15)^2 = 0.15 \times 0.15 = 0.0225$.

Question: Exercise 5.2, A: Find the square root of each of the following. (1) 25

Answer: $\sqrt{25} = 5$, since $5^2 = 25$.

Question: Exercise 5.2, A: (2) $(4)^2$

Answer: $\sqrt{((4)^2)} = 4$.

Question: Exercise 5.2, A: (3) 81

Answer: $\sqrt{81} = 9$, since $9^2 = 81$.

Question: Exercise 5.2, A: (4) $(36)^2$

Answer: $\sqrt{((36)^2)} = 36$.

Question: Exercise 5.2, A: (5) a^2

Answer: $\sqrt{(a^2)} = a$.

Question: Exercise 5.2, A: (6) y^2

Answer: $\sqrt{(y^2)} = y$.

Question: Exercise 5.2, A: (7) $(49)^2$

Answer: $\sqrt{((49)^2)} = 49$.

Question: Exercise 5.2, A: (8) 64

Answer: $\sqrt{64} = 8$, since $8^2 = 64$.

Question: Exercise 5.2, B: Find the square root by the factor and division methods. (1) 676

Answer: $676 = 2 \times 2 \times 13 \times 13 = 2^2 \times 13^2$, so $\sqrt{676} = 2 \times 13 = 26$. Check: $26^2 = 676$. ✓

Question: Exercise 5.2, B: (2) 169

Answer: $169 = 13 \times 13 = 13^2$, so $\sqrt{169} = 13$. Check: $13^2 = 169$. ✓

Question: Exercise 5.2, B: (3) 484

Answer: $484 = 2 \times 2 \times 11 \times 11 = 2^2 \times 11^2$, so $\sqrt{484} = 2 \times 11 = 22$. Check: $22^2 = 484$. ✓

Question: Exercise 5.2, B: (4) 961

Answer: $961 = 31 \times 31 = 31^2$, so $\sqrt{961} = 31$. Check: $31^2 = 961$. ✓

Question: Exercise 5.2, B: (5) 4900

Answer: $4900 = 2 \times 2 \times 5 \times 5 \times 7 \times 7 = 2^2 \times 5^2 \times 7^2$, so $\sqrt{4900} = 2 \times 5 \times 7 = 70$. Check: $70^2 = 4900$. ✓

Question: Exercise 5.2, B: (6) 1089

Answer: $1089 = 3 \times 3 \times 11 \times 11 = 3^2 \times 11^2$, so $\sqrt{1089} = 3 \times 11 = 33$. Check: $33^2 = 1089$. ✓

Question: Exercise 5.2, B: (7) 1600

Answer: $1600 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5$, so $\sqrt{1600} = 2 \times 2 \times 2 \times 5 = 40$. Check: $40^2 = 1600$. ✓

Question: Exercise 5.2, B: (8) 2304

Answer: $2304 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3$, so $\sqrt{2304} = 2 \times 2 \times 2 \times 2 \times 3 = 48$. Check: $48^2 = 2304$. ✓

Question: Exercise 5.2, B: (9) 3136

Answer: 3136 = 2 × 2 × 2 × 2 × 2 × 2 × 7 × 7, so √3136 = 2 × 2 × 2 × 7 = 56. Check: $56^2 = 3136$. ✓

Question: Exercise 5.2, B: (10) 1681

Answer: 1681 = 41 × 41 = 41^2, so √1681 = 41. Check: $41^2 = 1681$. ✓

Question: Exercise 5.2, B: (11) 1369

Answer: 1369 = 37 × 37 = 37^2, so √1369 = 37. Check: $37^2 = 1369$. ✓

Question: Exercise 5.2, B: (12) 2025

Answer: 2025 = 3 × 3 × 3 × 3 × 5 × 5 = $3^4 \times 5^2$, so √2025 = 3 × 3 × 5 = 45. Check: $45^2 = 2025$. ✓

Question: Exercise 5.2, C: Find the square root of the following fractions by the factor and division methods. (1) 144/225

Answer: √144 = 12 and √225 = 15, so √(144/225) = 12/15 = 4/5. Check: $(4/5)^2 = 16/25 = 144/225$ (both divide by 9).

Question: Exercise 5.2, C: (2) 324/441

Answer: √324 = 18 and √441 = 21, so √(324/441) = 18/21 = 6/7. Check: $(6/7)^2 = 36/49 = 324/441$ (both divide by 9).

Question: Exercise 5.2, C: (3) 1225/9801

Answer: √1225 = 35 and √9801 = 99, so √(1225/9801) = 35/99. Check: $35^2 = 1225$ and $99^2 = 9801$. ✓

Question: Exercise 5.2, C: (4) 1 25/144

Answer: 1 25/144 = 169/144, √169 = 13 and √144 = 12, so √(1 25/144) = 13/12 = 1 1/12. Check: $(13/12)^2 = 169/144$. ✓

Question: Exercise 5.2, C: (5) 1 48/121

Answer: 1 48/121 = 169/121, √169 = 13 and √121 = 11, so √(1 48/121) = 13/11 = 1 2/11. Check: $(13/11)^2 = 169/121$. ✓

Question: Exercise 5.2, C: (6) 3 13/36

Answer: 3 13/36 = 121/36, √121 = 11 and √36 = 6, so √(3 13/36) = 11/6 = 1 5/6. Check: $(11/6)^2 = 121/36$. ✓

Question: Exercise 5.2, C: (7) 961/1681

Answer: √961 = 31 and √1681 = 41, so √(961/1681) = 31/41. Check: 31^2 = 961 and 41^2 = 1681. ✓

Question: Exercise 5.2, C: (8) 1024/1225

Answer: √1024 = 32 and √1225 = 35, so √(1024/1225) = 32/35. Check: 32^2 = 1024 and 35^2 = 1225. ✓

Question: Exercise 5.2, C: (9) 2 217/576

Answer: 2 217/576 = 1369/576, √1369 = 37 and √576 = 24, so √(2 217/576) = 37/24 = 1 13/24. Check: $(37/24)^2$ = 1369/576. ✓

Question: Exercise 5.2, C: (10) 5 55/169

Answer: 5 55/169 = 900/169, √900 = 30 and √169 = 13, so √(5 55/169) = 30/13 = 2 4/13. Check: $(30/13)^2$ = 900/169. ✓

Question: Exercise 5.2, C: (11) 1 63/81

Answer: 1 63/81 = 144/81, √144 = 12 and √81 = 9, so √(1 63/81) = 12/9 = 4/3 = 1 1/3. Check: $(4/3)^2$ = 16/9 = 144/81 (both divide by 9).

Question: Exercise 5.2, C: (12) 3 325/900

Answer: 3 325/900 = 3025/900, √3025 = 55 and √900 = 30, so √(3 325/900) = 55/30 = 11/6 = 1 5/6. Check: $(11/6)^2$ = 121/36 = 3025/900 (both divide by 25).

Question: Exercise 5.2, D: Find the square root of the following decimals by the factor and division methods. (1) 3.24

Answer: 3.24 = 324/100, √324 = 18 and √100 = 10, so √3.24 = 18/10 = 1.8. Check: 1.8^2 = 3.24. ✓

Question: Exercise 5.2, D: (2) 4.41

Answer: 4.41 = 441/100, √441 = 21 and √100 = 10, so √4.41 = 21/10 = 2.1. Check: 2.1^2 = 4.41. ✓

Question: Exercise 5.2, D: (3) 5.29

Answer: 5.29 = 529/100, √529 = 23 and √100 = 10, so √5.29 = 23/10 =

2.3. Check: $2.3^2 = 5.29$. ✓

Question: Exercise 5.2, D: (4) 7.29

Answer: 7.29 = 729/100, √729 = 27 and √100 = 10, so √7.29 = 27/10 = 2.7. Check: $2.7^2 = 7.29$. ✓

Question: Exercise 5.2, D: (5) 6.25

Answer: 6.25 = 625/100, √625 = 25 and √100 = 10, so √6.25 = 25/10 = 2.5. Check: $2.5^2 = 6.25$. ✓

Question: Exercise 5.2, D: (6) 37.21

Answer: 37.21 = 3721/100, √3721 = 61 and √100 = 10, so √37.21 = 61/10 = 6.1. Check: $6.1^2 = 37.21$. ✓

Question: Exercise 5.2, D: (7) 7.84

Answer: 7.84 = 784/100, √784 = 28 and √100 = 10, so √7.84 = 28/10 = 2.8. Check: $2.8^2 = 7.84$. ✓

Question: Exercise 5.2, D: (8) 10.24

Answer: 10.24 = 1024/100, √1024 = 32 and √100 = 10, so √10.24 = 32/10 = 3.2. Check: $3.2^2 = 10.24$. ✓

Question: Exercise 5.2, D: (9) 30.25

Answer: 30.25 = 3025/100, √3025 = 55 and √100 = 10, so √30.25 = 55/10 = 5.5. Check: $5.5^2 = 30.25$. ✓

Question: Exercise 5.2, D: (10) 100.00

Answer: 100.00 = 100, √100 = 10. Check: $10^2 = 100$. ✓

Question: Exercise 5.2, D: (11) 33.64

Answer: 33.64 = 3364/100, √3364 = 58 and √100 = 10, so √33.64 = 58/10 = 5.8. Check: $5.8^2 = 33.64$. ✓

Question: Exercise 5.2, D: (12) 34.81

Answer: 34.81 = 3481/100, √3481 = 59 and √100 = 10, so √34.81 = 59/10 = 5.9. Check: $5.9^2 = 34.81$. ✓

Question: Exercise 5.3, Question 1: In an examination hall 784 chairs are

arranged so that the number of chairs in each row equals the number of rows. Find the number of rows and the number of chairs in each row.

Answer: Number of rows = $\sqrt{784} = 28$ (since $28^2 = 784$). So there are 28 rows and 28 chairs in each row.

Question: Exercise 5.3, Question 2: A square plot has an area of 240.25 square metres. Find the length of each of its sides.

Answer: Side length = $\sqrt{240.25} = 15.5$ metres (since $15.5^2 = 240.25$).

Question: Exercise 5.3, Question 3: The length of a room is three times its width and its area is 720.75 square metres. Find its length and width.

Answer: Let the width be x, then the length is 3x. Area: $3x \times x = 3x^2 = 720.75$, so $x^2 = 240.25$ and $x = 15.5$. So the width is 15.5 metres and the length is $3 \times 15.5 = 46.5$ metres.

Question: Exercise 5.3, Question 4: 2116 biscuit boxes are placed in rows. The number of boxes in each row equals the number of rows. Find the number of rows and the number of boxes in each row.

Answer: Number of rows = $\sqrt{2116} = 46$ (since $46^2 = 2116$). So there are 46 rows and 46 boxes in each row.

Question: Exercise 5.3, Question 5: 1681 students are to take part in a school function. Find the number of chairs in each row for seating, when the number of students in each row equals the number of columns.

Answer: Number of rows = number of students in each row = $\sqrt{1681} = 41$ (since $41^2 = 1681$). So there are 41 rows and 41 students will sit in each row.

Question: Exercise 5.3, Question 6: There are 625 trucks. They are standing in rows. The number of trucks in each row is the same as the number of rows. Find the number of rows and the number of trucks in each row.

Answer: Number of rows = $\sqrt{625} = 25$ (since $25^2 = 625$). So there are 25

rows and 25 trucks in each row.

Question: Exercise 5.3, Question 7: A physical-training teacher arranged 1024 students in rows. The number of students in each row is the same as the number of rows. Tell how many rows there are in all and what the number of students in each row is.

Answer: Number of rows = $\sqrt{1024} = 32$ (since $32^2 = 1024$). So there are 32 rows in all and 32 students in each row.

Question: Exercise 5.3, Question 8: The length of a rectangular plot is three times its width. If the area of the plot is 995.9052 square metres, find its length and width.

Answer: Let the width be x, then the length is 3x. Area: $3x^2 = 995.9052$, so $x^2 = 331.9684$ and $x = 18.22$. So the width is 18.22 metres and the length is $3 \times 18.22 = 54.66$ metres.

Question: Exercise 5.3, Question 9: Find the length of the side of a square-shaped school boundary whose area is 3387.24 square metres.

Answer: Side length = $\sqrt{3387.24} = 58.2$ metres (since $58.2^2 = 3387.24$).

Question: Exercise 5.3, Question 10: In a comb factory, 15,198 spools of thread are to be arranged in the form of a square, but 69 spools are left over. Tell how many spools there are in all in each row?

Answer: Spools placed in the square = $15198 - 69 = 15129$. Spools in each row = $\sqrt{15129} = 123$ (since $123^2 = 15129$). So there are 123 spools in each row.

Question: Review Exercise 5, Question 1: Give short answers to the following. (i) What are the different methods of finding the square root of a number?

Answer: There are two methods of finding a square root: (1) the factor method, (2) the division method.

Question: Review Exercise 5, Question 1: (ii) What is a perfect square

number? Give an example.

Answer: A number that is the square of a number is called a perfect square number. Example: 25 = 5², so 25 is a perfect square number.

Question: Review Exercise 5, Question 1: (iii) State any two properties of a perfect square number and give examples.

Answer: (1) The square of an even number is even, e.g. 6² = 36. (2) The square of an odd number is odd, e.g. 7² = 49.

Review Exercise 5, Question 2: Fill in the blanks. (i) The square of 31 is _____.

Answer: 961 (31² = 961).

Review Exercise 5, Question 2: (ii) The square root of 49 is 7 (7² = 49).

Review Exercise 5, Question 2: (iii) 1, 4, 9, 16... are called _____ numbers.

Answer: perfect square numbers (1², 2², 3², 4², ...).

Review Exercise 5, Question 2: (iv) The square of 12 is 144.

Review Exercise 5, Question 2: (v) √(25/36) = √25/√36 = 5/6.

Review Exercise 5, Question 2: (vi) The square of 1 is 1 (1² = 1).

✗ False: Review Exercise 5, Question 3: Write 'T' for true and 'F' for false. (i) The prime factors of 235 are 2, 3 and 5.

✓ True: Review Exercise 5, Question 3: (ii) 19² = 361.

✗ False: Review Exercise 5, Question 3: (iii) The square of an odd number is an even number.

✗ False: Review Exercise 5, Question 3: (iv) 25/49 > 5/7.

✓ True: Review Exercise 5, Question 3: (v) (4.5)² > 4.5.

✗ False: Review Exercise 5, Question 3: (vi) $0.7 < (0.7)^2$.

Review Exercise 5, Question 4: Choose the correct answer. (i) If $p^2 = q$, then p is called the _____ of q.

✓ (a) square root

(b) square

(c) prime factor

(d) base of the root

Review Exercise 5, Question 4: (ii) The numbers whose square and square root are equal are _____.

(a) 0 and 2

(b) 1 and 3

✓ (c) 0 and 1

(d) 2 and 3

Review Exercise 5, Question 4: (iii) _____ is a perfect square number.

(a) 14

(b) 15

✓ (c) 16

(d) 17

Question: Review Exercise 5, Question 5: Find the square root of the following. (i) 729

Answer: $\sqrt{729} = 27$ ($27^2 = 729$).

Question: Review Exercise 5, Question 5: (ii) 7056

Answer: $\sqrt{7056} = 84$ ($84^2 = 7056$).

Question: Review Exercise 5, Question 5: (iii) 26569

Answer: $\sqrt{26569} = 163$ ($163^2 = 26569$).

Question: Review Exercise 5, Question 5: (iv) 42025

Answer: $\sqrt{42025} = 205$ ($205^2 = 42025$).

Question: Review Exercise 5, Question 5: (v) 5 71/121

Answer: 5 71/121 = 676/121, √676 = 26 and √121 = 11, so √(5 71/121) = 26/11 = 2 4/11.

Question: Review Exercise 5, Question 5: (vi) 2116/9604

Answer: √2116 = 46 and √9604 = 98, so √(2116/9604) = 46/98 = 23/49.

Question: Review Exercise 5, Question 5: (vii) 1 984/14641

Answer: 1 984/14641 = 15625/14641, √15625 = 125 and √14641 = 121, so √(1 984/14641) = 125/121 = 1 4/121.

Question: Review Exercise 5, Question 5: (viii) 0.0256

Answer: 0.0256 = 256/10000, √256 = 16 and √10000 = 100, so √0.0256 = 16/100 = 0.16. Check: $0.16^2 = 0.0256$. ✓

Question: Review Exercise 5, Question 5: (ix) 344569

Answer: √344569 = 587 ($587^2 = 344569$).

Question: Review Exercise 5, Question 5: (x) 6400/9801

Answer: √6400 = 80 and √9801 = 99, so √(6400/9801) = 80/99.

Question: Review Exercise 5, Question 5: (xi) 5 41/64

Answer: 5 41/64 = 361/64, √361 = 19 and √64 = 8, so √(5 41/64) = 19/8 = 2 3/8.

Question: Review Exercise 5, Question 5: (xii) 131.1025

Answer: 131.1025 = 1311025/10000, √1311025 = 1145 and √10000 = 100, so √131.1025 = 1145/100 = 11.45. Check: $11.45^2 = 131.1025$. ✓

Question: Review Exercise 5, Question 6: A square garden has an area of 12100 square metres. Find the length of its side.

Answer: Side length = √12100 = 110 metres ($110^2 = 12100$).

Question: Review Exercise 5, Question 7: The floor of a square room has an area of 144 square metres. Find the length of the floor.

Answer: Length of the floor = √144 = 12 metres ($12^2 = 144$).

Question: Review Exercise 5, Question 8: A rectangular plot has an area of 7688 square metres. Find the width, if the length of the plot is twice its

width.

Answer: Let the width be x, then the length is 2x. Area: $2x^2 = 7688$, so $x^2 = 3844$ and $x = 62$. So the width is 62 metres (and the length is 124 metres).

Question: Review Exercise 5, Question 9: The length of a carpet is three times its width. Find the width and length of the carpet, if its area is 18.75 square metres.

Answer: Let the width be x, then the length is 3x. Area: $3x^2 = 18.75$, so $x^2 = 6.25$ and $x = 2.5$. So the width is 2.5 metres and the length is $3 \times 2.5 = 7.5$ metres.

Question: Review Exercise 5, Question 10: Najib spent 900 rupees in the month of June. Each day he spent as many rupees as there are days in the month. Tell how many rupees he spent each day?

Answer: June has 30 days. One day's spending = $\sqrt{900} = 30$ rupees.

Check: 30 days × 30 rupees = 900 rupees. ✓

6. Direct and Inverse Proportion

Question: Exercise 6.1, Question 1: Find A : B : C, if: (i) A : B = 2 : 5 and B : C = 5 : 4

Answer: The number for B is 5 in both ratios, so it stays the same: A : B : C = 2 : 5 : 4.

Question: Exercise 6.1, Question 1: (ii) A : B = 3 : 7 and B : C = 8 : 9

Answer: To make B the same: A : B = 3 : 7 = 24 : 56 and B : C = 8 : 9 = 56 : 63, so A : B : C = 24 : 56 : 63.

Question: Exercise 6.1, Question 1: (iii) A : B = 10 : 11 and B : C = 20 : 21

Answer: To make B the same: A : B = 10 : 11 = 200 : 220 and B : C = 20 : 21 = 220 : 231, so A : B : C = 200 : 220 : 231.

Question: Exercise 6.1, Question 2: Find X : Y : Z, if: (i) X : Y = 3 : 5 and X : Z = 4 : 9

Answer: To make X the same: X : Y = 3 : 5 = 12 : 20 and X : Z = 4 : 9 = 12 : 27, so X : Y : Z = 12 : 20 : 27.

Question: Exercise 6.1, Question 2: (ii) X : Y = 5 : 7 and X : Z = 11 : 13

Answer: To make X the same: X : Y = 5 : 7 = 55 : 77 and X : Z = 11 : 13 = 55 : 65, so X : Y : Z = 55 : 77 : 65.

Question: Exercise 6.1, Question 2: (iii) X : Z = 3 : 5 and Y : Z = 7 : 8

Answer: To make Z the same: X : Z = 3 : 5 = 24 : 40 and Y : Z = 7 : 8 = 35 : 40, so X : Y : Z = 24 : 35 : 40.

Question: Exercise 6.1, Question 3: Find A : B : C : D, if: (i) A : B = 2 : 3, B : C = 4 : 5 and C : D = 7 : 10

Answer: First A : B : C: make B the same: A : B = 8 : 12, B : C = 12 : 15, so A : B : C = 8 : 12 : 15. Then make C the same: A : B : C = 56 : 84 : 105 and C : D = 105 : 150, so A : B : C : D = 56 : 84 : 105 : 150.

Question: Exercise 6.1, Question 3: (ii) A : B = 4 : 3, B : C = 5 : 7 and A : D = 7 : 9

Answer: First A : B : C: make B the same: A : B = 20 : 15, B : C = 15 : 21, so A : B : C = 20 : 15 : 21. From A : D = 7 : 9: D = 20 × 9/7 = 180/7. Multiplying all by 7: A : B : C : D = 140 : 105 : 147 : 180. Check: A : D = 140 : 180 = 7 : 9. ✓

Question: Exercise 6.1, Question 4: Find the unknown number x, if: (i) x : 2 = 5 : 6

Answer: From the proportion: 6 × x = 2 × 5, so 6x = 10, i.e. x = 10/6 = 5/3.

Question: Exercise 6.1, Question 4: (ii) 3 : x = 9 : 7

Answer: From the proportion: 9 × x = 3 × 7, so 9x = 21, i.e. x = 21/9 = 7/3.

Question: Exercise 6.1, Question 4: (iii) 4 : 5 = x : 7

Answer: From the proportion: 5 × x = 4 × 7, so 5x = 28, i.e. x = 28/5.

Question: Exercise 6.1, Question 5: The ratio of Shazia's weight to Nazia's weight is 5 : 9. The ratio of Nazia's weight to Madiha's weight is 8 : 11. Find their continued ratio of weights.

Answer: To make Nazia the same: Shazia : Nazia = 5 : 9 = 40 : 72 and Nazia : Madiha = 8 : 11 = 72 : 99, so Shazia : Nazia : Madiha = 40 : 72 : 99.

Question: Exercise 6.1, Question 6: Divide 2000 rupees among Saleem, Irfan and Imran in the ratio 2 : 3 : 5.

Answer: Sum of the ratio terms = 2 + 3 + 5 = 10. One share = 2000/10 = 200 rupees. So Saleem's share = 2 × 200 = 400 rupees, Irfan's share = 3 × 200 = 600 rupees, Imran's share = 5 × 200 = 1000 rupees.

Question: Exercise 6.1, Question 7: In the annual examination the ratio of the marks of three friends is as follows: the ratio of Akhtar's marks to

Aziz's marks is 3 : 4 and the ratio of Aziz's marks to Anas's marks is 5 : 7. Find their continued ratio of marks.

Answer: To make Aziz the same: Akhtar : Aziz = 3 : 4 = 15 : 20 and Aziz : Anas = 5 : 7 = 20 : 28, so Akhtar : Aziz : Anas = 15 : 20 : 28.

Question: Exercise 6.1, Question 8: The ratio of the property of the members of a household is as follows: the ratio of the father's property to the mother's property is 5 : 2. The ratio of the mother's property to the daughter's property is 3 : 4. While the ratio of the daughter's property to the son's property is 6 : 7. Find their continued ratio of property.

Answer: First Father : Mother : Daughter: make Mother the same: Father : Mother = 15 : 6, Mother : Daughter = 6 : 8, so Father : Mother : Daughter = 15 : 6 : 8. Then make Daughter the same: Father : Mother : Daughter = 45 : 18 : 24 and Daughter : Son = 24 : 28, so Father : Mother : Daughter : Son = 45 : 18 : 24 : 28.

Question: Exercise 6.1, Question 9: A person's amount of 27,400 rupees is to be spent on utility bills. The ratio of gas to water is 5 : 2, the ratio of water to electricity is 4 : 3, the ratio of electricity to phone is 7 : 6. Find the actual amount of each bill.

Answer: Continued ratio: Gas : Water : Electricity : Phone = 70 : 28 : 21 : 18 (make Water the same: Gas : Water : Electricity = 10 : 4 : 3; then make Electricity the same: 70 : 28 : 21 : 18). Sum of the ratio terms = 137. One share = 27400/137 = 200 rupees. So Gas = 70 × 200 = 14,000 rupees, Water = 28 × 200 = 5,600 rupees, Electricity = 21 × 200 = 4,200 rupees, Phone = 18 × 200 = 3,600 rupees. Check: 14000 + 5600 + 4200 + 3600 = 27,400. ✓

Question: Exercise 6.1, Question 10: A student bought one book each of Class 7 Mathematics, Science, English and Urdu, for 675 rupees in all. The ratio of the prices of the books is as follows: the ratio of the price of the

English book to the Urdu book is 2 : 3, the ratio of the price of the Urdu book to the Science book is 2 : 5, the ratio of the price of the Science book to the Mathematics book is 3 : 4. Find the price of each book.

Answer: Continued ratio: English : Urdu : Science : Mathematics = 4 : 6 : 15 : 20 (make Urdu the same: 4 : 6 : 15; then make Science the same: 4 : 6 : 15 : 20). Sum of the ratio terms = 45. One share = 675/45 = 15 rupees. So English = 4 × 15 = 60 rupees, Urdu = 6 × 15 = 90 rupees, Science = 15 × 15 = 225 rupees, Mathematics = 20 × 15 = 300 rupees. Check: 60 + 90 + 225 + 300 = 675. ✓

Question: Exercise 6.2, Question 1: Solve the following sum by the unitary method and the proportion method. The price of 15 chairs is 4,500 rupees. Find the price of 20 such chairs.

Answer: Unitary method: price of 1 chair = 4500/15 = 300 rupees, so price of 20 chairs = 20 × 300 = 6,000 rupees. Proportion method: 15 : 20 = 4500 : x, so 15x = 20 × 4500, i.e. x = 90000/15 = 6,000 rupees.

Question: Exercise 6.2, Question 2: Six identical taps can fill a water tank in 3 hours. How many such taps will fill the same tank in 2 hours?

Answer: This is inverse proportion (more taps means less time). Unitary method: 1 tap will fill the tank in 6 × 3 = 18 hours, so to fill it in 2 hours 18/2 = 9 taps are needed. Proportion method: 6 : x = 2 : 3, so 2x = 18, i.e. x = 9 taps.

Question: Exercise 6.2, Question 3: If 5 men finish a stock of supplies in 8 days, how many men will be able to finish the same stock of supplies in 4 days?

Answer: This is inverse proportion (fewer days means more men). Unitary method: 1 man will finish the stock in 5 × 8 = 40 days, so to finish it in 4 days 40/4 = 10 men are needed. Proportion method: 5 : x = 4 : 8, so 4x = 40, i.e. x = 10 men.

Question: Exercise 6.2, Question 4: The price of 4 kilograms of mangoes is 300 rupees. Find the price of 10 kilograms of mangoes.

Answer: Unitary method: price of 1 kilogram of mangoes = 300/4 = 75 rupees, so price of 10 kilograms = 10 × 75 = 750 rupees. Proportion method: 4 : 10 = 300 : x, so 4x = 3000, i.e. x = 750 rupees.

Question: Exercise 6.2, Question 5: A student reads 30 pages of a book in 4 days. How many pages of the same book will he read in 10 days?

Answer: Unitary method: in 1 day 30/4 = 7.5 pages, so in 10 days 10 × 7.5 = 75 pages. Proportion method: 4 : 10 = 30 : x, so 4x = 300, i.e. x = 75 pages.

Question: Exercise 6.2, Question 6: 20 soldiers finish a stock of supplies in 8 days. How many soldiers will finish the same stock in 5 days?

Answer: This is inverse proportion. Unitary method: 1 soldier will finish the stock in 20 × 8 = 160 days, so to finish it in 5 days 160/5 = 32 soldiers are needed. Proportion method: 20 : x = 5 : 8, so 5x = 160, i.e. x = 32 soldiers.

Question: Exercise 6.2, Question 7: If 4 labourers complete a job in 8 hours. In how many hours will 6 labourers complete the same job?

Answer: This is inverse proportion. Unitary method: 1 labourer will complete the job in 4 × 8 = 32 hours, so 6 labourers will complete the job in 32/6 = 16/3 = 5 1/3 hours (i.e. 5 hours and 20 minutes). Proportion method: 4 : 6 = x : 8, so 6x = 32, i.e. x = 16/3 hours.

Question: Exercise 6.2, Question 8: Find the price of 18 kilograms of spinach, if the price of 6 kilograms of spinach is 105 rupees.

Answer: Unitary method: price of 1 kilogram of spinach = 105/6 = 17.5 rupees, so price of 18 kilograms = 18 × 17.5 = 315 rupees. Proportion method: 6 : 18 = 105 : x, so 6x = 1890, i.e. x = 315 rupees.

Question: Exercise 6.2, Question 9: If the weight of 4/7 part of a piece of

metal is 28 kilograms, find the weight of the 8/9 part of the same piece of metal.

Answer: Unitary method: weight of the whole piece = 28 × 7/4 = 49 kilograms, so weight of the 8/9 part = 49 × 8/9 = 392/9 = 43 5/9 kilograms. Proportion method: 4/7 : 8/9 = 28 : x, so (4/7)x = 28 × 8/9, i.e. x = 392/9 kilograms.

Question: Exercise 6.2, Question 10: The cost of a party of 35 guests is 5250 rupees, find the cost of a party of 85 guests.

Answer: Unitary method: cost per guest = 5250/35 = 150 rupees, so cost of 85 guests = 85 × 150 = 12,750 rupees. Proportion method: 35 : 85 = 5250 : x, so 35x = 446250, i.e. x = 12,750 rupees.

Question: Exercise 6.3, Question 1: 20 labourers can complete half of a job in a certain time. How many labourers are needed to complete a quarter of the job in the same time?

Answer: This is direct proportion (less work means fewer labourers). 1/2 of the job → 20 labourers, so 1/4 of the job → 20 × (1/4)/(1/2) = 20 × 1/2 = 10 labourers.

Question: Exercise 6.3, Question 2: If 6 labourers complete a job in 10 days. Tell how many labourers are needed to complete the same job in 15 days?

Answer: This is inverse proportion (more days means fewer labourers). 6 × 10 = 60 labourer-days, so labourers needed in 15 days = 60/15 = 4 labourers.

Question: Exercise 6.3, Question 3: If 5 labourers complete a job in 8 days. Tell how many labourers are needed to complete the same job in 4 days?

Answer: This is inverse proportion. 5 × 8 = 40 labourer-days, so labourers needed in 4 days = 40/4 = 10 labourers.

Question: Exercise 6.3, Question 4: If 3 weavers weave a piece of cloth in

four days. Tell in how many days will five such weavers weave the same cloth?

Answer: This is inverse proportion (more weavers means fewer days). 3 × 4 = 12 weaver-days, so days needed by 5 weavers = 12/5 = 2 2/5 days (i.e. 2 days, 9 hours and 36 minutes).

Question: Exercise 6.3, Question 5: If 5 men complete a third part of a job. Tell how much part of the job will 7 men complete in the same time?

Answer: This is direct proportion (more men means more work). 5 men → 1/3 part, so 1 man → 1/15 part, and 7 men → 7/15 part.

Question: Exercise 6.3, Question 6: 12 labourers are needed to build a bungalow in 64 days. Tell how many days will 18 labourers take to build the same bungalow, working at the same speed?

Answer: This is inverse proportion. 12 × 64 = 768 labourer-days, so days needed by 18 labourers = 768/18 = 128/3 = 42 2/3 days (i.e. 42 days and 16 hours).

Question: Exercise 6.3, Question 7: Fodder (grass) for 1500 sheep lasts 75 days. If the sheep use the fodder at the same rate, find: (i) how many sheep are there that will eat up the same fodder (grass) completely in 90 days.

Answer: This is inverse proportion (more days means fewer sheep). 1500 × 75 = 112500 sheep-days, so sheep that will finish the fodder in 90 days = 112500/90 = 1250 sheep.

Question: Exercise 6.3, Question 7: (ii) The number of days in which 1200 sheep will eat up the same fodder (grass) completely.

Answer: 1500 × 75 = 112500 sheep-days, so days needed by 1200 sheep = 112500/1200 = 93.75 days (i.e. 93 days and 18 hours).

Question: Exercise 6.4, Question 1: Convert the following into metres per second. (i) 50 kilometres per hour

Answer: 50 kilometres per hour = 50 × 1000/3600 = 500/36 = 125/9 = 13 8/9 metres per second (about 13.89 metres per second).

Question: Exercise 6.4, Question 1: (ii) 75 kilometres per hour

Answer: 75 kilometres per hour = 75 × 1000/3600 = 750/36 = 125/6 = 20 5/6 metres per second (about 20.83 metres per second).

Question: Exercise 6.4, Question 1: (iii) 80 kilometres per hour

Answer: 80 kilometres per hour = 80 × 1000/3600 = 800/36 = 200/9 = 22 2/9 metres per second (about 22.22 metres per second).

Question: Exercise 6.4, Question 1: (iv) 505 kilometres per hour

Answer: 505 kilometres per hour = 505 × 1000/3600 = 5050/36 = 2525/18 = 140 5/18 metres per second (about 140.28 metres per second).

Question: Exercise 6.4, Question 2: Convert the following into kilometres per hour. (i) 30 metres per second

Answer: 30 metres per second = 30 × 3600/1000 = 108 kilometres per hour.

Question: Exercise 6.4, Question 2: (ii) 12 metres per second

Answer: 12 metres per second = 12 × 3600/1000 = 43.2 kilometres per hour.

Question: Exercise 6.4, Question 2: (iii) 42 metres per second

Answer: 42 metres per second = 42 × 3600/1000 = 151.2 kilometres per hour.

Question: Exercise 6.4, Question 2: (iv) 25.5 metres per second

Answer: 25.5 metres per second = 25.5 × 3600/1000 = 91.8 kilometres per hour.

Question: Exercise 6.4, Question 3: A motorcycle covers a distance of 80 kilometres in 2 hours. Find its speed.

Answer: Speed = distance/time = 80/2 = 40 kilometres per hour.

Question: Exercise 6.4, Question 4: A van, at a speed of 60 kilometres per

hour, will take how much time to cover a distance of 180 kilometres?

Answer: Time = distance/speed = 180/60 = 3 hours.

Question: Exercise 6.4, Question 5: If a cyclist covers a distance of 150 metres in 30 seconds, tell how much distance he will cover in 45 seconds?

Answer: Speed = 150/30 = 5 metres per second, so distance in 45 seconds = 5 × 45 = 225 metres.

Question: Exercise 6.4, Question 6: A bus takes 3 hours to reach its destination. If its speed is 60 kilometres per hour, in how much time will it cover the same distance at a speed of 70 kilometres per hour?

Answer: Distance = speed × time = 60 × 3 = 180 kilometres. New time = distance/speed = 180/70 = 18/7 = 2 4/7 hours (i.e. 2 hours, 34 minutes and about 17 seconds).

Question: Exercise 6.4, Question 7: A car takes 3 hours to cover a distance of 210 kilometres. (i) What will its speed be, calculated per hour?

Answer: Speed = distance/time = 210/3 = 70 kilometres per hour.

Question: Exercise 6.4, Question 7: (ii) What will its speed be in kilometres per minute?

Answer: 70 kilometres per hour = 70/60 = 7/6 = 1 1/6 kilometres per minute (about 1.17 kilometres per minute).

Question: Exercise 6.4, Question 7: (iii) What will its speed be in metres per second?

Answer: 70 kilometres per hour = 70 × 1000/3600 = 700/36 = 175/9 = 19 4/9 metres per second (about 19.44 metres per second).

Question: Exercise 6.4, Question 7: (iv) How much distance will it cover in 4 1/2 hours at a speed of 70 kilometres per hour?

Answer: Distance = speed × time = 70 × 4 1/2 = 70 × 9/2 = 315 kilometres.

Question: Exercise 6.4, Question 7: (v) How much time will it take to cover

a distance of 350 kilometres?

Answer: Time = distance/speed = 350/70 = 5 hours.

Question: Review Exercise 6, Question 1: If x : y = 2 : 5 and x : z = 4 : 7, find x : y : z.

Answer: To make x the same: x : y = 2 : 5 = 4 : 10 and x : z = 4 : 7, so x : y : z = 4 : 10 : 7.

Question: Review Exercise 6, Question 2: If P : Q = 3 : 4, Q : R = 5 : 6 and R : S = 7 : 8, find P : Q : R : S.

Answer: First P : Q : R: make Q the same: P : Q = 15 : 20, Q : R = 20 : 24, so P : Q : R = 15 : 20 : 24. Then make R the same: P : Q : R = 105 : 140 : 168 and R : S = 168 : 192, so P : Q : R : S = 105 : 140 : 168 : 192.

Question: Review Exercise 6, Question 3: The price of 10 pens is 250 rupees. Find the price of 17 such pens.

Answer: Price of 1 pen = 250/10 = 25 rupees, so price of 17 pens = 17 × 25 = 425 rupees.

Question: Review Exercise 6, Question 4: If 10 labourers complete a job in 7 days, how many labourers will be needed to complete the same job in 5 days?

Answer: This is inverse proportion. 10 × 7 = 70 labourer-days, so labourers needed in 5 days = 70/5 = 14 labourers.

Question: Review Exercise 6, Question 5: If x : 5 = 6 : 15, find x.

Answer: From the proportion: 15 × x = 5 × 6, so 15x = 30, i.e. x = 2.

Question: Review Exercise 6, Question 6: If 7 : 8 = y : 24, find y.

Answer: From the proportion: 8 × y = 7 × 24, so 8y = 168, i.e. y = 21.

Question: Review Exercise 6, Question 7: Convert 60 kilometres per hour into metres per second.

Answer: 60 kilometres per hour = 60 × 1000/3600 = 600/36 = 50/3 = 16 2/3 metres per second (about 16.67 metres per second).

Question: Review Exercise 6, Question 8: Convert 105 metres per second into kilometres per hour.

Answer: 105 metres per second = 105 × 3600/1000 = 378 kilometres per hour.

Question: Review Exercise 6, Question 9: A car covers a distance of 30 kilometres in 25 minutes. Tell how much distance it will cover in 55 minutes? And find its speed: (i) in kilometres per hour (ii) in kilometres per minute (iii) in metres per second

Answer: Distance = 30/25 × 55 = 66 kilometres. (i) Speed = 30 kilometres/25 minutes = 30 × 60/25 = 72 kilometres per hour. (ii) Speed = 30/25 = 1.2 kilometres per minute. (iii) Speed = 72 × 1000/3600 = 20 metres per second.

Question: Review Exercise 6, Question 10: The ratio of the petrol consumption of three cars A, B and C is 7 : 9 : 11. Tell, if the total consumption of all three cars is 162 litres, what is the petrol consumption of each car?

Answer: Sum of the ratio terms = 7 + 9 + 11 = 27. One share = 162/27 = 6 litres. So car A = 7 × 6 = 42 litres, car B = 9 × 6 = 54 litres, car C = 11 × 6 = 66 litres. Check: 42 + 54 + 66 = 162. ✓

Question: Review Exercise 6, Question 11: The cost of carrying goods up to 40 kilometres is 100 rupees. Find: (i) What will the cost of carrying the goods up to 150 kilometres be?

Answer: (i) Cost = 100/40 × 150 = 375 rupees.

Question: Review Exercise 6, Question 11: (ii) If the cost of carrying the goods is 560 rupees, what will the distance be?

Answer: (ii) Distance = 40 × 560/100 = 224 kilometres.

Question: Mental Store: Take any three-digit number whose digits are in decreasing order (first digit larger than the last). Reverse it, subtract the

smaller number from the larger, reverse the answer, and add the two. What answer do you get every time? Show with two examples.

Answer: You get 1089 every time. Example 1: Take 732 (7 > 2). Reverse: 237. Subtract: 732 − 237 = 495. Reverse the answer: 594. Add: 495 + 594 = 1089. Example 2: Take 321. Reverse: 123. Subtract: 321 − 123 = 198. Reverse the answer: 891. Add: 198 + 891 = 1089. (This works only when the middle digit of the first subtraction is 9 and the first and last digits are not equal.)

7. Financial Arithmetic

Question: Exercise 7.1, Part A, Question 1: The value of a property is Rs 286,400 and the tax rate is 5%. How much is the property tax?

Answer: Property tax = value × rate ÷ 100 = 286,400 × 5 ÷ 100 = 1,432,000 ÷ 100 = Rs 14,320.

Question: Exercise 7.1, Part A, Question 2: The value of a property is Rs 1,657,900 and the tax rate is 2%. How much is the property tax?

Answer: Property tax = 1,657,900 × 2 ÷ 100 = 3,315,800 ÷ 100 = Rs 33,158.

Question: Exercise 7.1, Part A, Question 3: The value of a property is Rs 814,500 and the tax rate is 3%. How much is the property tax?

Answer: Property tax = 814,500 × 3 ÷ 100 = 2,443,500 ÷ 100 = Rs 24,435.

Question: Exercise 7.1, Part A, Question 4: The value of a property is Rs 234,000 and the tax rate is 1.4%. How much is the property tax?

Answer: Property tax = 234,000 × 1.4 ÷ 100 = 327,600 ÷ 100 = Rs 3,276.

Question: Exercise 7.1, Part A, Question 5: The value of a property is Rs 2,050,000 and the tax rate is 1.6%. How much is the property tax?

Answer: Property tax = 2,050,000 × 1.6 ÷ 100 = 3,280,000 ÷ 100 = Rs 32,800.

Question: Exercise 7.1, Part B, Question 1: Property tax of Rs 25,500 was paid at the rate of 2%. What was the value of the property?

Answer: Value of property = tax × 100 ÷ rate = 25,500 × 100 ÷ 2 = 2,550,000 ÷ 2 = Rs 1,275,000.

Question: Exercise 7.1, Part B, Question 2: Property tax of Rs 35,900 was paid at the rate of 4%. What was the value of the property?

Answer: Value of property = 35,900 × 100 ÷ 4 = 3,590,000 ÷ 4 = Rs 897,500.

Question: Exercise 7.1, Part B, Question 3: Property tax of Rs 4,800 was paid at the rate of 5%. What was the value of the property?

Answer: Value of property = 4,800 × 100 ÷ 5 = 480,000 ÷ 5 = Rs 96,000.

Question: Exercise 7.1, Part B, Question 4: Property tax of Rs 8,500 was paid at the rate of 3.5%. What was the value of the property?

Answer: Value of property = 8,500 × 100 ÷ 3.5 = 850,000 ÷ 3.5 = Rs 242,857.14 (approximately).

Question: Exercise 7.1, Part B, Question 5: Property tax of Rs 11,700 was paid at the rate of 1.5%. What was the value of the property?

Answer: Value of property = 11,700 × 100 ÷ 1.5 = 1,170,000 ÷ 1.5 = Rs 780,000.

Question: Exercise 7.1, Part C, Question 1: Property tax of Rs 30,000 was paid and the value of the property is Rs 1,800,000. Find the tax rate.

Answer: Rate = tax ÷ value × 100 = 30,000 ÷ 1,800,000 × 100 = 1.67% (approximately).

Question: Exercise 7.1, Part C, Question 2: Property tax of Rs 33,000 was paid and the value of the property is Rs 3,300,000. Find the tax rate.

Answer: Rate = 33,000 ÷ 3,300,000 × 100 = 1%.

Question: Exercise 7.1, Part C, Question 3: Property tax of Rs 12,000 was paid and the value of the property is Rs 600,000. Find the tax rate.

Answer: Rate = 12,000 ÷ 600,000 × 100 = 2%.

Question: Exercise 7.1, Part C, Question 4: Property tax of Rs 8,676 was paid and the value of the property is Rs 433,800. Find the tax rate.

Answer: Rate = 8,676 ÷ 433,800 × 100 = 2%.

Question: Exercise 7.1, Part C, Question 5: Property tax of Rs 12,000 was paid and the value of the property is Rs 600,000. Find the tax rate.

Answer: Rate = 12,000 ÷ 600,000 × 100 = 2%. (This question is the same as Question 3.)

Question: Exercise 7.1, Part D, Question 1: What is a tax?

Answer: A tax is the amount that the government collects from people's income, production or services, so that the country's expenses can be met.

Question: Exercise 7.1, Part D, Question 2: What is the purpose of levying a tax?

Answer: The purpose of levying a tax is to meet the government's expenses, such as defence, police, education, health and public services.

Question: Exercise 7.1, Part D, Question 3: Explain property tax.

Answer: Property tax is the tax that the government levies on a person's property or personal estate. It is paid every year, and its rate is generally about 2% of the value of the property.

Question: Exercise 7.1, Part D, Question 4: Name any four properties that are exempt from tax.

Answer: Properties exempt from tax: (1) government properties, (2) a residential building on 120 square yards of land or less, (3) a building with a covered area of 600 square feet or less, (4) properties such as a library, garden, place of worship, orphanage and cemetery.

Question: Exercise 7.2, Question 1: A Parker pen costs Rs 875. Find the GST on it at the rate of 17%.

Answer: GST = price of item × 17% = 875 × 0.17 = Rs 148.75.

Question: Exercise 7.2, Question 2: A mobile phone costs Rs 4,850. Find the GST on it at the rate of 17%.

Answer: GST = 4,850 × 0.17 = Rs 824.50.

Question: Exercise 7.2, Question 3: The total price of cosmetics is Rs 14,976, while the actual price of the goods is Rs 12,800. Find the amount

of GST.

Answer: GST = total price − price of item = 14,976 − 12,800 = Rs 2,176. (Check: 12,800 × 17% = 2,176.)

Question: Exercise 7.2, Question 4: A microwave oven costs Rs 15,200. The GST charged on it is at the rate of 17%. Find the amount of GST and the total price of the oven.

Answer: GST = 15,200 × 0.17 = Rs 2,584. Total price = 15,200 + 2,584 = Rs 17,784.

Question: Exercise 7.2, Question 5: Abdullah bought a watch for Rs 7,880. Find the amount of GST in the total price at the rate of 17%.

Answer: Here Rs 7,880 is the total price (including GST). Price of item = total price ÷ 1.17 = 7,880 ÷ 1.17 = Rs 6,735.04. GST = 7,880 − 6,735.04 = Rs 1,144.96 (about Rs 1,145).

Question: Exercise 7.2, Question 6: A refrigerator was sold for Rs 35,375 including 17% GST. Find the actual price of the refrigerator.

Answer: Actual price = total price ÷ 1.17 = 35,375 ÷ 1.17 = Rs 30,235.04 (approximately).

Question: Exercise 7.2, Question 7: A person bought a dishwasher for Rs 61,000. Find the price of the dishwasher without GST. Also find the amount of GST.

Answer: Price without GST = 61,000 ÷ 1.17 = Rs 52,136.75 (approximately). GST = 61,000 − 52,136.75 = Rs 8,863.25 (approximately).

Question: Exercise 7.2, Question 8: Alia paid Rs 2,720 GST at the rate of 17% on the purchase of decorative items. Find the price of the decorative items without GST.

Answer: Price of item = GST ÷ rate = 2,720 ÷ 0.17 = Rs 16,000.

Question: Exercise 7.2, Question 9: An electric company charges Rs 1,547

GST on electricity usage of Rs 9,100. Find the rate of GST.

Answer: GST rate = GST ÷ price of item × 100 = 1,547 ÷ 9,100 × 100 = 17%.

Question: Exercise 7.2, Question 10: The price of a shirt without tax is Rs 2,560 and the GST on it is Rs 436. Find the rate of GST.

Answer: GST rate = 436 ÷ 2,560 × 100 = 17.03% (approximately, i.e. about 17%).

Question: Practical Work 1, (i): Find the unknown quantities using the formula. Cost price Rs 1,600, selling price Rs 1,750, formula: profit = selling price − cost price. What is the total profit?

Answer: Total profit = 1,750 − 1,600 = Rs 150.

Question: Practical Work 1, (ii): Cost price Rs 2,000, total profit Rs 600, formula: selling price = cost price + profit. What is the selling price?

Answer: Selling price = 2,000 + 600 = Rs 2,600.

Question: Practical Work 1, (iii): Selling price Rs 10,450, total profit Rs 950, formula: cost price = selling price − profit. What is the cost price?

Answer: Cost price = 10,450 − 950 = Rs 9,500.

Question: Practical Work 2, (i): Selling price Rs 1,000, cost price Rs 800, total profit Rs 200. Formula: profit rate = profit ÷ selling price × 100. Find the profit rate.

Answer: Profit rate = 200 ÷ 1,000 × 100 = 20%.

Question: Practical Work 2, (ii): Cost price Rs 1,000, total profit Rs 100, profit rate 10%. Formula: selling price = cost price × (1 + profit rate). Find the selling price.

Answer: Selling price = 1,000 × (1 + 0.10) = 1,000 × 1.10 = Rs 1,100.

Question: Practical Work 2, (iii): Selling price Rs 2,000, cost price Rs 1,600, total profit Rs 400. Formula: profit rate = profit ÷ selling price × 100. Find the profit rate.

Answer: Profit rate = 400 ÷ 2,000 × 100 = 20%.

Question: Practical Work 2, (iv): Selling price Rs 1,200, cost price Rs 1,000, profit rate given as 20%. Find the total profit.

Answer: Total profit = selling price − cost price = 1,200 − 1,000 = Rs 200. (Note: Rs 200 is 20% of the cost price, but 16.67% of the selling price — the given 20% fits the cost-price base, not the selling-price formula in the table.)

Question: Practical Work 2, (v): Selling price Rs 2,750, cost price Rs 2,500, total profit Rs 250. Find the profit rate.

Answer: By the table's formula, profit rate = profit ÷ selling price × 100 = 250 ÷ 2,750 × 100 = 9.09% (approximately). (On the cost price this rate would be 250 ÷ 2,500 × 100 = 10%.)

Question: Practical Work 3, (i): Principal Rs 500, period 2 years, mark-up rate 12%. Formula: mark-up = principal × rate × time ÷ 100. Find the mark-up.

Answer: Mark-up = 500 × 12 × 2 ÷ 100 = 12,000 ÷ 100 = Rs 120.

Question: Practical Work 3, (ii): Mark-up Rs 205, period 1 year, mark-up rate 8%. Formula: principal = mark-up × 100 ÷ (rate × time). Find the principal.

Answer: Principal = 205 × 100 ÷ (8 × 1) = 20,500 ÷ 8 = Rs 2,562.50.

Question: Practical Work 3, (iii): Principal Rs 1,650, mark-up Rs 528, period 10 years. Formula: rate = mark-up × 100 ÷ (principal × time). Find the rate.

Answer: Rate = 528 × 100 ÷ (1,650 × 10) = 52,800 ÷ 16,500 = 3.2%.

Question: Practical Work 3, (iv): Principal Rs 3,500, mark-up Rs 350, mark-up rate 2.5%. Formula: time = mark-up × 100 ÷ (principal × rate). Find the time.

Answer: Time = 350 × 100 ÷ (3,500 × 2.5) = 35,000 ÷ 8,750 = 4 years.

Question: Practical Work 3, (v): Principal Rs 100,000, period 3 years, mark-up rate 1.25%. Find the mark-up.

Answer: Mark-up = 100,000 × 1.25 × 3 ÷ 100 = 375,000 ÷ 100 = Rs 3,750.

Question: Practical Work 3, (vi): Mark-up Rs 1,050, period 5 years, mark-up rate 4.5%. Find the principal.

Answer: Principal = 1,050 × 100 ÷ (4.5 × 5) = 105,000 ÷ 22.5 = Rs 4,666.67 (approximately).

Question: Exercise 7.3, Question 1: A shopkeeper bought a helmet for Rs 1,860 and sold it for Rs 2,650. Find his profit and the profit rate (on the cost price).

Answer: Profit = 2,650 − 1,860 = Rs 790. Profit rate = 790 ÷ 1,860 × 100 = 42.47% (approximately).

Question: Exercise 7.3, Question 2: A shopkeeper bought a microwave oven for Rs 6,250 and sold it for Rs 6,700. Find the profit rate (on the cost price).

Answer: Profit = 6,700 − 6,250 = Rs 450. Profit rate = 450 ÷ 6,250 × 100 = 7.2%.

Question: Exercise 7.3, Question 3: A shopkeeper uses a 65% mark-up rate on computer goods. The cost price of a particular item is Rs 1,520. Find the selling price of that item.

Answer: Selling price = cost price × (1 + rate) = 1,520 × (1 + 0.65) = 1,520 × 1.65 = Rs 2,508.

Question: Exercise 7.3, Question 4: The cost price of a novel is Rs 1,100. Find the selling price of the novel at a 66% mark-up rate.

Answer: Selling price = 1,100 × (1 + 0.66) = 1,100 × 1.66 = Rs 1,826.

Question: Exercise 7.3, Question 5: A mobile phone company uses a 55% mark-up rate. The cost price of each mobile is Rs 2,260. Find the selling

price of that mobile.

Answer: Selling price = 2,260 × (1 + 0.55) = 2,260 × 1.55 = Rs 3,503.

Question: Exercise 7.3, Question 6: A mobile company uses a 45% mark-up rate. Find the cost price of a mobile whose selling price is Rs 4,785.

Answer: Cost price = selling price ÷ (1 + rate) = 4,785 ÷ 1.45 = Rs 3,300.

Question: Exercise 7.3, Question 7: The selling price of a calculator is Rs 1,230 and the mark-up rate on it is 50%. Find the cost price of the calculator.

Answer: Cost price = 1,230 ÷ (1 + 0.50) = 1,230 ÷ 1.5 = Rs 820.

Question: Exercise 7.3, Question 8: A stationery seller uses a 60% mark-up on everything he sells. If the selling price of an item is Rs 144, find the cost price of that item.

Answer: Cost price = 144 ÷ (1 + 0.60) = 144 ÷ 1.6 = Rs 90.

Question: Exercise 7.3, Question 9: In a small home business selling children's clothes, the monthly income is Rs 25,000. The cost of goods and other expenses together are Rs 10,550. Find the percentage of total profit.

Answer: Total profit = 25,000 − 10,550 = Rs 14,450. Profit percentage (on total sales) = 14,450 ÷ 25,000 × 100 = 57.8%.

Question: Exercise 7.3, Question 10: In a particular business the monthly earnings are Rs 2,500,000. Goods worth Rs 15,000 that were sold were returned. The cost of the goods sold is Rs 505,000 and other expenses are Rs 580,000. Find the profit percentage.

Answer: Net sales = 2,500,000 − 15,000 = Rs 2,485,000. Total profit = 2,485,000 − 505,000 − 580,000 = Rs 1,400,000. Profit percentage = 1,400,000 ÷ 2,485,000 × 100 = 56.34% (approximately).

Question: Exercise 7.3, Question 11: A company's total monthly earnings are Rs 1,850,000. Goods worth Rs 85,000 that were sold were returned.

The cost of the goods sold is Rs 850,000. Administrative expenses are Rs 550,000. Find the percentage of the company's total profit.

Answer: Net sales = 1,850,000 − 85,000 = Rs 1,765,000. Total profit = 1,765,000 − 850,000 − 550,000 = Rs 365,000. Profit percentage = 365,000 ÷ 1,765,000 × 100 = 20.68% (approximately).

Question: Exercise 7.4 (mark-up), Question 1: Qasim bought a car for Rs 450,800 with a bank loan. The annual mark-up rate is 12%. Find the selling price of the car, if the period is 3 years.

Answer: Mark-up = 450,800 × 12 × 3 ÷ 100 = Rs 162,288. Selling price = 450,800 + 162,288 = Rs 613,088.

Question: Exercise 7.4 (mark-up), Question 2: Najma bought a jewellery set for Rs 175,500 in instalments. The annual mark-up rate is 10%. Find the selling price of the jewellery set, if the period is 9 months.

Answer: Time = 9 ÷ 12 = 0.75 years. Mark-up = 175,500 × 10 × 0.75 ÷ 100 = Rs 13,162.50. Selling price = 175,500 + 13,162.50 = Rs 188,662.50.

Question: Exercise 7.4 (mark-up), Question 3: Shazia sold a mobile phone for Rs 16,900, while the mark-up rate is 15%. Find her mark-up, if the period is 219 days. (Here Rs 16,900 is taken as the amount on which the mark-up was charged.)

Answer: Time = 219 ÷ 365 = 0.6 years. Mark-up = 16,900 × 15 × 0.6 ÷ 100 = 16,900 × 0.09 = Rs 1,521.

Question: Exercise 7.4 (mark-up), Question 4: If the mark-up amount on a colour TV set is Rs 8,500 for 1 ½ years at 15%, find the purchase price of the TV.

Answer: Principal = mark-up × 100 ÷ (rate × time) = 8,500 × 100 ÷ (15 × 1.5) = 850,000 ÷ 22.5 = Rs 37,777.78 (approximately).

Question: Exercise 7.4 (mark-up), Question 5, (i): Find the unknown

quantity from the table. Principal Rs 800,000, period 1 year, mark-up rate 12%. What is the mark-up?

Answer: Mark-up = 800,000 × 12 × 1 ÷ 100 = Rs 96,000.

Question: Exercise 7.4 (mark-up), Question 5, (ii): Find the unknown quantity from the table. Mark-up Rs 10,000, mark-up rate 10%, principal Rs 100,000 (the first digit is unclear in print). What is the period?

Answer: Time = 10,000 × 100 ÷ (100,000 × 10) = 1,000,000 ÷ 1,000,000 = 1 year. (Note: the principal is printed as "00,000" in the book, with the first digit missing; it is taken here as 100,000, which gives a period of 1 year.)

Question: Exercise 7.4 (mark-up), Question 5, (iii): Find the unknown quantity from the table. Mark-up Rs 6,600, period 2 years, mark-up rate 11%. What is the principal?

Answer: Principal = 6,600 × 100 ÷ (11 × 2) = 660,000 ÷ 22 = Rs 30,000.

Question: Exercise 7.4 (mark-up), Question 5, (iv): Find the unknown quantity from the table. Principal Rs 200,000, period 1 ½ years, mark-up rate 8%. What is the mark-up?

Answer: Mark-up = 200,000 × 8 × 1.5 ÷ 100 = 2,400,000 ÷ 100 = Rs 24,000.

Question: Exercise 7.4 (mark-up), Question 5, (v): Find the unknown quantity from the table. Principal Rs 4,000, mark-up Rs 800, period 2 ½ years. Find the mark-up rate.

Answer: Rate = 800 × 100 ÷ (4,000 × 2.5) = 80,000 ÷ 10,000 = 8%.

Question: Exercise 7.4 (mark-up), Question 5, (vi): Find the unknown quantity from the table. Mark-up Rs 45,000, period 73 days, mark-up rate 10%. What is the principal?

Answer: Time = 73 ÷ 365 = 0.2 years. Principal = 45,000 × 100 ÷ (10 × 0.2) = 4,500,000 ÷ 2 = Rs 2,250,000.

Question: Exercise 7.4 (Zakat and Ushr), Question 1: A person keeps Rs 50,000 for a whole year. Is he sahib-e-nisab? If so, find the amount of Zakat he has to pay.

Answer: Whether he is sahib-e-nisab depends on the current value of the nisab in gold or silver, which is not given in the book. If this amount is above the nisab, then Zakat = 50,000 × 2.5% = 50,000 × 0.025 = Rs 1,250.

Question: Exercise 7.4 (Zakat and Ushr), Question 2: A salaried person has Rs 200,000 and 3 tolas of pure gold. Find the Zakat he has to pay. The price of gold is Rs 47,400 per tola.

Answer: Value of gold = 3 × 47,400 = Rs 142,200. Total amount = 200,000 + 142,200 = Rs 342,200. Zakat = 342,200 × 2.5% = Rs 8,555.

Question: Exercise 7.4 (Zakat and Ushr), Question 3: A lady doctor has jewellery worth Rs 500,000 and she saved Rs 150,000 during the whole year. Find the amount of Zakat she has to pay.

Answer: Total amount = 500,000 + 150,000 = Rs 650,000. Zakat = 650,000 × 2.5% = Rs 16,250.

Question: Exercise 7.4 (Zakat and Ushr), Question 4: A trader paid Rs 20,500 as Zakat. Find his savings for the whole year.

Answer: Annual savings = Zakat ÷ 2.5% = 20,500 ÷ 0.025 = Rs 820,000.

Question: Exercise 7.4 (Zakat and Ushr), Question 5: A farmer pays Rs 10,000 as Ushr. Find the value of his produce, which he produced.

Answer: The question does not state the method of irrigation. At the 5% rate, the value of the produce = 10,000 ÷ 0.05 = Rs 200,000. (If the rate were 10%, the value would be 10,000 ÷ 0.10 = Rs 100,000.)

Question: Exercise 7.4 (Zakat and Ushr), Question 6: A farmer produces 15,000 kg of wheat. The rate of Ushr on it is 5%. Find the amount of Ushr. The price of wheat is Rs 5.25 per kg.

Answer: Total value of wheat = 15,000 × 5.25 = Rs 78,750. Ushr = 78,750 × 5% = 78,750 × 0.05 = Rs 3,937.50.

Question: Exercise 7.4 (Zakat and Ushr), Question 7: Juman sold the orange produce of his orchard for Rs 5,000,000. Find the amount of Ushr at the rate of 5%.

Answer: Ushr = 5,000,000 × 5% = 5,000,000 × 0.05 = Rs 250,000.

Question: Exercise 7.4 (Zakat and Ushr), Question 8: A farmer produces 1,500 kg of wheat in one part of his land and 1,200 kg of rice in the other part. Find the Ushr at the rate of 5%. The price of wheat is Rs 30 per kg and the price of rice is Rs 65 per kg.

Answer: Value of wheat = 1,500 × 30 = Rs 45,000. Value of rice = 1,200 × 65 = Rs 78,000. Total value = Rs 123,000. Ushr = 123,000 × 5% = Rs 6,150.

Question: Review Exercise 7, Question 1: The price of a mobile phone is Rs 8,500. The seller charges 17% GST. Find the tax charged on the sale.

Answer: GST = 8,500 × 17% = 8,500 × 0.17 = Rs 1,445. Selling price = 8,500 + 1,445 = Rs 9,945.

Question: Review Exercise 7, Question 2: An imported LCD TV is sold for Rs 55,800 including GST. Find the price of the LCD without tax. The GST rate is 17%.

Answer: Price without tax = 55,800 ÷ 1.17 = Rs 47,692.31 (approximately). GST = 55,800 − 47,692.31 = Rs 8,107.69 (approximately).

Question: Review Exercise 7, Question 3: An electric company charges a consumer Rs 16,750 for electricity. Find the amount of GST the consumer will pay at the rate of 17%.

Answer: GST = 16,750 × 0.17 = Rs 2,847.50.

Question: Review Exercise 7, Question 4: A person paid Rs 16,989 GST at

the rate of 17% on the purchase of a room set. Find the price of the room set without GST.

Answer: Price of the room set = GST ÷ rate = 16,989 ÷ 0.17 = Rs 99,935.29 (approximately).

Question: Review Exercise 7, Question 5: At an annual mark-up rate of 12.5%, a trader took a loan of Rs 1,700,000 from a bank. The period for him is 1 ½ years. Find the mark-up.

Answer: Mark-up = 1,700,000 × 12.5 × 1.5 ÷ 100 = Rs 318,750.

Question: Review Exercise 7, Question 6: A lady doctor paid Rs 2,850 mark-up to the bank, for a period of 6 months. The mark-up rate is 12%. Find the principal.

Answer: Time = 6 ÷ 12 = 0.5 years. Principal = 2,850 × 100 ÷ (12 × 0.5) = 285,000 ÷ 6 = Rs 47,500.

Question: Review Exercise 7, Question 7: An elderly citizen deposits Rs 500,000 in a bank, if the mark-up rate is 10% and the period is 2 ½ years. Find his mark-up.

Answer: Mark-up = 500,000 × 10 × 2.5 ÷ 100 = Rs 125,000.

Question: Review Exercise 7, Question 8: A car dealer uses a 35% mark-up rate. Find the cost price of a car which is sold for Rs 1,051,000.

Answer: Cost price = 1,051,000 ÷ (1 + 0.35) = 1,051,000 ÷ 1.35 = Rs 778,518.52 (approximately).

Question: Review Exercise 7, Question 9: A trader saves Rs 95,000 in a year and has goods worth Rs 500,000. Find the Zakat due on him.

Answer: Total amount = 95,000 + 500,000 = Rs 595,000. Zakat = 595,000 × 2.5% = Rs 14,875.

Question: Review Exercise 7, Question 10: A teacher has gold jewellery worth Rs 400,000. Her annual savings are Rs 45,000. Find the Zakat due on her.

Answer: Total amount = 400,000 + 45,000 = Rs 445,000. Zakat = 445,000 × 2.5% = Rs 11,125.

Question: Review Exercise 7, Question 11: A person pays Rs 12,500 as Zakat. Find the amount on which he paid the Zakat.

Answer: Amount = Zakat ÷ 2.5% = 12,500 ÷ 0.025 = Rs 500,000.

Question: Review Exercise 7, Question 12: A farmer produced 5,000 kg of wheat, worth Rs 48,000. Find the Ushr due on it at the rate of 5%. The price of wheat is Rs 35 per kg.

Answer: There is a contradiction in the print here: 5,000 kg × Rs 35 = Rs 175,000, not Rs 48,000. On the given total of Rs 48,000, Ushr = 48,000 × 5% = Rs 2,400. By the per-kg price the total is Rs 175,000, on which Ushr = 175,000 × 5% = Rs 8,750.

Question: Review Exercise 7, Question 13: A farmer pays Rs 2,000 as Ushr. Find the value of his produce, which he produced. The Ushr rate is 5%.

Answer: Value of produce = 2,000 ÷ 0.05 = Rs 40,000.

8. Algebraic Expressions

Question: Practical Work, (1): Identify whether this expression is a monomial, binomial or trinomial: 3.4 + 3.4x

Answer: This expression has two terms (3.4 and 3.4x), so it is a binomial expression.

Question: Practical Work, (2): Identify whether this expression is a monomial, binomial or trinomial: $z^2 + 5z + 6$

Answer: This expression has three terms (z^2, 5z and 6), so it is a trinomial expression.

Question: Practical Work, (3): Identify whether this expression is a monomial, binomial or trinomial: -8

Answer: This expression has only one term (-8), so it is a monomial expression.

Question: Practical Work, (4): Identify whether this expression is a monomial, binomial or trinomial: 2c + 5b + 6

Answer: This expression has three terms (2c, 5b and 6), so it is a trinomial expression.

Question: Practical Work, (5): Identify whether this expression is a monomial, binomial or trinomial: 14 + x

Answer: This expression has two terms (14 and x), so it is a binomial expression.

Question: Practical Work, (6): Identify whether this expression is a monomial, binomial or trinomial: $5x - 2$

Answer: This expression has two terms (5x and -2), so it is a binomial expression.

Question: Practical Work, (7): Identify whether this expression is a

monomial, binomial or trinomial: 4b − 2a

Answer: This expression has two terms (4b and −2a), so it is a binomial expression.

Question: Practical Work, (8): Identify whether this expression is a monomial, binomial or trinomial: $f^2 + 5f + 6$

Answer: This expression has three terms (f^2, 5f and 6), so it is a trinomial expression.

Question: Exercise 8.1, Question 1, (1): Add the following: 5a, 8a

Answer: 5a + 8a = (5 + 8)a = 13a.

Question: Exercise 8.1, Question 1, (2): Add the following: a, a

Answer: a + a = 2a.

Question: Exercise 8.1, Question 1, (3): Add the following: −12x, −2x

Answer: −12x + (−2x) = −(12 + 2)x = −14x.

Question: Exercise 8.1, Question 1, (4): Add the following: 9ab, −7ab

Answer: 9ab + (−7ab) = (9 − 7)ab = 2ab.

Question: Exercise 8.1, Question 1, (5): Add the following: 8y, 3y, −4y

Answer: 8y + 3y + (−4y) = (8 + 3 − 4)y = 7y.

Question: Exercise 8.1, Question 1, (6): Add the following: $6x^2 + 2x + 4$, $3x^2 + 5x$

Answer: Collect like terms: $(6x^2 + 3x^2) + (2x + 5x) + 4 = 9x^2 + 7x + 4$.

Question: Exercise 8.1, Question 2, (1): Find the sum of the following expressions: 2a + 3b, 6a + 5b

Answer: (2a + 3b) + (6a + 5b) = (2a + 6a) + (3b + 5b) = 8a + 8b.

Question: Exercise 8.1, Question 2, (2): Find the sum of the following expressions: 3x − 4y, 4x + 6y

Answer: (3x − 4y) + (4x + 6y) = (3x + 4x) + (−4y + 6y) = 7x + 2y.

Question: Exercise 8.1, Question 2, (3): Find the sum of the following expressions: 2a − 3b, 4a + 5b, 5a − 8b

Answer: (2a + 4a + 5a) + (−3b + 5b − 8b) = 11a − 6b.

Question: Exercise 8.1, Question 2, (4): Find the sum of the following expressions: ab + 4cd + 6ad, 8ab + 5cd + 6ad

Answer: (ab + 8ab) + (4cd + 5cd) + (6ad + 6ad) = 9ab + 9cd + 12ad.

Question: Exercise 8.1, Question 2, (5): Find the sum of the following expressions: 4.5a + 5.7b + 3.2c, 3.5a + 1.3b + 2.8c

Answer: (4.5a + 3.5a) + (5.7b + 1.3b) + (3.2c + 2.8c) = 8a + 7b + 6c.

Question: Exercise 8.1, Question 2, (6): Find the sum of the following expressions: 1.7a + 3.2b + 4.9c, 4.3a + 3.2b + 3.1c, 6.0a + 9.6b + 3.0c

Answer: (1.7a + 4.3a + 6.0a) + (3.2b + 3.2b + 9.6b) + (4.9c + 3.1c + 3.0c) = 12a + 16b + 11c.

Question: Exercise 8.1, Question 2, (7): Find the sum of the following expressions: 6a + 11b + 10c, 13a + 9b + 5c, 8a + 7b + 12c

Answer: (6a + 13a + 8a) + (11b + 9b + 7b) + (10c + 5c + 12c) = 27a + 27b + 27c.

Question: Exercise 8.1, Question 2, (8): Find the sum of the following expressions: 4a + 6b + 5c, 7a + 5b + 5c, 3a + 7b + 7c

Answer: (4a + 7a + 3a) + (6b + 5b + 7b) + (5c + 5c + 7c) = 14a + 18b + 17c.

Question: Exercise 8.1, Question 2, (9): Find the sum of the following expressions: 2a + 3b + 4c, 6a + 8b + 4c, 3a + 2b + 3c

Answer: (2a + 6a + 3a) + (3b + 8b + 2b) + (4c + 4c + 3c) = 11a + 13b + 11c.

Question: Exercise 8.1, Question 2, (10): Find the sum of the following expressions: 14e − 15f + 10g, 13e + 14f − g, −5e + 3g

Answer: (14e + 13e − 5e) + (−15f + 14f) + (10g − g + 3g) = 22e − f + 12g.

Question: Exercise 8.2, Question (1): Subtract the second expression from

the first: 8a, 5a

Answer: 8a − 5a = 3a.

Question: Exercise 8.2, Question (2): Subtract the second expression from the first: −7ab, 12ab

Answer: −7ab − 12ab = −19ab.

Question: Exercise 8.2, Question (3): Subtract the second expression from the first: 4a, −8a

Answer: 4a − (−8a) = 4a + 8a = 12a.

Question: Exercise 8.2, Question (4): Subtract the second expression from the first: 6a + 7b, 2a + 3b

Answer: (6a + 7b) − (2a + 3b) = (6a − 2a) + (7b − 3b) = 4a + 4b.

Question: Exercise 8.2, Question (5): Subtract the second expression from the first: 9a + 11b + 15c, 5a + 3b − 6c

Answer: (9a + 11b + 15c) − (5a + 3b − 6c) = (9a − 5a) + (11b − 3b) + (15c + 6c) = 4a + 8b + 21c.

Question: Exercise 8.2, Question (6): Subtract the second expression from the first: 9c + 8d + 6e, 3e − 5c + 4d

Answer: (9c + 8d + 6e) − (3e − 5c + 4d) = (9c + 5c) + (8d − 4d) + (6e − 3e) = 14c + 4d + 3e.

Question: Exercise 8.2, Question (7): Subtract the second expression from the first: 18x + 16y + 12z, 10x + 9y + 8z

Answer: (18x − 10x) + (16y − 9y) + (12z − 8z) = 8x + 7y + 4z.

Question: Exercise 8.2, Question (8): Subtract the second expression from the first: 14a + 15b + 16c, 8a + 10b − 2c

Answer: (14a − 8a) + (15b − 10b) + (16c + 2c) = 6a + 5b + 18c.

Question: Exercise 8.2, Question (9): Subtract the second expression from the first: $3x^2 + 25ab + 21bc$, $4x^2 + 18ab − 6bc$

Answer: $(3x^2 − 4x^2) + (25ab − 18ab) + (21bc + 6bc) = −x^2 + 7ab + 27bc$.

Question: Exercise 8.2, Question (10): Subtract the second expression from the first: 5a + 4ab + 2b, 4a + 3ab + b

Answer: (5a − 4a) + (4ab − 3ab) + (2b − b) = a + ab + b.

Question: Exercise 8.2, Question 11: From 13a + 8b + 12c subtract the sum of 9a + 3b + 5c and 10a + 10b + 17c.

Answer: Sum of the two expressions = (9a + 10a) + (3b + 10b) + (5c + 17c) = 19a + 13b + 22c. Then (13a + 8b + 12c) − (19a + 13b + 22c) = −6a − 5b − 10c.

Question: Exercise 8.2, Question 12: What should be added to 2x + 5y − 12 to get 6x + 8y?

Answer: Required expression = (6x + 8y) − (2x + 5y − 12) = (6x − 2x) + (8y − 5y) + 12 = 4x + 3y + 12.

Question: Exercise 8.3, Part I, (1): Multiply the following expressions together (monomial by monomial): 15a, 8a

Answer: $15a \times 8a = (15 \times 8)(a \times a) = 120a^2$.

Question: Exercise 8.3, Part I, (2): Multiply the following expressions together: 9a, 6b

Answer: $9a \times 6b = (9 \times 6)ab = 54ab$.

Question: Exercise 8.3, Part I, (3): Multiply the following expressions together: $10a^2$, $2a^5$

Answer: $10a^2 \times 2a^5 = (10 \times 2)a^{2+5} = 20a^7$.

Question: Exercise 8.3, Part I, (4): Multiply the following expressions together: $4c^5$, $8cd^2$

Answer: $4c^5 \times 8cd^2 = (4 \times 8)c^{5+1}d^2 = 32c^6d^2$.

Question: Exercise 8.3, Part I, (5): Multiply the following expressions together: $6a^2b$, $-5b^2$

Answer: $6a^2b \times (-5b^2) = (6 \times -5)a^2b^{1+2} = -30a^2b^3$.

Question: Exercise 8.3, Part I, (6): Multiply the following expressions

together: $-3a$, $2b$, -3

Answer: $-3a \times 2b \times (-3) = (-3 \times 2 \times -3)ab = 18ab$.

Question: Exercise 8.3, Part I, (7): Multiply the following expressions together: $4m^2$, $-5m^3$, $3m^4$

Answer: $4m^2 \times (-5m^3) \times 3m^4 = (4 \times -5 \times 3)m^{2+3+4} = -60m^9$.

Question: Exercise 8.3, Part I, (8): Multiply the following expressions together: $-6xy$, $3x^3y^2$, $2xy$

Answer: $-6xy \times 3x^3y^2 \times 2xy = (-6 \times 3 \times 2)x^{1+3+1}y^{1+2+1} = -36x^5y^4$.

Question: Exercise 8.3, Part II, (1): Multiply the following expressions together (binomial by monomial): $a + b$, 9

Answer: $(a + b) \times 9 = 9a + 9b$.

Question: Exercise 8.3, Part II, (2): Multiply the following expressions together: $x - y$, $-4y$

Answer: $(x - y) \times (-4y) = -4xy + 4y^2$.

Question: Exercise 8.3, Part II, (3): Multiply the following expressions together: $2m^2 - 3n^3$, $-5mn$

Answer: $(2m^2 - 3n^3) \times (-5mn) = 2m^2 \times (-5mn) - 3n^3 \times (-5mn) = -10m^3n + 15mn^4$.

Question: Exercise 8.3, Part II, (4): Multiply the following expressions together: $3xy + 2xy^2$, $-6xy^2$

Answer: $(3xy + 2xy^2) \times (-6xy^2) = 3xy \times (-6xy^2) + 2xy^2 \times (-6xy^2) = -18x^2y^3 - 12x^2y^4$.

Question: Exercise 8.3, Part II, (5): Multiply the following expressions together: $6lm - 4mn$, $3l^2m^2n$

Answer: $(6lm - 4mn) \times 3l^2m^2n = 6lm \times 3l^2m^2n - 4mn \times 3l^2m^2n = 18l^3m^3n - 12l^2m^3n^2$.

Question: Exercise 8.3, Part II, (6): Multiply the following expressions together: $-5bc + 3cd$, $-2bc^2d$

Answer: $(-5bc + 3cd) \times (-2bc^2d) = (-5bc) \times (-2bc^2d) + 3cd \times (-2bc^2d) = 10b^2c^3d - 6bc^3d^2$.

Question: Exercise 8.3, Part III, (1): Multiply the following expressions together (binomial by binomial): $8c + 12d$, $-3c + 2d$

Answer: $(8c + 12d)(-3c + 2d) = -24c^2 + 16cd - 36cd + 24d^2 = -24c^2 - 20cd + 24d^2$.

Question: Exercise 8.3, Part III, (2): Multiply the following expressions together: $a + b$, $a - b$

Answer: $(a + b)(a - b) = a^2 - ab + ab - b^2 = a^2 - b^2$.

Question: Exercise 8.3, Part III, (3): Multiply the following expressions together: $a^2 + b^2$, $a^2 - b^2$

Answer: $(a^2 + b^2)(a^2 - b^2) = a^4 - a^2b^2 + a^2b^2 - b^4 = a^4 - b^4$.

Question: Exercise 8.3, Part III, (4): Multiply the following expressions together: $3x - 2y$, $2x + 5y$

Answer: $(3x - 2y)(2x + 5y) = 6x^2 + 15xy - 4xy - 10y^2 = 6x^2 + 11xy - 10y^2$.

Question: Exercise 8.3, Part III, (5): Multiply the following expressions together: $2lm + 3mn$, $-2m + 3n$

Answer: $(2lm + 3mn)(-2m + 3n) = -4lm^2 + 6lmn - 6m^2n + 9mn^2$.

Question: Exercise 8.3, Part III, (6): Multiply the following expressions together: $8pq^2 - 3p^2q$, $2pq$

Answer: $(8pq^2 - 3p^2q) \times 2pq = 8pq^2 \times 2pq - 3p^2q \times 2pq = 16p^2q^3 - 6p^3q^2$.

Question: Exercise 8.3, Part IV, (1): Multiply the following expressions together (trinomial by monomial): $2a - 3b + 4c$, $2abc$

Answer: $(2a - 3b + 4c) \times 2abc = 4a^2bc - 6ab^2c + 8abc^2$.

Question: Exercise 8.3, Part IV, (2): Multiply the following expressions together: $a^2 + b^2 - c^2$, ab

Answer: $(a^2 + b^2 - c^2) \times ab = a^3b + ab^3 - abc^2$.

Question: Exercise 8.3, Part IV, (3): Multiply the following expressions together: $m^2n + mn^2 - m^2n^2$, $-mn$

Answer: $(m^2n + mn^2 - m^2n^2) \times (-mn) = -m^3n^2 - m^2n^3 + m^3n^3$.

Question: Exercise 8.3, Part IV, (4): Multiply the following expressions together: $x^2 - 6xy + 9y^2$, $-3xy$

Answer: $(x^2 - 6xy + 9y^2) \times (-3xy) = -3x^3y + 18x^2y^2 - 27xy^3$.

Question: Exercise 8.3, Part IV, (5): Multiply the following expressions together: $2p^2 - 3pq + q^2$, $4pqr$

Answer: $(2p^2 - 3pq + q^2) \times 4pqr = 8p^3qr - 12p^2q^2r + 4pq^2r$.

Question: Exercise 8.3, Part IV, (6): Multiply the following expressions together: $xyz + x^2y^2 - y^2z^2$, xyz

Answer: $(xyz + x^2y^2 - y^2z^2) \times xyz = x^2y^2z^2 + x^3y^3z - xy^3z^3$.

Question: Exercise 8.3, Part V, (1): Multiply the following expressions together (trinomial by binomial): $1 + 2x - 3x^2$, $1 - x$

Answer: $(1 + 2x - 3x^2)(1 - x) = 1 - x + 2x - 2x^2 - 3x^2 + 3x^3 = 1 + x - 5x^2 + 3x^3$.

Question: Exercise 8.3, Part V, (2): Multiply the following expressions together: $8c + 12d - 6e$, $c^2 - d^2$

Answer: $(8c + 12d - 6e)(c^2 - d^2) = 8c^3 - 8cd^2 + 12c^2d - 12d^3 - 6c^2e + 6d^2e$.

Question: Exercise 8.3, Part V, (3): Multiply the following expressions together: $a^2 - ab - b^2$, $a - b$

Answer: $(a^2 - ab - b^2)(a - b) = a^3 - a^2b - a^2b + ab^2 - ab^2 + b^3 = a^3 - 2a^2b + b^3$.

Question: Exercise 8.3, Part V, (4): Multiply the following expressions together: $x^2 - 5x + 4$, $3x - 4$

Answer: $(x^2 - 5x + 4)(3x - 4) = 3x^3 - 4x^2 - 15x^2 + 20x + 12x - 16 = 3x^3$

− 19x² + 32x − 16.

Question: Exercise 8.3, Part V, (5): Multiply the following expressions together: a² − 2a²b + b², a² − b²

Answer: (a² − 2a²b + b²)(a² − b²) = a⁴ − a²b² − 2a⁴b + 2a²b³ + a²b² − b⁴ = a⁴ − 2a⁴b + 2a²b³ − b⁴.

Question: Exercise 8.3, Part V, (6): Multiply the following expressions together: 3p²q + 2pq² − q²r, p² − q²

Answer: (3p²q + 2pq² − q²r)(p² − q²) = 3p⁴q − 3p²q³ + 2p³q² − 2pq⁴ − p²q²r + q⁴r.

Question: Exercise 8.4, Question 1, (1): Simplify: 5a − 2a(3 + 4b)

Answer: 5a − 2a(3 + 4b) = 5a − 6a − 8ab = −a − 8ab.

Question: Exercise 8.4, Question 1, (2): Simplify: 7m − 4m(5 − 6n)

Answer: 7m − 4m(5 − 6n) = 7m − 20m + 24mn = −13m + 24mn.

Question: Exercise 8.4, Question 1, (3): Simplify: 6x{2x(3x² − 4) − (6x² + 8y)}

Answer: Open the inner bracket: 2x(3x² − 4) = 6x³ − 8x. Then 6x³ − 8x − 6x² − 8y. Multiply outside: 6x × (6x³ − 6x² − 8x − 8y) = 36x⁴ − 36x³ − 48x² − 48xy.

Question: Exercise 8.4, Question 1, (4): Simplify: 3x − {(2x + y)(2x − y) − (4x² + 2y²)}

Answer: (2x + y)(2x − y) = 4x² − y². Then (4x² − y²) − (4x² + 2y²) = −3y². So 3x − (−3y²) = 3x + 3y².

Question: Exercise 8.4, Question 1, (5): Simplify: 4lm{3lm(d² − m²) − 2mn(m² − n²)}

Answer: Inner products: 3lm(d² − m²) = 3lmd² − 3lm³; 2mn(m² − n²) = 2m³n − 2mn³. Then 4lm × (3lmd² − 3lm³ − 2m³n + 2mn³) = 12l²m²d² − 12l²m⁴ − 8lm⁴n + 8lm²n³.

Question: Exercise 8.4, Question 1, (6): Simplify: 3pq{2pq(p² + q²) + 4qr(p

− r)}

Answer: Inner products: $2pq(p^2 + q^2) = 2p^3q + 2pq^3$; $4qr(p - r) = 4pqr - 4qr^2$. Then $3pq \times (2p^3q + 2pq^3 + 4pqr - 4qr^2) = 6p^4q^2 + 6p^2q^4 + 12p^2q^2r - 12pq^2r^2$.

Question: Exercise 8.4, Question 2, (1): Prove the following: $(x + 3)(x + 4) = x^2 + 7x + 12$

Answer: LHS = $(x + 3)(x + 4) = x^2 + 4x + 3x + 12 = x^2 + 7x + 12$ = RHS. So the result is proved.

Question: Exercise 8.4, Question 2, (2): Prove the following: $(x + 5)(x + 6) = x^2 + 11x + 30$

Answer: LHS = $(x + 5)(x + 6) = x^2 + 6x + 5x + 30 = x^2 + 11x + 30$ = RHS. So the result is proved.

Question: Exercise 8.4, Question 2, (3): Prove the following: $(x + 2)(x - 4) = x^2 - 2x - 8$

Answer: LHS = $(x + 2)(x - 4) = x^2 - 4x + 2x - 8 = x^2 - 2x - 8$ = RHS. So the result is proved.

Question: Exercise 8.4, Question 2, (4): Prove the following: $(x - 3)(x + 4) = x^2 + x - 12$

Answer: LHS = $(x - 3)(x + 4) = x^2 + 4x - 3x - 12 = x^2 + x - 12$ = RHS. So the result is proved.

Question: Exercise 8.4, Question 2, (5): Prove the following: $(x - 5)(x - 4) = x^2 - 9x + 20$

Answer: LHS = $(x - 5)(x - 4) = x^2 - 4x - 5x + 20 = x^2 - 9x + 20$ = RHS. So the result is proved.

Question: Exercise 8.4, Question 2, (6): Prove the following: $(x - 7)(x + 8) = x^2 + x - 56$

Answer: LHS = $(x - 7)(x + 8) = x^2 + 8x - 7x - 56 = x^2 + x - 56$ = RHS. So the result is proved.

Question: Exercise 8.4, Question 2, (7): Prove the following: $(x - 9)(x + 2) = x^2 - 7x - 18$

Answer: LHS = $(x - 9)(x + 2) = x^2 + 2x - 9x - 18 = x^2 - 7x - 18$ = RHS. So the result is proved.

Question: Exercise 8.4, Question 2, (8): Prove the following: $(x - 1)(x + 9) = x^2 + 8x - 9$

Answer: LHS = $(x - 1)(x + 9) = x^2 + 9x - x - 9 = x^2 + 8x - 9$ = RHS. So the result is proved.

Question: Exercise 8.4, Question 2, (9): Prove the following: $(x - 2)(x + 5) = x^2 + 3x - 10$

Answer: LHS = $(x - 2)(x + 5) = x^2 + 5x - 2x - 10 = x^2 + 3x - 10$ = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (1): Prove the following results by using the formula: $(a + 7)^2 = a^2 + 14a + 49$

Answer: Use the formula $(a + b)^2 = a^2 + 2ab + b^2$. LHS = $(a + 7)^2 = a^2 + 2 \times a \times 7 + 7^2 = a^2 + 14a + 49$ = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (2): Prove the following results by using the formula: $(a + 11)^2 = a^2 + 22a + 121$

Answer: LHS = $(a + 11)^2 = a^2 + 2 \times a \times 11 + 11^2 = a^2 + 22a + 121$ = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (3): Prove the following results by using the formula: $(2x + 6)^2 = 4x^2 + 24x + 36$

Answer: LHS = $(2x + 6)^2 = (2x)^2 + 2 \times 2x \times 6 + 6^2 = 4x^2 + 24x + 36$ = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (4): Prove the following results by using the formula: $(3a + 5)^2 = 9a^2 + 30a + 25$

Answer: LHS = $(3a + 5)^2 = (3a)^2 + 2 \times 3a \times 5 + 5^2 = 9a^2 + 30a + 25$ = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (5): Prove the following results by using the formula: $(7x + 3)^2 = 49x^2 + 42x + 9$

Answer: LHS = $(7x + 3)^2 = (7x)^2 + 2 \times 7x \times 3 + 3^2 = 49x^2 + 42x + 9$ = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (6): Prove the following results by using the formula: $(6x + 9)^2 = 36x^2 + 108x + 81$

Answer: LHS = $(6x + 9)^2 = (6x)^2 + 2 \times 6x \times 9 + 9^2 = 36x^2 + 108x + 81$ = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (7): Prove the following results by using the formula: $(5a + 4)^2 = 25a^2 + 40a + 16$

Answer: LHS = $(5a + 4)^2 = (5a)^2 + 2 \times 5a \times 4 + 4^2 = 25a^2 + 40a + 16$ = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (8): Prove the following results by using the formula: $(9x + 10)^2 = 81x^2 + 180x + 100$

Answer: LHS = $(9x + 10)^2 = (9x)^2 + 2 \times 9x \times 10 + 10^2 = 81x^2 + 180x +$ 100 = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (9): Prove the following results by using the formula: $(5x + 11)^2 = 25x^2 + 110x + 121$

Answer: LHS = $(5x + 11)^2 = (5x)^2 + 2 \times 5x \times 11 + 11^2 = 25x^2 + 110x +$ 121 = RHS. So the result is proved.

Question: Exercise 8.5, Question 1, (10): Prove the following results by using the formula: $(6x + 7)^2 = 36x^2 + 84x + 49$

Answer: LHS = $(6x + 7)^2 = (6x)^2 + 2 \times 6x \times 7 + 7^2 = 36x^2 + 84x + 49$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (1): Prove the following results by using the formula: $(a - 3)^2 = a^2 - 6a + 9$

Answer: Use the formula $(a - b)^2 = a^2 - 2ab + b^2$. LHS = $(a - 3)^2 = a^2 - 2 \times a \times 3 + 3^2 = a^2 - 6a + 9$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (2): Prove the following results by using the formula: $(a - 7)^2 = a^2 - 14a + 49$

Answer: LHS = $(a - 7)^2 = a^2 - 2 \times a \times 7 + 7^2 = a^2 - 14a + 49$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (3): Prove the following results by using the formula: $(2a - 5)^2 = 4a^2 - 20a + 25$

Answer: LHS = $(2a - 5)^2 = (2a)^2 - 2 \times 2a \times 5 + 5^2 = 4a^2 - 20a + 25$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (4): Prove the following results by using the formula: $(6x - 8)^2 = 36x^2 - 96x + 64$

Answer: LHS = $(6x - 8)^2 = (6x)^2 - 2 \times 6x \times 8 + 8^2 = 36x^2 - 96x + 64$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (5): Prove the following results by using the formula: $(4a - 7)^2 = 16a^2 - 56a + 49$

Answer: LHS = $(4a - 7)^2 = (4a)^2 - 2 \times 4a \times 7 + 7^2 = 16a^2 - 56a + 49$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (6): Prove the following results by using the formula: $(3a - 8)^2 = 9a^2 - 48a + 64$

Answer: LHS = $(3a - 8)^2 = (3a)^2 - 2 \times 3a \times 8 + 8^2 = 9a^2 - 48a + 64$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (7): Prove the following results by using the formula: $(7a - 5)^2 = 49a^2 - 70a + 25$

Answer: LHS = $(7a - 5)^2 = (7a)^2 - 2 \times 7a \times 5 + 5^2 = 49a^2 - 70a + 25$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (8): Prove the following results by using the formula: $(9y - 10)^2 = 81y^2 - 180y + 100$

Answer: LHS = $(9y - 10)^2 = (9y)^2 - 2 \times 9y \times 10 + 10^2 = 81y^2 - 180y + 100$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (9): Prove the following results by using the formula: $(8a - 9)^2 = 64a^2 - 144a + 81$

Answer: LHS = $(8a - 9)^2 = (8a)^2 - 2 \times 8a \times 9 + 9^2 = 64a^2 - 144a + 81$ = RHS. So the result is proved.

Question: Exercise 8.6, Question (10): Prove the following results by using the formula: $(9a - 11)^2 = 81a^2 - 198a + 121$

Answer: LHS = $(9a - 11)^2 = (9a)^2 - 2 \times 9a \times 11 + 11^2 = 81a^2 - 198a +$ 121 = RHS. So the result is proved.

Question: Exercise 8.7, Part I, (1): Prove: $(3x + 2)(3x - 2) = 9x^2 - 4$

Answer: Use the formula $(a + b)(a - b) = a^2 - b^2$. LHS = $(3x)^2 - 2^2 = 9x^2$ $- 4$ = RHS. So the result is proved.

Question: Exercise 8.7, Part I, (2): Prove: $(4x + 3)(4x - 3) = 16x^2 - 9$

Answer: LHS = $(4x)^2 - 3^2 = 16x^2 - 9$ = RHS. So the result is proved.

Question: Exercise 8.7, Part I, (3): Prove: $(5x + 4)(5x - 4) = 25x^2 - 16$

Answer: LHS = $(5x)^2 - 4^2 = 25x^2 - 16$ = RHS. So the result is proved.

Question: Exercise 8.7, Part I, (4): Prove: $(3x + 7)(3x - 7) = 9x^2 - 49$

Answer: LHS = $(3x)^2 - 7^2 = 9x^2 - 49$ = RHS. So the result is proved.

Question: Exercise 8.7, Part I, (5): Prove: $(6x + 5)(6x - 5) = 36x^2 - 25$

Answer: LHS = $(6x)^2 - 5^2 = 36x^2 - 25$ = RHS. So the result is proved.

Question: Exercise 8.7, Part I, (6): Prove: $(7x + 6)(7x - 6) = 49x^2 - 36$

Answer: LHS = $(7x)^2 - 6^2 = 49x^2 - 36$ = RHS. So the result is proved.

Question: Exercise 8.7, Part II, (1): Prove the following: $4a^2 - 9b^2 = (2a +$ $3b)(2a - 3b)$

Answer: RHS = $(2a + 3b)(2a - 3b) = (2a)^2 - (3b)^2 = 4a^2 - 9b^2$ = LHS. So the result is proved.

Question: Exercise 8.7, Part II, (2): Prove the following: $64m^2 - 81 = (8m +$ $9)(8m - 9)$

Answer: RHS = $(8m + 9)(8m - 9) = (8m)^2 - 9^2 = 64m^2 - 81$ = LHS. So

the result is proved.

Question: Exercise 8.7, Part II, (3): Prove the following: $9p^2 - 100q^2 = (3p + 10q)(3p - 10q)$

Answer: RHS = $(3p + 10q)(3p - 10q) = (3p)^2 - (10q)^2 = 9p^2 - 100q^2$ = LHS. So the result is proved.

Question: Exercise 8.7, Part II, (4): Prove the following: $100y^2 - 169 = (10y + 13)(10y - 13)$

Answer: RHS = $(10y + 13)(10y - 13) = (10y)^2 - 13^2 = 100y^2 - 169$ = LHS. So the result is proved.

Question: Exercise 8.8, Question (1): Factorise the following expressions: $5a^2x - 15a$

Answer: The common factor in both terms is 5a. $5a^2x - 15a = 5a(ax - 3)$.

Question: Exercise 8.8, Question (2): Factorise the following expressions: $m^4 - m^3 + m^2$

Answer: The common factor in all three terms is m^2. $m^4 - m^3 + m^2 = m^2(m^2 - m + 1)$.

Question: Exercise 8.8, Question (3): Factorise the following expressions: $7a^3 + 14a^2 - 7$

Answer: The common factor in all three terms is 7. $7a^3 + 14a^2 - 7 = 7(a^3 + 2a^2 - 1)$.

Question: Exercise 8.8, Question (4): Factorise the following expressions: $mx^2 - m^2x - mx$

Answer: The common factor in all three terms is mx. $mx^2 - m^2x - mx = mx(x - m - 1)$.

Question: Exercise 8.8, Question (5): Factorise the following expressions: $13n - 2n^3 + 39n^5$

Answer: The common factor in all three terms is n. $13n - 2n^3 + 39n^5 = n(13 - 2n^2 + 39n^4)$.

Question: Exercise 8.8, Question (6): Factorise the following expressions:

$a^2bc + ab^2c + abc^2$

Answer: The common factor in all three terms is abc. $a^2bc + ab^2c + abc^2 = abc(a + b + c)$.

Question: Exercise 8.8, Question (7): Factorise the following expressions:

$3a(b + c) + 6d(b + c)$

Answer: The common factor in both parts is (b + c). $3a(b + c) + 6d(b + c) = (b + c)(3a + 6d) = 3(b + c)(a + 2d)$.

Question: Exercise 8.8, Question (8): Factorise the following expressions:

$xy(a + b) + y(a + b)^2 + x(a + b)$

Answer: The common factor in all three parts is (a + b). $= (a + b)[xy + y(a + b) + x] = (a + b)(xy + ay + by + x)$.

Question: Exercise 8.8, Question (9): Factorise the following expressions:

$4lm(x - y) + 8mn(x - y)^2 + 12nl(x - y)$

Answer: The common factor in all three parts is $4(x - y)$. $= 4(x - y)[lm + 2mn(x - y) + 3nl] = 4(x - y)(lm + 2mnx - 2mny + 3nl)$.

Question: Exercise 8.8, Question (10): Factorise the following expressions:

$pq(p^2 - q)^2 + pr(p^2 - q)^2 + qr(p^2 - q)^2$

Answer: The common factor in all three parts is $(p^2 - q)^2$. $= (p^2 - q)^2(pq + pr + qr)$.

Question: Exercise 8.9, Question (1): Factorise the following expressions:

$x^2 + 12x + 36$

Answer: This is a perfect-square pattern: the first term is the square of x, the last term is the square of 6, and the middle term is $2 \times x \times 6 = 12x$. So $x^2 + 12x + 36 = (x + 6)^2 = (x + 6)(x + 6)$.

Question: Exercise 8.9, Question (2): Factorise the following expressions:

$a^2 + 4a + 4$

Answer: The first term is the square of a, the last term is the square of 2,

and the middle term is 2 × a × 2 = 4a. So $a^2 + 4a + 4 = (a + 2)^2 = (a + 2)(a + 2)$.

Question: Exercise 8.9, Question (3): Factorise the following expressions:

$4a^2 + 12a + 9$

Answer: The first term is the square of 2a, the last term is the square of 3, and the middle term is 2 × 2a × 3 = 12a. So $4a^2 + 12a + 9 = (2a + 3)^2 = (2a + 3)(2a + 3)$.

Question: Exercise 8.9, Question (4): Factorise the following expressions:

$9x^2 + 42xy + 49y^2$

Answer: The first term is the square of 3x, the last term is the square of 7y, and the middle term is 2 × 3x × 7y = 42xy. So $9x^2 + 42xy + 49y^2 = (3x + 7y)^2 = (3x + 7y)(3x + 7y)$.

Question: Exercise 8.9, Question (5): Factorise the following expressions:

$25a^2 + 80ab + 64b^2$

Answer: The first term is the square of 5a, the last term is the square of 8b, and the middle term is 2 × 5a × 8b = 80ab. So $25a^2 + 80ab + 64b^2 = (5a + 8b)^2 = (5a + 8b)(5a + 8b)$.

Question: Exercise 8.9, Question (6): Factorise the following expressions:

$16b^2 + 40b + 25$

Answer: The first term is the square of 4b, the last term is the square of 5, and the middle term is 2 × 4b × 5 = 40b. So $16b^2 + 40b + 25 = (4b + 5)^2 = (4b + 5)(4b + 5)$.

Question: Exercise 8.9, Question (7): Factorise the following expressions:

$4x^2 + 2x + 1/4$

Answer: The first term is the square of 2x, the last term is the square of 1/2, and the middle term is 2 × 2x × 1/2 = 2x. So $4x^2 + 2x + 1/4 = (2x + 1/2)^2 = (2x + 1/2)(2x + 1/2)$.

Question: Exercise 8.9, Question (8): Factorise the following expressions:

$144x^2 - 4x + 1/36$

Answer: The first term is the square of 12x, the last term is the square of 1/6, and the middle term is $2 \times 12x \times 1/6 = 4x$. So $144x^2 - 4x + 1/36 = (12x - 1/6)^2 = (12x - 1/6)(12x - 1/6)$.

Question: Exercise 8.9, Question (9): Factorise the following expressions:

$18c^2 + 60cd + 50cd^2$

Answer: As printed, only the common factor 2c can be taken out: $18c^2 + 60cd + 50cd^2 = 2c(9c + 30d + 25d^2)$, and the inner part is not a perfect square. The book very likely intends $50d^2$, which would give the answer $2(3c + 5d)^2$.

Question: Exercise 8.9, Question (10): Factorise the following expressions:

$4a^2 + 48a + 144$

Answer: Take out the common factor 4: $4(a^2 + 12a + 36) = 4(a + 6)^2 = 4(a + 6)(a + 6)$.

Question: Exercise 8.9, Question (11): Factorise the following expressions:

$5y^2 + 40y + 80$

Answer: Take out the common factor 5: $5(y^2 + 8y + 16) = 5(y + 4)^2 = 5(y + 4)(y + 4)$.

Question: Exercise 8.9, Question (12): Factorise the following expressions:

$3s^2 - 48st + 192t^2$

Answer: Take out the common factor 3: $3(s^2 - 16st + 64t^2) = 3(s - 8t)^2 = 3(s - 8t)(s - 8t)$.

Question: Exercise 8.9, Question (13): Factorise the following expressions:

$9a^2t^2 - 42ats + 49s^2$

Answer: The first term is the square of 3at, the last term is the square of 7s, and the middle term is $2 \times 3at \times 7s = 42ats$. So $9a^2t^2 - 42ats + 49s^2 = (3at - 7s)^2 = (3at - 7s)(3at - 7s)$.

Question: Exercise 8.9, Question (14): Factorise the following expressions:

$4a^4 - 12a^2 + 9$

Answer: The first term is the square of $2a^2$, the last term is the square of 3, and the middle term is $2 \times 2a^2 \times 3 = 12a^2$. So $4a^4 - 12a^2 + 9 = (2a^2 - 3)^2 = (2a^2 - 3)(2a^2 - 3)$.

Question: Exercise 8.9, Question (15): Factorise the following expressions:

$50p^2q^2 - 70pqr + 98r^2$

Answer: Take out the common factor 2: $2(25p^2q^2 - 35pqr + 49r^2)$. Inside: the first term is the square of $5pq$, the last term is the square of $7r$, and the middle term is $2 \times 5pq \times 7r = 70pqr$. So $= 2(5pq - 7r)^2 = 2(5pq - 7r)(5pq - 7r)$.

Question: Exercise 8.10, Question (1): Factorise the following expressions:

$b^2 - c^2$

Answer: This is a difference of two squares: $b^2 - c^2 = (b + c)(b - c)$.

Question: Exercise 8.10, Question (2): Factorise the following expressions:

$a^2 - 36$

Answer: $a^2 - 36 = a^2 - 6^2 = (a + 6)(a - 6)$.

Question: Exercise 8.10, Question (3): Factorise the following expressions:

$a^2 - 49$

Answer: $a^2 - 49 = a^2 - 7^2 = (a + 7)(a - 7)$.

Question: Exercise 8.10, Question (4): Factorise the following expressions:

$25 - y^2$

Answer: $25 - y^2 = 5^2 - y^2 = (5 + y)(5 - y)$.

Question: Exercise 8.10, Question (5): Factorise the following expressions:

$4x^2 - 9$

Answer: $4x^2 - 9 = (2x)^2 - 3^2 = (2x + 3)(2x - 3)$.

Question: Exercise 8.10, Question (6): Factorise the following expressions:

$9x^2 - 16y^2$

Answer: $9x^2 - 16y^2 = (3x)^2 - (4y)^2 = (3x + 4y)(3x - 4y)$.

Question: Exercise 8.10, Question (7): Factorise the following expressions: $6x^2 - 24$

Answer: Take out the common factor 6: $6(x^2 - 4) = 6(x + 2)(x - 2)$.

Question: Exercise 8.10, Question (8): Factorise the following expressions: $200 - 18z^2$

Answer: Take out the common factor 2: $2(100 - 9z^2) = 2(10 + 3z)(10 - 3z)$.

Question: Exercise 8.10, Question (9): Factorise the following expressions: $81a^2 - 121b^2$

Answer: $81a^2 - 121b^2 = (9a)^2 - (11b)^2 = (9a + 11b)(9a - 11b)$.

Question: Exercise 8.10, Question (10): Factorise the following expressions: $a^2c^2 - 16c^2d^2$

Answer: Take out the common factor c^2: $c^2(a^2 - 16d^2) = c^2(a + 4d)(a - 4d)$.

Question: Exercise 8.10, Question (11): Factorise the following expressions: $36 - (3a - 2b)^2$

Answer: $36 - (3a - 2b)^2 = 6^2 - (3a - 2b)^2 = [6 + (3a - 2b)][6 - (3a - 2b)] = (6 + 3a - 2b)(6 - 3a + 2b)$.

Question: Exercise 8.10, Question (12): Factorise the following expressions: $(2x + 3y)^2 - 100z^2$

Answer: $(2x + 3y)^2 - (10z)^2 = (2x + 3y + 10z)(2x + 3y - 10z)$.

Question: Exercise 8.10, Question (13): Factorise the following expressions: $64b^2 - (6b + d)^2$

Answer: $64b^2 - (6b + d)^2 = (8b)^2 - (6b + d)^2 = (8b + 6b + d)(8b - 6b - d) = (14b + d)(2b - d)$.

Question: Exercise 8.10, Question (14): Factorise the following expressions: $(3a - 4b)^2 - (3a - b)^2$

Answer: Difference of two squares: $[(3a - 4b) + (3a - b)][(3a - 4b) - (3a$

− b)] = (6a − 5b)(−3b) = −3b(6a − 5b).

Question: Exercise 8.10, Question (15): Factorise the following expressions: $49(2a - 3b)^2 - 16(a + 2b)^2$

Answer: $[7(2a - 3b)]^2 - [4(a + 2b)]^2 = [7(2a - 3b) + 4(a + 2b)][7(2a - 3b) - 4(a + 2b)] = (18a - 13b)(10a - 29b)$.

Question: Exercise 8.10, Question (16): Factorise the following expressions: $225(p - q)^2 - 49(p + q)^2$

Answer: $[15(p - q)]^2 - [7(p + q)]^2 = [15(p - q) + 7(p + q)][15(p - q) - 7(p + q)] = (22p - 8q)(8p - 22q) = 4(11p - 4q)(4p - 11q)$.

Question: Exercise 8.10, Question (17): Factorise the following expressions: $72(c + d)^2 - 162(c - d)^2$

Answer: Take out the common factor 18: $18[4(c + d)^2 - 9(c - d)^2] = 18[2(c + d) + 3(c - d)][2(c + d) - 3(c - d)] = 18(5c - d)(5d - c)$.

Question: Exercise 8.10, Question (18): Factorise the following expressions: $48l(m - n)^2 - 75l(p - q)^2$

Answer: Take out the common factor 3l: $3l[16(m - n)^2 - 25(p - q)^2] = 3l[4(m - n) + 5(p - q)][4(m - n) - 5(p - q)] = 3l(4m - 4n + 5p - 5q)(4m - 4n - 5p + 5q)$.

Question: Exercise 8.11, Part I, (1): Factorise the following expressions by grouping: $5x^2 + 9x + 4$

Answer: $5x^2 + 9x + 4 = 5x^2 + 5x + 4x + 4 = 5x(x + 1) + 4(x + 1) = (x + 1)(5x + 4)$.

Question: Exercise 8.11, Part I, (2): Factorise the following expressions by grouping: $ax + ay + bx + by$

Answer: $ax + ay + bx + by = a(x + y) + b(x + y) = (x + y)(a + b)$.

Question: Exercise 8.11, Part I, (3): Factorise the following expressions by grouping: $b^2 + bc^2 + 4b + 4c^2$

Answer: $b^2 + bc^2 + 4b + 4c^2 = b(b + c^2) + 4(b + c^2) = (b + 4)(b + c^2)$.

Question: Exercise 8.11, Part I, (4): Factorise the following expressions by grouping: $p^2 - 6pq - pq + 6q^2$

Answer: $p^2 - 6pq - pq + 6q^2 = p(p - 6q) - q(p - 6q) = (p - q)(p - 6q)$.

Question: Exercise 8.11, Part I, (5): Factorise the following expressions by grouping: $a^2 + ab^2 + 5a + 5b^2$

Answer: $a^2 + ab^2 + 5a + 5b^2 = a(a + b^2) + 5(a + b^2) = (a + 5)(a + b^2)$.

Question: Exercise 8.11, Part I, (6): Factorise the following expressions by grouping: $7x^2 + 7x + pxy + py$

Answer: $7x^2 + 7x + pxy + py = 7x(x + 1) + py(x + 1) = (x + 1)(7x + py)$.

Question: Exercise 8.11, Part I, (7): Factorise the following expressions by grouping: $2c^2d - 2c^2d^2 + 4cd^3 - 4cd^4$

Answer: Take out the common factor $2d$: $2d(c^2 - c^2d + 2cd^2 - 2cd^3) = 2d[c^2(1 - d) + 2cd^2(1 - d)] = 2d(1 - d)(c^2 + 2cd^2) = 2cd(1 - d)(c + 2d^2)$.

Question: Exercise 8.11, Part I, (8): Factorise the following expressions by grouping: $2y^2 - 6y - 20$

Answer: Take out the common factor 2: $2(y^2 - 3y - 10) = 2(y - 5)(y + 2)$.

Question: Exercise 8.11, Part I, (9): Factorise the following expressions by grouping: $x^2y^2 + 4xy - 5$

Answer: Treat xy as one letter: $(xy)^2 + 4(xy) - 5$. Split the middle term: $5 - 1 = 4$. $= (xy + 5)(xy - 1)$.

Question: Exercise 8.11, Part I, (10): Factorise the following expressions by grouping: $p^2 - pq - p + q$

Answer: $p^2 - pq - p + q = p(p - q) - (p - q) = (p - q)(p - 1)$.

Question: Exercise 8.11, Part II, (1): Factorise the following expressions by splitting the middle term: $x^2 + 6x + 8$

Answer: Split the middle term: $6 = 2 + 4$ and $2 \times 4 = 8$. $x^2 + 6x + 8 = x^2 + 2x + 4x + 8 = x(x + 2) + 4(x + 2) = (x + 2)(x + 4)$.

Question: Exercise 8.11, Part II, (2): Factorise the following expressions by

splitting the middle term: $y^2 + 8y + 12$

Answer: $8 = 2 + 6$ and $2 \times 6 = 12$. $y^2 + 8y + 12 = y^2 + 2y + 6y + 12 = y(y + 2) + 6(y + 2) = (y + 2)(y + 6)$.

Question: Exercise 8.11, Part II, (3): Factorise the following expressions by splitting the middle term: $a^2 + 7a + 10$

Answer: $7 = 2 + 5$ and $2 \times 5 = 10$. $a^2 + 7a + 10 = a^2 + 2a + 5a + 10 = a(a + 2) + 5(a + 2) = (a + 2)(a + 5)$.

Question: Exercise 8.11, Part II, (4): Factorise the following expressions by splitting the middle term: $c^2 + 5c + 6$

Answer: $5 = 2 + 3$ and $2 \times 3 = 6$. $c^2 + 5c + 6 = c^2 + 2c + 3c + 6 = c(c + 2) + 3(c + 2) = (c + 2)(c + 3)$.

Question: Exercise 8.11, Part II, (5): Factorise the following expressions by splitting the middle term: $d^2 + 6d + 5$

Answer: $6 = 1 + 5$ and $1 \times 5 = 5$. $d^2 + 6d + 5 = d^2 + d + 5d + 5 = d(d + 1) + 5(d + 1) = (d + 1)(d + 5)$.

Question: Exercise 8.11, Part II, (6): Factorise the following expressions by splitting the middle term: $p^2 - 4p + 3$

Answer: $-4 = (-1) + (-3)$ and $(-1) \times (-3) = 3$. $p^2 - 4p + 3 = p^2 - p - 3p + 3 = p(p - 1) - 3(p - 1) = (p - 1)(p - 3)$.

Question: Exercise 8.11, Part II, (7): Factorise the following expressions by splitting the middle term: $r^2 + 7r + 12$

Answer: $7 = 3 + 4$ and $3 \times 4 = 12$. $r^2 + 7r + 12 = r^2 + 3r + 4r + 12 = r(r + 3) + 4(r + 3) = (r + 3)(r + 4)$.

Question: Exercise 8.11, Part II, (8): Factorise the following expressions by splitting the middle term: $a^2 + 3a - 10$

Answer: $3 = 5 + (-2)$ and $5 \times (-2) = -10$. $a^2 + 3a - 10 = a^2 + 5a - 2a - 10 = a(a + 5) - 2(a + 5) = (a + 5)(a - 2)$.

Question: Exercise 8.11, Part II, (9): Factorise the following expressions by

splitting the middle term: $m^2 - 5m + 14$

Answer: This expression cannot be factorised over integers, because there are no two integers whose sum is -5 and whose product is 14 (the discriminant $25 - 56 = -31$ is less than 0). So $m^2 - 5m + 14$ has no integer factors.

Question: Exercise 8.11, Part II, (10): Factorise the following expressions by splitting the middle term: $x^2 - 2x - 15$

Answer: $-2 = (-5) + 3$ and $(-5) \times 3 = -15$. $x^2 - 2x - 15 = x^2 - 5x + 3x - 15 = x(x - 5) + 3(x - 5) = (x - 5)(x + 3)$.

Question: Exercise 8.11, Part II, (11): Factorise the following expressions by splitting the middle term: $y^2 + 7y - 18$

Answer: $7 = 9 + (-2)$ and $9 \times (-2) = -18$. $y^2 + 7y - 18 = y^2 + 9y - 2y - 18 = y(y + 9) - 2(y + 9) = (y + 9)(y - 2)$.

Question: Exercise 8.11, Part II, (12): Factorise the following expressions by splitting the middle term: $x^2 + 10x + 21$

Answer: $10 = 3 + 7$ and $3 \times 7 = 21$. $x^2 + 10x + 21 = x^2 + 3x + 7x + 21 = x(x + 3) + 7(x + 3) = (x + 3)(x + 7)$.

Question: Review Exercise 8, Question 1, (i): Write the definition of the following: Numerical coefficient

Answer: A numerical coefficient is the fixed number that multiplies a variable, for example in $5x$ the coefficient is 5.

Question: Review Exercise 8, Question 1, (ii): Write the definition of the following: Constant

Answer: A constant is a symbol that has a fixed numerical value, for example 3, -5.

Question: Review Exercise 8, Question 1, (iii): Write the definition of the following: Binomial expression

Answer: A binomial expression is an expression with two terms, for

example 2x + 5.

Question: Review Exercise 8, Question 1, (iv): Write the definition of the following: Algebraic identity / formula

Answer: An algebraic identity / formula is a standard ready-made result, for example $(a + b)^2 = a^2 + 2ab + b^2$, which is used to find products easily.

Question: Review Exercise 8, Question 1, (v): Write the definition of the following: Factorisation

Answer: Factorisation is writing an expression as a product of its factors, for example $x^2 + 5x + 6 = (x + 3)(x + 2)$.

Review Exercise 8, Question 2, (i): Fill in the blank: $(a + b)^2 =$ $a^2 + 2ab + b^2$.

Review Exercise 8, Question 2, (ii): Fill in the blank: $(a - b)^2 =$ $a^2 - 2ab + b^2$.

Review Exercise 8, Question 2, (iii): Fill in the blank: $(x + a)(x + b) =$ _____

Answer: $x^2 + (a + b)x + ab$.

Review Exercise 8, Question 2, (iv): Fill in the blank: $a^2 - b^2 =$ $(a + b)(a - b)$.

Review Exercise 8, Question 2, (v): Fill in the blank: literal _____

Answer: Literal numbers are symbols used for numerical values (literal numbers).

Review Exercise 8, Question 2, (vi): Fill in the blank: monomial _____

Answer: is an expression with only one term.

Question: Review Exercise 8, Question 3, (i): Choose the correct answer: The factors of $x^2 - x$ are: (a) $x(x + 1)$ (b) $x(x - 1)$ (c) $(x - 1)(x + 1)$ (d) $x^2(x - 1)$

Answer: The correct answer is (b) $x(x - 1)$, because $x(x - 1) = x^2 - x$.

Question: Review Exercise 8, Question 3, (ii): Choose the correct answer: What kind of expression is 3x + 5? (a) Monomial (b) Trinomial (c) Binomial (d) None

Answer: The correct answer is (c) Binomial, because it has two terms.

Question: Review Exercise 8, Question 3, (iii): Choose the correct answer: In 5y, what is 5 called? (a) Variable (b) Coefficient (c) Constant (d) Expression

Answer: The correct answer is (c) Constant (in the book's wording, 5 is called the numerical coefficient / constant).

Question: Review Exercise 8, Question 3, (iv): Choose the correct answer: The factors of $a^2 - 9$ are: (a) $(a + 3)(a - 3)$ (b) $(a + 9)(a - 9)$ (c) $(a + 3)^2$ (d) $(a - 3)^2$

Answer: The correct answer is (a) $(a + 3)(a - 3)$, because $a^2 - 9 = a^2 - 3^2 = (a + 3)(a - 3)$.

Question: Review Exercise 8, Question 3, (v): Choose the correct answer: $(x - y)^2$ equals: (a) $x^2 + 2xy + y^2$ (b) $x^2 - 2xy - y^2$ (c) $x^2 - 2xy + y^2$ (d) $x^2 + y^2$

Answer: The correct answer is (c) $x^2 - 2xy + y^2$, because $(x - y)^2 = x^2 - 2xy + y^2$.

Question: Review Exercise 8, Question 4, (i): For given $A = 2(x^2 + y^2 + z^2)$, $B = -x^2 + 3y^2 + 2z^2$, $C = x^2 - y^2 - 3z^2$, find A + B + C.

Answer: $A + B + C = 2x^2 + 2y^2 + 2z^2 + (-x^2 + 3y^2 + 2z^2) + (x^2 - y^2 - 3z^2) = 2x^2 + 4y^2 + z^2$.

Question: Review Exercise 8, Question 4, (ii): For given A, B, C, find B + C − A.

Answer: $B + C - A = (-x^2 + 3y^2 + 2z^2) + (x^2 - y^2 - 3z^2) - (2x^2 + 2y^2 + 2z^2) = -2x^2 - 3z^2$.

Question: Review Exercise 8, Question 4, (iii): For given A, B, C, find A − B

+ C.

Answer: A − B + C = (2x² + 2y² + 2z²) − (−x² + 3y² + 2z²) + (x² − y² − 3z²) = 4x² − 2y² − 3z².

Question: Review Exercise 8, Question 4, (iv): For given A, B, C, find A + B − C.

Answer: A + B − C = (2x² + 2y² + 2z²) + (−x² + 3y² + 2z²) − (x² − y² − 3z²) = 6y² + 7z².

Question: Review Exercise 8, Question 4, (v): For given A, B, C, find A − B − C.

Answer: A − B − C = (2x² + 2y² + 2z²) − (−x² + 3y² + 2z²) − (x² − y² − 3z²) = 2x² + 3z².

Question: Review Exercise 8, Question 4, (vi): For given A, B, C, find B − C − A.

Answer: B − C − A = (−x² + 3y² + 2z²) − (x² − y² − 3z²) − (2x² + 2y² + 2z²) = −4x² + 2y² + 3z².

Question: Review Exercise 8, Question 5, (i): Simplify: (x − 2y)(x + 2y)

Answer: Use the formula (a + b)(a − b) = a² − b²: (x − 2y)(x + 2y) = x² − (2y)² = x² − 4y².

Question: Review Exercise 8, Question 5, (ii): Simplify: (4x²)(3x + 1)

Answer: (4x²)(3x + 1) = 4x² × 3x + 4x² × 1 = 12x³ + 4x².

Question: Review Exercise 8, Question 5, (iii): Simplify: 2x(x + y) − 2y(x − y)

Answer: 2x(x + y) = 2x² + 2xy; 2y(x − y) = 2xy − 2y². Difference = (2x² + 2xy) − (2xy − 2y²) = 2x² + 2y².

Question: Review Exercise 8, Question 5, (iv): Simplify: (a²b³)(2a − 3b)

Answer: (a²b³)(2a − 3b) = a²b³ × 2a − a²b³ × 3b = 2a³b³ − 3a²b⁴.

Question: Review Exercise 8, Question 5, (v): Simplify: (a² − b²)(a² + b²)

Answer: (a² − b²)(a² + b²) = (a²)² − (b²)² = a⁴ − b⁴.

Question: Review Exercise 8, Question 5, (vi): Simplify: $(a^2 + 1)(a^2 - a - 1)$

Answer: $(a^2 + 1)(a^2 - a - 1) = a^4 - a^3 - a^2 + a^2 - a - 1 = a^4 - a^3 - a - 1$.

Question: Review Exercise 8, Question 5, (vii): Simplify: $x(y + 1) - y(x + 1) - (x - y)$

Answer: $x(y + 1) = xy + x$; $y(x + 1) = xy + y$. Then $(xy + x) - (xy + y) - (x - y) = xy + x - xy - y - x + y = 0$.

Question: Review Exercise 8, Question 5, (viii): Simplify: $a^2(b^2 - c^2) + b^2(c^2 - a^2) + c^2(a^2 - b^2)$

Answer: $= a^2b^2 - a^2c^2 + b^2c^2 - a^2b^2 + a^2c^2 - b^2c^2 = 0$.

Question: Review Exercise 8, Question 6, (i): Prove: $(3x - 5)(3x + 5) = 9x^2 - 25$

Answer: LHS $= (3x)^2 - 5^2 = 9x^2 - 25 =$ RHS. So the result is proved.

Question: Review Exercise 8, Question 6, (ii): Prove: $(2a - 5b)^2 = 4a^2 - 20ab + 25b^2$

Answer: LHS $= (2a)^2 - 2 \times 2a \times 5b + (5b)^2 = 4a^2 - 20ab + 25b^2 =$ RHS. So the result is proved.

Question: Review Exercise 8, Question 6, (iii): Prove: $(3x^2 + 4y^2)^2 = 9x^4 + 24x^2y^2 + y^4$

Answer: LHS $= (3x^2 + 4y^2)^2 = 9x^4 + 2 \times 3x^2 \times 4y^2 + (4y^2)^2 = 9x^4 + 24x^2y^2 + 16y^4$. This differs from the printed RHS $9x^4 + 24x^2y^2 + y^4$, because the 16 is missing in print. The correct RHS is $9x^4 + 24x^2y^2 + 16y^4$.

Question: Review Exercise 8, Question 6, (iv): Prove: $9x^2 - 25y^2 = (3x - 5y)(3x + 5y)$

Answer: RHS $= (3x)^2 - (5y)^2 = 9x^2 - 25y^2 =$ LHS. So the result is proved.

Question: Review Exercise 8, Question 7, (i): Factorise the following expressions: $10a^2 - 200a^4b$

Answer: Take out the common factor $10a^2$: $10a^2(1 - 20a^2b)$.

Question: Review Exercise 8, Question 7, (ii): Factorise the following expressions: $36x^3y^3z^3 - 27x^2y^4z + 63xyz^4$

Answer: Take out the common factor 9xyz: $9xyz(4x^2y^2z^2 - 3xy^3 + 7z^3)$.

Question: Review Exercise 8, Question 7, (iii): Factorise the following expressions: $15x^4y + 21x^3y^2 - 27x^2y^2 - 33xy^4$

Answer: Take out the common factor 3xy: $3xy(5x^3 + 7x^2y - 9xy - 11y^3)$.

Question: Review Exercise 8, Question 7, (iv): Factorise the following expressions: $x(a^2 + 11) - 16(a^2 + 11)$

Answer: The common factor in both parts is $(a^2 + 11)$. $x(a^2 + 11) - 16(a^2 + 11) = (a^2 + 11)(x - 16)$.

Question: Review Exercise 8, Question 7, (v): Factorise the following expressions: $x^2(ab + c) + xy(ab + c) + z^2(ab + c)$

Answer: The common factor in all three parts is $(ab + c)$. $= (ab + c)(x^2 + xy + z^2)$.

Question: Review Exercise 8, Question 8, (i): Factorise the following expressions: $a^2 - 26a + 169$

Answer: The first term is the square of a, the last term is the square of 13, and the middle term is $2 \times a \times 13 = 26a$. So $a^2 - 26a + 169 = (a - 13)^2 = (a - 13)(a - 13)$.

Question: Review Exercise 8, Question 8, (ii): Factorise the following expressions: $1 - 6x^2y^2z + 9x^4y^4z^2$

Answer: The first term is the square of 1, the last term is the square of $3x^2y^2z$, and the middle term is $2 \times 1 \times 3x^2y^2z = 6x^2y^2z$. So $= (1 - 3x^2y^2z)^2 = (1 - 3x^2y^2z)(1 - 3x^2y^2z)$.

Question: Review Exercise 8, Question 8, (iii): Factorise the following expressions: $7ab^2 - 343a$

Answer: Take out the common factor 7a: $7a(b^2 - 49) = 7a(b + 7)(b - 7)$.

Question: Review Exercise 8, Question 8, (iv): Factorise the following

expressions: $75 - 3(x - y)^2$

Answer: Take out the common factor 3: $3[25 - (x - y)^2] = 3[5 + (x - y)][5 - (x - y)] = 3(5 + x - y)(5 - x + y)$.

Question: Review Exercise 8, Question 8, (v): Factorise the following expressions: $49(x + y)^2 - 16(x - y)^2$

Answer: $[7(x + y)]^2 - [4(x - y)]^2 = [7(x + y) + 4(x - y)][7(x + y) - 4(x - y)] = (11x + 3y)(3x + 11y)$.

Question: Review Exercise 8, Question 8, (vi): Factorise the following expressions: $9/16\ a^2 + ab + 4/9\ b^2$

Answer: The first term is the square of (3/4 a), the last term is the square of (2/3 b), and the middle term is $2 \times (3/4\ a) \times (2/3\ b) = ab$. So $= (3/4\ a + 2/3\ b)^2 = (3/4\ a + 2/3\ b)(3/4\ a + 2/3\ b)$.

Question: Review Exercise 8, Question 8, (vii): Factorise the following expressions: $a^2l^2/b^2 - 2aclm/bd + c^2m^2/d^2$

Answer: The first term is the square of (al/b), the last term is the square of (cm/d), and the middle term is $2 \times (al/b) \times (cm/d) = 2aclm/bd$. So $= (al/b - cm/d)^2 = (al/b - cm/d)(al/b - cm/d)$.

Question: Review Exercise 8, Question 8, (viii): Factorise the following expressions: $(a - 9/5)^2 - 36/25\ m^2$

Answer: $(a - 9/5)^2 - (6m/5)^2 = (a - 9/5 + 6m/5)(a - 9/5 - 6m/5)$.

9. Linear Equations

✓ True: Practical Work, (1): State whether this is a linear equation in one variable or not: $4x + 5 = 2$

✗ False: Practical Work, (2): State whether this is a linear equation in one variable or not: $y^2 + 3$

✗ False: Practical Work, (3): State whether this is a linear equation in one variable or not: $x^2 + 3x + 4 = 0$

✓ True: Practical Work, (4): State whether this is a linear equation in one variable or not: $x + 5 = 20$

✗ False: Practical Work, (5): State whether this is a linear equation in one variable or not: $3x^2 + 7 = 0$

✓ True: Practical Work, (6): State whether this is a linear equation in one variable or not: $x + x = 20$

✗ False: Practical Work, (7): State whether this is a linear equation in one variable or not: $x^2 + y^2 = 9$

✗ False: Practical Work, (8): State whether this is a linear equation in one variable or not: $m = n + 5$

✗ False: Practical Work, (9): State whether this is a linear equation in one variable or not: $x + 7 = 11y$

✓ True: Practical Work, (10): State whether this is a linear equation in one variable or not: $7v - 3 = 4$

✗ False: Practical Work, (11): State whether this is a linear equation in one variable or not: $a^2 + 2ab + b^2 = 0$

✓ True: Practical Work, (12): State whether this is a linear equation in one variable or not: $x = 13 - 4$

Question: Exercise 9.1, Part A, (1): Solve the following equation: $x + 5 = 9$

Answer: Subtract 5 from both sides: x + 5 − 5 = 9 − 5, so x = 4.

Question: Exercise 9.1, Part A, (2): Solve the following equation: 12 − x = 7

Answer: Subtract 12 from both sides: −x = 7 − 12 = −5, so x = 5.

Question: Exercise 9.1, Part A, (3): Solve the following equation: 5x = 3

Answer: Divide both sides by 5: x = 3/5.

Question: Exercise 9.1, Part A, (4): Solve the following equation: 4x + 10 = 2

Answer: Subtract 10 from both sides: 4x = 2 − 10 = −8. Then divide both sides by 4: x = −2.

Question: Exercise 9.1, Part A, (5): Solve the following equation: 5 − 3x = −4

Answer: Subtract 5 from both sides: −3x = −4 − 5 = −9. Then divide both sides by −3: x = 3.

Question: Exercise 9.1, Part A, (6): Solve the following equation: 2 + 14x = 30

Answer: Subtract 2 from both sides: 14x = 28. Then divide both sides by 14: x = 2.

Question: Exercise 9.1, Part A, (7): Solve the following equation: 9 + 5x = 3x + 13

Answer: Subtract 3x from both sides: 9 + 2x = 13. Then subtract 9 from both sides: 2x = 4, so x = 2.

Question: Exercise 9.1, Part A, (8): Solve the following equation: 4 − 3x = 8 + x

Answer: Subtract x from both sides: 4 − 4x = 8. Then subtract 4 from both sides: −4x = 4, so x = −1.

Question: Exercise 9.1, Part A, (9): Solve the following equation: 4x + 5/2 = x + 4 1/2

Answer: Write 4 1/2 = 9/2. Subtract x from both sides: 3x + 5/2 = 9/2. Then subtract 5/2 from both sides: 3x = 4/2 = 2, so x = 2/3.

Question: Exercise 9.1, Part B, (1): Solve the following equation: 5 + 3(x − 1) = 5x − 6

Answer: Open the bracket: 5 + 3x − 3 = 5x − 6, i.e. 3x + 2 = 5x − 6. Subtract 3x from both sides: 2 = 2x − 6. Add 6 to both sides: 8 = 2x, so x = 4.

Question: Exercise 9.1, Part B, (2): Solve the following equation: 6 − 4(x + 3) = 2(x − 1)

Answer: Open the brackets: 6 − 4x − 12 = 2x − 2, i.e. −4x − 6 = 2x − 2. Add 4x to both sides: −6 = 6x − 2. Add 2 to both sides: −4 = 6x, so x = −2/3.

Question: Exercise 9.1, Part B, (3): Solve the following equation: 5(3 − x) − 2(4 − 3x) = 11 − 2(x − 1)

Answer: Open the brackets: 15 − 5x − 8 + 6x = 11 − 2x + 2, i.e. x + 7 = 13 − 2x. Add 2x to both sides: 3x + 7 = 13. Subtract 7 from both sides: 3x = 6, so x = 2.

Question: Exercise 9.1, Part B, (4): Solve the following equation: 5(1 − 2x) + 2(3 − x) = 3(x + 4) + 14

Answer: Open the brackets: 5 − 10x + 6 − 2x = 3x + 12 + 14, i.e. −12x + 11 = 3x + 26. Add 12x to both sides: 11 = 15x + 26. Subtract 26 from both sides: −15 = 15x, so x = −1.

Question: Exercise 9.1, Part B, (5): Solve the following equation: (x + 2)(x + 3) = (x − 3)(x − 2) + 20

Answer: Expand both sides: $x^2 + 5x + 6 = x^2 − 5x + 6 + 20$. Subtract x^2 + 6 from both sides: 5x = −5x + 20. Add 5x to both sides: 10x = 20, so x = 2.

Question: Exercise 9.1, Part C, (1): Solve the following equation: 6x + 2 =

29 − 3x

Answer: Add 3x to both sides: 9x + 2 = 29. Subtract 2 from both sides: 9x = 27, so x = 3.

Question: Exercise 9.1, Part C, (2): Solve the following equation: (1/3)x + 4 = (4x − 1)/5

Answer: Multiply both sides by 15: 5x + 60 = 12x − 3. Subtract 5x from both sides: 60 = 7x − 3. Add 3 to both sides: 63 = 7x, so x = 9.

Question: Exercise 9.1, Part C, (3): Solve the following equation: 3x/4 = 2/5

Answer: Multiply both sides by 20: 15x = 8, so x = 8/15.

Question: Exercise 9.1, Part C, (4): Solve the following equation: 8/x = 2

Answer: Multiply both sides by x: 8 = 2x, so x = 4.

Question: Exercise 9.1, Part C, (5): Solve the following equation: 7/(3x) = 2

Answer: Multiply both sides by 3x: 7 = 6x, so x = 7/6.

Question: Exercise 9.1, Part C, (6): Solve the following equation: 3/(x + 1) = 6/(5x − 1)

Answer: Cross-multiply: 3(5x − 1) = 6(x + 1), i.e. 15x − 3 = 6x + 6. Subtract 6x from both sides: 9x − 3 = 6. Add 3 to both sides: 9x = 9, so x = 1.

Question: Exercise 9.1, Part D, (1): Solve the following equation: 5 + x/3 = 7

Answer: Subtract 5 from both sides: x/3 = 2. Multiply both sides by 3: x = 6.

Question: Exercise 9.1, Part D, (2): Solve the following equation: x/2 − 1 = 5

Answer: Add 1 to both sides: x/2 = 6. Multiply both sides by 2: x = 12.

Question: Exercise 9.1, Part D, (3): Solve the following equation: 3x/4 − 2 = x/3 + 3

Answer: Multiply both sides by 12: 9x − 24 = 4x + 36. Subtract 4x from

both sides: 5x − 24 = 36. Add 24 to both sides: 5x = 60, so x = 12.

Question: Exercise 9.1, Part D, (4): Solve the following equation: 4 − 2x/3 = (x + 2)/5

Answer: Multiply both sides by 15: 60 − 10x = 3x + 6. Add 10x to both sides: 60 = 13x + 6. Subtract 6 from both sides: 54 = 13x, so x = 54/13.

Question: Exercise 9.1, Part D, (5): Solve the following equation: (x + 2)/3 = (1 − 2x)/5

Answer: Cross-multiply: 5(x + 2) = 3(1 − 2x), i.e. 5x + 10 = 3 − 6x. Add 6x to both sides: 11x + 10 = 3. Subtract 10 from both sides: 11x = −7, so x = −7/11.

Question: Exercise 9.1, Part D, (6): Solve the following equation: (4x + 3)/3 = (x + 7)/2

Answer: Cross-multiply: 2(4x + 3) = 3(x + 7), i.e. 8x + 6 = 3x + 21. Subtract 3x from both sides: 5x + 6 = 21. Subtract 6 from both sides: 5x = 15, so x = 3.

Question: Exercise 9.1, Part D, (7): Solve the following equation: (5x + 1)/2 − (x + 2)/6 = (2x + 4)/3

Answer: Multiply both sides by 6: 3(5x + 1) − (x + 2) = 2(2x + 4), i.e. 14x + 1 = 4x + 8. Subtract 4x from both sides: 10x + 1 = 8. Subtract 1 from both sides: 10x = 7, so x = 7/10.

Question: Exercise 9.1, Part D, (8): Solve the following equation: (x − 3)/3 − 7/2 = (4x − 3)/3 − 2/3

Answer: Multiply both sides by 6: 2(x − 3) − 21 = 2(4x − 3) − 4, i.e. 2x − 27 = 8x − 10. Subtract 2x from both sides: −27 = 6x − 10. Add 10 to both sides: −17 = 6x, so x = −17/6.

Question: Exercise 9.2, Question (1): Solve the following equation and check your answer: 3x + 9 = 18

Answer: Subtract 9 from both sides: 3x = 9, so x = 3. Check: LHS = 3(3) +

9 = 18 = RHS. ✓

Question: Exercise 9.2, Question (2): Solve the following equation and check your answer: 2x − 5 = 9

Answer: Add 5 to both sides: 2x = 14, so x = 7. Check: LHS = 2(7) − 5 = 9 = RHS. ✓

Question: Exercise 9.2, Question (3): Solve the following equation and check your answer: 4x − 5 = 11

Answer: Add 5 to both sides: 4x = 16, so x = 4. Check: LHS = 4(4) − 5 = 11 = RHS. ✓

Question: Exercise 9.2, Question (4): Solve the following equation and check your answer: 5x − 20 = 30

Answer: Add 20 to both sides: 5x = 50, so x = 10. Check: LHS = 5(10) − 20 = 30 = RHS. ✓

Question: Exercise 9.2, Question (5): Solve the following equation and check your answer: 3x + 17 = 2

Answer: Subtract 17 from both sides: 3x = −15, so x = −5. Check: LHS = 3(−5) + 17 = 2 = RHS. ✓

Question: Exercise 9.2, Question (6): Solve the following equation and check your answer: 13 − 3x = −2

Answer: Subtract 13 from both sides: −3x = −15, so x = 5. Check: LHS = 13 − 3(5) = −2 = RHS. ✓

Question: Exercise 9.2, Question (7): Solve the following equation and check your answer: 4x + 8 = 2

Answer: Subtract 8 from both sides: 4x = −6, so x = −3/2. Check: LHS = 4(−3/2) + 8 = 2 = RHS. ✓

Question: Exercise 9.2, Question (8): Solve the following equation and check your answer: (x + 3)/(x − 1) = 6/7

Answer: Cross-multiply: 7(x + 3) = 6(x − 1), i.e. 7x + 21 = 6x − 6.

Subtract 6x from both sides: x + 21 = −6, so x = −27. Check: LHS = (−27 + 3)/(−27 − 1) = (−24)/(−28) = 6/7 = RHS. ✓

Question: Exercise 9.2, Question (9): Solve the following equation and check your answer: 5x + 1 = 6/5

Answer: Subtract 1 from both sides: 5x = 1/5, so x = 1/25. Check: LHS = 5(1/25) + 1 = 6/5 = RHS. ✓

Question: Exercise 9.2, Question (10): Solve the following equation and check your answer: (x − 1)/(x + 2) = 4/3

Answer: Cross-multiply: 3(x − 1) = 4(x + 2), i.e. 3x − 3 = 4x + 8. Subtract 3x from both sides: −3 = x + 8, so x = −11. Check: LHS = (−11 − 1)/(−11 + 2) = (−12)/(−9) = 4/3 = RHS. ✓

Question: Exercise 9.2, Question (11): Solve the following equation and check your answer: (2x + 3)/(7x + 1) = 7/15

Answer: Cross-multiply: 15(2x + 3) = 7(7x + 1), i.e. 30x + 45 = 49x + 7. Subtract 30x from both sides: 45 = 19x + 7. Subtract 7 from both sides: 38 = 19x, so x = 2. Check: LHS = (4 + 3)/(14 + 1) = 7/15 = RHS. ✓

Question: Exercise 9.2, Question (12): Solve the following equation and check your answer: (3x − 8)/(5x + 2) = 1

Answer: Multiply both sides by (5x + 2): 3x − 8 = 5x + 2. Subtract 3x from both sides: −8 = 2x + 2. Subtract 2 from both sides: −10 = 2x, so x = −5. Check: LHS = (−15 − 8)/(−25 + 2) = (−23)/(−23) = 1 = RHS. ✓

Question: Exercise 9.3, Question 1: Aslam had Rs 35. He bought some pencils and Rs 5 were left with him. Tell how much one pencil costs?

Answer: Let the cost of one pencil be x. The number of pencils is not clear in the printed question; taking three pencils, form the equation: 3x + 5 = 35. Subtract 5 from both sides: 3x = 30, so x = 10. One pencil costs Rs 10. (If the number of pencils is different, use the same method.)

Question: Exercise 9.3, Question 2: Ali had some rupees. If Rs 7 are added

to that amount, the total becomes Rs 22. Tell how many rupees Ali had?

Answer: Let Ali's amount be x. Equation: x + 7 = 22. Subtract 7 from both sides: x = 15. Ali had Rs 15.

Question: Exercise 9.3, Question 3: The length of a room is 1.5 metres more than its width. If the perimeter of the room is 63 metres, find the length and the width.

Answer: Let the width be x, so the length = x + 1.5. Perimeter: 2(x + x + 1.5) = 63, i.e. 2x + 1.5 = 31.5, i.e. 2x = 30, so x = 15. The width is 15 m and the length is 16.5 m.

Question: Exercise 9.3, Question 4: The perimeter of a triangle ABC is 11 cm. Side AB is 1 cm longer than BC, and side AC is 3 cm shorter than BC. Tell the length of each side of the triangle.

Answer: Let BC be x, so AB = x + 1 and AC = x − 3. Perimeter: x + (x + 1) + (x − 3) = 11, i.e. 3x − 2 = 11, i.e. 3x = 13, so x = 13/3 = 4 1/3. So BC = 4 1/3 cm, AB = 5 1/3 cm, AC = 1 1/3 cm.

Question: Exercise 9.3, Question 5: A 32 cm long wire is bent to make a square. Tell how long each side of the square is?

Answer: Let the side of the square be x. Perimeter: 4x = 32, so x = 8. Each side of the square is 8 cm.

Question: Exercise 9.3, Question 6: Hassan and Asghar together have Rs 800. Hassan has four times as many rupees as Asghar. Tell how many rupees Hassan and Asghar have separately?

Answer: Let Asghar's amount be x, so Hassan's amount = 4x. Equation: x + 4x = 800, i.e. 5x = 800, so x = 160. Asghar has Rs 160 and Hassan has Rs 640.

Question: Exercise 9.3, Question 7: Ask your friend to think of a number, add 3 to it, then multiply that result by 2, then subtract 6 from it, and then divide that result by 2. Tell what relation the original thought number and

the final answer have?

Answer: Let the original number be x. Steps: x → x + 3 → 2(x + 3) = 2x + 6 → 2x + 6 − 6 = 2x → 2x ÷ 2 = x. The final answer equals the original number, i.e. both are equal.

Question: Exercise 9.3, Question 8: Tell two consecutive numbers whose sum is 43.

Answer: Let the first number be x, so the second = x + 1. Equation: x + (x + 1) = 43, i.e. 2x + 1 = 43, i.e. 2x = 42, so x = 21. The two numbers are 21 and 22.

Question: Exercise 9.3, Question 9: Tell three consecutive numbers whose sum is 48.

Answer: Let the first number be x, so the others = x + 1, x + 2. Equation: x + (x + 1) + (x + 2) = 48, i.e. 3x + 3 = 48, i.e. 3x = 45, so x = 15. The three numbers are 15, 16 and 17.

Question: Exercise 9.3, Question 10: The sum of 1/3 part and 1/4 part of some amount is 14; tell that amount.

Answer: Let the amount be x. Equation: x/3 + x/4 = 14. Multiply both sides by 12: 4x + 3x = 168, i.e. 7x = 168, so x = 24. That amount is 24.

Question: Exercise 9.3, Question 11: The sum of two numbers is 9. If four times the second number is subtracted from five times the first number, the answer is 9. Tell what those two numbers are?

Answer: Let the first number be x, so the second = 9 − x. Equation: 5x − 4(9 − x) = 9, i.e. 5x − 36 + 4x = 9, i.e. 9x = 45, so x = 5. The second number = 9 − 5 = 4. The two numbers are 5 and 4.

Question: Exercise 9.3, Question 12: The sum of four consecutive numbers is 120; find those numbers.

Answer: Let the first number be x. Equation: x + (x + 1) + (x + 2) + (x + 3) = 120, i.e. 4x + 6 = 120, i.e. 4x = 114, i.e. x = 28.5. This is not a whole

number, so the sum of four consecutive whole numbers cannot be 120 (we get 28.5, 29.5, 30.5, 31.5, which are not whole numbers). The printed question contains a contradiction.

Question: Review Exercise 9, Question 1, (i): Write the linear equations of the following: The sum of two consecutive numbers is 42.

Answer: Let the first number be x. Equation: x + (x + 1) = 42, i.e. 2x + 1 = 42, i.e. 2x = 41, i.e. x = 20.5. This is not a whole number, so the sum of two consecutive whole numbers cannot be 42 (we get 20.5 and 21.5). The printed question contains a contradiction.

Question: Review Exercise 9, Question 1, (ii): Write the linear equations of the following: The sum of three consecutive numbers is 21.

Answer: Let the first number be x. Equation: x + (x + 1) + (x + 2) = 21, i.e. 3x + 3 = 21, i.e. 3x = 18, so x = 6. The three numbers are 6, 7 and 8.

Question: Review Exercise 9, Question 1, (iii): Write the linear equations of the following: The sum of four consecutive numbers is 46.

Answer: Let the first number be x. Equation: x + (x + 1) + (x + 2) + (x + 3) = 46, i.e. 4x + 6 = 46, i.e. 4x = 40, so x = 10. The four numbers are 10, 11, 12 and 13.

Question: Review Exercise 9, Question 2: Write any three linear equations with different variables and different forms.

Answer: This is an open-ended question; the student may write any three correct linear equations. Examples: (1) x + 5 = 12; (2) (2x + 3)/5 = 7; (3) 3x − 4 = 2x + 6.

Question: Review Exercise 9, Question 3: What is meant by the solution of an equation? Explain with an example.

Answer: The solution of an equation is the value of the variable that makes both sides of the equation equal. Example: the solution of x + 3 = 8 is x = 5, because 5 + 3 = 8. Another example: the solution of 2x = 10 is x = 5,

because 2 × 5 = 10.

Review Exercise 9, Question 4, (i): Fill in the blanks: An equation in which one variable is of power one is called a linear equation in _____.

Answer: one variable.

Review Exercise 9, Question 4, (ii): Fill in the blanks: The solution of an equation is also called its _____.

Answer: root.

Review Exercise 9, Question 4, (iii): Fill in the blanks: The method of finding the value of any variable that makes the equation true is called the _____ of the equation.

Answer: solution

Review Exercise 9, Question 4, (iv): Fill in the blanks: Adding _____ to both sides of a linear equation does not affect the equality of the equation.

Answer: same number

Question: Review Exercise 9, Question 5, (i): Tick (✓) the correct answer: The lowest degree, i.e. power, of a linear equation is: (a) 1 (b) 2 (c) 3 (d) 4

Answer: The correct answer is (a) 1, because in a linear equation the power of the variable is 1.

Question: Review Exercise 9, Question 5, (ii): Tick (✓) the correct answer: From the equation 1 1/2 x = 1, the value of x is: (a) 1/2 (b) 3/2 (c) 1 (d) 2/3

Answer: The correct answer is (d) 2/3, because 1 1/2 x = 1, i.e. (3/2)x = 1, so x = 2/3.

Question: Review Exercise 9, Question 5, (iii): Tick (✓) the correct answer: The equation 2x + 1 = 6 is a linear equation, whose degree is: (a) 1 (b) 2 (c) 3 (d) 4

Answer: The correct answer is (a) 1, because the power of x is 1.

Question: Review Exercise 9, Question 5, (iv): Tick (✓) the correct answer: The solution of the equation 1 − 6x = 7 is: (a) x = 0 (b) x = 1 (c) x = 2 (d)

x = 3

Answer: As printed, none of the given options is correct. Solving 1 − 6x = 7: −6x = 6, so x = −1, which is not among the options. The printed question contains an error.

Question: Review Exercise 9, Question 6, (i): Solve the following equations: x/2 − 4 = 2

Answer: Add 4 to both sides: x/2 = 6. Multiply both sides by 2: x = 12.

Question: Review Exercise 9, Question 6, (ii): Solve the following equations: (x − 7)/4 = 3

Answer: Multiply both sides by 4: x − 7 = 12. Add 7 to both sides: x = 19.

Question: Review Exercise 9, Question 6, (iii): Solve the following equations: (x − 5)/2

Answer: This expression is incomplete as printed, because it has no equality sign and no right-hand-side value, so it cannot be solved as an equation.

Question: Review Exercise 9, Question 6, (iv): Solve the following equations: (x + 1)/(x − 3) = 5/3

Answer: Cross-multiply: 3(x + 1) = 5(x − 3), i.e. 3x + 3 = 5x − 15. Subtract 3x from both sides: 3 = 2x − 15. Add 15 to both sides: 18 = 2x, so x = 9.

Question: Review Exercise 9, Question 6, (v): Solve the following equations: (x − 2)/(x − 1) = 3/4

Answer: Cross-multiply: 4(x − 2) = 3(x − 1), i.e. 4x − 8 = 3x − 3. Subtract 3x from both sides: x − 8 = −3. Add 8 to both sides: x = 5.

Question: Review Exercise 9, Question 6, (vi): Solve the following equations: (x + 3)/3 = (x + 6)/2

Answer: Cross-multiply: 2(x + 3) = 3(x + 6), i.e. 2x + 6 = 3x + 18. Subtract 2x from both sides: 6 = x + 18. Subtract 18 from both sides: x =

−12.

Question: Review Exercise 9, Question 7: A 120 cm long wire is bent into a rectangle so that its length is 10 cm more than its width. Tell what the length and width of the rectangle are?

Answer: Let the width be x, so the length = x + 10. Perimeter: 2(x + x + 10) = 120, i.e. 2x + 10 = 60, i.e. 2x = 50, so x = 25. The width is 25 cm and the length is 35 cm.

10. Basic Concepts of Geometry

Practical Work: In the given figure two straight lines cut each other at point O, where m∠HOG = 120° and m∠EOH = 60°. Fill in the blank: m∠FOG = _____

Practical Work: In the same figure fill in the blank: m∠EOF = m∠HOG = 120°.

Practical Work: From the given figure of triangle DEF fill in the blank: m∠DEF = _____

Answer: 65°.

Practical Work: In the same figure DG and EF cut each other at point F. Using vertically opposite angles fill in the blank: m∠DFE = _____ (i) (In the figure m∠GFH = 40° is given.)

Practical Work: In the same triangle, using the sum of the three angles of a triangle, fill in the blank: m∠EDF = _____ (ii)

Answer: 180°, i.e. 65° + 40° + m∠EDF = 180°. So m∠EDF = 180° − (65° + 40°) = 180° − 105° = 75°.

Question: Exercise 10.1, Question 1: Looking at figure (1), write the names of all the adjacent angles.

Answer: In figure (1) four rays OP, OS, OR and OQ start from O. The pairs of adjacent angles are: (∠POQ, ∠QOS), (∠QOS, ∠SOR), (∠POS, ∠SOR) and (∠POQ, ∠QOR). Each pair has a common vertex O and a common side.

Question: Exercise 10.1, Question 2: Looking at figure (2), write the pairs of all supplementary angles. (In the figure m∠AOD = 50°, m∠DOC = 90°, m∠COB = 40°.)

Answer: On the straight line AB the angles at O sum to 180°. m∠DOB =

m∠DOC + m∠COB = 90° + 40° = 130°, and 50° + 130° = 180°, so (∠AOD, ∠DOB) are supplementary. m∠AOC = m∠AOD + m∠DOC = 50° + 90° = 140°, and 140° + 40° = 180°, so (∠AOC, ∠COB) are also supplementary.

Question: Exercise 10.1, Question 3: Look at figure (3), in which m∠MON = 130° and m∠MOP = 50°. Find the measure of the remaining angle ∠PON.

Answer: From the figure ∠MON = ∠MOP + ∠PON, i.e. 130° = 50° + m∠PON. So m∠PON = 130° − 50° = 80°.

Question: Exercise 10.1, Question 4: Find the measure of the complementary angle of each of the following angle measures. (i) 49°

Answer: Complementary angles sum to 90°, so the complementary angle = 90° − 49° = 41°.

Question: Exercise 10.1, Question 4: (ii) 62°

Answer: The complementary angle = 90° − 62° = 28°.

Question: Exercise 10.1, Question 4: (iii) 54°

Answer: The complementary angle = 90° − 54° = 36°.

Question: Exercise 10.1, Question 4: (iv) 36°

Answer: The complementary angle = 90° − 36° = 54°.

Question: Exercise 10.1, Question 4: (v) 67°

Answer: The complementary angle = 90° − 67° = 23°.

Question: Exercise 10.1, Question 4: (vi) 81°

Answer: The complementary angle = 90° − 81° = 9°.

Question: Exercise 10.1, Question 4: (vii) 15°

Answer: The complementary angle = 90° − 15° = 75°.

Question: Exercise 10.1, Question 5: Find the measure of the supplementary angle of each of the following angle measures. (i) 80°

Answer: Supplementary angles sum to 180°, so the supplementary angle = 180° − 80° = 100°.

Question: Exercise 10.1, Question 5: (ii) 132°

Answer: The supplementary angle = 180° − 132° = 48°.

Question: Exercise 10.1, Question 5: (iii) 49°

Answer: The supplementary angle = 180° − 49° = 131°.

Question: Exercise 10.1, Question 5: (iv) 125°

Answer: The supplementary angle = 180° − 125° = 55°.

Question: Exercise 10.1, Question 5: (v) 76°

Answer: The supplementary angle = 180° − 76° = 104°.

Question: Exercise 10.1, Question 5: (vi) 103°

Answer: The supplementary angle = 180° − 103° = 77°.

Question: Exercise 10.1, Question 5: (vii) 68°

Answer: The supplementary angle = 180° − 68° = 112°.

Question: Exercise 10.1, Question 6: Look at the given figure (4), where two straight lines cut each other and one angle is 55°. Write the names and measures of the vertically opposite angles.

Answer: The vertically opposite angle to the 55° angle is also 55°. The other two angles are vertically opposite to each other and each is 180° − 55° = 125°. So the four angles measure 55°, 125°, 55°, 125°.

Question: Exercise 10.1, Question 7: Write the measures of the unknown angles in the following triangles. (i) Triangle ABC, in which m∠B = 90° and the exterior angle at C is 125°.

Answer: ∠ACB and the 125° exterior angle are on a straight line, so m∠ACB = 180° − 125° = 55°. The sum of the three angles of a triangle is 180°, so m∠BAC = 180° − 90° − 55° = 35°.

Question: Exercise 10.1, Question 7: (ii) Triangle PQR, in which m∠P = 62° and the vertically opposite angle at R is m∠TRS = 38°.

Answer: ∠PRQ and ∠TRS are vertically opposite angles, so m∠PRQ = 38°. The sum of the three angles of a triangle is 180°, so m∠PQR = 180° − 62°

− 38° = 80°.

Exercise 10.1, Question 8: Choose the correct answer from the given material and fill in the blank. (i) If the measure of an angle is equal to its complement, then the measure of that angle is _____. (90°, 45°, 180°)

Answer: complement gives x = 90° − x, i.e. 2x = 90°, so x = 45°. Correct answer: 45°.

Exercise 10.1, Question 8: (ii) If the measure of an angle is equal to its supplement, then the measure of that angle will be _____. (45°, 90°, 180°)

Answer: supplement gives x = 180° − x, i.e. 2x = 180°, so x = 90°. Correct answer: 90°.

Exercise 10.1, Question 8: (iii) The supplement of an acute angle is a _____ angle. (an acute, a right, an obtuse)

Answer: less than 90°. Its supplement = 180° − (less than 90°) = more than 90° and less than 180°, i.e. an obtuse

Exercise 10.1, Question 8: (iv) In a pair of supplementary angles, if one angle is a right angle then the other supplementary angle will be _____. (an acute, a right, an obtuse)

Answer: angle. Correct answer:

Exercise 10.1, Question 8: (v) In a pair of complementary angles, each angle will be _____. (acute angle, right angle, obtuse angle)

Answer: is less than 90°, i.e. acute. Correct answer: acute angle.

Exercise 10.1, Question 8: (vi) Two straight lines cut each other at one point, so that of the four angles formed, one is a right angle; then each of the remaining angles will be _____. (acute angle, right angle, obtuse angle)

Answer: angle. Correct answer:

Question: Exercise 10.2, Question 1: Observe the following pairs of figures

and tell whether they are congruent or similar. (i) Two triangles, both of the same shape and measure.

Answer: Both triangles have the same shape and measure, so they are congruent.

Question: Exercise 10.2, Question 1: (ii) Two squares, both of the same shape but different measures.

Answer: The shape is the same and the measure is different, so they are similar (not congruent).

Question: Exercise 10.2, Question 1: (iii) Two rectangles, both of the same shape but different measures.

Answer: The shape is the same and the measure is different, so they are similar.

Question: Exercise 10.2, Question 1: (iv) Two circles, both of the same shape but different measures.

Answer: The shape is the same and the measure is different, so they are similar.

Question: Exercise 10.2, Question 1: (v) Two line segments, one 3 cm and the other 5 cm long.

Answer: Both have the same shape (straight line segments) but different lengths, so they are similar.

Question: Exercise 10.2, Question 1: (vi) Two rhombuses, both of the same shape but different measures.

Answer: The shape is the same and the measure is different, so they are similar.

Question: Exercise 10.2, Question 1: (vii) Two trapeziums, both of the same shape but different measures.

Answer: The shape is the same and the measure is different, so they are similar.

Question: Exercise 10.2, Question 1: (viii) Two parallelograms, both of the same shape and measure.

Answer: Both the shape and the measure are the same, so they are congruent.

Question: Exercise 10.2, Question 1: (ix) Two pentagons, both of the same shape and measure.

Answer: Both the shape and the measure are the same, so they are congruent.

Question: Exercise 10.2, Question 1: (x) Two semicircles, both of the same shape and measure.

Answer: Both the shape and the measure are the same, so they are congruent.

Question: Exercise 10.2, Part B: Draw two rectangles whose measures are different and tell whether they are similar or not.

Answer: Both rectangles have the same shape but different measures, so they are similar (not congruent).

Question: Exercise 10.2, Part B: Draw two circles whose radii are different and tell whether they are congruent or not.

Answer: Both circles have the same shape but different radii, so they are not congruent; they are only similar.

Question: Exercise 10.2, Part B: Draw two triangles whose measures are the same and tell whether they are similar or not.

Answer: Both triangles have the same measures, so they are congruent and also similar.

Question: Exercise 10.2, Part B: Draw two squares whose measures are different and tell whether they are congruent or not.

Answer: Both squares have the same shape but different measures, so they are not congruent; they are only similar.

Question: Exercise 10.2, Part B: Draw two rectangles whose measures are the same and tell whether they are congruent or not.

Answer: Both rectangles have the same measures, so they are congruent.

Question: Exercise 10.2, Part C: Observe the figures P, Q and R given on the grid. (i) Are figure P and figure R both of the same type of figures?

Answer: Yes. Both figures are of the same type — both are step (L-shaped) figures.

Question: Exercise 10.2, Part C: (ii) Are figure P and figure R figures of the same measure?

Answer: No. Counting the grid cells, P is 5 cells long and 2 cells wide, while R is 6 cells long and 3 cells wide. So the measures are different.

Question: Exercise 10.2, Part C: (iii) Are figure P and figure R both congruent?

Answer: No. Because the measures are different, they are not congruent.

Question: Exercise 10.2, Part C: (iv) Are figure P and figure R both similar figures?

Answer: No. In similar figures the ratio of corresponding sides is the same, but the length-to-width ratio of P is 5:2 while that of R is 6:3, which is not the same. So they are not similar either.

Question: Exercise 10.2, Part C: (v) What is the relation between figure P and figure Q?

Answer: P and Q are congruent. Q is P rotated through 180° — both cover 6 grid cells and have the same shape and measure. So they are also similar.

✓ True: Exercise 10.2, Part D: Tell whether each of the following statements is correct or incorrect. (i) The 10-rupee note and the 100-rupee note have the same shape, so they are both similar.

False: Exercise 10.2, Part D: (ii) The 20-rupee note and the 50-rupee

note do not have equal measures, so they are not similar.

False: Exercise 10.2, Part D: (iii) The 500-rupee note and the 1,000-rupee note are congruent.

True: Exercise 10.2, Part D: (iv) The 100-rupee note and the 50-rupee note are not congruent.

True: Exercise 10.2, Part D: (v) The 20-rupee note and the 500-rupee note are similar.

Question: Exercise 10.3, Question 1: Observe the following pairs of figures and tell whether they are congruent, similar, or neither. (i) Two triangles, both of the same type and measure.

Answer: Congruent. Both are of the same type and measure.

Question: Exercise 10.3, Question 1: (ii) Two rectangles, both of the same type but different measures.

Answer: Similar. The type is the same but the measure is different.

Question: Exercise 10.3, Question 1: (iii) An oval shape and a circle, both of different types.

Answer: Neither similar nor congruent. The types themselves are different.

Question: Exercise 10.3, Question 1: (iv) Two rhombuses, both of the same type but different measures.

Answer: Similar. The type is the same but the measure is different.

Question: Exercise 10.3, Question 1: (v) Two step (T-like) figures, both of the same type but different measures.

Answer: Similar. The type is the same but the measure is different.

Question: Exercise 10.3, Question 1: (vi) Two right triangles, both of the same type but different measures.

Answer: Similar. The type is the same but the measure is different.

Question: Exercise 10.3, Question 1: (vii) Two quadrilaterals (trapezium-like), both of the same type but different measures.

Answer: Similar. The type is the same but the measure is different.

Question: Exercise 10.3, Question 1: (viii) Two triangles, one acute-angled and the other right-angled — both of different types.

Answer: Neither similar nor congruent. One triangle is acute-angled and the other is right-angled, i.e. the types are different.

Question: Exercise 10.3, Question 1: (ix) A circle and a semicircle — both of different types.

Answer: Neither similar nor congruent. A circle and a semicircle are figures of different types.

Question: Exercise 10.3, Question 2: Considering the following data, write the pairs of congruent line segments using symbols. mAB = 3.2 cm, mDE = 5.1 cm, mMN = 4.5 cm, mYZ = 4.8 cm, mRS = 5.4 cm, mYZ = 3.9 cm; mCD = 4.5 cm, mGH = 4.8 cm, mXZ = 5.1 cm, mPQ = 5.4 cm, mEF = 3.9 cm, mXY = 3.2 cm.

Answer: Line segments of equal measure are congruent: AB ≅ XY (3.2 cm), DE ≅ XZ (5.1 cm), MN ≅ CD (4.5 cm), YZ ≅ GH (4.8 cm), RS ≅ PQ (5.4 cm), YZ ≅ EF (3.9 cm). Note: in the printed book the measure of YZ is given twice — once as 4.8 cm and once as 3.9 cm — whereas the same line segment cannot have two different measures; this is a printing error. The pairs are formed according to the printed measures.

Question: Exercise 10.3, Question 3: Considering the following data, write the pairs of congruent angles using symbols. m∠ABC = 60°, m∠XYZ = 110°, m∠COD = 45°, m∠AOB = 70°; m∠DEF = 70°, m∠PQR = 45°, m∠PRQ = 60°, m∠MNS = 110°.

Answer: Angles of equal measure are congruent: ∠ABC ≅ ∠PRQ (60°); ∠XYZ ≅ ∠MNS (110°); ∠COD ≅ ∠PQR (45°); ∠AOB ≅ ∠DEF (70°).

Exercise 10.3, Question 4: Fill in the blank. (i) If m∠ABC = 100° and ∠ABC ≅ ∠DEF then m∠DEF = _____

Answer: 100°.

Exercise 10.3, Question 4: (ii) If m∠PQR = m∠XYZ then ∠PQR _____ ∠XYZ.

Answer: ≅

Exercise 10.3, Question 4: (iii) Line segments of equal measure are called _____ line segments.

Answer: congruent

Exercise 10.3, Question 4: (iv) Angles equal in measure are called _____ angles.

Answer: equal

Exercise 10.3, Question 4: (v) Two circles are congruent when their _____ are equal.

Answer: diameters (or radii)

Question: Exercise 10.3, Question 5: In each pair of the following similar figures find the length of the unknown sides. (i) Right triangles ABC and DEF are similar. In triangle ABC, AB = 12 cm, BC = 5 cm, AC = x, m∠B = 90°, m∠C = 50°. In triangle DEF, DE = y, EF = 3 cm, DF = 5 cm, m∠E = 90°.

Answer: In similar triangles the ratio of corresponding sides is the same: BC/EF = AC/DF = AB/DE, i.e. 5/3 = x/5 = 12/y. From x/5 = 5/3, x = 25/3 = 8.33 cm (approximately). From 12/y = 5/3, y = 36/5 = 7.2 cm.

Question: Exercise 10.3, Question 5: (ii) Triangles PQE and XYZ are similar. In triangle PQE, PQ = 4 cm, PE = 6 cm, QE = x, m∠P = 90°, m∠E = 40°. In triangle XYZ, XY = 8 cm, YZ = 14 cm, XZ = y, m∠X = 90°, m∠Y = 50°.

Answer: Corresponding angles: X with P, Y with Q and Z with E. Ratio of corresponding sides: XY/PQ = 8/4 = 2. So QE = YZ/2 = 14/2 = 7 cm, i.e. x = 7 cm. XZ = 2 × PE = 2 × 6 = 12 cm, i.e. y = 12 cm.

Question: Exercise 10.3, Question 5: (iii) Trapezium ABCD and the smaller

trapezium are similar. In the larger trapezium the top side AB = y, side BC = 10 cm, bottom side CD = 12 cm, side DA = 10 cm. In the smaller trapezium the top side is 4 cm, the sides are 5 cm and the bottom side is x.

Answer: Ratio of corresponding sides: 10/5 = 2, i.e. the larger trapezium is twice the smaller. So y = 2 × 4 = 8 cm and x = 12/2 = 6 cm.

Question: Exercise 10.4, Question 1: In the given figure triangles APC and BPD are congruent. Which congruence property applies? (In the figure AP ≅ BP, CP ≅ DP and the angles at P are vertically opposite.)

Answer: ΔAPC ≅ ΔBPD (by the SAS property), because AP ≅ BP, CP ≅ DP and the included angles ∠APC and ∠BPD are equal because they are vertically opposite.

Question: Exercise 10.4, Question 2: In the given figure triangles PQO and MNO are congruent. Which congruence property applies? (Both are right triangles, QO ≅ NO and PO ≅ MO.)

Answer: ΔPQO ≅ ΔMNO (by the HS property), because both are right triangles, their hypotenuses PO and MO are equal and one side QO and NO is equal.

Question: Exercise 10.4, Question 3: In the given figure triangles PQR and PQS are congruent. Which congruence property applies? (RQ ≅ SQ, both are right triangles and PQ is a common side.)

Answer: ΔPQR ≅ ΔPQS (by the SAS property), because RQ ≅ SQ, both have right angles at Q and PQ is a common side of both.

Question: Exercise 10.4, Question 4: In the given rectangle PQRS, PR is a diagonal. Triangles PQR and RSP are congruent. Which congruence property applies?

Answer: ΔPQR ≅ ΔRSP (by the SSS property), because in the rectangle PQ ≅ RS, QR ≅ SP and PR is a common side of both.

Question: Exercise 10.4, Question 5: In the given figure triangles ABC and DEC are congruent. Which congruence property applies? (∠A ≅ ∠D, ∠B ≅ ∠E and AC ≅ DC.)

Answer: ΔABC ≅ ΔDEC (by the ASA property), because ∠A ≅ ∠D, ∠B ≅ ∠E and the included sides AC and DC are equal.

Question: Exercise 10.4, Question 6: In the given figure triangles ACB and DCE are congruent. Which congruence property applies? (AC ≅ CD, BC ≅ CE and the angles at C are vertically opposite.)

Answer: ΔACB ≅ ΔDCE (by the SAS property), because AC ≅ CD, BC ≅ CE and the included angles ∠ACB and ∠DCE are equal because they are vertically opposite.

Question: Exercise 10.4, Question 7: In the given figure triangles PQR and POR are congruent. Which congruence property applies? (PQ ≅ PO, QR ≅ OR and PR is a common side.)

Answer: ΔPQR ≅ ΔPOR (by the SSS property), because PQ ≅ PO, QR ≅ OR and PR is a common side of both.

Question: Exercise 10.4, Question 8: In the given parallelogram ABCD, BD is a diagonal. Triangles ABD and CDB are congruent. Which congruence property applies? (∠ABD ≅ ∠CDB, ∠ADB ≅ ∠CBD and BD is a common side.)

Answer: ΔABD ≅ ΔCDB (by the ASA property), because ∠ABD ≅ ∠CDB, ∠ADB ≅ ∠CBD and the included side BD is common.

Question: Exercise 10.4, Question 9: In the given figure triangles PQO and SRO are congruent. Which congruence property applies? (Both are right triangles, QO ≅ RO and PO ≅ SO.)

Answer: ΔPQO ≅ ΔSRO (by the HS property), because both are right triangles, their hypotenuses PO and SO are equal and one side QO and RO is equal.

Question: Exercise 10.4, Question 10: In the given figure triangles CDE and FED are congruent. Which congruence property applies? (CD ≅ FE, there are right angles at D and E and DE is a common side.)

Answer: ΔCDE ≅ ΔFED (by the SAS property), because CD ≅ FE, there are right angles at D and E and DE is a common side of both.

Exercise 10.5, Question 1: Fill in the blank. A line segment joining two points on a circle is the _____ of the circle.

Answer: called a chord.

Exercise 10.5, Question 2: Fill in the blank. The distance between the centre of a circle and any point on the circle is called the _____.

Answer: radius.

Exercise 10.5, Question 3: Fill in the blank. The line segment which joins the centre of a circle to any point on the circle is called the _____.

Answer: radius.

Exercise 10.5, Question 4: Fill in the blank. All points of a circle are at an equal distance from the _____ of the circle.

Answer: centre

Exercise 10.5, Question 5: Fill in the blank. The chord which passes through the centre of a circle is called the _____.

Answer: diameter.

Exercise 10.5, Question 6: Fill in the blank. The diameter of a circle passes through the _____ of the circle. Its length is _____ the _____ of the radius.

Answer: centre, twice, length

Exercise 10.5, Question 7: Fill in the blank. The major and minor chord parts of a circle are the _____ of the circle.

Answer: major segment and

Exercise 10.5, Question 8: Fill in the blank. The radii of any circle are many

but are equal to each other in _____ and _____.

Answer: measure, length.

Exercise 10.5, Question 9: Fill in the blank. The radius (Radius) of a circle is the _____ of the diameter of the circle.

Answer: half

Exercise 10.5, Question 10: Fill in the blank. The diameters of any circle are many but are equal to each other in _____ and _____.

Answer: measure, length.

Question: Exercise 10.6, Question 1: Draw two semicircles of radii 2.5 cm and 3 cm. In each semicircle draw any one angle and measure it. Observe whether the angle in a semicircle is a right angle.

Answer: Method: With a compass draw two semicircles of radii 2.5 cm and 3 cm. Join any point on the arc of each semicircle to both ends of the diameter and measure the angle with a protractor. Observation: In both semicircles the angle will be 90°, i.e. the angle in a semicircle is a right angle. The student will check this with their own measurements on their own figures.

Question: Exercise 10.6, Question 2: Draw a semicircle of radius 3.3 cm. Observe whether every angle in the semicircle is 90°.

Answer: Method: With a compass draw a semicircle of radius 3.3 cm. Choose two or three different points on the arc, join each to both ends of the diameter and measure the angles with a protractor. Observation: Every angle will be 90°, i.e. every angle in a semicircle is a right angle. The student will check this with their own measurements on their own figure.

Question: Exercise 10.6, Question 3: Draw a circle of radius 2.7 cm. Observe that the angles in the same segment are equal.

Answer: Method: With a compass draw a circle of radius 2.7 cm. Draw a

chord on the circle which divides the circle into two segments. In the same segment choose two or three different points on the arc, join each to both ends of the chord and measure the angles with a protractor. Observation: The measures of all angles formed in the same segment will be equal. The student will check this with their own measurements on their own figure.

Question: Exercise 10.6, Question 4: Draw a circle of radius 2.8 cm. In each segment of the circle (major and minor) draw two angles each and check whether they are equal.

Answer: Method: With a compass draw a circle of radius 2.8 cm. Draw a chord on the circle. Choose two points on the arc in the major segment and two points on the arc in the minor segment, join each to both ends of the chord and measure the angles with a protractor. Observation: The two angles of the major segment will be equal to each other and the two angles of the minor segment will be equal to each other. The student will check this with their own measurements on their own figure.

Question: Exercise 10.6, Question 5: Draw a circle of radius 3.4 cm. Draw two angles in the same segments with a protractor and check whether they are equal.

Answer: Method: With a compass draw a circle of radius 3.4 cm. Draw a chord on the circle. Choose two points on the arc in the same segment, join each to both ends of the chord and measure the angles with a protractor. Observation: The two angles will be equal. The student will check this with their own measurements on their own figure.

Question: Review Exercise 10, Question 1: Write the definition of adjacent angles and show it with a figure.

Answer: Definition: Two angles which have a common vertex and a common side, and whose other two sides lie on the two sides of the common side, are called adjacent angles. Example: In the figure, at the

rays OE, OF, OG, OH starting from O, (∠EOF, ∠FOG), (∠FOG, ∠GOH), etc. are pairs of adjacent angles.

Question: Review Exercise 10, Question 2: Write the definitions of complementary and supplementary angles and show one pair of each with a figure.

Answer: Complementary angles: two angles whose sum is 90° are called complementary angles (example: 30° and 60°). Supplementary angles: two angles whose sum is 180° are called supplementary angles (example: 50° and 130°).

Question: Review Exercise 10, Question 3: Write the definition of vertically opposite angles and show with a figure how they are.

Answer: When two straight lines cut each other, the angles opposite to each other are called vertically opposite angles. They are equal to each other. Example: In the figure, the vertically opposite angle to the 55° angle is also 55°.

Question: Review Exercise 10, Question 4: What is meant by similar figures? Show it with a figure and write its symbol.

Answer: Figures whose shape is the same but whose measure is different are called similar figures. Example: two circles of different radii or two squares of different sides are similar. Symbol: ~ (example: figure A ~ figure B).

Question: Review Exercise 10, Question 5: What is meant by congruent figures? Show it with a figure and write its symbol.

Answer: Figures whose shape and measure are both the same are called congruent figures. Example: two triangles of the same measures are congruent. Symbol: ≅ (example: ΔABC ≅ ΔDEF).

Question: Review Exercise 10, Question 6: Write the definition of a circle and show its parts with a figure: centre, radius, diameter, chord, arc, major

arc, minor arc, semicircle and segment.

Answer: Circle: the closed curved figure whose points are all at an equal distance (radius) from a fixed point (centre). In the figure the centre O, the radius from the centre to the circle, the diameter passing through the centre, the chord joining two points on the circle, a part of the circle as an arc (minor arc and major arc), the semicircle formed by a diameter and the segment formed by a chord are shown.

Question: Review Exercise 10, Question 7: ABCD is a parallelogram in which AC is a diagonal. Prove that the two triangles formed are congruent.

Answer: The diagonal AC divides the parallelogram into two triangles ΔABC and ΔCDA. In a parallelogram the opposite sides are equal: AB ≅ CD and BC ≅ DA, and AC is a common side of both. So by the SSS property ΔABC ≅ ΔCDA.

Question: Review Exercise 10, Question 8: Draw a semicircle of radius 3.3 cm. Observe that the angle in the semicircle is a right angle.

Answer: Method: With a compass draw a semicircle of radius 3.3 cm. Choose a point on the arc, join it to both ends of the diameter and measure the angle with a protractor. Observation: The angle will be 90°, i.e. the angle in the semicircle is a right angle. The student will check this with their own measurements on their own figure.

Question: Review Exercise 10, Question 9: Draw a semicircle of radius 4.4 cm and draw triangle ABC as in the figure. Measure the three angles and tell whether the angle at C is a right angle.

Answer: Method: With a compass draw a semicircle of radius 4.4 cm, take diameter AB and point C on the arc to form triangle ABC, and measure the three angles with a protractor. Observation: The angle at C, ∠ACB, will be 90°, i.e. it is a right angle. Note: the printed book writes ∠ABC here, which appears to be a printing error — the angle formed in the semicircle

opposite the diameter AB is at C, i.e. ∠ACB; the angle at B, ∠ABC, is acute.

Question: Review Exercise 10, Question 10: Draw a circle of radius 3.6 cm. Observe that the angles in the minor segment of the circle are equal to each other.

Answer: Method: With a compass draw a circle of radius 3.6 cm. Draw a chord on the circle. Choose two points on the arc in the minor segment, join each to both ends of the chord and measure the angles with a protractor. Observation: The two angles of the minor segment will be equal to each other. The student will check this with their own measurements on their own figure.

Question: Review Exercise 10, Question 11: Draw a circle of radius 4 cm. Draw a chord on the circle which is not a diameter. Observe that the angles in the major segment of the circle are equal to each other.

Answer: Method: With a compass draw a circle of radius 4 cm. Draw a chord on the circle which does not pass through the centre (i.e. is not a diameter). Choose two points on the arc in the major segment, join each to both ends of the chord and measure the angles with a protractor. Observation: The two angles of the major segment will be equal to each other. The student will check this with their own measurements on their own figure.

Review Exercise 10, Question 12: Fill in the following blanks. (i) Angles which have one vertex and one common side are _____ angles.

Answer: adjacent

Review Exercise 10, Question 12: (ii) If the sum of two angles is 90° then those angles are called _____.

Answer: complementary.

Review Exercise 10, Question 12: (iii) Two similar figures are called similar

when _____ and _____.

Answer: their shape is the same, their measure is different.

Review Exercise 10, Question 12: (iv) If two straight lines cut each other at one point then the _____ angles formed from them are _____ and are _____ to each other.

Answer: vertically opposite, equal, congruent

Review Exercise 10, Question 12: (v) In a right triangle the _____ and one side (base or perpendicular) are congruent, then the _____ and _____ of the other right triangle are _____ then the triangles will be _____.

Answer: hypotenuse, hypotenuse, corresponding side, equal to them,, congruent (HS property).

11. Practical Geometry

Question: Exercise 11.1, Question 1: Divide a 6 cm long line segment into 4 equal parts.

Answer: Method (the book's method): First draw a 6 cm long line segment. From one end draw a ray at an acute angle (about 40°). With a compass mark 4 equal arcs on the ray. Join the fourth mark to the other end of the line segment and draw parallels to it through the other marks to cut the line segment. Result: each part = 6 ÷ 4 = 1.5 cm. All four parts will be 1.5 cm each.

Question: Exercise 11.1, Question 2: Divide the line segment mAB = 7.5 cm into 5 equal parts.

Answer: Method: First draw the line segment AB 7.5 cm long. From one end draw a ray at an acute angle, mark 5 equal arcs on the ray with a compass, join the fifth mark to B and draw parallels to it through the other marks to cut AB. Result: each part = 7.5 ÷ 5 = 1.5 cm. All five parts will be 1.5 cm each.

Question: Exercise 11.1, Question 3: Divide the line segment mCD = 9 cm into 4 equal parts.

Answer: Method: First draw the line segment CD 9 cm long. From one end draw a ray at an acute angle, mark 4 equal arcs on the ray with a compass, join the fourth mark to D and draw parallels to it through the other marks to cut CD. Result: each part = 9 ÷ 4 = 2.25 cm. All four parts will be 2.25 cm each.

Question: Exercise 11.1, Question 4: Divide a 5 cm long line segment in the ratio 2:3.

Answer: Method: First draw a 5 cm long line segment. From one end draw

a ray at an acute angle, mark 2 + 3 = 5 equal arcs on the ray with a compass, join the fifth mark to the other end of the line segment and draw a parallel to it through the second mark to cut the line segment. Result: first part = (2/5) × 5 = 2 cm, second part = (3/5) × 5 = 3 cm.

Question: Exercise 11.1, Question 5: Divide an 8.4 cm long line segment in the ratio 1:5.

Answer: Method: First draw an 8.4 cm long line segment. From one end draw a ray at an acute angle, mark 1 + 5 = 6 equal arcs on the ray with a compass, join the sixth mark to the other end of the line segment and draw a parallel to it through the first mark to cut the line segment. Result: first part = (1/6) × 8.4 = 1.4 cm, second part = (5/6) × 8.4 = 7 cm.

Question: Exercise 11.1, Question 6: Divide the line segment mMN = 7 cm in the ratio 3:5.

Answer: Method: First draw the line segment MN 7 cm long. From one end draw a ray at an acute angle, mark 3 + 5 = 8 equal arcs on the ray with a compass, join the eighth mark to N and draw a parallel to it through the third mark to cut MN. Result: first part = (3/8) × 7 = 2.625 cm, second part = (5/8) × 7 = 4.375 cm.

Question: Exercise 11.1, Question 7: Divide a 6.6 cm long line segment in the ratio 1:2:3.

Answer: Method: First draw a 6.6 cm long line segment. From one end draw a ray at an acute angle, mark 1 + 2 + 3 = 6 equal arcs on the ray with a compass, join the sixth mark to the other end of the line segment and draw parallels to it through the first and third marks to cut the line segment. Result: first part = (1/6) × 6.6 = 1.1 cm, second part = (2/6) × 6.6 = 2.2 cm, third part = (3/6) × 6.6 = 3.3 cm.

Question: Exercise 11.1, Question 8: Divide the line segment mXY = 7 cm in the ratio 2:2:3.

Answer: Method: First draw the line segment XY 7 cm long. From one end draw a ray at an acute angle, mark 2 + 2 + 3 = 7 equal arcs on the ray with a compass, join the seventh mark to Y and draw parallels to it through the second and fourth marks to cut XY. Result: first part = (2/7) × 7 = 2 cm, second part = (2/7) × 7 = 2 cm, third part = (3/7) × 7 = 3 cm.

Question: Exercise 11.1, Question 9: Divide an 8.4 cm long line segment in the ratio 4:2:1.

Answer: Method: First draw an 8.4 cm long line segment. From one end draw a ray at an acute angle, mark 4 + 2 + 1 = 7 equal arcs on the ray with a compass, join the seventh mark to the other end of the line segment and draw parallels to it through the fourth and sixth marks to cut the line segment. Result: first part = (4/7) × 8.4 = 4.8 cm, second part = (2/7) × 8.4 = 2.4 cm, third part = (1/7) × 8.4 = 1.2 cm.

Question: Exercise 11.2, Question 1: Construct triangle ABC, whose sides are in the ratio 1:2:3 and whose perimeter is 9.6 cm.

Answer: Method: First draw a 9.6 cm long line segment and divide it in the ratio 1:2:3 (the method of Exercise 11.1). The sides will be: (1/6) × 9.6 = 1.6 cm, (2/6) × 9.6 = 3.2 cm, (3/6) × 9.6 = 4.8 cm. Then try to construct the triangle with these three lengths: from one end of the base draw an arc of 3.2 cm and from the other end an arc of 1.6 cm. Note: there is a contradiction in the printed question — 1.6 + 3.2 = 4.8, i.e. the sum of the two smaller sides equals the largest side, so the triangle inequality (the sum of two sides must be greater than the third) is not satisfied and no triangle can be formed. The method is the same, but with the given numbers a triangle is impossible.

Question: Exercise 11.2, Question 2: Construct triangle DEF, whose perimeter is 8.4 cm and whose sides are in the ratio 2:3:3.

Answer: Method: First draw an 8.4 cm long line segment and divide it in

the ratio 2:3:3. The sides will be: (2/8) × 8.4 = 2.1 cm, (3/8) × 8.4 = 3.15 cm, (3/8) × 8.4 = 3.15 cm. Then draw the base 2.1 cm, draw arcs of 3.15 cm from one end and 3.15 cm from the other end, and join their intersection point to both ends — triangle DEF will be formed.

Question: Exercise 11.2, Question 3: Construct triangle LMN, whose perimeter is 90 mm and whose sides are in the ratio 2:3:4.

Answer: Method: First draw a 90 mm long line segment and divide it in the ratio 2:3:4. The sides will be: (2/9) × 90 = 20 mm, (3/9) × 90 = 30 mm, (4/9) × 90 = 40 mm. Then draw the base 20 mm, draw arcs of 30 mm from one end and 40 mm from the other end, and join their intersection point to both ends — triangle LMN will be formed.

Question: Exercise 11.2, Question 4: Construct triangle PQR, whose sides are in the ratio 5:3:2 and whose perimeter is 10 cm.

Answer: Method: First draw a 10 cm long line segment and divide it in the ratio 5:3:2. The sides will be: (5/10) × 10 = 5 cm, (3/10) × 10 = 3 cm, (2/10) × 10 = 2 cm. Note: there is a contradiction in the printed question — 3 + 2 = 5, i.e. the sum of the two smaller sides equals the largest side, so the triangle inequality is not satisfied and no triangle can be formed. The method is the same, but with the given numbers a triangle is impossible.

Question: Exercise 11.2, Question 5: Construct triangle XYZ, whose sides are in the ratio 3:3:2 and whose perimeter is 88 mm.

Answer: Method: First draw an 88 mm long line segment and divide it in the ratio 3:3:2. The sides will be: (3/8) × 88 = 33 mm, (3/8) × 88 = 33 mm, (2/8) × 88 = 22 mm. Then draw the base 22 mm, draw arcs of 33 mm from one end and 33 mm from the other end, and join their intersection point to both ends — triangle XYZ will be formed.

Question: Exercise 11.3, Question 1: Construct equilateral triangles with

the following sides. (i) 2.9 cm

Answer: Method: Draw a base 2.9 cm long. Open the compass to 2.9 cm and draw arcs from both ends of the base. Join the intersection point of the arcs to both ends of the base — an equilateral triangle will be formed, all three sides of which are 2.9 cm.

Question: Exercise 11.3, Question 1: (ii) 3.2 cm

Answer: Method: Draw a base 3.2 cm long. Open the compass to 3.2 cm and draw arcs from both ends of the base. Join the intersection point of the arcs to both ends of the base — an equilateral triangle will be formed, all three sides of which are 3.2 cm.

Question: Exercise 11.3, Question 1: (iii) 4.1 cm

Answer: Method: Draw a base 4.1 cm long. Open the compass to 4.1 cm and draw arcs from both ends of the base. Join the intersection point of the arcs to both ends of the base — an equilateral triangle will be formed, all three sides of which are 4.1 cm.

Question: Exercise 11.3, Question 1: (iv) 3.8 cm

Answer: Method: Draw a base 3.8 cm long. Open the compass to 3.8 cm and draw arcs from both ends of the base. Join the intersection point of the arcs to both ends of the base — an equilateral triangle will be formed, all three sides of which are 3.8 cm.

Question: Exercise 11.3, Question 1: (v) 2.8 cm

Answer: Method: Draw a base 2.8 cm long. Open the compass to 2.8 cm and draw arcs from both ends of the base. Join the intersection point of the arcs to both ends of the base — an equilateral triangle will be formed, all three sides of which are 2.8 cm.

Question: Exercise 11.3, Question 1: (vi) 4.3 cm

Answer: Method: Draw a base 4.3 cm long. Open the compass to 4.3 cm and draw arcs from both ends of the base. Join the intersection point of

the arcs to both ends of the base — an equilateral triangle will be formed, all three sides of which are 4.3 cm.

Question: Exercise 11.3, Question 2: Construct equilateral triangles with the following altitudes. (i) 3.6 cm

Answer: Method (the book's method): Draw a base line and draw a perpendicular at its midpoint. Mark the altitude of 3.6 cm on the perpendicular — that will be the vertex of the equilateral triangle. At the vertex make 30°–30° angles on both sides of the perpendicular (because the altitude bisects the 60° vertex angle); those rays will cut the base line at the two ends of the triangle. Result: side = 2 × altitude ÷ √3 = 2 × 3.6 ÷ 1.732 ≈ 4.16 cm.

Question: Exercise 11.3, Question 2: (ii) 3.9 cm

Answer: Method: The same method — base line, perpendicular at the middle, altitude 3.9 cm on the perpendicular, 30°–30° angles on both sides at the vertex. Result: side = 2 × 3.9 ÷ 1.732 ≈ 4.50 cm.

Question: Exercise 11.3, Question 2: (iii) 4.3 cm

Answer: Method: The same method — base line, perpendicular at the middle, altitude 4.3 cm on the perpendicular, 30°–30° angles on both sides at the vertex. Result: side = 2 × 4.3 ÷ 1.732 ≈ 4.97 cm.

Question: Exercise 11.3, Question 2: (iv) 4.5 cm

Answer: Method: The same method — base line, perpendicular at the middle, altitude 4.5 cm on the perpendicular, 30°–30° angles on both sides at the vertex. Result: side = 2 × 4.5 ÷ 1.732 ≈ 5.20 cm.

Question: Exercise 11.3, Question 2: (v) 4.7 cm

Answer: Method: The same method — base line, perpendicular at the middle, altitude 4.7 cm on the perpendicular, 30°–30° angles on both sides at the vertex. Result: side = 2 × 4.7 ÷ 1.732 ≈ 5.43 cm.

Question: Exercise 11.3, Question 2: (vi) 4.8 cm

Answer: Method: The same method — base line, perpendicular at the middle, altitude 4.8 cm on the perpendicular, 30°–30° angles on both sides at the vertex. Result: side = 2 × 4.8 ÷ 1.732 ≈ 5.54 cm.

Question: Exercise 11.3, Question 3: Construct an equilateral triangle PQR, whose side is 50 mm.

Answer: Method: Draw a base QR 50 mm long. Open the compass to 50 mm and draw arcs from Q and R. Join the intersection point of the arcs to Q and R — triangle PQR will be formed, all three sides of which are 50 mm.

Question: Exercise 11.3, Question 4: Construct an equilateral triangle MNO, whose altitude is 36 mm.

Answer: Method: Draw a base line and draw a perpendicular at its midpoint. Mark the altitude of 36 mm on the perpendicular — that will be vertex M. At the vertex make 30°–30° angles on both sides of the perpendicular; those rays will cut the base line at N and O. Result: side = 2 × 36 ÷ 1.732 ≈ 41.57 mm (about 41.6 mm).

Question: Exercise 11.4, Question 1: Construct isosceles triangles with the following base and base angle. (i) Base 4.2 cm, base angle 42°

Answer: Method: Draw a base 4.2 cm long. At both ends of the base make 42°–42° angles with a protractor. Join the intersection point of the two rays to the ends — an isosceles triangle will be formed. Vertex angle = 180° − 2 × 42° = 96°.

Question: Exercise 11.4, Question 1: (ii) Base 4.8 cm, base angle 48°

Answer: Method: Draw a base 4.8 cm long. At both ends of the base make 48°–48° angles with a protractor and join the intersection point of the rays to the ends — an isosceles triangle will be formed. Vertex angle = 180° − 2 × 48° = 84°.

Question: Exercise 11.4, Question 1: (iii) Base 38 mm, base angle 38°

Answer: Method: Draw a base 38 mm long. At both ends of the base make

38°–38° angles with a protractor and join the intersection point of the rays to the ends — an isosceles triangle will be formed. Vertex angle = 180° − 2 × 38° = 104°.

Question: Exercise 11.4, Question 1: (iv) Base 45 mm, base angle 45°

Answer: Method: Draw a base 45 mm long. At both ends of the base make 45°–45° angles with a protractor and join the intersection point of the rays to the ends — an isosceles triangle will be formed. Vertex angle = 180° − 2 × 45° = 90°.

Question: Exercise 11.4, Question 2: Construct isosceles triangles with the following vertex angle and altitude. (i) Vertex angle 76°, altitude 4 cm

Answer: Method: Draw a base line and draw a perpendicular at its midpoint. Mark the altitude of 4 cm on the perpendicular — that will be the vertex. At the vertex make 76° ÷ 2 = 38°–38° angles on both sides of the perpendicular (the altitude bisects the vertex angle); those rays will cut the base line at the two ends of the triangle. The base angles will be (180° − 76°) ÷ 2 = 52°.

Question: Exercise 11.4, Question 2: (ii) Vertex angle 80°, altitude 5 cm

Answer: Method: Base line, perpendicular at the middle, altitude 5 cm on the perpendicular, 80° ÷ 2 = 40°–40° angles on both sides of the perpendicular at the vertex. The base angles will be (180° − 80°) ÷ 2 = 50°.

Question: Exercise 11.4, Question 2: (iii) Vertex angle 90°, altitude 5.3 cm

Answer: Method: Base line, perpendicular at the middle, altitude 5.3 cm on the perpendicular, 90° ÷ 2 = 45°–45° angles on both sides of the perpendicular at the vertex. The base angles will be (180° − 90°) ÷ 2 = 45°.

Question: Exercise 11.4, Question 2: (iv) Vertex angle 100°, altitude 60 mm

Answer: Method: Base line, perpendicular at the middle, altitude 60 mm on the perpendicular, 100° ÷ 2 = 50°–50° angles on both sides of the perpendicular at the vertex. The base angles will be (180° − 100°) ÷ 2 = 40°.

Question: Exercise 11.4, Question 3: Construct isosceles triangles with the following altitude and base angle. (i) Altitude 4.5 cm, base angle 54°

Answer: Method: Draw a base line and draw a perpendicular at its midpoint. Mark the altitude of 4.5 cm on the perpendicular — that will be the vertex. From the vertex draw rays towards both ends of the base line so that 54°–54° angles are formed at the base. The vertex angle will be 180° − 2 × 54° = 72°.

Question: Exercise 11.4, Question 3: (ii) Altitude 39 mm, base angle 65°

Answer: Method: Base line, perpendicular at the middle, altitude 39 mm on the perpendicular, from the vertex draw rays towards both ends of the base line so that 65°–65° angles are formed at the base. The vertex angle will be 180° − 2 × 65° = 50°.

Question: Exercise 11.4, Question 3: (iii) Altitude 4.7 cm, base angle 70°

Answer: Method: Base line, perpendicular at the middle, altitude 4.7 cm on the perpendicular, from the vertex draw rays towards both ends of the base line so that 70°–70° angles are formed at the base. The vertex angle will be 180° − 2 × 70° = 40°.

Question: Exercise 11.4, Question 3: (iv) Altitude 59 mm, base angle 75°

Answer: Method: Base line, perpendicular at the middle, altitude 59 mm on the perpendicular, from the vertex draw rays towards both ends of the base line so that 75°–75° angles are formed at the base. The vertex angle will be 180° − 2 × 75° = 30°.

Question: Exercise 11.5, Part A, Question 1: Construct parallelogram PQRS, in which mPQ = 5.9 cm, mQR = 4.1 cm and m∠PQR = 65°.

Answer: Method: Draw side PQ 5.9 cm long. At Q make a 65° angle with a protractor and mark side QR 4.1 cm long on that ray. Draw an arc from P equal to QR (4.1 cm) and an arc from R equal to PQ (5.9 cm); join their intersection point to P and R — parallelogram PQRS will be formed.

Question: Exercise 11.5, Part A, Question 2: Construct parallelogram CDEF, in which mCD = 63 mm, mDE = 39 mm and m∠CDE = 70°.

Answer: Method: Draw side CD 63 mm long. At D make a 70° angle with a protractor and mark side DE 39 mm long on that ray. Draw an arc from C equal to DE (39 mm) and an arc from E equal to CD (63 mm); join their intersection point to C and E — parallelogram CDEF will be formed.

Question: Exercise 11.5, Part A, Question 3: Construct parallelogram ABCD, in which mAB = 6.6 cm, mAD = 4 cm and m∠BAD = 50°.

Answer: Method: Draw side AB 6.6 cm long. At A make a 50° angle with a protractor and mark side AD 4 cm long on that ray. Draw an arc from B equal to AD (4 cm) and an arc from D equal to AB (6.6 cm); join their intersection point to B and D — parallelogram ABCD will be formed.

Question: Exercise 11.5, Part A, Question 4: Construct parallelogram RSTU, in which mRS = 70 mm, mST = 45 mm and m∠RST = 80°.

Answer: Method: Draw side RS 70 mm long. At S make an 80° angle with a protractor and mark side ST 45 mm long on that ray. Draw an arc from R equal to ST (45 mm) and an arc from T equal to RS (70 mm); join their intersection point to R and T — parallelogram RSTU will be formed.

Question: Exercise 11.5, Part A, Question 5: Construct parallelogram ACEG, in which mAC = 7.5 cm, mCE = 3.4 cm and m∠ACE = 75°.

Answer: Method: Draw side AC 7.5 cm long. At C make a 75° angle with a protractor and mark side CE 3.4 cm long on that ray. Draw an arc from A equal to CE (3.4 cm) and an arc from E equal to AC (7.5 cm); join their intersection point to A and E — parallelogram ACEG will be formed.

Question: Exercise 11.5, Part A, Question 6: Construct parallelogram LMNO, in which mON = 6 cm, mMN = 3.5 cm and m∠MNO = 55°.

Answer: Method: Draw side ON 6 cm long. At N make a 55° angle with a protractor and mark side MN 3.5 cm long on that ray. Draw an arc from O equal to MN (3.5 cm) and an arc from M equal to ON (6 cm); join their intersection point to O and M — parallelogram LMNO will be formed.

Question: Exercise 11.5, Part B, Question 1: Construct parallelogram ABCD, in which mBC = 6 cm, mCD = 4 cm and diagonal mBD = 7.3 cm.

Answer: Method: Draw the diagonal BD 7.3 cm long. Draw arcs of 6 cm from B and 4 cm from D; join their intersection point to B and D — triangle BCD will be formed. Draw arcs of 4 cm from B and 6 cm from D on the other side; join their intersection point to B and D — that will be A. Parallelogram ABCD will be formed. (Check: 6 + 4 > 7.3 and 7.3 + 4 > 6, so the triangle is possible.)

Question: Exercise 11.5, Part B, Question 2: Construct parallelogram BDFH, in which mBD = 50 mm, mBH = 30 mm and mHD = 64 mm.

Answer: Method: Draw the diagonal HD 64 mm long. Draw arcs of 30 mm from H and 50 mm from D; join their intersection point to H and D — triangle HDB will be formed. Draw arcs of 50 mm from H and 30 mm from D on the other side; join their intersection point to H and D — that will be F. Parallelogram BDFH will be formed. (Check: 30 + 50 > 64, so the triangle is possible.)

Question: Exercise 11.5, Part B, Question 3: Construct parallelogram ACEG, in which mAC = 6.6 cm, mAG = 3.3 cm and mCG = 7.9 cm.

Answer: Method: Draw the diagonal CG 7.9 cm long. Draw arcs of 6.6 cm from C and 3.3 cm from G; join their intersection point to C and G — triangle CGA will be formed. Draw arcs of 3.3 cm from C and 6.6 cm from G on the other side; join their intersection point to C and G — that will be

E. Parallelogram ACEG will be formed. (Check: 6.6 + 3.3 > 7.9, so the triangle is possible.)

Question: Exercise 11.5, Part B, Question 4: Construct parallelogram PQRS, in which mRS = 4.8 cm, mRQ = 2.9 cm and mSQ = 7 cm.

Answer: Method: Draw the diagonal SQ 7 cm long. Draw arcs of 4.8 cm from S and 2.9 cm from Q; join their intersection point to S and Q — triangle SQR will be formed. Draw arcs of 2.9 cm from S and 4.8 cm from Q on the other side; join their intersection point to S and Q — that will be P. Parallelogram PQRS will be formed. (Check: 4.8 + 2.9 > 7, so the triangle is possible.)

Question: Exercise 11.5, Part B, Question 5: Construct parallelogram RSTU, in which mRS = 59 mm, mST = 36 mm and mTR = 7.5 cm.

Answer: Method: Draw the diagonal TR 7.5 cm = 75 mm long. Draw arcs of 36 mm from T and 59 mm from R; join their intersection point to T and R — triangle TRS will be formed. Draw arcs of 59 mm from T and 36 mm from R on the other side; join their intersection point to T and R — that will be U. Parallelogram RSTU will be formed. (Check: 59 + 36 > 75, so the triangle is possible.)

Question: Exercise 11.5, Part B, Question 6: Construct parallelogram UVWX, in which mUX = 7 cm, mUV = 5 cm and mVX = 9 cm.

Answer: Method: Draw the diagonal VX 9 cm long. Draw arcs of 5 cm from V and 7 cm from X; join their intersection point to V and X — triangle VXU will be formed. Draw arcs of 7 cm from V and 5 cm from X on the other side; join their intersection point to V and X — that will be W. Parallelogram UVWX will be formed. (Check: 5 + 7 > 9, so the triangle is possible.)

Question: Exercise 11.6, Question I: Measure the three angles of each of the given four triangles (ABC, DEF, MNL, PQR) with a protractor and check that their sum is 180°.

Answer: Method: Measure the three angles of each triangle with a protractor and add them. Observation: In each triangle the sum of the three angles will be 180°. Example: if the angles of a triangle measure 60°, 70° and 50°, the sum = 60° + 70° + 50° = 180°. The student will check this for the four triangles with their own measured values.

Question: Exercise 11.6, Question II: Draw two or more triangles of different types, measure their angles with a protractor and check that their sum is 180°.

Answer: Method: Draw two or more triangles of different types (for example one right, one acute and one obtuse). Measure the three angles of each triangle with a protractor and add them. Observation: In each triangle the sum of the three angles will be 180°. Example: if the angles of a right triangle measure 90°, 60° and 30°, the sum = 90° + 60° + 30° = 180°. The student will check this with their own measurements on their own triangles.

Question: Exercise 11.6, Question III: Measure the four angles of each of the given six quadrilaterals (ABCD, PQRS, OLNM, CDEF, TWXY, MQOP) with a protractor and check that their sum is 360°.

Answer: Method: Measure the four angles of each quadrilateral with a protractor and add them. Observation: In each quadrilateral the sum of the four angles will be 360°. Example: if the angles of a quadrilateral measure 90°, 90°, 90° and 90°, the sum = 360°. The student will check this for the six quadrilaterals with their own measured values.

Question: Exercise 11.6, Question IV: Draw two or more quadrilaterals, measure their angles with a protractor and check that their sum is 360°.

Answer: Method: Draw two or more quadrilaterals (for example a rectangle and a parallelogram). Measure the four angles of each quadrilateral with a protractor and add them. Observation: In each quadrilateral the sum of

the four angles will be 360°. Example: if the angles of a parallelogram measure 70°, 110°, 70° and 110°, the sum = 360°. The student will check this with their own measurements on their own quadrilaterals.

Question: Review Exercise 11, Part A, Question 1: What is meant by a line segment? Write its properties.

Answer: Line segment: the part of a straight line which has two ends (end points). Properties: it has a fixed length; it can be measured; it has two end points.

Question: Review Exercise 11, Part A, Question 2: What is meant by a triangle? Write its properties.

Answer: Triangle: a simple figure closed by three straight sides, which has three vertices, three sides and three angles. Properties: the sum of its three angles is 180°; the sum of two sides is always greater than the third side.

Question: Review Exercise 11, Part A, Question 3: Write the names of the types of triangles.

Answer: By sides: equilateral triangle, isosceles triangle and scalene triangle. By angles: acute-angled triangle, right-angled triangle and obtuse-angled triangle.

Question: Review Exercise 11, Part A, Question 4: When are two line segments called congruent?

Answer: When the measures (lengths) of two line segments are equal, then they are called congruent.

Question: Review Exercise 11, Part A, Question 5: What is the sum of the angles of a triangle? Prove it with a protractor.

Answer: The sum of the three angles of a triangle is 180°. Proof: Measure the three angles of any triangle with a protractor and add them — the sum will be 180°. Example: 60° + 70° + 50° = 180°.

Question: Review Exercise 11, Part A, Question 6: What is meant by a quadrilateral? Write the names of its types.

Answer: Quadrilateral: a simple figure closed by four straight sides, which has four vertices, four sides and four angles. Types: square, rectangle, parallelogram, rhombus, trapezium and kite.

Question: Review Exercise 11, Part A, Question 7: Write the properties of an equilateral triangle.

Answer: Properties of an equilateral triangle: all three sides are equal; all three angles are equal and each is 60°; the altitude bisects the base and also bisects the vertex angle.

Question: Review Exercise 11, Part A, Question 8: Is a square a rectangle, a parallelogram or a rhombus?

Answer: A square is all three — a rectangle, a parallelogram and a rhombus — because all four of its angles are right angles, its opposite sides are parallel and equal, and all four sides are equal.

Question: Review Exercise 11, Part A, Question 9: What is the sum of the angles of a quadrilateral? Prove it by observation.

Answer: The sum of the four angles of a quadrilateral is 360°. Proof: Measure the four angles of any quadrilateral with a protractor and add them — the sum will be 360°. Example: 90° + 90° + 90° + 90° = 360°.

Question: Review Exercise 11, Part A, Question 10: Define altitude (Altitude) and vertex angle.

Answer: Altitude: the perpendicular distance drawn from any vertex of a triangle to the opposite side is called the altitude. Vertex angle: the angle formed between the two equal sides of a triangle is called the vertex angle.

Question: Review Exercise 11, Part B, Question 1: Divide a 10.5 cm long line segment in the ratio 3:4.

Answer: Method: Draw a 10.5 cm long line segment. From one end draw a ray at an acute angle, mark 3 + 4 = 7 equal arcs on the ray with a compass, join the seventh mark to the other end of the line segment and draw a parallel to it through the third mark to cut the line segment. Result: first part = (3/7) × 10.5 = 4.5 cm, second part = (4/7) × 10.5 = 6 cm.

Question: Review Exercise 11, Part B, Question 2: Divide an 11.9 cm long line segment in the ratio 2:4:2.

Answer: Method: Draw an 11.9 cm long line segment. From one end draw a ray at an acute angle, mark 2 + 4 + 2 = 8 equal arcs on the ray with a compass, join the eighth mark to the other end of the line segment and draw parallels to it through the second and sixth marks to cut the line segment. Result: first part = (2/8) × 11.9 = 2.975 cm, second part = (4/8) × 11.9 = 5.95 cm, third part = (2/8) × 11.9 = 2.975 cm.

Question: Review Exercise 11, Part B, Question 3: Construct an isosceles triangle, whose base is 52 mm and base angle is 70°.

Answer: Method: Draw a base 52 mm long. At both ends of the base make 70°–70° angles with a protractor and join the intersection point of the rays to the ends — an isosceles triangle will be formed. Vertex angle = 180° − 2 × 70° = 40°.

Question: Review Exercise 11, Part B, Question 4: Construct parallelogram PQRS, in which mSR = 4.1 cm, mQR = 3.1 cm and m∠QRS = 70°.

Answer: Method: Draw side SR 4.1 cm long. At R make a 70° angle with a protractor and mark side QR 3.1 cm long on that ray. Draw an arc from S equal to QR (3.1 cm) and an arc from Q equal to SR (4.1 cm); join their intersection point to S and Q — parallelogram PQRS will be formed.

Question: Review Exercise 11, Part B, Question 5: Construct an equilateral triangle, whose altitude is 6 cm.

Answer: Method: Draw a base line and draw a perpendicular at its

midpoint. Mark the altitude of 6 cm on the perpendicular — that will be the vertex. At the vertex make 30°–30° angles on both sides of the perpendicular; those rays will cut the base line at the two ends of the triangle. Result: side = 2 × 6 ÷ 1.732 ≈ 6.93 cm.

Question: Review Exercise 11, Part B, Question 6: Construct a triangle, whose perimeter is 129 mm and whose sides are in the ratio 3:4:5.

Answer: Method: Draw a 129 mm long line segment and divide it in the ratio 3:4:5. The sides will be: (3/12) × 129 = 32.25 mm, (4/12) × 129 = 43 mm, (5/12) × 129 = 53.75 mm. Then draw the base 32.25 mm, draw arcs of 43 mm from one end and 53.75 mm from the other end, and join their intersection point to both ends — the triangle will be formed.

Question: Review Exercise 11, Part B, Question 7: Construct an isosceles triangle, whose altitude is 6.2 cm and vertex angle is 40°.

Answer: Method: Draw a base line and draw a perpendicular at its midpoint. Mark the altitude of 6.2 cm on the perpendicular — that will be the vertex. At the vertex make 40° ÷ 2 = 20°–20° angles on both sides of the perpendicular; those rays will cut the base line at the two ends of the triangle. The base angles will be (180° − 40°) ÷ 2 = 70°.

Question: Review Exercise 11, Part B, Question 8: Construct a parallelogram, whose two sides are 43 mm and 32 mm and one diagonal is 6.4 cm.

Answer: Method: Draw the diagonal 6.4 cm = 64 mm long. Draw arcs of 43 mm from one end of the diagonal and 32 mm from the other end; join their intersection point to both ends — a triangle will be formed. Draw arcs of 32 mm and 43 mm on the other side of the diagonal and join the intersection point to both ends — a parallelogram will be formed. (Check: 43 + 32 > 64, so the triangle is possible.)

Question: Review Exercise 11, Part C, Question 1: The figure formed by

three non-collinear points is called: (a) quadrilateral (b) line segment (c) triangle (d) circle.

Answer: Correct answer: (c) triangle. Joining three non-collinear points forms a triangle.

Question: Review Exercise 11, Part C, Question 2: In an isosceles triangle the vertex angle is 56°. Its base angle is: (a) 67° (b) 70° (c) 60° (d) 90°.

Answer: Note: there is a contradiction in the printed question. In an isosceles triangle the base angle = (180° − 56°) ÷ 2 = 124° ÷ 2 = 62°, but 62° is not among the given options (67°, 70°, 60°, 90°). The correct answer is 62°, which is not present in the options.

Question: Review Exercise 11, Part C, Question 3: A quadrilateral whose four angles are right angles and whose opposite sides are equal is: (a) square (b) rectangle (c) rhombus (d) trapezium.

Answer: Correct answer: (b) rectangle. A rectangle has all four angles right angles and opposite sides equal.

Question: Review Exercise 11, Part C, Question 4: If mOP = mQR then: (a) mOP = mQR (b) mOP ≠ mQR (c) mOP > mQR (d) mOP < mQR.

Answer: Correct answer: (a) mOP = mQR. Line segments of equal measure are equal.

Question: Review Exercise 11, Part C, Question 5: The sum of the four angles of a quadrilateral is: (a) 180° (b) 270° (c) 90° (d) 360°.

Answer: Correct answer: (d) 360°. The sum of the four angles of a quadrilateral is 360°.

Question: Review Exercise 11, Part C, Question 6: A quadrilateral whose opposite sides are parallel and equal is: (a) square (b) parallelogram (c) rectangle (d) rhombus.

Answer: Correct answer: (b) parallelogram. Its opposite sides are parallel and equal.

Question: Review Exercise 11, Part C, Question 7: In an isosceles triangle the base angle is 45°. Its vertex angle is: (a) 180° (b) 90° (c) 60° (d) 45°.

Answer: Correct answer: (b) 90°. Vertex angle = 180° − 2 × 45° = 180° − 90° = 90°.

Question: Review Exercise 11, Part C, Question 8: Each angle of an equilateral triangle is: (a) 90° (b) 45° (c) 60° (d) 180°.

Answer: Correct answer: (c) 60°. Each angle of an equilateral triangle is 180° ÷ 3 = 60°.

12. Circumference, Area and Volume of a Circle

Question: Practical Work (Example 1): What is the distance around a circular disc called? How many times the diameter is the distance around the circle, approximately?

Answer: The distance around a circle is called the circumference of the circle. The circumference of a circle is approximately 3 times (exactly 22/7 ≈ 3.14 times) its diameter.

Question: Practical Work (Example 1, table): Measure the circumference and diameter of the following circles and find the ratio circumference ÷ diameter. (i) Diameter = 3.5 cm, circumference = 11 cm.

Answer: Ratio = circumference ÷ diameter = 11 ÷ 3.5 = 110/35 = 22/7 ≈ 3.14. So circumference = π × diameter, where π = 22/7.

Question: Practical Work (Example 1, table): (ii) Diameter = 28 mm, circumference = 88 mm.

Answer: Ratio = circumference ÷ diameter = 88 ÷ 28 = 22/7 ≈ 3.14. So circumference = π × diameter, where π = 22/7.

Question: Practical Work (Example 1, table): (iii) Diameter = 4.2 cm, circumference = 13.2 cm.

Answer: Ratio = circumference ÷ diameter = 13.2 ÷ 4.2 = 132/42 = 22/7 ≈ 3.14. So circumference = π × diameter, where π = 22/7.

Question: Practical Work (Example 1, table): (iv) Diameter = 21 mm, circumference = 66 mm.

Answer: Ratio = circumference ÷ diameter = 66 ÷ 21 = 22/7 ≈ 3.14. So circumference = π × diameter, where π = 22/7.

Question: Practical Work 12.1.2: Measure the circumference of a cylinder (for example a round box) by wrapping a thread around it.

Answer: Method: Mark one end of a thread on the cylinder, wrap it once around the cylinder and mark again. Straighten the thread and measure the distance between the marks with a scale — that is the circumference of the cylinder. Example: if the thread measures 44 cm, the circumference = 44 cm. Check: the circumference can also be found from circumference = 2π × radius. The student will take their own measurement with their own cylinder.

Question: Exercise 12.1, Part A, Question 1: Find the circumference of a circle from the diameter. (1) Diameter = 28 cm.

Answer: Formula: circumference = π × diameter. Circumference = (22/7) × 28 = 22 × 4 = 88 cm.

Question: Exercise 12.1, Part A, Question 1: (2) Diameter = 35 cm.

Answer: Circumference = π × diameter = (22/7) × 35 = 22 × 5 = 110 cm.

Question: Exercise 12.1, Part A, Question 1: (3) Diameter = 42 mm.

Answer: Circumference = π × diameter = (22/7) × 42 = 22 × 6 = 132 mm.

Question: Exercise 12.1, Part A, Question 1: (4) Diameter = 56 mm.

Answer: Circumference = π × diameter = (22/7) × 56 = 22 × 8 = 176 mm.

Question: Exercise 12.1, Part B, Question 1: Find the circumference of a circle from the radius. (1) Radius = 10.5 cm.

Answer: Formula: circumference = 2π × radius. Circumference = 2 × (22/7) × 10.5 = 2 × 22 × 1.5 = 66 cm.

Question: Exercise 12.1, Part B, Question 1: (2) Radius = 28 cm.

Answer: Circumference = 2π × radius = 2 × (22/7) × 28 = 2 × 22 × 4 = 176 cm.

Question: Exercise 12.1, Part B, Question 1: (3) Radius = 38.5 cm.

Answer: Circumference = 2π × radius = 2 × (22/7) × 38.5 = 2 × 22 × 5.5

= 242 cm.

Question: Exercise 12.1, Part B, Question 1: (4) Radius = 49 cm.

Answer: Circumference = 2π × radius = 2 × (22/7) × 49 = 2 × 22 × 7 = 308 cm.

Question: Exercise 12.1, Part B, Question 1: (5) Radius = 59.5 cm.

Answer: Circumference = 2π × radius = 2 × (22/7) × 59.5 = 2 × 22 × 8.5 = 374 cm.

Question: Exercise 12.1, Part B, Question 1: (6) Radius = 63 cm.

Answer: Circumference = 2π × radius = 2 × (22/7) × 63 = 2 × 22 × 9 = 396 cm.

Question: Exercise 12.1, Part B, Question 1: (7) Radius = 80.5 cm.

Answer: Circumference = 2π × radius = 2 × (22/7) × 80.5 = 2 × 22 × 11.5 = 506 cm.

Question: Exercise 12.1, Part B, Question 1: (8) Radius = 77 cm.

Answer: Circumference = 2π × radius = 2 × (22/7) × 77 = 2 × 22 × 11 = 484 cm.

Question: Exercise 12.1, Part C, Question 1: Find the radius of a circle from the circumference. (1) Circumference = 22 cm.

Answer: Formula: radius = circumference × 7 / 44. Radius = 22 × 7 / 44 = 3.5 cm.

Question: Exercise 12.1, Part C, Question 1: (2) Circumference = 66 cm.

Answer: Radius = circumference × 7 / 44 = 66 × 7 / 44 = 10.5 cm.

Question: Exercise 12.1, Part C, Question 1: (3) Circumference = 88 cm.

Answer: Radius = circumference × 7 / 44 = 88 × 7 / 44 = 14 cm.

Question: Exercise 12.1, Part C, Question 1: (4) Circumference = 110 cm.

Answer: Radius = circumference × 7 / 44 = 110 × 7 / 44 = 17.5 cm.

Question: Exercise 12.1, Part C, Question 1: (5) Circumference = 132 cm.

Answer: Radius = circumference × 7 / 44 = 132 × 7 / 44 = 21 cm.

Question: Exercise 12.1, Part C, Question 1: (6) Circumference = 176 cm.

Answer: Radius = circumference × 7 / 44 = 176 × 7 / 44 = 28 cm.

Question: Exercise 12.1, Part C, Question 1: (7) Circumference = 220 cm.

Answer: Radius = circumference × 7 / 44 = 220 × 7 / 44 = 35 cm.

Question: Exercise 12.1, Part C, Question 1: (8) Circumference = 198 cm.

Answer: Radius = circumference × 7 / 44 = 198 × 7 / 44 = 31.5 cm.

Question: Exercise 12.1, Part D, Question 1: Find the diameter of a circle from the circumference. (1) Circumference = 44 cm.

Answer: Formula: diameter = circumference × 7 / 22. Diameter = 44 × 7 / 22 = 14 cm.

Question: Exercise 12.1, Part D, Question 1: (2) Circumference = 154 cm.

Answer: Diameter = circumference × 7 / 22 = 154 × 7 / 22 = 49 cm.

Question: Exercise 12.1, Part D, Question 1: (3) Circumference = 242 mm.

Answer: Diameter = circumference × 7 / 22 = 242 × 7 / 22 = 77 mm.

Question: Exercise 12.1, Part D, Question 1: (4) Circumference = 264 mm.

Answer: Diameter = circumference × 7 / 22 = 264 × 7 / 22 = 84 mm.

Question: Exercise 12.1, Part D, Question 1: (5) Circumference = 48.4 mm.

Answer: Diameter = circumference × 7 / 22 = 48.4 × 7 / 22 = 15.4 mm.

Question: Exercise 12.1, Part D, Question 1: (6) Circumference = 30.8 cm.

Answer: Diameter = circumference × 7 / 22 = 30.8 × 7 / 22 = 9.8 cm.

Question: Exercise 12.1, Part D, Question 1: (7) Circumference = 52.8 cm.

Answer: Diameter = circumference × 7 / 22 = 52.8 × 7 / 22 = 16.8 cm.

Question: Exercise 12.1, Part D, Question 1: (8) Circumference = 39.6 mm.

Answer: Diameter = circumference × 7 / 22 = 39.6 × 7 / 22 = 12.6 mm.

Question: Exercise 12.1, Part E, Question 1: Complete the table of the length of the minute hand of a clock and the distance covered in one hour. (1) Length of the minute hand (radius) = 1.4 cm.

Answer: In one hour the hand completes one full round, so the distance =

circumference = 2π × radius = 2 × (22/7) × 1.4 = 8.8 cm.

Question: Exercise 12.1, Part E, Question 1: (2) Length of the minute hand = 21 mm.

Answer: Distance = 2π × radius = 2 × (22/7) × 21 = 132 mm.

Question: Exercise 12.1, Part E, Question 1: (3) Length of the minute hand = 35 mm.

Answer: Distance = 2π × radius = 2 × (22/7) × 35 = 220 mm.

Question: Exercise 12.1, Part E, Question 1: (4) Length of the minute hand = 2.8 cm.

Answer: Distance = 2π × radius = 2 × (22/7) × 2.8 = 17.6 cm.

Question: Exercise 12.1, Part E, Question 1: (5) Length of the minute hand = 42 mm.

Answer: Distance = 2π × radius = 2 × (22/7) × 42 = 264 mm.

Question: Exercise 12.1, Part F, Question 1: What is the minimum length of thread needed to wrap around each of the following things? (1) A plate, whose diameter is 21 cm.

Answer: Minimum thread = circumference = π × diameter = (22/7) × 21 = 66 cm.

Question: Exercise 12.1, Part F, Question 1: (2) A tyre, whose diameter is 56 cm.

Answer: Minimum thread = circumference = π × diameter = (22/7) × 56 = 176 cm.

Question: Exercise 12.1, Part F, Question 1: (3) A table, whose diameter is 1.4 m.

Answer: Minimum thread = circumference = π × diameter = (22/7) × 1.4 = 4.4 m.

Question: Practical Work 12.1.3: Cut a circle into 16 equal sectors, arrange them into a nearly rectangular shape, and derive the formula for the area

of a circle.

Answer: Method: (i) Draw a circle of any suitable radius. (ii) Cut the circle into 16 equal sectors. (iii) Arrange the sectors in a row so that they form a nearly rectangular shape. (iv) The length of this shape = half the circumference = π × radius and the width = radius. So area = length × width = π × radius × radius = πr^2. The student will do this practical work with their own circle.

Question: Exercise 12.2, Part A, Question 1: Find the area of a circle from the radius. (1) Radius = 14 cm.

Answer: Formula: area = πr^2. Area = (22/7) × 14 × 14 = 22 × 2 × 14 = 616 cm^2.

Question: Exercise 12.2, Part A, Question 1: (2) Radius = 10.5 cm.

Answer: Area = πr^2 = (22/7) × 10.5 × 10.5 = 22 × 1.5 × 10.5 = 346.5 cm^2.

Question: Exercise 12.2, Part A, Question 1: (3) Radius = 21 cm.

Answer: Area = πr^2 = (22/7) × 21 × 21 = 22 × 3 × 21 = 1386 cm^2.

Question: Exercise 12.2, Part A, Question 1: (4) Radius = 17.5 mm.

Answer: Area = πr^2 = (22/7) × 17.5 × 17.5 = 22 × 2.5 × 17.5 = 962.5 mm^2.

Question: Exercise 12.2, Part A, Question 1: (5) Radius = 35 mm.

Answer: Area = πr^2 = (22/7) × 35 × 35 = 22 × 5 × 35 = 3850 mm^2.

Question: Exercise 12.2, Part A, Question 1: (6) Radius = 24.5 cm.

Answer: Area = πr^2 = (22/7) × 24.5 × 24.5 = 22 × 3.5 × 24.5 = 1886.5 cm^2.

Question: Exercise 12.2, Part A, Question 1: (7) Radius = 71.4 cm.

Answer: Area = πr^2 = (22/7) × 71.4 × 71.4 = 22 × 10.2 × 71.4 = 16022.16 cm^2.

Question: Exercise 12.2, Part A, Question 1: (8) Radius = 38.5 cm.

Answer: Area = πr² = (22/7) × 38.5 × 38.5 = 22 × 5.5 × 38.5 = 4658.5 cm².

Question: Exercise 12.2, Part B, Question 1: Find the radius of a circle from the area. (1) Area = 154 mm².

Answer: Formula: radius = √(area × 7 / 22). Radius = √(154 × 7 / 22) = √49 = 7 mm.

Question: Exercise 12.2, Part B, Question 1: (2) Area = 2464 cm².

Answer: Radius = √(area × 7 / 22) = √(2464 × 7 / 22) = √784 = 28 cm.

Question: Exercise 12.2, Part B, Question 1: (3) Area = 7546 mm².

Answer: Radius = √(area × 7 / 22) = √(7546 × 7 / 22) = √2401 = 49 mm.

Question: Exercise 12.2, Part B, Question 1: (4) Area = 5544 mm².

Answer: Radius = √(area × 7 / 22) = √(5544 × 7 / 22) = √1764 = 42 mm.

Question: Exercise 12.2, Part B, Question 1: (5) Area = 6.16 mm².

Answer: Radius = √(area × 7 / 22) = √(6.16 × 7 / 22) = √1.96 = 1.4 mm.

Question: Exercise 12.2, Part B, Question 1: (6) Area = 962.5 cm².

Answer: Radius = √(area × 7 / 22) = √(962.5 × 7 / 22) = √306.25 = 17.5 cm.

Question: Exercise 12.2, Part B, Question 1: (7) Area = 38.5 m².

Answer: Radius = √(area × 7 / 22) = √(38.5 × 7 / 22) = √12.25 = 3.5 m.

Question: Exercise 12.2, Part B, Question 1: (8) Area = 75.46 mm².

Answer: Radius = √(area × 7 / 22) = √(75.46 × 7 / 22) = √24.01 = 4.9 mm.

Question: Exercise 12.2, Part B, Question 1: (9) Area = 13.86 m².

Answer: Radius = √(area × 7 / 22) = √(13.86 × 7 / 22) = √4.41 = 2.1 m.

Question: Exercise 12.2, Part C, Question 1: Find the area of a circle from the diameter. (1) Diameter = 21 cm.

Answer: First radius = diameter ÷ 2 = 21 ÷ 2 = 10.5 cm. Area = πr² = (22/7) × 10.5 × 10.5 = 346.5 cm².

Question: Exercise 12.2, Part C, Question 1: (2) Diameter = 28 cm.

Answer: Radius = 28 ÷ 2 = 14 cm. Area = πr² = (22/7) × 14 × 14 = 616 cm².

Question: Exercise 12.2, Part C, Question 1: (3) Diameter = 42 cm.

Answer: Radius = 42 ÷ 2 = 21 cm. Area = πr² = (22/7) × 21 × 21 = 1386 cm².

Question: Exercise 12.2, Part C, Question 1: (4) Diameter = 56 cm.

Answer: Radius = 56 ÷ 2 = 28 cm. Area = πr² = (22/7) × 28 × 28 = 2464 cm².

Question: Exercise 12.2, Part C, Question 1: (5) Diameter = 8.4 cm.

Answer: Radius = 8.4 ÷ 2 = 4.2 cm. Area = πr² = (22/7) × 4.2 × 4.2 = 55.44 cm².

Question: Exercise 12.2, Part C, Question 1: (6) Diameter = 9.8 cm.

Answer: Radius = 9.8 ÷ 2 = 4.9 cm. Area = πr² = (22/7) × 4.9 × 4.9 = 75.46 cm².

Question: Exercise 12.2, Part C, Question 1: (7) Diameter = 11.2 cm.

Answer: Radius = 11.2 ÷ 2 = 5.6 cm. Area = πr² = (22/7) × 5.6 × 5.6 = 98.56 cm².

Question: Exercise 12.2, Part C, Question 1: (8) Diameter = 12.6 cm.

Answer: Radius = 12.6 ÷ 2 = 6.3 cm. Area = πr² = (22/7) × 6.3 × 6.3 = 124.74 cm².

Question: Exercise 12.2, Part D, Question 1: Find the diameter of a circle from the area. (1) Area = 154 cm².

Answer: First radius = √(area × 7 / 22) = √(154 × 7 / 22) = √49 = 7 cm. Diameter = 2 × radius = 2 × 7 = 14 cm.

Question: Exercise 12.2, Part D, Question 1: (2) Area = 1386 mm².

Answer: Radius = √(1386 × 7 / 22) = √441 = 21 mm. Diameter = 2 × 21 = 42 mm.

Question: Exercise 12.2, Part D, Question 1: (3) Area = 3850 mm².

Answer: Radius = √(3850 × 7 / 22) = √1225 = 35 mm. Diameter = 2 × 35 = 70 mm.

Question: Exercise 12.2, Part D, Question 1: (4) Area = 9856 mm².

Answer: Radius = √(9856 × 7 / 22) = √3136 = 56 mm. Diameter = 2 × 56 = 112 mm.

Question: Exercise 12.2, Part D, Question 1: (5) Area = 124.74 cm².

Answer: Radius = √(124.74 × 7 / 22) = √39.69 = 6.3 cm. Diameter = 2 × 6.3 = 12.6 cm.

Question: Exercise 12.2, Part D, Question 1: (6) Area = 186.34 cm².

Answer: Radius = √(186.34 × 7 / 22) = √59.29 = 7.7 cm. Diameter = 2 × 7.7 = 15.4 cm.

Question: Exercise 12.2, Part D, Question 1: (7) Area = 221.76 cm².

Answer: Radius = √(221.76 × 7 / 22) = √70.56 = 8.4 cm. Diameter = 2 × 8.4 = 16.8 cm.

Question: Exercise 12.2, Part D, Question 1: (8) Area = 260.26 cm².

Answer: Radius = √(260.26 × 7 / 22) = √82.81 = 9.1 cm. Diameter = 2 × 9.1 = 18.2 cm.

Question: Exercise 12.3, Part A, Question 1: Find the total surface area of a cylinder. (1) Radius = 14 cm, height = 26 cm.

Answer: Formula: total area = 2πr(r + h). Total area = 2 × (22/7) × 14 × (14 + 26) = 88 × 40 = 3520 cm².

Question: Exercise 12.3, Part A, Question 1: (2) Radius = 10 cm, height = 18 cm.

Answer: Total area = 2πr(r + h) = 2 × (22/7) × 10 × (10 + 18) = (440/7) × 28 = 1760 cm².

Question: Exercise 12.3, Part A, Question 1: (3) Radius = 21 cm, height = 29 cm.

Answer: Total area = 2πr(r + h) = 2 × (22/7) × 21 × (21 + 29) = 132 × 50 = 6600 cm².

Question: Exercise 12.3, Part A, Question 1: (4) Radius = 17.5 mm, height = 22.5 mm.

Answer: Total area = 2πr(r + h) = 2 × (22/7) × 17.5 × (17.5 + 22.5) = 110 × 40 = 4400 mm².

Question: Exercise 12.3, Part A, Question 1: (5) Radius = 30 cm, height = 40 cm.

Answer: Total area = 2πr(r + h) = 2 × (22/7) × 30 × (30 + 40) = (1320/7) × 70 = 13200 cm².

Question: Exercise 12.3, Part A, Question 1: (6) Radius = 25 mm, height = 41.5 mm.

Answer: Total area = 2πr(r + h) = 2 × (22/7) × 25 × (25 + 41.5) = (1100/7) × 66.5 = 10450 mm².

Question: Exercise 12.3, Part B, Question 1: Find the height of a cylinder from the total area. (1) Radius = 10.5 cm, total area = 1980 cm².

Answer: Formula: total area = 2πr(r + h), so h = total area ÷ (2πr) − r. h = 1980 ÷ (2 × (22/7) × 10.5) − 10.5 = 1980 ÷ 66 − 10.5 = 30 − 10.5 = 19.5 cm.

Question: Exercise 12.3, Part B, Question 1: (2) Radius = 17.5 cm, total area = 4400 cm².

Answer: h = total area ÷ (2πr) − r = 4400 ÷ (2 × (22/7) × 17.5) − 17.5 = 4400 ÷ 110 − 17.5 = 40 − 17.5 = 22.5 cm.

Question: Exercise 12.3, Part B, Question 1: (3) Radius = 12 mm, total area = 2112 mm².

Answer: h = total area ÷ (2πr) − r = 2112 ÷ (2 × (22/7) × 12) − 12 = 2112 ÷ (528/7) − 12 = 28 − 12 = 16 mm.

Question: Exercise 12.3, Part B, Question 1: (4) Radius = 15 cm, total area

= 3960 cm².

Answer: h = total area ÷ (2πr) − r = 3960 ÷ (2 × (22/7) × 15) − 15 = 3960 ÷ (660/7) − 15 = 42 − 15 = 27 cm.

Question: Exercise 12.3, Part B, Question 1: (5) Radius = 28 cm, total area = 10560 cm².

Answer: h = total area ÷ (2πr) − r = 10560 ÷ (2 × (22/7) × 28) − 28 = 10560 ÷ 176 − 28 = 60 − 28 = 32 cm.

Question: Exercise 12.3, Part C, Question 1: The sum of the radius and height and the total area are given. Find the radius and height. (1) Radius + height = 28 cm, total area = 1408 cm².

Answer: From total area = 2πr(r + h), 1408 = 2 × (22/7) × r × 28, so r = 1408 ÷ (2 × (22/7) × 28) = 1408 ÷ 176 = 8 cm. Height = 28 − 8 = 20 cm.

Question: Exercise 12.3, Part C, Question 1: (2) Radius + height = 35 cm, total area = 3300 cm².

Answer: From 3300 = 2 × (22/7) × r × 35, r = 3300 ÷ (2 × (22/7) × 35) = 3300 ÷ 220 = 15 cm. Height = 35 − 15 = 20 cm.

Question: Exercise 12.3, Part C, Question 1: (3) Radius + height = 49 cm, total area = 5852 cm².

Answer: From 5852 = 2 × (22/7) × r × 49, r = 5852 ÷ (2 × (22/7) × 49) = 5852 ÷ 308 = 19 cm. Height = 49 − 19 = 30 cm.

Question: Exercise 12.3, Part C, Question 1: (4) Radius + height = 56 mm, total area = 6864 mm².

Answer: From 6864 = 2 × (22/7) × r × 56, r = 6864 ÷ (2 × (22/7) × 56) = 6864 ÷ 352 = 19.5 mm. Height = 56 − 19.5 = 36.5 mm.

Question: Exercise 12.4, Part A, Question 1: Find the volume of a cylinder. (1) Radius = 14 cm, height = 20 cm.

Answer: Formula: volume = πr²h. Volume = (22/7) × 14 × 14 × 20 = 22 ×

2 × 14 × 20 = 12320 cm³.

Question: Exercise 12.4, Part A, Question 1: (2) Radius = 25 cm, height = 35 cm.

Answer: Volume = πr²h = (22/7) × 25 × 25 × 35 = 22 × 25 × 5 × 25 = 68750 cm³.

Question: Exercise 12.4, Part A, Question 1: (3) Radius = 28 cm, height = 40 cm.

Answer: Volume = πr²h = (22/7) × 28 × 28 × 40 = 22 × 4 × 28 × 40 = 98560 cm³.

Question: Exercise 12.4, Part A, Question 1: (4) Radius = 20 cm, height = 28 cm.

Answer: Volume = πr²h = (22/7) × 20 × 20 × 28 = 22 × 20 × 4 × 20 = 35200 cm³.

Question: Exercise 12.4, Part A, Question 1: (5) Radius = 1.05 m, height = 2.5 m.

Answer: Volume = πr²h = (22/7) × 1.05 × 1.05 × 2.5 = 8.6625 m³.

Question: Exercise 12.4, Part A, Question 1: (6) Radius = 63.5 cm, height = 75 cm.

Answer: Volume = πr²h = (22/7) × 63.5 × 63.5 × 75 = 950458.93 cm³ (approximately 950,459 cm³).

Question: Exercise 12.4, Part A, Question 1: (7) Diameter = 70 mm, height = 50 mm.

Answer: Radius = diameter ÷ 2 = 70 ÷ 2 = 35 mm. Volume = πr²h = (22/7) × 35 × 35 × 50 = 192500 mm³.

Question: Exercise 12.4, Part A, Question 1: (8) Diameter = 60 mm, height = 63 mm.

Answer: Radius = diameter ÷ 2 = 60 ÷ 2 = 30 mm. Volume = πr²h = (22/7) × 30 × 30 × 63 = 178200 mm³.

Question: Exercise 12.4, Part B, Question 1: Find the height of a cylinder from the volume. (1) Radius = 14 cm, volume = 15400 cm³.

Answer: Formula: h = volume ÷ (πr²). h = 15400 ÷ ((22/7) × 14 × 14) = 15400 ÷ 616 = 25 cm.

Question: Exercise 12.4, Part B, Question 1: (2) Radius = 20 cm, volume = 35200 cm³.

Answer: h = volume ÷ (πr²) = 35200 ÷ ((22/7) × 20 × 20) = 35200 ÷ (8800/7) = 28 cm.

Question: Exercise 12.4, Part B, Question 1: (3) Radius = 25 mm, volume = 82500 mm³.

Answer: h = volume ÷ (πr²) = 82500 ÷ ((22/7) × 25 × 25) = 82500 ÷ (13750/7) = 42 mm.

Question: Exercise 12.4, Part B, Question 1: (4) Radius = 21.5 mm, volume = 305085 mm³.

Answer: h = volume ÷ (πr²) = 305085 ÷ ((22/7) × 21.5 × 21.5) = 305085 ÷ 1452.5 = 210 mm.

Question: Exercise 12.4, Part B, Question 1: (5) Radius = 30 mm, volume = 99000 mm³.

Answer: h = volume ÷ (πr²) = 99000 ÷ ((22/7) × 30 × 30) = 99000 ÷ (19800/7) = 35 mm.

Question: Exercise 12.4, Part B, Question 1: (6) Radius = 28 cm, volume = 98560 cm³.

Answer: h = volume ÷ (πr²) = 98560 ÷ ((22/7) × 28 × 28) = 98560 ÷ 2464 = 40 cm.

Question: Exercise 12.4, Part C, Question 1: Find the radius of a cylinder from the height and volume. (1) Height = 25 cm, volume = 24650 cm³.

Answer: From volume = πr²h, r² = volume ÷ (πh) = 24650 ÷ ((22/7) × 25) = 24650 ÷ (550/7) = 313.73 (approximately). r = √313.73 ≈ 17.71 cm.

Note: with the printed numbers and π = 22/7 the radius does not come out to a whole number, so the answer is approximate.

Question: Exercise 12.4, Part C, Question 1: (2) Height = 20 m, volume = 12320 m³.

Answer: r² = volume ÷ (πh) = 12320 ÷ ((22/7) × 20) = 12320 ÷ (440/7) = 196. r = √196 = 14 m.

Question: Exercise 12.4, Part C, Question 1: (3) Height = 28 cm, volume = 35200 cm³.

Answer: r² = volume ÷ (πh) = 35200 ÷ ((22/7) × 28) = 35200 ÷ 88 = 400. r = √400 = 20 cm.

Question: Exercise 12.4, Part C, Question 1: (4) Height = 42 cm, volume = 118800 cm³.

Answer: r² = volume ÷ (πh) = 118800 ÷ ((22/7) × 42) = 118800 ÷ 132 = 900. r = √900 = 30 cm.

Question: Exercise 12.4, Part C, Question 1: (5) Height = 56 mm, volume = 281600 mm³.

Answer: r² = volume ÷ (πh) = 281600 ÷ ((22/7) × 56) = 281600 ÷ 176 = 1600. r = √1600 = 40 mm.

Question: Exercise 12.4, Part C, Question 1: (6) Height = 30 cm, volume = 41580 cm³.

Answer: r² = volume ÷ (πh) = 41580 ÷ ((22/7) × 30) = 41580 ÷ (660/7) = 441. r = √441 = 21 cm.

Question: Exercise 12.5, Question 1: A circular garden has a diameter of 70 m. Wire has to be wrapped 5 times around it. The wire costs Rs. 10 per metre. Find the total cost of the wire.

Answer: Circumference = π × diameter = (22/7) × 70 = 220 m. 5 rounds of wire = 5 × 220 = 1100 m. Total cost = 1100 × 10 = Rs. 11,000.

Question: Exercise 12.5, Question 2: A wire was bent into the shape of a

square of side 22 cm. If the same wire is bent into the shape of a circle, what will be the radius of the circle?

Answer: Perimeter of the square = 4 × 22 = 88 cm = length of the wire = circumference of the circle. From circumference = 2πr, radius = 88 × 7 / 44 = 14 cm.

Question: Exercise 12.5, Question 3: The radius of a bicycle wheel is 42 cm. How much distance will the bicycle cover in 1000 revolutions?

Answer: Distance in one revolution = circumference = 2π × radius = 2 × (22/7) × 42 = 264 cm = 2.64 m. Distance in 1000 revolutions = 1000 × 264 = 264000 cm = 2640 m = 2.64 km.

Question: Exercise 12.5, Question 4: Wrapping wire 4 times around a circular field cost Rs. 3960. The wire costs Rs. 5 per metre. Find the radius of the field.

Answer: Total length of wire = 3960 ÷ 5 = 792 m. Circumference for one round = 792 ÷ 4 = 198 m. Radius = circumference × 7 / 44 = 198 × 7 / 44 = 31.5 m.

Question: Exercise 12.5, Question 5: A wire was bent into the shape of an equilateral triangle of side 13.2 m. If the same wire is bent into the shape of a circle, what will be the diameter of the circle?

Answer: Perimeter of the triangle = 3 × 13.2 = 39.6 m = length of the wire = circumference of the circle. Diameter = circumference × 7 / 22 = 39.6 × 7 / 22 = 12.6 m.

Question: Exercise 12.5, Question 6: A circular field has a radius of 14 m. The cost of growing grass on it is Rs. 20 per square metre. Find the total cost.

Answer: Area = πr^2 = (22/7) × 14 × 14 = 616 m^2. Total cost = 616 × 20 = Rs. 12,320.

Question: Exercise 12.5, Question 7: A circular garden has an outer

diameter of 49 m and an inner diameter of 35 m. Find the area of the path between them and the cost of paving it if the cost is Rs. 100 per square metre.

Answer: Outer radius = 49 ÷ 2 = 24.5 m, inner radius = 35 ÷ 2 = 17.5 m. Outer area = (22/7) × 24.5 × 24.5 = 1886.5 m². Inner area = (22/7) × 17.5 × 17.5 = 962.5 m². Path area = 1886.5 − 962.5 = 924 m². Paving cost = 924 × 100 = Rs. 92,400.

Question: Exercise 12.5, Question 8: A circular table has a diameter of 2.8 m. The cost of polishing it is Rs. 100 per square metre. Find the total cost of polishing.

Answer: Radius = 2.8 ÷ 2 = 1.4 m. Area = πr² = (22/7) × 1.4 × 1.4 = 6.16 m². Total cost = 6.16 × 100 = Rs. 616.

Question: Exercise 12.5, Question 9: A circular fountain has a radius of 3.5 m. The cost of painting it is Rs. 75 per square metre. Find the total cost of painting.

Answer: Area = πr² = (22/7) × 3.5 × 3.5 = 38.5 m². Total cost = 38.5 × 75 = Rs. 2,887.50.

Question: Exercise 12.5, Question 10: A circular table cover has a diameter of 3.5 m. The cloth costs Rs. 80 per square metre. Find the total cost of the cloth.

Answer: Radius = 3.5 ÷ 2 = 1.75 m. Area = πr² = (22/7) × 1.75 × 1.75 = 9.625 m². Total cost = 9.625 × 80 = Rs. 770.

Question: Exercise 12.6, Question 1: A water tank has a radius of 10.5 m and a height of 8 m. Its bottom and curved surface are to be cemented. The cost of cementing is Rs. 50 per square metre. Find the total cost.

Answer: Bottom area = πr² = (22/7) × 10.5 × 10.5 = 346.5 m². Curved surface area = 2πrh = 2 × (22/7) × 10.5 × 8 = 528 m². Total area = 346.5 + 528 = 874.5 m². Total cost = 874.5 × 50 = Rs. 43,725.

Question: Exercise 12.6, Question 2: A steel tank has a height of 8.4 m and a radius of 7 m. The cost of steeling its total surface area is Rs. 100 per square metre. Find the total cost.

Answer: Total area = $2\pi r(r + h) = 2 \times (22/7) \times 7 \times (7 + 8.4) = 44 \times 15.4 = 677.6$ m². Total cost = 677.6 × 100 = Rs. 67,760.

Question: Exercise 12.6, Question 3: An open tank is in the shape of a cylinder, whose diameter is 2.8 m and height is 5 m. Find the area of steel needed for it and its cost if the steel costs Rs. 300 per square metre.

Answer: Radius = 2.8 ÷ 2 = 1.4 m. For an open tank, steel = bottom + curved surface = $\pi r^2 + 2\pi rh = (22/7) \times 1.4 \times 1.4 + 2 \times (22/7) \times 1.4 \times 5 = 6.16 + 44 = 50.16$ m². Cost = 50.16 × 300 = Rs. 15,048.

Question: Exercise 12.6, Question 4: A steel pipe whose length is 2.8 m and radius is 7 m. Find its surface area, if the pipe is open from both sides.

Answer: Surface area of a pipe open from both sides = curved surface = $2\pi rh = 2 \times (22/7) \times 7 \times 2.8 = 123.2$ m². Note: the printed question gives a radius of 7 m, which is even larger than the pipe's length of 2.8 m — this seems impossible and is probably a misprint for 7 cm. With the printed numbers the answer is 123.2 m²; if the radius were 7 cm the area would be 1.232 m².

Question: Exercise 12.6, Question 5: Find the surface area of an oil drum, whose length is 1.5 m and diameter is 70 cm.

Answer: Radius = 70 ÷ 2 = 35 cm = 0.35 m. Total surface area = $2\pi r(r + h) = 2 \times (22/7) \times 0.35 \times (0.35 + 1.5) = 2.2 \times 1.85 = 4.07$ m². (Curved surface only = $2\pi rh = 2 \times (22/7) \times 0.35 \times 1.5 = 3.3$ m².)

Question: Exercise 12.6, Question 6: A water tank has a height of 5 m and a diameter of 4.2 m. How many litres of water can it hold? (Hint: 1 litre = 1000 cm³.)

Answer: Radius = 4.2 ÷ 2 = 2.1 m = 210 cm, height = 5 m = 500 cm.

Volume = πr²h = (22/7) × 210 × 210 × 500 = 69,300,000 cm³. Litres = 69,300,000 ÷ 1000 = 69,300 litres.

Question: Exercise 12.6, Question 7: A container has a height of 63 cm and a radius of 25 cm. Find its capacity in litres.

Answer: Volume = πr²h = (22/7) × 25 × 25 × 63 = 123,750 cm³. Capacity = 123,750 ÷ 1000 = 123.75 litres.

Question: Exercise 12.6, Question 8: A well is 16 m deep and its diameter is 7 m. Find the volume of the earth dug out of the well.

Answer: Radius = 7 ÷ 2 = 3.5 m. Volume = πr²h = (22/7) × 3.5 × 3.5 × 16 = 616 m³. So the volume of the earth dug out is 616 m³.

Question: Exercise 12.6, Question 9: A soft-drink can has a length of 10.5 cm and a radius of 3.5 cm. Find its capacity.

Answer: Volume = πr²h = (22/7) × 3.5 × 3.5 × 10.5 = 404.25 cm³ (≈ 0.404 litres).

Question: Exercise 12.6, Question 10: A circular well has a diameter of 3.5 m and a depth of 24 m. The cost of digging it is Rs. 250 per cubic metre. Find the total cost of digging.

Answer: Radius = 3.5 ÷ 2 = 1.75 m. Volume = πr²h = (22/7) × 1.75 × 1.75 × 24 = 231 m³. Total cost = 231 × 250 = Rs. 57,750.

Question: Exercise 12.6, Question 11: An oil drum has a volume of 11 cubic metres. Find its radius.

Answer: Note: there is a contradiction in the printed question — the height of the drum is not given, so the radius cannot be determined. Method: from volume = πr²h, r = √(volume ÷ (πh)). The height of the drum is also needed to find the radius.

Question: Exercise 12.6, Question 12: An oil drum has a capacity of 3,850 litres. Find its height in metres, given that its radius is 70 cm.

Answer: Volume = 3,850 litres = 3,850 × 1000 = 3,850,000 cm³ = 3.85

m^3. Radius = 70 cm = 0.7 m. h = volume ÷ (πr^2) = 3.85 ÷ ((22/7) × 0.7 × 0.7) = 3.85 ÷ 1.54 = 2.5 m.

Question: Exercise 12.6, Question 13: A mosque in the shape of a circular cylinder has an inner diameter of 28 m and the wall height is 8.75 m. The cost of tiling the wall is Rs. 200 per square metre. Find the cost of tiling.

Answer: Radius = 28 ÷ 2 = 14 m. Wall area = curved surface = $2\pi rh$ = 2 × (22/7) × 14 × 8.75 = 770 m^2. Cost = 770 × 200 = Rs. 154,000.

Question: Exercise 12.6, Question 14: A swimming pool has a diameter of 42 m. Its depth is 5 m. The cost of tiling its wall is Rs. 100 per square metre. Find the cost of tiling.

Answer: Radius = 42 ÷ 2 = 21 m. Wall area = curved surface = $2\pi rh$ = 2 × (22/7) × 21 × 5 = 660 m^2. Cost = 660 × 100 = Rs. 66,000.

Question: Review Exercise 12, Part A, Question 1: What is the circumference of a circle? Write the formula for finding the circumference.

Answer: Circumference of a circle: the distance around a circle is called the circumference. Formula: circumference = π × diameter = 2π × radius.

Question: Review Exercise 12, Part A, Question 2: What is the relation between the circumference and the diameter of a circle?

Answer: The ratio of the circumference and the diameter of a circle is always constant: circumference ÷ diameter = π = 22/7 ≈ 3.14.

Question: Review Exercise 12, Part A, Question 3: What is a diameter? Write the relation between diameter and radius.

Answer: Diameter: the chord which passes through the centre of the circle. Relation: diameter = 2 × radius, i.e. radius = diameter ÷ 2.

Question: Review Exercise 12, Part A, Question 4: What is a radius? Write the formula for finding the area of a circle from the radius.

Answer: Radius: the distance between the centre of a circle and any point on the circle. Area formula: area = πr^2.

Question: Review Exercise 12, Part A, Question 5: Write the formula for finding the total surface area of a cylinder.

Answer: Total surface area of a cylinder = $2\pi r(r + h)$, where r is the radius and h is the height. (This includes the two circular ends $2\pi r^2$ and the curved surface $2\pi rh$.)

Question: Review Exercise 12, Part A, Question 6: What is the volume of a cylinder? Write the formula for finding the volume.

Answer: Volume of a cylinder: the space enclosed inside the cylinder.

Formula: volume = area of the base × height = $\pi r^2 h$.

Question: Review Exercise 12, Part A, Question 7: Write the formula for finding the circumference of a circle from the diameter.

Answer: Circumference = π × diameter.

Question: Review Exercise 12, Part A, Question 8: If the radius and height of a cylinder are equal, what will the volume formula become?

Answer: If r = h then volume = $\pi r^2 h = \pi r^2 \times r = \pi r^3$. (Or volume = πh^3.)

Review Exercise 12, Part B, Question 1: Fill in the blank. The formula for finding the curved surface area of a cylinder is _____.

Answer: = $2\pi rh$.

Review Exercise 12, Part B, Question 2: Fill in the blank. A closed box of circular shape is called a _____.

Answer: cylinder.

Review Exercise 12, Part B, Question 3: Fill in the blank. A cylinder has two circular ends, each of which is of the shape of a _____.

Answer: circle.

Review Exercise 12, Part B, Question 4: Fill in the blank. When the radius of a cylinder is one metre and the height is one metre, the volume will be _____.

Answer: printed line; the question is complete with those words.

Review Exercise 12, Part B, Question 5: Fill in the blank. The value of π up to two decimal places is _____.

Answer: 22/7).

Review Exercise 12, Part C, Question 1: Solve the following orally and mark (✓) on the correct answer. The circumference of a circle is _____ cm, when its diameter is 7 cm. (i) 22/7 (ii) 7/22 (iii) 7 (iv) 22

Answer: Correct answer: (iv) 22. Circumference = π ×

Review Exercise 12, Part C, Question 2: The area of a circle is _____ cm², when its radius is 1 cm. (i) 22/7 (ii) 7/22 (iii) 1/7 (iv) 1/22

Answer: Correct answer: (i) 22/7. Area = πr² = (22/7) ×

Review Exercise 12, Part C, Question 3: The area of the base circle of a cylinder is _____ cm². (i) 1/7 (ii) 7 (iii) 1/22 (iv) 22

Answer: not given, so the base area cannot be determined. Method: base area = πr². If the radius were 1 cm, the area would be 22/7 cm², which is not among the given options.

Review Exercise 12, Part C, Question 4: The curved surface area of a cylinder is _____ cm², when r = 7 cm and h = 1/22 cm. (i) 1/7 (ii) 2 (iii) 7 (iv) 22

Review Exercise 12, Part C, Question 5: The total surface area of a cylinder is _____ cm², when r = 1 cm and h = 1 cm. (i) 2π (ii) 3π (iii) 4π (iv) π

Review Exercise 12, Part C, Question 6: The volume of a cylinder is _____ cm³, when r = 1 cm and h = 1 cm. (i) 22/7 (ii) 7/22 (iii) 1/7 (iv) 1/22

Answer: Correct answer: (i) 22/7. Volume

Review Exercise 12, Part C, Question 7: The area of the two circular ends of a cylinder is _____ cm², when r = 1 cm. (i) 22 (ii) 44/7 (iii) 7 (iv) 1/7

Review Exercise 12, Part C, Question 8: The volume of a drum is _____ m³, when its height is 1 m and radius is 1 m. (i) 22 (ii) 7 (iii) 22/7 (iv) 1/7

Answer: Correct answer: (iii) 22/7. Volume = $\pi r^2 h$ = (22/7) ×

Question: Review Exercise 12, Part C, Question 9: The minute hand of a clock is 3.5 cm long. How much distance will its tip cover in one hour? (i) 22 cm (ii) 7 cm (iii) 11 cm (iv) 44 cm

Answer: Correct answer: (i) 22 cm. Distance = circumference = 2π × radius = 2 × (22/7) × 3.5 = 22 cm.

Question: Review Exercise 12, Part C, Question 10: If the radius of a circle is doubled, its area becomes: (i) double (ii) triple (iii) four times (iv) half.

Answer: Correct answer: (iii) four times. Area = πr^2; doubling the radius gives area = $\pi(2r)^2 = 4\pi r^2$, i.e. four times.

13. Data Handling

Question: Exercise 13.1, Question 1: The number of books sold daily in a bookshop during a week is given below: Monday 24, Tuesday 25, Wednesday 27, Thursday 28, Friday 22, Saturday 30, Sunday 21. Make a frequency distribution table (chart).

Answer: Method: Treat each day as a category, put tally marks against it and write the frequency. Table: Day | Tally marks | Frequency — Monday | | | 1; Tuesday | | | 1; Wednesday | | | 1; Thursday | | | 1; Friday | | | 1; Saturday | | | 1; Sunday | | | 1. Total frequency = 7. Total number of books sold = 24 + 25 + 27 + 28 + 22 + 30 + 21 = 177.

Question: Exercise 13.1, Question 2: The electricity bills paid by 12 electricity consumers are given below (Rs.): 5100, 7000, 3560, 6030, 4220, 6740, 4810, 5450, 7180, 5920, 6850, 5690. Make 5 class intervals of equal width and make a frequency distribution table (chart).

Answer: Method: Maximum = 7180, minimum = 3560. Range = 7180 − 3560 = 3620. Class width = 3620 ÷ 5 = 724 ≈ 725. Table (Class interval | Tally marks | Frequency): 3560–4284 | || | 2; 4285–5009 | | | 1; 5010–5734 | ||| | 3; 5735–6459 | || | 2; 6460–7184 | |||| | 4. Total frequency = 2 + 1 + 3 + 2 + 4 = 12.

Question: Exercise 13.1, Question 3: In an annual examination, 20 students of a school obtained the following marks out of 850: 551, 786, 678, 725, 788, 580, 720, 690, 750, 651, 599, 609, 719, 760, 625, 775, 646, 667, 753, 675. Keeping the class-interval width 10, make a frequency distribution table (chart).

Answer: Method: Starting from the minimum = 551, make intervals of width 10 and write each mark in the relevant interval with tally marks.

Table (Class interval | Frequency): 551–560 | 1; 561–570 | 0; 571–580 | 1; 581–590 | 0; 591–600 | 1; 601–610 | 1; 611–620 | 0; 621–630 | 1; 631–640 | 0; 641–650 | 1; 651–660 | 1; 661–670 | 1; 671–680 | 2; 681–690 | 1; 691–700 | 0; 701–710 | 0; 711–720 | 2; 721–730 | 1; 731–740 | 0; 741–750 | 1; 751–760 | 2; 761–770 | 0; 771–780 | 1; 781–790 | 2. Total frequency = 20. Note: with a class width of 10 there are 24 intervals, some of which are empty. If the class width were 100, the same data would fall into three intervals: 551–650 | 6; 651–750 | 9; 751–850 | 5.

Question: Exercise 13.1, Question 4: The following data is the number of people who came to visit a zoo: 134, 167, 145, 130, 155, 142, 130, 180, 162, 130, 120, 165, 170, 110, 200, 185, 132, 140, 110, 170, 100, 160, 133, 150, 125, 110, 145, 155, 160, 200. Keeping the class-interval width 10, make a frequency distribution table (chart).

Answer: Method: Starting from the minimum = 100, make intervals of width 10 and write each number in the relevant interval with tally marks. Table (Class interval | Tally marks | Frequency): 100–109 | | | 1; 110–119 | ||| | 3; 120–129 | || | 2; 130–139 | |||||| | 6; 140–149 | |||| | 4; 150–159 | ||| | 3; 160–169 | ||||| | 5; 170–179 | || | 2; 180–189 | || | 2; 190–199 | (none) | 0; 200–209 | || | 2. Total frequency = 30.

Question: Exercise 13.1, Question 5: The marks of 30 students in mathematics are given below: 45, 19, 49, 17, 37, 10, 25, 28, 50, 46, 43, 34, 25, 50, 48, 46, 42, 34, 33, 11, 28, 19, 30, 17, 25, 13, 11, 39, 41, 34. Make 5 class intervals of equal width and make a frequency distribution table (chart).

Answer: Method: Maximum = 50, minimum = 10. Range = 50 − 10 = 40. Class width = 40 ÷ 5 = 8. Table (Class interval | Tally marks | Frequency): 10–17 | |||||| | 6; 18–25 | ||||| | 5; 26–33 | |||| | 4; 34–41 | |||||| | 6; 42–50 | ||||||||| | 9. Total frequency = 6 + 5 + 4 + 6 + 9 = 30.

Question: Exercise 13.2, Question 1: In a class there are 80 students, of whom 60 passed and 20 failed. Draw a pie graph of this information.

Answer: Method: Total = 80. Angle of each part = (number in the part ÷ total) × 360°. Pass: (60/80) × 360° = 270°. Fail: (20/80) × 360° = 90°. Then draw a circle with a compass and make two sectors of 270° and 90°. Sum of angles = 270° + 90° = 360°.

Question: Exercise 13.2, Question 2: In a class there are 36 students. 16 students like red, 9 like blue, 7 like pink and 4 like white colour. Draw a pie graph of this information.

Answer: Method: Total = 36. Angle of each colour = (number ÷ 36) × 360°. Red: (16/36) × 360° = 160°. Blue: (9/36) × 360° = 90°. Pink: (7/36) × 360° = 70°. White: (4/36) × 360° = 40°. Then draw a circle and make four sectors of these angles. Sum of angles = 160° + 90° + 70° + 40° = 360°.

Question: Exercise 13.2, Question 3: At a fruit shop there are 12 apples, 24 bananas, 16 mangoes and 8 peaches. Draw a pie graph of this information.

Answer: Method: Total = 12 + 24 + 16 + 8 = 60. Angle of each fruit = (number ÷ 60) × 360°. Apple: (12/60) × 360° = 72°. Banana: (24/60) × 360° = 144°. Mango: (16/60) × 360° = 96°. Peach: (8/60) × 360° = 48°. Then draw a circle and make four sectors of these angles. Sum of angles = 72° + 144° + 96° + 48° = 360°.

Question: Exercise 13.2, Question 4: At a party there are 180 cold drinks, 124 sandwiches, 330 burgers and 86 samosas. Draw a pie graph of this information.

Answer: Method: Total = 180 + 124 + 330 + 86 = 720. Angle of each food = (number ÷ 720) × 360°. Cold drink: (180/720) × 360° = 90°. Sandwich: (124/720) × 360° = 62°. Burger: (330/720) × 360° = 165°. Samosa:

(86/720) × 360° = 43°. Then draw a circle and make four sectors of these angles. Sum of angles = 90° + 62° + 165° + 43° = 360°.

Question: Exercise 13.2, Question 5: The following data is the distance (in km) of 22 students from home to school: 90, 44, 55, 49, 28, 9, 92, 27, 18, 84, 50, 60, 79, 69, 24, 89, 63, 74, 35, 48, 39, 80. Make 6 class intervals of equal width, make a frequency distribution table and draw its pie graph.

Answer: Method: Maximum = 92, minimum = 9. Range = 92 − 9 = 83. Class width = 83 ÷ 6 = 13.83 ≈ 14. Table (Class interval | Frequency | Angle): 9–22 | 2 | (2/22) × 360° ≈ 32.7°; 23–36 | 4 | (4/22) × 360° ≈ 65.5°; 37–50 | 5 | (5/22) × 360° ≈ 81.8°; 51–64 | 3 | (3/22) × 360° ≈ 49.1°; 65–78 | 2 | (2/22) × 360° ≈ 32.7°; 79–92 | 6 | (6/22) × 360° ≈ 98.2°. Total frequency = 22, sum of angles ≈ 360°. Then draw a circle and make six sectors of these angles.

Question: Exercise 13.2, Question 6: In a pie graph the favourite foods of 180 students are shown: chicken 180°, fish and chips 90°, noodles 50°, pachiyon 40°. (i) Which is the most favourite food and how many students like it? (ii) Which is the least favourite food and how many students like it? (iii) How many students like noodles? (iv) Complete the table (chart) using the pie graph.

Answer: Method: number = (angle ÷ 360°) × 180. (i) The largest sector is chicken (180°): number = (180/360) × 180 = 90 students. (ii) The smallest sector is pachiyon (40°): number = (40/360) × 180 = 20 students. (iii) Noodles: (50/360) × 180 = 25 students. (iv) Completed table (Food | Number | Angle): chicken | 90 | 180°; noodles | 25 | 50°; pachiyon | 20 | 40°; fish and chips | 45 | 90°; total | 180 | 360°. Note: in the printed table the number against fish and chips is written as 40, whereas its angle in the graph is 90°, which gives a number of (90/360) × 180 = 45 — there is a contradiction between the two. Here the graph has been taken as

correct, because the question asks to use the graph.

Review Exercise 13, Question 1: Fill in the following blanks. (i) Collected information is called _____.

Answer: data.

Review Exercise 13, Question 1: (ii) The number of observations falling in one category is called the _____.

Answer: frequency.

Review Exercise 13, Question 1: (iii) A _____ is made to present data in an arranged form.

Answer: table (chart)

Review Exercise 13, Question 1: (iv) The smaller number of a class interval is called the _____.

Answer: lower limit.

Review Exercise 13, Question 1: (v) The table in which data is divided into class intervals and presented with frequencies is called a _____.

Answer: frequency table.

Review Exercise 13, Question 1: (vi) Dividing data into class intervals is called _____ the data.

Answer: organising

Review Exercise 13, Question 1: (vii) The marks put against each observation are called _____.

Answer: tally marks.

Review Exercise 13, Question 1: (viii) The larger number of a class interval is called the _____.

Answer: upper limit.

Review Exercise 13, Question 1: (ix) The difference between the upper and lower limits of a class interval is called the _____.

Answer: class interval width.

Review Exercise 13, Question 1: (x) The graph in which data is shown in the form of a circle is called a _____.

Answer: pie graph.

Question: Review Exercise 13, Question 2: Choose the correct answer. (i) The sum of the angles of all sectors in a pie graph is: (a) 90° (b) 180° (c) 270° (d) 360°.

Answer: Correct answer: (d) 360°. The sum of the angles of all sectors in a pie graph is 360°.

Question: Review Exercise 13, Question 2: (ii) In the class interval (10–50) the upper limit is: (a) 10 (b) 30 (c) 50 (d) 60.

Answer: Correct answer: (c) 50. In the class interval (10–50) the upper limit is 50.

Question: Review Exercise 13, Question 2: (iii) In the class interval (9–12) the lower limit is: (a) 9 (b) 10 (c) 11 (d) 12.

Answer: Correct answer: (a) 9. In the class interval (9–12) the lower limit is 9.

Question: Review Exercise 13, Question 2: (iv) The maximum number is 21 and the minimum number is 3. If 3 class intervals are to be made, the class interval width will be: (a) 3 (b) 6 (c) 9 (d) 18.

Answer: Correct answer: (b) 6. Range = 21 − 3 = 18. Class interval width = 18 ÷ 3 = 6.

Question: Review Exercise 13, Question 3: The ages of visitors who came to a park are given below: 25, 50, 49, 47, 26, 10, 2, 1, 15, 17, 18, 19, 27, 28, 30, 35, 17, 32, 31, 3, 4, 9, 10, 15, 12, 13, 17, 24, 20, 22, 24, 26, 30, 17, 35, 40, 36, 32, 31, 37. Keeping the class-interval width 10, make a frequency distribution table (chart).

Answer: Method: Starting from the minimum = 1, make intervals of width 10 and write each age in the relevant interval with tally marks. Table

(Class interval | Tally marks | Frequency): 1–10 | ||||||| | 7; 11–20 | |||||||||| | 11; 21–30 | ||||||||| | 10; 31–40 | ||||||||| | 9; 41–50 | ||| | 3. Total frequency = 7 + 11 + 10 + 9 + 3 = 40.